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PROVISIONS FOR ELDERLY AND HANDICAPPED PEDESTRIANS

Vol. 3. The Development and
Evaluation of Countermeasures

May 1980

Final Report



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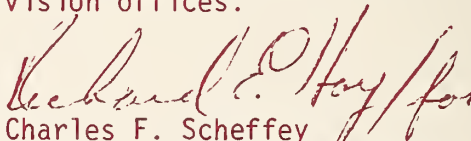
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FOREWORD

This report describes the investigation of the problems and hazards experienced by elderly and handicapped pedestrians. From these investigations, countermeasures were developed and field tested. Another product of this research is a manual for "Development of Priority Accessible Networks," (Implementation Package, FHWA-IP-80-8). This manual presents design information and methodology for creating a barrier free pedestrian facility.

Research in pedestrian safety is included in the Federally Coordinated Program of Highway Research and Development as Task 1 of Project 1E, "Safety of Pedestrians and Abutting Property Occupants." Mr. John C. Fegan is the Project Manager, Office of Research, and Mr. Richard Richter is the Implementation Manager, Office of Development.

Sufficient copies of this report are being distributed to provide a minimum of two copies to each regional office, one copy to each division office and one copy for each State highway agency. Direct distribution is being made to the division offices.


Charles F. Scheffey
Director, Office of Research
Federal Highway Administration

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introduction

The idea of a built environment that is free from barriers in the sense that it is accessible to the handicapped as well as the rest of society is now widely accepted. And the federal government and all of the states have legislation mandating some degree of accessibility.

The major thrust of changes directed at improving the environment from this point of view has been focussed on making buildings accessible. Accessible towns do not yet exist, nor even accessible districts.

The Federal Highway Administration of the U.S. Department of Transportation in 1974 initiated a research study, "Provisions for Elderly and Handicapped Pedestrians" to examine the pedestrian environment, and to make proposals for improving accessibility. The Pedestrian Research Laboratory of Georgia Institute of Technology has carried out this contract and the work is described in a series of technical reports.

The research included three major tasks, firstly, a survey to identify the nature of the hazards and problems experienced by the target groups; secondly, the development and field testing of countermeasures to resolve some of these problems; and finally, the production of a manual designed to assist cities wishing to improve accessibility. The report that follows describes the second of these tasks.

The report is in two parts: Part one discusses Laboratory studies and Part two deals with the testing of some design solutions in various United States cities.

The Laboratory studies were used to develop criteria for the design of short ramps, curb ramps and curb cuts. Secondly, as part of an exploration into the feasibility of providing orientation information and warning signals to severely visually handicapped people, various walkway surfacing materials were tested for detectability by subjects using long cane techniques. Finally, wheelchair manufacturers were surveyed in order to gather useable design information on the dimensional range of models generally available.

For the field studies in Part two, fifteen countermeasures to some of the most taxing accessibility problems were selected, and thirteen were constructed and tested in six cities. The report describes these countermeasures, the tests that were developed to evaluate them, and the conclusions that were drawn.

a study of short ramps

Background

Most cities have programs to install curb ramps and many have developed their own design guidelines for these ramps. However, design criteria founded on reliable and replicable research have not been available.

From available reports, curb ramps can create problems as well as resolving them. During interviews conducted in several cities (as discussed in Part 2 of Volume 2, "Hazards, Barriers, Problems and the Law") it was reported to us that:

- curb ramps are often so steep that they are not useable by some of the wheelchair users for whom they are intended, and even able bodied people avoid them,
- the side slopes, or flares, are usually too steep, and accidents attributable to them have been reported,
- a visually impaired person may fail to detect curb ramps, and may inadvertently stray into the street; and that a small step or lip at the bottom of ramps will obviate this danger,
- curb ramps are sometimes located (particularly on corners) so that there is insufficient space at the top for a wheelchair user to stop or turn, and insufficient space at the bottom for wheelchairs to maneuver without leaving the crosswalk.

Several additional problems and hazards resulting from unsatisfactory curb ramp design were reported, but most of these concerned the location of the ramps. And generally the problem was reported in cities that use a standard curb ramp design for all situations. As discussed in The Implementation Package, "A Manual for Developing a Priority Accessible Network", a single curb ramp design that is universally applicable does not exist. The design must be tailored to the environmental conditions, and the design will be affected by:

- the height of the curb,
- the slope of the sidewalk towards the gutter,
- the slope of the sidewalk if it is on a hill,
- the slope of the street towards the gutter,
- the width of the sidewalk,
- the location of the drainage catch basin,
- the slope and direction of the gutter,
- the location of existing street furniture—lighting poles, signs, mail boxes, etc.,
- the layout and geometry of the streets,

- the width and location of the crosswalks,
- the radius of the sidewalk at the corner,
- the materials of the surfaces of the sidewalk, the street, and the curb ramp,
- the finances available, etc.

Several types of curb ramps can be identified, each of which has advantages and disadvantages, and each of which is appropriate for a particular set of environmental conditions. These typical designs, and a checklist of environmental and design variables, are set out in The Implementation Package, "A Manual for Developing a Priority Accessible Network". This volume describes the field testing of these and other measures in several cities.

The study that follows is directed at producing answers to the following questions:

- how steep should curb ramps be so that they can be used by most handicapped people without too much difficulty? As space for curb ramps is usually restricted, what is sought is the reasonable maximum gradient. (Of course, if there are no space nor money constraints, then in terms of gradient, a more gentle slope will usually be preferred to a steeper slope.)
- should the gradient of curb ramps be reduced for longer ramps to make them easier to use? On the other hand, a short steep ramp may be the simplest solution for some locations. Are steeper ramps useable if they are short?
- how steep should the flares at the side of curb ramps be? The steeper the flare, the more hazardous and difficult they become for people to travel over. On the other hand, the gentler the slope, the more space they occupy and the more difficult it becomes to find sufficient space within which to locate the curb ramp.
- how much room at the top and bottom of curb ramps is necessary to permit wheelchair users to maneuver, take a run at the ramp, and stop in safety? The momentum of a wheelchair traveling down a ramp will carry it some distance beyond the bottom of the ramp. Even at the top of the ramp, the wheelchair will need space to slow down and stop.
- is it a fact that visually impaired people cannot detect ramps easily, or do they only have difficulty with some ramp gradients?
- is it a fact that a small step or lip at the bottom of curb ramps will make it easier for the visually impaired to detect curb ramps? Do these steps adversely affect the use of the ramp by people in wheelchairs, and others?

Previous Studies

Very little published research has been directed at these questions. Only the gradient of ramps has been evaluated methodically, and none of the published studies is directly applicable to curb ramps.

Dixon,¹ in a 1957 unpublished dissertation, examined ramps intended for use by people in wheelchairs. By means of a questionnaire, he surveyed the opinions of people who used a group of interior and exterior ramps on the campus of the University of Illinois. This survey, *inter alia*, provided subjective responses to the gradient of the ramps. Using these data, he constructed a laboratory experimental ramp which could be adjusted for gradients ranging from 1:8.6 to 1:5 (these are substantially steeper than the 1:12 maximum recommended by the present and the proposed revisions to the ANSI² standards). The 61 subjects used the ramp as a single 30 foot continuous run, or in two 15 foot sections separated by a landing. The subjects rated each of the gradients on a five point scale--very easy, easy, moderately hard, hard, very hard. And the testers, independently rated the degree of difficulty experienced by the subjects using the same scale. A maximum gradient of 1:6 was recommended, and a minimum of 1:8.6. An examination of Dixon's data, however, indicates that with a gradient of 1:6, 47% of the subjects rated the ramp as hard or very hard, and only 10% of the subjects rated the ramp as easy or very easy. For the 1:8.6 gradient, 5% of the subjects rated the ramp as hard or very hard, and 70% of the subjects rated the ramp as easy or very easy. And, it should be noted that the subjects were all university students. Dixon made no recommendations that related the gradients to the length of the ramp.

Walter³ reported, in a 1971 study, on the use of an adjustable laboratory ramp indoors. Two lengths were used--20 feet (6.1 m) at gradients from 1:8.6 to 1:16, and 10 feet (3.05 m) at gradients of 1:5.5 to 1:10. Subjects rated the ramps using a four point scale--easy, comfortable, difficult and impossible. Gradients that were rated by a high percentage of the subjects to be 'easy' or 'comfortable' were recommended--for self-propelled wheelchairs, a slope of 1:9 for the 10 foot (3.05 m) ramp, and 1:16 for the 20 foot (6.1 m) ramp; for assistant-propelled wheelchairs, a slope of 1:9 for the 10 foot (6.1 m) ramp, and 1:12 for the 20 foot (6.1 m) ramp.

Neither of these two studies, nor a more extensive study currently underway at Syracuse University, provides us with enough data to answer the criteria questions listed earlier. Therefore, the experiments which will now be described were initiated.

Methodology

1. The test site. To avoid the potential hazards and logistic problems inherent in using curb ramps in traffic in urban settings, an outdoor laboratory was constructed as a test site at Georgia Institute of Technology. The test site was designed to simulate closely, typical environmental conditions. Included in the test area were 12 curb ramps varying in gradient from 1:5.3 to 1:20. The ramps varied in length from 2 feet (.61 m) to 10 feet (3.05 m), and traversed curbs of 3" (7.6 cm), 6" (15 cm), and 9" (22.9 cm) (see Table 1), dimensions that are typical for curbs in cities. Fig. 1 illustrates the methods used in this report for calculating

ramp and flare gradients.

Two of these ramps (numbers 2 and 11) were built with a step, or lip, at the bottom; one of 1" (2.54 cm), the other $\frac{1}{2}$ " (1.27 cm) respectively. The test site also includes three flares for evaluation, and one mountable curb. Each ramp has an approach that has at least 6 feet (1.83 m) of level concrete at the top, and 8 feet (2.44 m) at the bottom, to provide for adequate starting and stopping distances. Ramps and sidewalk approaches are all 48" (121.92 cm) wide. All the ramps and approaches were constructed of concrete, and finished with a light broom finish to simulate typical city sidewalks. Guardrails were constructed at critical points to ensure the safety of the subjects.

The test site was designed to carry out both the curb ramp evaluations and the texture evaluations described in this report. A drawing of the layout is shown in Figure 2, and Figure 3 is a view of the site from the west. The site lies half in shade and half in full sunlight, during normal testing sessions, with the light and shade pattern reversing during the day. Testing was conducted from early August 1976 to early November 1976, with ambient temperatures ranging from 45 degrees Fahrenheit to more than 90 degrees Fahrenheit, and from full sunlight to cloudy.

Table 1 Dimensions of Ramps

Ramp No.	Gradient*	Length	Curb Height	Lip Height	*Flare
1	1:20	10' (3.05 m)	6" (15.24 cm)	1" (2.54 cm)	1:8 1:12
2	1:12	6' (1.83 m)	6" (15.24 cm)		
3	1:8	4' (1.22 m)	6" (15.24 cm)		
**4	1:12	6' (1.83 m)	6" (15.24 cm)		
5	1:13:3	10' (3.05 m)	9" (22.85 cm)		
6	1:10.67	8' (2.44 m)	9" (22.85 cm)		
7	1:5.3	4' (1.22 m)	9" (22.85 cm)		
8	1:8	6' (1.83 m)	9" (22.85 cm)		
***9	1:8	2' (.61 m)	3" (7.63 cm)	$\frac{1}{2}$ " (1.27 cm)	1:16
10	1:16	4' (1.22 m)	3" (7.62 cm)		
11	1:8	2' (.61 m)	3" (7.62 cm)		
12	1:16	8' (2.44 m)	6" (15.24 cm)		
13	mountable curb	1' (.305 m)	1 3/4" (4.45 cm)		
* See Figure 1 for illustration of these terms ** Identical to #2 except for the lip *** Identical to #11 except for the lip					

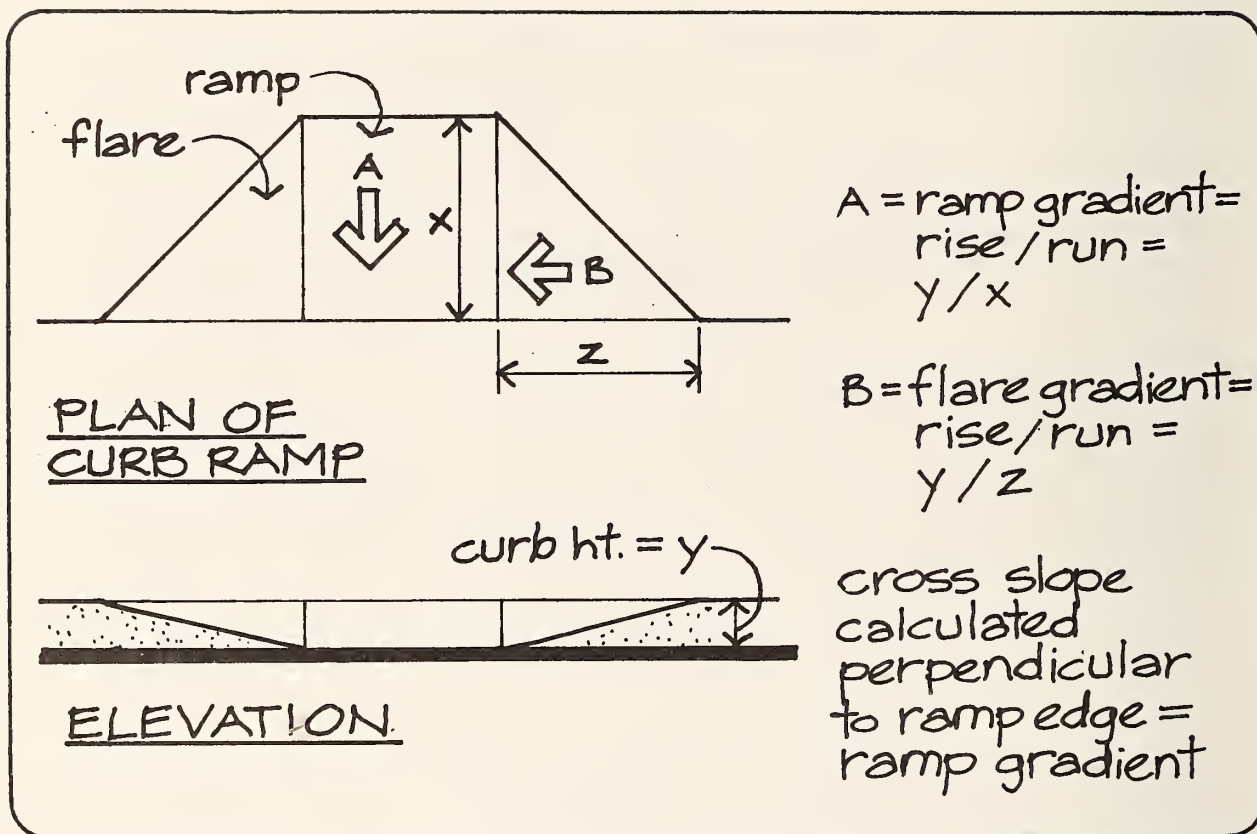


Figure 1 Ramp and Flare Gradient

2. The test subjects. Volunteers for testing were solicited through newspaper advertisements, radio spot messages, television and radio public service announcements, presentations to community groups, and through direct contact with agencies and associations that work with the elderly and handicapped. The sample tested included subjects with a variety of physical impairments; elderly people; and a group of able bodied subjects for control.

All of the subjects were experienced travellers. The subjects with physical impairments provided a range of disabilities representative of the kind of disabilities typical of handicapped pedestrians (as discussed in Volume 2, "Hazards, Barriers, Problems and the Law"). These were pedestrians:

- that use a self propelled wheelchair
- that use a powered wheelchair
- that use canes
- that use crutches

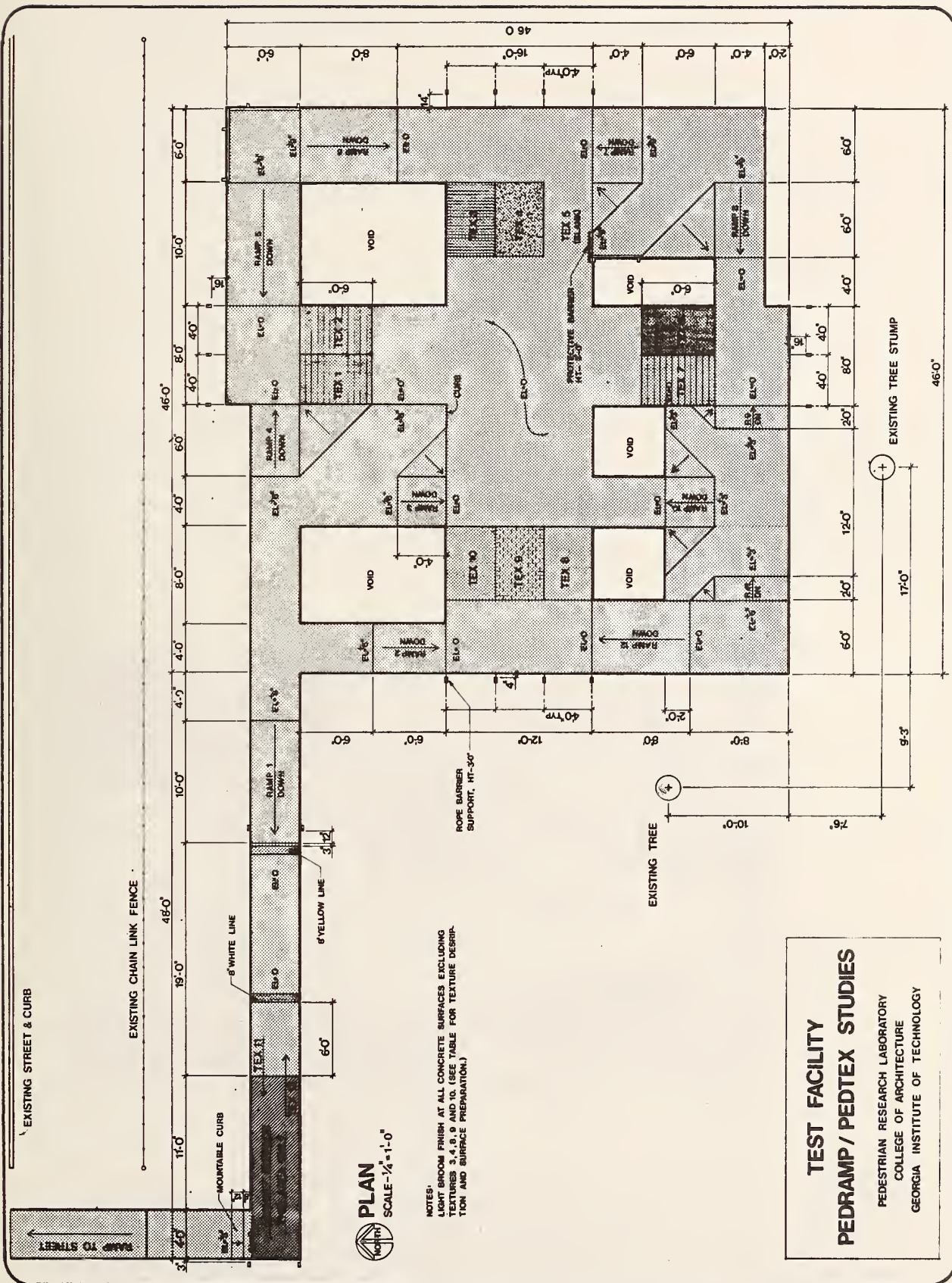


Figure 2 Layout of Ramp and Texture Test Site

- with prosthetic lower limbs
- that walk with difficulty but use no aids
- that are elderly
- that are able-bodied
- that are severely visually impaired

No attempt was made to ensure that the percentage of subjects in each of the categories reflects the percentages found in the general population. This was not considered to be necessary, since the aim of the study was to identify acceptable design criteria for ramps for the various categories of subjects. Furthermore, as will be apparent from Table 2 (test subjects by age, sex and diagnosis), within each of the disability categories there is represented a variety of diagnoses. Clearly then, with the total of 120 subjects, the sample is somewhat limited so the results must be considered as indicative, rather than conclusive.



Figure 3 View of the Test Site

3. Procedure. Upon arrival at the test site, each subject was briefed on the experiment and asked to sign a statement of informed consent (see Appendix A). The subjects were escorted, individually, around the course, to experience and evaluate each ramp in both directions. To avoid order effects, the sequence of ramps tested was randomized. Since the course was designed with more than one ramp sharing a landing, randomizing was by landing, rather than individual ramp. No ramp was negotiated immediately prior to being rated.

Table 2 Subject Description by Handicapped Group - Sex - Age

SUBJECT GROUP AND DISABILITY SUBDIVISIONS	NO. OF SUBJECTS	SEX	16-25	26-35	36-45	46-55	56-65	66-75	76-85	AGE NOT GIVEN
1. Manual Wheelchair	18	M 12 F 6	8 3	4 3						
a. Cerebral Palsy	5	M 3 F 2	3 2							
b. Quad.-Spinal Cord Injury	1	M 1 F 1	1							
c. Multi. Birth Defects	1	M 1 F 1	1							
d. Residual of Polio	4	M 3 F 1	1	2 4	1		1			
e. Paraplegic	6	M 3 F 3	1	4						
f. Spina-Bifida	1	M 1 F 1		1						
2. Electric Wheelchair	6	M 5 F 1	1	3	1					
a. M.S.	1	M 1 F 1				1				
b. Quad-Spinal Cord Injury	5	M 5 F 1	1	3	1	1				
3. Cane Users	8	M 5 F 3			1 1	3 1	1		1	
a. Unilateral Ampu- tee - Above knee Left side	1	M 1 F 1					1			
b. Hemiparises-Left side	1	M 1 F 1			1					
c. Paralysis from Stroke-Left side	1	M 1 F 1				1				
d. Paralysis-Partial Right side	1	M 1 F 1							1	
e. Residual of Polio	1	M 1 F 1				1				
f. Osteo-Arthritis	1	M 1 F 1				1				
g. M.S.	2	M 1 F 1			1	1				

Table 2 Subject Description by Handicapped Group - Sex - Age (Continued)

SUBJECT GROUP AND DISABILITY SUBDIVISIONS	NO. OF SUBJECTS	SEX	61-65	66-70	71-75	76-80	81-85	86-90	AGE NOT GIVEN
4. Elderly	15	M 3 F 12	2 1	1 3	3 3	3		2	
SUBJECT NO.		SEX	16-25	26-35	36-45	46-55	56-65	66-75	AGE NOT GIVEN
5. Prosthetic Limbs	7	M 7 F		5	2				
a. Unilateral above knee - amputee right side	1	M 1 F		1					
b. Unilateral above knee - amputee left side	1	M 1 F			1				
c. Unilateral below knee-amputee right side	1	M 1 F		1					
d. Bilateral below knee-amputee	2	M 2 F		1	1				
e. Osteo-Militus right heel - Unilateral b.k. amputee-left side	1	M 1 F		1					
f. Bilateral above knee - amputee	1	M 1 F		1					
6. Crutch Users	9	M 4 F 5	1 2	2 1	1 1	1	1		
a. Residual of Polio	6	M 2 F 4		1 1		1			
b. Hemophila	1	M 1 F	1	1	1		1		
c. Arthritis-Chon- dromalazia - arms & Legs	1	M 1 F	1						
d. On crutches temporarily	1	M 1 F	1						

Table 2 Subject Description by Handicapped Group - Sex - Age (Continued)

SUBJECT GROUP AND DISABILITY SUBDIVISIONS	NO. OF SUBJECTS	SEX	16-25	26-35	36-45	46-55	56-65	66-75	76-85	AGE NOT GIVEN
7. Visually handi- capped	31	M 13 F 18	5 4	2 4	4 3	1 3	1 2	2		
a. Totally blind	18	M 8 F 10	4 3	3 1	3 2	1 1	1 1	1		
b. High Partials	8	M 3 F 5	1 1	1 1	1		1 1	1		1
c. Low Partials	5	M 1 F 4		1 1	1	2				
8. Able-bodied	21	M 9 F 12	6 5	1 4	1	1 1	2			
9. Walking w/Diffi- culty	5	M 3 F 2	1		1	1	1 1			
a. Ataxia	1	M 1 F 1	1							
b. Stroke-one leg longer than the other	1	M 1 F 1					1			
c. Stroke-partial paralysis left side	1	M 1 F 1					1			
d. Mental Retarda- tion	1	M 1 F 1				1				
e. Polio (one leg)	1	M 1 F 1			1					

The subjects were assisted to the beginning of each ramp, and allowed to position themselves at what they considered to be an appropriate distance from the ramp. They were instructed to stop as soon as possible after negotiating the ramp. Starting and stopping distances were recorded for subjects using wheelchairs. If the ramp was considered by the subject to be too difficult, they could refuse to try a ramp, or they could break off the test at any time. After negotiating a ramp, the subject rated the ramp for ease of use, using a four-point Likert scale. The investigator also rated the effect of the ramp on the subject's performance, using an identical scale (the scales are described in the section on Scoring).

Various other evaluative measures were considered and rejected before this subjective rating system was chosen. Measurement of work and effort using heart rate, oxygen consumption, etc., were found to be inadequate descriptors of the physical and emotional stress demanded by the ramps. Therefore, success or failure, and comparative difficulty were chosen as the most appropriate measures, with additional corroboration provided by the ratings of the observers.

Three curb ramp flares were also tested by each subject. Each subject was asked to travel over each flare and rate it for its effect on stability and balance, using the four-point Likert scale. The investigator also rated the effect of the flare on the subject's performance, using the same four-point Likert scale.

Finally, each subject was asked to negotiate a mountable curb.

4. Scoring.

- a. Gradient. Subjects rated each ramp gradient, both after ascending and after descending. A subject was asked how difficult it was to ascend or descend the ramp, using the following scale:

1. not at all difficult
2. a little difficult
3. difficult
4. very difficult

The tester used the same scale to develop an independent rating.

- b. Stopping distance. Stopping distance is defined as the distance from the tip of the toe of a wheelchair user to the top or bottom of the ramp. Stopping distances were recorded only for subjects using wheelchairs. Index-marks were placed every six inches (15.24 cm) along the upper and lower approaches to each ramp. The tester recorded starting and stopping distances in terms of the number of six-inch (15.24 cm) increments, to the nearest index mark.

- c. Flares. Subjects rated the ramp flares in a manner similar to that used for ramps. After negotiating the flare, subjects were asked how it affected their comfort and stability, using the following scale:

1. not at all
2. a little
3. moderately
4. very much

The tester noted the subject's stability and balance while negotiating the cross slope and gave a rating using the following scale:

1. no noticeable affect
2. noticeable affect
3. a marked loss of control
4. complete loss of control

- d. Half curbs. A half curb is defined as that portion of a curb that must be mounted to reach the side flares. It is that portion of the curb that slopes down from the full curb height to merge into the curb ramp. Normally, one would not choose to mount the curbs at that point. However, the purpose of the test was to examine what happens if elderly and handicapped people, by intention or inadvertence, ascend the curb at that point.

After negotiating each half curb, each subject rated its effect on comfort and stability, using the same four-point scale used for flares. The tester noted the subject's stability on the half curb, using the same scale:

1. no noticeable affect
2. a noticeable but minor gait change
3. a marked loss of balance
4. a complete loss of balance

- e. Data collection format. The informed consent form, and a general information form are given in Appendix A, and the ramp evaluation coding forms are presented in Appendix B.

Results, Conclusions and Recommendations

1. Ramp gradient. The responses of the subjects to the ramp gradients are presented in the following sixteen tables--eight tables for ratings of difficulty in ascent, and

eight for descent. The eight tables reflect the responses of the eight respondent categories that evaluated the ramps.

In the tables showing the results, six pieces of information are given.

The subject rating of the ramp is the mean of their ratings (on the 1 to 4 scale) of the ramp; similarly, the tester rating is the mean. In the discussion on the results that follows, a ramp is considered to be acceptable if about 80% of the subjects rated the ramp as 1 or 2; that is, 'not at all difficult' or 'a little difficult', rather than 'difficult' or 'very difficult.' A ramp will always present some difficulty, but if the ramp is 'difficult' or 'very difficult,' it may be unwise, or even hazardous, for handicapped pedestrians, when they must use the ramp to enter or leave a street close to moving traffic. Similarly, for the ramp to be acceptable to a particular group, it should not be 'difficult' or 'very difficult' for a large percentage of the group.

a. Manual Wheelchair Users.

From Table 3, the curb ramps to the right of the heavy line are acceptable for ascent:

Ramps #1, 4, 5, 9, 10, 11, 12 and 13 are acceptable

Ramps #2, 3, 6, 7, 8 are unacceptable

It is worth noting that ramp #2 (which is otherwise identical to ramp #4) was rendered unacceptable by the addition of a 1" (2.54 cm) lip. The addition of a $\frac{1}{2}$ " (1.27 cm) lip to ramp #11 (otherwise identical to #9) makes it more difficult, but it remains acceptable.

From Table 4, for descent, several more curb ramps (numbers 3, 6, 8) are acceptable. But ramp #2, with the 1" (2.54 cm) lip, remains unacceptable.

b. Electric Wheelchairs.

From Table 5, for ascent:

Ramps #1, 4, 5, 6, 9, 10, 11, 12, 13, are acceptable

Ramp 11, with the $\frac{1}{2}$ " (1.27 cm) lip is barely acceptable; the ramp with the 1" (2.54 cm) lip was impossible for two of the six subjects.

From Table 6, for descent:

Ramps #1, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, are acceptable

Ramps #2, 7 are unacceptable

All of the ramps, except the steepest, are acceptable, but the ramp with the 1" (2.54 cm) lip remains unacceptable.

c. Cane Users

From Table 7 for ascent:

Ramps #1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13 are acceptable

Ramps #7, 8, are unacceptable

From Table 8 for descent:

Ramps #1, 2, 4, 5, 6, 9, 10, 11, 12, 13 are acceptable

Ramps #3, 7, 8, are unacceptable

d. Crutch Users

From Table 9, for ascent:

Ramps #1, 2, 4, 5, 6, 9, 10, 11, 12, 13, are acceptable

Ramps #3, 7, 8, are unacceptable

From Table 10, for descent:

Ramps #1, 2, 4, 5, 6, 9, 10, 11, 12, 13, are acceptable

Ramps #3, 7, 8, are unacceptable

e. Prosthetic Lower Limbs

From Table 11, for ascent:

Ramps #1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, are acceptable

Ramp #7 is not acceptable

From Table 12, for descent:

Ramps #1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, are acceptable

Ramps #7 is not acceptable

f. Elderly

From Table 13, for ascent:

Ramps #1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, are acceptable

Ramp #7 is unacceptable

From Table 14, for descent:

Ramps #1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, are acceptable

None are unacceptable

g. People Walking with Difficulties but Without Prostheses

From Table 15, for ascent:

Ramps #1, 3, 4, 5, 6, 9, 10, 12, 13 are acceptable

Ramps #2, 7, 8, 11, are unacceptable

From Table 16, for descent:

Ramps #1, 4, 5, 6, 10, 12, 13 are acceptable

Ramps #2, 3, 7, 8, 9, 11, are unacceptable

h. Able-Bodied

From Table 17 for ascent:

Ramps #1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, are acceptable

None are unacceptable

From Table 18, for descent:

Ramps #1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, are acceptable

Ramps #7, 8, are unacceptable

i. Summary of Results of Response to Ramp Gradients

From Table 19, for ascent:

Ramps #1, 4, 5, 9, 10, 12, 13, are acceptable

Ramps #2, 3, 6, 7, 8, 11, are unacceptable

From Table 19, for descent:

Ramps #1, 4, 5, 6, 10, 12, 13 are acceptable

Ramps #2, 3, 7, 8, 9, 11 are unacceptable

Table 3 Gradient Ratings by Manual Wheelchair Users: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth .61m 1.32 95% (1.74)				4' lgth 1.22m 1.06 100% (1.33)	
6" 15.2 cm		4' lgth 1.22m 2.44 44% (2.61)		6' lgth 1.83m 1.26 100% (1.84)		8' lgth 2.44m 1.39 84% (1.34)	10' lgth 3.05m 1.32 95% (1.93)
9" 22.9 cm	4' lgth 1.22m 2.88 32% (3.06) 2 unable	6' lgth 1.83m 2.68 47% (3.00)	8' lgth 2.44m 2.16 58% (2.42)		10' lgth 3.05m 1.63 84% (2.05)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth .61m 1.58 89% (1.84)	
6" 15.2 cm		6' lgth 1.83m 2.00 68% (2.58)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth .305m (16) yes (0) no Completion 100%

Only ramps to the right of the heavy line are considered acceptable

KEY

Ramp Number	
Ramp Length	4' lgth 2
Subject Rating	1.22m
Percent of Subjects rating a 1 or 2	2.88 32%
Tester Rating	(3.06)
No. of Subjects unable to complete test	2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 4 Gradient Ratings by Manual Wheelchair Users: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth .61m 1.21 100% (1.53)				4' lgth 1.22m 1.00 100% (1.17)	
6" 15.2 cm		4' lgth 1.22m 1.53 84% (1.74)		6' lgth 1.83m 1.26 100% (1.47)		8' lgth 2.44m 1.05 100% (1.32)	10' lgth 3.05m 1.05 100% (1.21)
9" 22.9 cm	4' lgth 1.22m 2.55 42% (2.83) 1 unable	6' lgth 1.83m 1.56 89% (2.11)	8' lgth 2.44m 1.37 89% (1.79)		10' lgth 3.05m 1.16 100% (1.21)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth .61m 1.42 95% (1.53)	
6" 15.2 cm		6' lgth 1.83m 1.58 74% (1.79)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth .305m (17) yes (0) no Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	
Ramp Length	4' lgth 1.22m
Subject Rating	2.88
Percent of Subjects rating a 1 or 2	32% (3.06)
Tester Rating	2
No. of Subjects unable to complete test	2 unable

RATINGS:

- Not at all difficult
- A little difficult
- Difficult
- Very difficult

Table 5 Gradient Ratings by Electric Wheelchair Users: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth .61m 1.17 100% (1.00)				4' lgth 1.22m 1.00 100% (1.00)	
6" 15.2 cm		4' lgth 1.22m 2.0 67% (1.83)		6' lgth 1.83m 1.33 100% (1.17)		8' lgth 2.44m 1.00 100% (1.17)	10' lgth 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 1.22m 3.0 40% (2.8) 1 unable	6' lgth 1.83m 2.17 67% (2.00)	8' lgth 2.44m 1.5 100% (1.83)		10' lgth 3.05m 1.00 100% (1.17)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth .61m 2.0 83% (1.8)	
6" 15.2 cm		6' lgth 1.83m 2.25 75% (1.75) 2 unable

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth .305m (5) yes (0) no Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	4' lgth 1.22m 2
Ramp Length	1.22m
Subject Rating	2.88 32%
Percent of Subjects rating a 1 or 2	(3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 6 Gradient Ratings by Electric Wheelchair Users: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3' 7.62 cm		2' lgth 9 .61m 1.00 100% (1.17)				4' lgth 10 1.22m 1.00 100% (1.00)	
6" 15.2 cm		4' lgth 3 1.22m 1.33 100% (1.67)		6' lgth 4 1.83m 1.17 100% (1.00)		8' lgth 12 2.44m 1.00 100% (1.00)	10' lgth 1 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 7 1.22m 2.67 33% (2.67)	6' lgth 8 1.83m 1.33 100% (1.83)	8' lgth 6 2.44m 1.00 100% (1.17)		10' lgth 5 3.05m 1.17 100% (1.17)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.6 cm	2' lgth 11 .61m 2.00 100% (1.67)		1/2" lip 2.54 cm
6" 15.2 cm		6' lgth 2 1.83m 2.5 50% (2.5)	1" lip 1.27 cm

MOUNTABLE CURB

1' lgth 13
.305m
(5) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number —
Ramp Length — 4' lgth 2
Subject Rating — 1.22m
Percent of Subjects rating a 1 or 2 — 2.88 32%
Tester Rating — (3.06)
No. of Subjects unable to complete test — 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 7 Gradient Ratings by Cane Users: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.14 100% (1.14) 1 refused				4' lgth 10 1.22m 1.00 100% (1.00)	
6" 15.2 cm		4' lgth 3 1.22m 1.75 86% (2.00)		6' lgth 4 1.83m 1.29 86% (1.29)		8' lgth 12 2.44m 1.14 100% (1.29)	10' lgth 1 3.05m 1.13 100% (1.38)
9" 22.9 cm	4' lgth 7 1.22m 2.25 75% (2.50)	6' lgth 8 1.83m 2.00 63% (2.25)	8' lgth 6 2.44m 1.38 86% (1.50)		10' lgth 5 3.05m 1.13 100% (1.50)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 1.00 100% (1.29) 1 refused	
6" 15.2 cm		6' lgth 2 1.83m 1.57 86% (1.38)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13
.305m
(8) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number —
Ramp Length — 4' lgth 2
Subject Rating — 1.22m
Percent of Subjects rating a 1 or 2 — 2.88 32%
Tester Rating — (3.06)
No. of Subjects unable to complete test — 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 8 Gradient Ratings by Cane Users: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.50 100% (1.63)				4' lgth 10 1.22m 1.13 100% (1.29)	
6" 15.2 cm		4' lgth 3 1.22m 2.13 75% (2.25)		6' lgth 4 1.83m 1.38 88% (1.50)		8' lgth 12 2.44m 1.13 100% (1.25)	10' lgth 1 3.05m 1.13 100% (1.13)
9" 22.9 cm	4' lgth 7 1.22m 2.75 38% (3.00)	6' lgth 8 1.83m 2.38 50% (2.38)	8' lgth 6 2.44m 1.75 88% (1.88)		10' lgth 5 3.05m 1.25 100% (1.38)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.6 cm	2' lgth 11 .61m 1.38 100% (1.28)		1/2" lip 1.27cm
6" 15.2 cm		6' lgth 2 1.83m 1.25 100% (1.63)	1" lip 2.54cm

MOUNTABLE CURB

1' lgth 13 .305m (8) yes (0) no Completion 100 %
--

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	4' lgth 2
Ramp Length	1.22m
Subject Rating	2.88 32%
Percent of Subjects rating a 1 or 2	(3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 9 Gradient Ratings by Crutch Users: Ascent

curb height	G R A D I E N T S						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.22 100% (1.33)				4' lgth 10 1.22m 1.22 100% (1.33)	
6" 15.2 cm		4' lgth 3 1.22m 2.33 56% (2.44)		6' lgth 4 1.83m 1.44 89% (1.44)		8' lgth 12 2.44m 1.00 100% (1.22)	10' lgth 1 3.05m 1.11 100% (1.22)
9" 22.9 cm	4' lgth 7 1.22m 2.89 22% (3.00)	6' lgth 8 1.83m 2.44 44% (2.67)	8' lgth 6 2.44m 1.44 100% (1.78)		10' lgth 5 3.05m 1.44 100% (1.56)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 1.33 100% (1.44)	
6" 15.2 cm		6' lgth 2 1.83m 1.78 89% (1.89)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13
.305m
(8) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number —
Ramp Length — 4' lgth 2
Subject Rating — 1.22m
Percent of Subjects rating a 1 or 2 — 2.88 32%
Tester Rating — (3.06)
No. of Subjects unable to complete test — 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 10 Gradient Ratings by Crutch Users: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.44 89% (1.44)				4' lgth 10 1.22m 1.11 100% (1.00)	
6" 15.2 cm		4' lgth 3 1.22m 2.33 44% (2.11)		6' lgth 4 1.83m 1.44 100% (1.44)		8' lgth 12 2.44m 1.00 100% (1.11)	10' lgth 1 3.05m 1.22 100% (1.22)
9" 22.9 cm	4' lgth 7 1.22m 2.67 33% (2.89)	6' lgth 8 1.83m 2.00 67% (1.67)	8' lgth 6 2.44m 1.67 89% (1.56)		10' lgth 5 3.05m 1.33 89% (1.56)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 1.33 100% (1.22)	
6" 15.2 cm		6' lgth 2 1.83m 1.67 89% (1.67)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13 .305m
(8) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number ———
 Ramp Length ——— 4' lgth 2
 Subject Rating ——— 1.22m
 Percent of Subjects rating a 1 or 2 ——— 2.88 32%
 Tester Rating ——— (3.06)
 No. of Subjects unable to complete test ——— 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 11 Gradient Ratings by those with Prosthetic Lower Limbs: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.00 100% (1.00)				4' lgth 10 1.22m 1.00 100% (1.00)	
6" 15.2 cm		4' lgth 3 1.22m 1.22 86% (1.43)		6' lgth 4 1.83m 1.29 100% (1.14)		8' lgth 12 2.44m 1.00 100% (1.00)	10' lgth 1 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 7 1.22m 2.14 71% (1.86)	6' lgth 8 1.83m 2.00 86% (1.71)	8' lgth 6 2.44m 1.57 100% (1.29)		10' lgth 5 3.05m 1.14 100% (1.29)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.6 cm	2' lgth 11 .61m 1.57 86% (1.29)		1/2" lip 1.27cm
6" 15.2 cm		6' lgth 2 1.83m 1.43 86% (1.00)	1" lip 2.54cm

MOUNTABLE CURB

1' lgth 13 .305m
(7) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	
Ramp Length	4' lgth 2 1.22m
Subject Rating	2.88 32%
Percent of Subjects rating a 1 or 2	(3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 12 Gradient Ratings by those with Prosthetic Lower Limbs: Descent

curb height	G R A D I E N T S						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.29 100% (1.14)				4' lgth 10 1.22m 1.00 100% (1.14)	
6" 15.2 cm		4' lgth 3 1.22m 1.86 86% (1.43)		6' lgth 4 1.83m 1.29 100% (1.14)		8' lgth 12 2.44m 1.00 100% (1.00)	10' lgth 1 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 7 1.22m 2.00 71% (1.71) 1 unable	6' lgth 8 1.83m 2.00 86% (1.83)	8' lgth 6 2.44m 1.57 100% (1.57)		10' lgth 5 3.05m 1.29 100% (1.43)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 1.71 86% (1.29)	
6" 15.2 cm		6' lgth 2 1.83m 1.43 86% (1.29)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13
.305m
(7) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number —
Ramp Length — 4' lgth 2
Subject Rating — 1.22m
Percent of Subjects rating a 1 or 2 — 2.88, 32%
Tester Rating — (3.06)
No. of Subjects unable to complete test — 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 13 Gradient Ratings by Elderly Persons: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth .61m 1.33 100% (1.20)	9			4' lgth 1.22m 1.00 100% (1.00)	10
6" 15.2 cm		4' lgth 1.22m 1.40 93% (1.47)	3	6' lgth 1.83m 1.27 100% (1.27)	4	8' lgth 2.44m 1.13 100% (1.07)	12 10' lgth 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 1.22m 1.64 93% (2.07) 1 unable	7 6' lgth 1.83m 1.50 93% (1.43)	8 8' lgth 2.44m 1.33 93% (1.40)	6	10' lgth 3.05m 1.27 93% (1.40)	5	

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth .61m 1.53 87% (1.40)	11
6" 15.2 cm		6' lgth 1.83m 1.47 93% (1.26)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth .305m (11) yes (0) no Completion 100%	13
--	----

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	4' lgth 1.22m 2.88 32% (3.06) 2 unable
Ramp Length	2
Subject Rating	
Percent of Subjects rating a 1 or 2	
Tester Rating	
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 14 Gradient Ratings by Elderly Persons: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.6 cm		2' lgth 9 .61m 1.27 100% (1.27)				4' lgth 10 1.22m 1.07 100% (1.00)	
6" 15.2 cm		4' lgth 3 1.22m 1.53 87% (1.53)		6' lgth 4 1.83m 1.20 100% (1.33)		8' lgth 12 2.44m 1.00 100% (1.07)	10' lgth 1 3.05m 1.00 100% (1.00)
9" 22.9 cm	4' lgth 7 1.22m 1.86 85% (2.28)	6' lgth 8 1.83m 1.43 93% (1.50)	8' lgth 6 2.44m 1.20 100% (1.53)		10' lgth 5 3.05m 1.27 100% (1.47)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.62 cm	2' lgth 11 .61m 1.47 93% (1.27)		1/2" lip 1.27cm
6" 15.2 cm		6' lgth 2 1.83m 1.60 87% (1.60)	1" lip 2.54cm

MOUNTABLE CURB

1' lgth 13 .305m (12) yes (0) no Completion 100%
--

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	
Ramp Length	4' lgth 2
Subject Rating	1.22m
Percent of Subjects rating a 1 or 2	2.88 32% (3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 15 Gradient Ratings by People Walking with Difficulty but without
Prostheses: Ascent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.5 100% (1.8)				4' lgth 10 1.22m 1.25 100% (1.6)	
6" 15.2 cm		4' lgth 3 1.22m 1.2 100% (2.2)		6' lgth 4 1.83m 1.6 75% (2.0)		8' lgth 12 2.44m 1.25 100% (1.6)	10' lgth 1 3.05m 1.25 100% (1.8)
9" 22.9 cm	4' lgth 7 1.22m 2.2 50% (2.4)	6' lgth 8 1.83m 2.25 75% (2.4)	8' lgth 6 2.44m 1.75 100% (2.2)		10' lgth 5 3.05m 2.25 50% (2.2)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 2.0 75% (2.0)	
6" 15.2 cm		6' lgth 2 1.83m 2.0 50% (2.6)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13 .305m
(2) yes
(0) no
Completion 100%

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number —
Ramp Length — 4' lgth 2
Subject Rating — 1.22m
Percent of Subjects rating a 1 or 2 — 2.88 32%
Tester Rating — (3.06)
No. of Subjects unable to complete test — 2 unable

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 16 Gradient Ratings by People Walking with Difficulty but without Prostheses: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth .61m 2.25 75% (2.4)				4' lgth 1.22m 1.25 100% (1.4)	
6" 15.2 cm		4' lgth 1.22m 2.25 50% (2.2)		6' lgth 1.83m 1.25 100% (2.0)		8' lgth 2.44m 1.25 100% (1.8)	10' lgth 3.05m 1.75 75% (1.8)
9" 22.9 cm	4' lgth 1.22m 2.0 75% (2.6)	6' lgth 1.83m 1.5 100% (2.2)	8' lgth 2.44m 1.75 100% (2.2)		10' lgth 3.05m 1.5 100% (1.8)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.6 cm	2' lgth .61m 2.0 75% (2.0)		1/2" lip 1.27cm
6" 15.2 cm		6' lgth 1.83m 2.0 75% (2.4)	1" lip 2.54cm

MOUNTABLE CURB

1' lgth .305m (3) yes (0) no Completion 100%	13
--	----

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	2
Ramp Length	4' lgth 1.22m
Subject Rating	2.88 32%
Percent of Subjects rating a 1 or 2	(3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

- Not at all difficult
- A little difficult
- Difficult
- Very difficult

Table 17 Gradient Ratings from Able Bodied People: Ascent

curb height	G R A D I E N T S						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.6 cm		2' lgth 9 .61m 1.19 100% (1.14)				4' lgth 10 1.22m 1.14 100% (1.09)	
6" 15.2 cm		4' lgth 3 1.22m 1.29 90% (1.14)		6' lgth 4 1.83m 1.14 100% (1.00)		8' lgth 12 2.44m 1.05 100% (1.00)	10' lgth 1 3.05m 1.10 100% (1.00)
9" 22.9 cm	4' lgth 7 1.22m 1.85 80% (1.50)	6' lgth 8 1.83m 1.57 90% (1.24)	8' lgth 6 2.44m 1.24 95% (1.14)		10' lgth 5 3.05m 1.24 95% (1.00)		

RAMPS WITH LIPS

c.h.	1:8	1:12	
3" 7.62 cm	2' lgth 11 .61m 1.48 90% (1.33)		1/2" lip 1.27cm
6" 15.2 cm		6' lgth 2 1.83m 1.48 90% (1.14)	1" lip 2.54cm

MOUNTABLE CURB

1' lgth 13 .305m (14) yes (0) no Completion 100%
--

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	
Ramp Length	4' lgth 2
Subject Rating	1.22m
Percent of Subjects rating a 1 or 2	2.88 32% (3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 18 Gradient Ratings from Able Bodied People: Descent

curb height	GRADIENTS						
	1:5.3	1:8	1:10.67	1:12	1:13.3	1:16	1:20
3" 7.62 cm		2' lgth 9 .61m 1.33 100% (1.14)				4' lgth 10 1.22m 1.10 100% (1.05)	
6" 15.2 cm		4' lgth 3 1.22m 1.71 90% (1.48)		6' lgth 4 1.83m 1.10 100% (1.00)		8' lgth 12 2.44m 1.05 100% (1.00)	10' lgth 1 3.05m 1.00 100% (1.09)
9" 22.9 cm	4' lgth 7 1.22m 2.19 71% (1.48)	6' lgth 8 1.83m 1.67 76% (1.29)	8' lgth 6 2.44m 1.48 90% (1.29)		10' lgth 5 3.05m 1.24 95% (1.10)		

RAMPS WITH LIPS

c.h.	1:8	1:12
3" 7.6 cm	2' lgth 11 .61m 1.33 86% (1.19)	
6" 15.2 cm		6' lgth 2 1.83m 1.48 86% (1.38)

1/2" lip
1.27cm

1" lip
2.54cm

MOUNTABLE CURB

1' lgth 13 .305m (14) yes (0) no Completion 100%
--

Only ramps to the right of the heavy line are considered acceptable.

KEY

Ramp Number	
Ramp Length	4' lgth 2
Subject Rating	1.22m
Percent of Subjects rating a 1 or 2	2.88 32% (3.06)
Tester Rating	2 unable
No. of Subjects unable to complete test	

RATINGS:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

Table 19 Summary of Responses to Ramp Gradients

SUBJECT GROUP	<u>Ascent</u> <u>Descent</u>	RAMP NUMBER												
		1	2	3	4	5	6	7	8	9	10	11	12	13
a) Manual Wheelchairs	ascent	O	X	X	O	O	X	X	X	O	O	O	O	O
	descent	O	X	O	O	O	O	X	O	O	O	O	O	O
b) Electric Wheelchairs	ascent	O	X	X	O	O	O	X	X	O	O	O	O	O
	descent	O	X	O	O	O	O	X	O	O	O	O	O	O
c) Cane Users	ascent	O	O	O	O	O	O	X	X	O	O	O	O	O
	descent	O	O	X	O	O	O	X	X	O	O	O	O	O
d) Crutch Users	ascent	O	O	X	O	O	O	X	X	O	O	O	O	O
	descent	O	O	X	O	O	O	X	X	O	O	O	O	O
e) Prosthetic Lower Limbs	ascent	O	O	O	O	O	O	X	O	O	O	O	O	O
	descent	O	O	O	O	O	O	X	O	O	O	O	O	O
f) Elderly	ascent	O	O	O	O	O	O	X	O	O	O	O	O	O
	descent	O	O	O	O	O	O	O	O	O	O	O	O	O
g) Walking Difficulties	ascent	O	X	O	O	O	O	X	X	O	O	X	O	O
	descent	O	X	X	O	O	O	X	X	X	O	X	O	O
h) Able Bodied	ascent	O	O	O	O	O	O	O	O	O	O	O	O	O
	descent	O	O	O	O	O	O	X	X	O	O	O	O	O
O = acceptable		X = not acceptable												

Conclusions

From the tests it is clear that ramps 2, 3, 7, 8 are unacceptable to several handicapped groups and should not be used for curb ramps. Ramps 6 and 9 were found to be unacceptable to one group, either for ascent or for descent, respectively.

Secondly, it is clear that steeper ramps are acceptable if they are short.

Thirdly, the ramp with a one inch (2.54 cm) lip (ramp 2) is unequivocally unacceptable; ramp 11 with the $\frac{1}{2}$ " (1.27 cm) lip is unacceptable to those with walking difficulties but marginally acceptable to other groups.

Recommendations

1. Based on these findings, Table 20 shows recommendations for maximum acceptable gradients for three different curb heights.

2. Lips on the bottom of ramps cannot be recommended; and lips greater than $\frac{1}{2}$ " (1.27 cm) must not be used.

Table 20 Ramp Gradient Recommendations

Curb Heights Not Exceeding	Maximum Gradient*				
	Gradients Steeper Than 1:8	1:8	1:10	1:12	1:16
3" (7.62 cm)	X	O	O	O	O
6" (15.2 cm)	X	X	O	O	O
9" (22.9 cm)	X	X	X	O	O
X Not Recommended O Recommended					
*Whenever possible slopes less than the maximum should be employed.					

2. Ramp detection by the visually impaired. The purpose of this component of the study was to examine the accuracy of the statement reported earlier, "that a visually impaired person may fail to detect curb ramps, and may inadvertently stray into the street; and that a small step or lip at the bottom of ramps will obviate this danger."

In this test, six ramps were used; two had lips, four had no lips and one of the latter ramps was a mountable curb.

For the first test, the visually impaired subjects were asked to imagine that they were crossing a street. They were told that the pavement in front of them might continue as a level surface, or they might encounter a curb ramp or a curb. They were then asked to walk forward and to stop immediately if they detected a curb or a ramp. If they detected a ramp they were then asked to indicate the exact bottom and then the top of the ramp.

For the second test, they were asked to imagine that they were on a sidewalk approaching a street crossing that they wished to use. They were told that before the street crossing they might encounter a wheelchair ramp, or the sidewalk might continue on the level. Again, they were asked to walk forward and to stop immediately if they detected a ramp; if they did, then they were asked to indicate the exact top and then the bottom of the ramp.

As discussed earlier, the legally blind cannot be considered as a homogeneous group; their abilities to see vary considerably. So for the purposes of this study, they were treated within the three broad categories that are most frequently used—totally blind, high partials, and low partials.

Table 21 shows, for the three groups, the results of the ramp detection tests. Almost all of the subjects detected all the ramps in ascent. Only the 1:20 ramp was missed, and by one low partial and one high partial. In descent, the results

are similar. Three percent of the subjects failed to detect the 1:8 ramp, the 1:16 ramp, and the 1:20 ramp; five percent of the subjects failed to detect the mountable curb.

Table 21 Ramp Detection by Visually Impaired

Ramp No.	Grade	Lip	Direction	Totally Blind			Low Partial			High Partial		
				Yes	No	%Fail	Yes	No	%Fail	Yes	No	%Fail
11	1:8	½" (1.27 cm)	up	18	0	0	5	0	0	8	0	0
2	1:12	1" (2.54 cm)	up	18	0	0	5	0	0	8	0	0
4	1:12	—	up	18	0	0	5	0	0	8	0	0
10	1:16	—	up	18	0	0	5	0	0	8	0	0
1	1:20	—	up	17	0	0	4	1	20.0	7	1	12.5
13	Mountable Curb		up	13	0	0	3	0	0	6	0	0
	Length 12"											
	(30.48 cm)											
	Curb Ht. 1 ¾"											
	(4.45 cm)											
11	1:8	½" (1.27 cm)	dn	18	0	0	4	1	20.0	7	0	0
2	1:12	1" (2.54 cm)	dn	18	0	0	5	0	0	8	0	0
4	1:12	—	dn	18	0	0	5	0	0	8	0	0
10	1:16	—	dn	17	1	5.5	5	0	0	8	0	0
1	1:20	—	dn	17	1	5.5	4	0	0	8	0	0
13	Mountable Curb		dn	13	0	0	2	1	33.3	6	0	0
	Length 12"											
	(30.48 cm)											
	Curb Ht. 1 ¾"											
	(4.45 cm)											

Table 22 presents the stopping distances--the distance from the top or bottom of the ramp to the point within which 90% of the visually impaired people stopped after detecting the ramp. The variation in these distances perhaps gives some measure of the difficulty in detection. Ramps that descended from the walkway were generally detected later than those ascending. And the smaller gradients were generally (but not uniformly) detected later than the steeper grades.

The ramp with the 1" (2.54 cm) lip was detected more quickly in ascent than a ramp of the same gradients without the lip. However, 8% of the totally blind subjects tripped at the lips and a further 17% commented adversely on the lips.

Table 23 shows the ability of the subjects to indicate the top and bottom of the ramps. This measure is important. If curb ramps must be used by the visually impaired, then to avoid vehicular traffic they must know where it is safe to stand and wait, preparatory to crossing the street; and likewise to know with some confidence, that they have reached the curb ramp on the other side of the street.

Table 22 Stopping Distances: Ramp Detection

Ramp No.	Grade	Lip	Travel	Stopping distance 90th percentile	Travel Direct	Stopping distance 90th percentile
11	1:8	$\frac{1}{2}$ " (1.27 cm)	up	6" (15.24 cm)	dn	18" (45.72 cm)
2	1:12	1" (2.54 cm)	up	6" (15.24 cm)	dn	30" (76.2 cm)
4	1:12	-	up	24" (60.95 cm)	dn	30" (76.2 cm)
10	1:16	-	up	18" (45.72 cm)	dn	36" (91.44 cm)
1	1:20	-	up	36" (91.44 cm)	dn	42" (or more) (106.68 cm)

From Table 23, it can be seen that in ascent, those with partial vision had more difficulty in locating the ends of the ramps than those with no vision; and not unexpectedly, the low partial had more difficulty than the high partials. Overall, the percentage of detection failures were low, ranging from 3% to 10%, and the failures were greatest for the two smallest gradients, 1:16 and 1:20. The presence of the lips did not alter the overall detection rate.

In descent, the detection failures were slightly greater, ranging from 5% to 11%. Once again, it was the low partials who had the greatest difficulty in detecting the ends of the ramps. And again, the two ramps with the lowest gradients seemed to be the most difficult (however, the steepest ramp also presented some difficulties).

Conclusions

The overwhelming majority of the visually impaired subjects had little difficulty in detecting the ramps and accurately determining the top and the bottom of the ramps. These are, of course, laboratory results, and these results may not be matched on curb ramps in cities. However, it does indicate that the expressed fear of inadvertently straying into the street on a curb ramp may not be a substantial risk in reality for those who are exercising normal caution. But the results do suggest that the gentler ramps, 1:16 and 1:20, are somewhat more difficult to detect.

Providing a lip on the ramps at the bottom was shown to have more disadvantages than advantages, in that the lip becomes a trip hazard, and does not materially improve the ramp detectability.

Table 23 Landing Detection by Visually Impaired

Ramp No.	Grade	Lip	Dir	Totally Blind				Low Partial				High Partial				Total % Failed						
				Bottom		Top		Bottom		Top		Bottom		Top								
				Yes	No Fail	Yes	No Fail	Yes	No Fail	Yes	No Fail	Yes	No Fail	Yes	No Fail							
11	1:8	½" (1.27 cm)	up	18	0	0.0	16	2	11.1	4	1	20.0	5	0	0.0	7	0	0.0	6	1	14.3	7
2	1:12	1" (2.54 cm)	up	18	0	0.0	18	0	0.0	5	0	0.0	4	1	20.0	8	0	0.0	7	1	12.5	3
4	1:12	-	up	18	0	0.0	18	0	0.0	5	0	0.0	4	1	20.0	8	0	0.0	7	1	12.5	3
10	1:16	-	up	16	2	11.1	18	0	0.0	4	1	20.0	3	2	40.0	8	0	0.0	8	0	0.0	8
1	1:20	-	up	17	0	0.0	17	0	0.0	2	3	60.0	4	1	20.0	7	1	12.5	7	1	12.5	9.6
11	1:8	½" (1.27 cm)	dn	17	1	5.5	18	0	0.0	3	2	40.0	3	2	40.0	7	0	0.0	6	1	14.2	10
2	1:12	1" (2.54 cm)	dn	18	0	0.0	16	2	11.1	5	0	0.0	5	0	0.0	8	0	0.0	7	1	12.5	5
4	1:12	-	dn	17	1	5.5	18	0	0.0	5	0	0.0	4	1	20.0	8	0	0.0	7	1	12.5	5
10	1:16	-	dn	15	3	16.7	16	2	11.1	4	1	20.0	4	1	20.0	8	0	0.0	7	0	0.0	11
1	1:20	-	dn	14	3	17.6	16	1	5.9	3	1	25.0	4	0	0.0	8	0	0.0	8	0	0.0	9

Recommendations

1. Wherever possible, curb ramps should not be located in the direct path that visually impaired people will take.
2. Where the junction between the curb ramp and the street or walkway is not obvious in that the gradient differential is 1:20 (or less steep), then the ramp should be finished with a surface that is detectable by the visually impaired (it would be advantageous to treat all walkway ramps in this way).
3. A lip, or step, at the bottom of the ramp should not be provided.
3. Ramp flares. The responses of the subjects to crossing over the flare of a ramp and across the ramp itself are shown in Table 24. As might be expected, people who use electric and manual wheelchairs are most adversely affected. Both of these two groups are naturally very concerned about the dangers of overbalancing; and both groups are concerned about the unusually high and unbalanced power demands that are needed to cross the combination of slopes that are presented when traversing a curb ramp in this way. Other groups share these concerns and the flare at ramp 3 was even rejected by able-bodied subjects.

Conclusions

Only the flare (1:16) on ramp 10 was acceptable to all groups. The flare (1:12) on ramp 4 was barely acceptable to most groups with the exception of those in wheelchairs. The flare (1:8) on ramp 3 was unacceptable to most groups.

Recommendations

- a. Ramp flares that will normally be used by wheelchairs, in order to enter or exit from curb ramps, should not be steeper than 1:16 in the direction at right angles to the slope of the curb ramp.
- b. Ramp flares that lie within the paths that pedestrians must use, should not be steeper than 1:12 in the direction at right angles to the slope of the curb ramp.
- c. Ramp flares that do not lie within the path that pedestrians must use should not be steeper than 1:10 in the direction at right angles to the slope of the curb ramp.
4. Half Curbs. The responses of the subjects to ascending two of the curb ramps at the flares are shown in Table 25. It should be noted that subjects with prosthetic limbs, canes and in wheelchairs were not tested.

Those on crutches and those with walking difficulties are affected by the flares most.

Table 24 Ramp Flare Ratings—All Categories of Subjects

Flare No.	Totally Blind	Low Partial	High Partial	Prosthetic Limbs	Canes	Crutches	Elderly	Manual Wheelchair	Electric Wheelchair	Able Bodied	Walking with Difficulty
10cs 1:16	Subject (Avg.) Rating	1.00	1.00	1.00	1.38	1.22	1.20	1.37	1.83	1.00	1.50
	Percent Sub.Rate 1 or 2	100%	100%	100%	100%	88%	100%	95%	83%	100%	100%
	Evaluators (Avg.) Rating	1.13	1.00	1.14	1.63	1.33	1.15	1.79	1.83	1.10	1.60
4cs 1:12	Subject (Avg.) Rating	1.56	1.33	1.57	2.00	1.89	1.15	2.37	2.83	1.48	2.25
	Percent Sub.Rate 1 or 2	94%	100%	100%	78%	78%	100%	53%	33%	100%	75%
	Evaluators (Avg.) Rating	1.50	1.83	1.43	2.55	1.78	1.38	2.58	2.83	1.33	2.4
3cs 1:18	Subject (Avg.) Rating	1.63	1.80	2.29	2.33	2.44	1.31	2.78	3.6	1.86	2.5
	Percent Sub Rate 1 or 2	84%	100%	71%	56%	56%	100%	39%	0%	76%	50%
	Evaluators (Avg.) Rating	1.32	1.60	2.00	2.67	2.22	1.77	2.83	3.4	1.62	2.2
Circled are ratings that show the flares that are unacceptable.											

Table 25 Half-Curb Ratings—Categories of Subjects

Half Curb No.	Totally Blind	Low Partial	High Partial	Crutches	Elderly	Able Bodied	Walking with Difficulty
3HC 1:8	1.58	1.40	1.86	3.33	1.36	1.33	3.00
	89%	100%	71%	33%	91%	93%	50%
	1.53	1.80	2.00	2.67	1.73	1.40	3.30
4HC 1:12	1.41	1.60	1.33	3.00	1.27	1.33	2.00
	94%	100%	100%	33%	100%	93%	100%
	1.35	1.80	2.00	3.00	1.64	1.47	3.00

Conclusions

The 1:8 flare was unacceptable to several of the group; the 1:12 flare was acceptable to all the groups except for those on crutches.

Recommendations

In general, flares steeper than 1:12 should be avoided.

5. Stopping Distances--Manual Wheelchairs.

To establish wheelchair stopping distances, the subjects were asked to stop as soon as possible after negotiating the ramps. Then the distance from the tip of the toe of the wheelchair user to the top (or bottom) of the ramp was measured. Figures 4 and 5 show the frequency distributions for the stopping distances--based on six-inch (15.24 cm) intervals.

It should be noted that because a minimum of 72" (182.88 cm) of overrun distance was available at the top or bottom of all the ramps, the test subjects were not forced to stop in the shortest possible distance.

Conclusions

From the distributions, it can be seen that a very small number of subjects were able to stop immediately after traversing a ramp. The mean distance travelled before stopping, after ascending the ramps, was 54.4" (138.2 cm); and 52.9" (134.37 cm) after descending the ramps. Of course, the length of the chair is included in these figures, and from the discussion on wheelchair dimensions which is discussed later in this report, the mean length of wheelchairs was found to be 44.7" (113.5 cm). So the actual distance travelled was 9.9" (25.146 cm) in ascent and 8.2" (20.8 cm) in descent.

It is interesting to note that the wheelchairs stopped slightly sooner in descent than in ascent. This can probably be attributed to the different physical motions required from the users of manual wheelchairs. In descent, the manual wheelchair user will usually apply the appropriate braking action continuously, and therefore, will be able to control the stopping distance with greater precision.

In ascent, the manual wheelchair user is using considerable effort in a series of actions to propel the chair up the ramp; stopping the chair will then be a new action. Furthermore, in ascent, the wheelchair user will frequently 'take a run' at the ramp to gather momentum.

Recommendations

To allow sufficient space at the top or bottom of ramps for most wheelchair users, a distance of 66" (167.6 cm) from the top or bottom of the ramps should be provided. Sixty-six inches (167.6 cm) will be adequate for 85% of descending wheelchairs, and 91% of ascending wheelchairs.

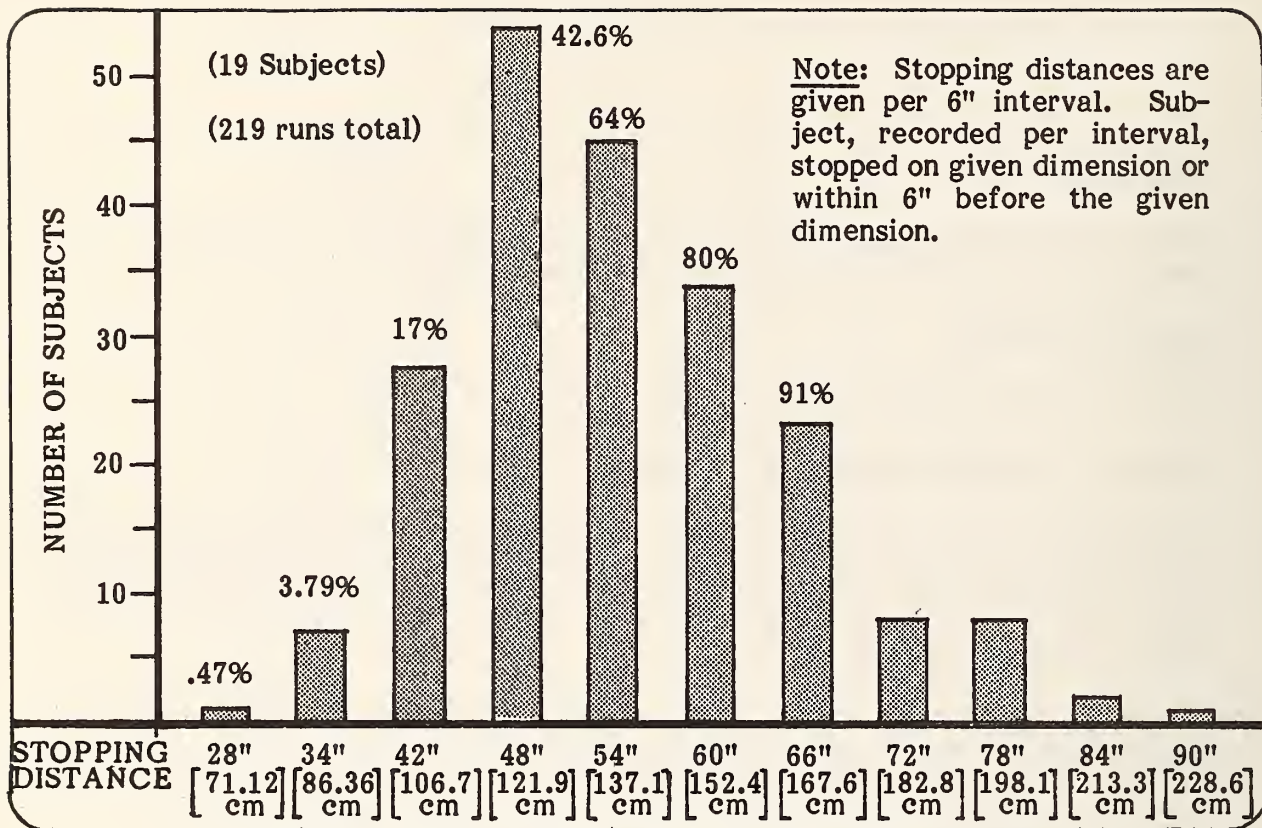


Figure 4 Stopping Distances—Manual Wheelchairs: Ascent

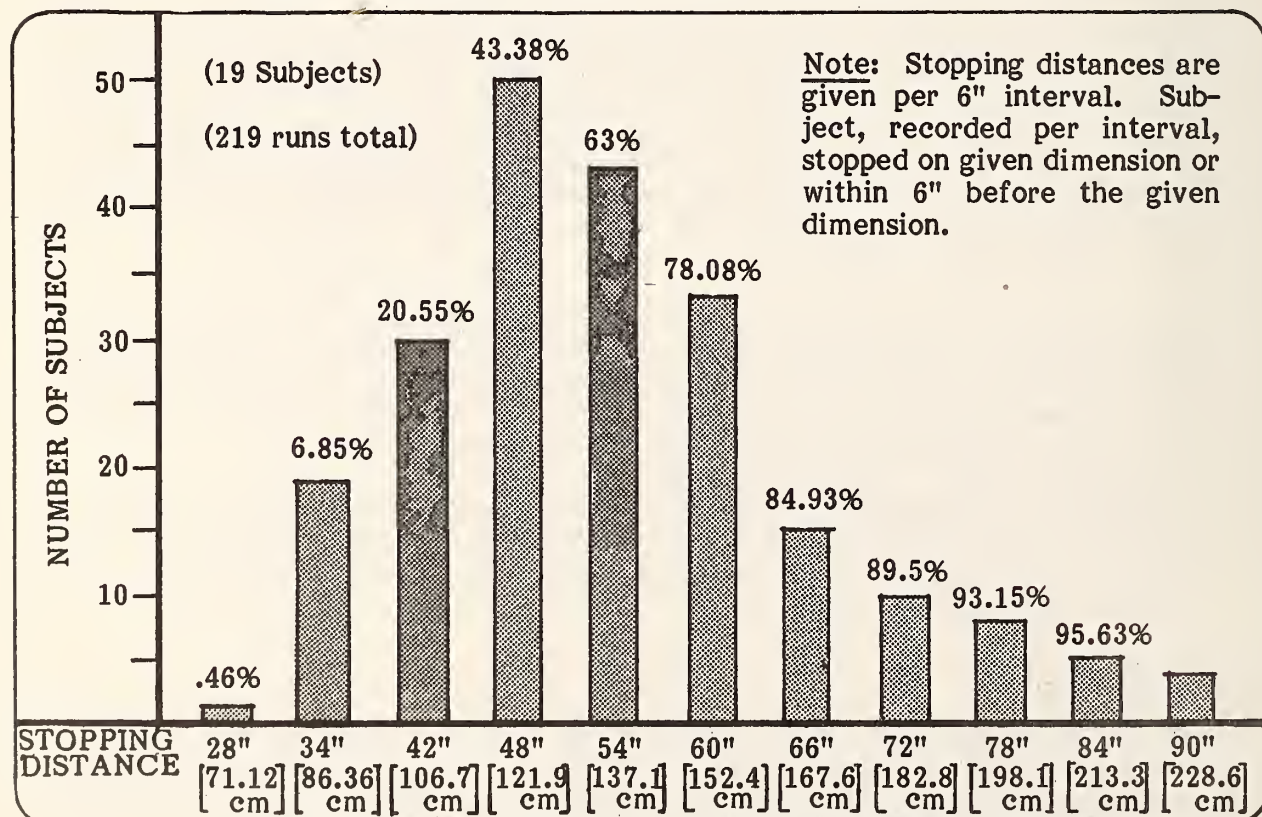


Figure 5 Stopping Distances—Manual Wheelchairs: Descent

a study of tactile surfaces

Background

Visually impaired pedestrians differ greatly in their abilities to travel. Some have confidence, skill and experience that is comparable to similar sighted people; they walk quickly, avoid obstacles, cross streets with safety, and find their way to their destination with apparent ease. At the other end of the mobility scale are those who for various reasons cannot be considered to be competent travellers; they may lack confidence, experience, training, intelligence, cognitive and perceptual skills, or emotional maturity. The travel abilities of most visually impaired pedestrians lie somewhere between that of extraordinary competence to marginal competence.

Visually impaired pedestrians also vary considerably in their ability to see. Even those that are legally blind may be able to see sufficiently well to perceive most of the visual travel cues used by sighted pedestrians. Other partially sighted people may be barely able to distinguish light from darkness.

Regardless of their travel abilities there are some environmental conditions that tax the skills of visually impaired pedestrians to the limit. These are situations where there is too little pertinent information to be sensed, or where the information is confusing, or where they are unfamiliar with the terrain and there are unforeseen hazards. Some examples of these were reported to us by visually impaired people during interviews conducted in five cities and described in Volume 2 of this series, "Hazards, Barriers, Problems and the Law".

For instance, some of the respondents stated that they had considerable difficulty in finding their way safely at:

- street crossings at oblique angle road intersections
- parking lots (particularly at shopping centers)
- large plazas and paved open spaces
- airports and other transit terminals
- sidewalks that have extensive street furnishings--mail boxes, planters, vending machines, parking meters, lamp and sign poles, etc.
- locations where the sidewalk level blends imperceptibly into that of the street; where there are curb ramps and curb cuts or flush junctions
- stairways within walkways.

There are well tested ways of dealing with all these conditions adequately, if not always conveniently, and these techniques are included in the vocabulary of the visually impaired mobility training programs. However, these problems are still perceived by many visually impaired people as being the object of apprehension or, at least, uncertainty. And this has led to the suggestion that the cognitive mapping process that is one of the techniques used by most visually impaired people can be reinforced by the

provision of active environmental cues in addition to the passive cues that are routinely used. For example, in negotiating a sidewalk that has an abundance of street furniture, a visually impaired person using the long cane technique will navigate in part by remembering the location of the furniture. However, these furniture locations are not chosen with the visually impaired in mind. If on the other hand, environmental clues that are usable by the visually impaired could be intentionally provided, it is hypothesized, then some of the difficulties and dangers facing these pedestrians would be reduced.

Several types of active environmental cues have been suggested and some of these are discussed in Part 2 of this Volume. One of these suggested cues is in the use of detectable, special walkway surfaces to inform or alert visually impaired pedestrians. These surfaces would be selected for their ease of detection, and located where they would provide visually impaired people with information such as the presence of a hazard (such as some steps in the immediate vicinity); or as a directional pathway across an oblique angled crosswalk, for example, or even as a guideway strip through transit terminals, college campuses, etc.

It is implicit in the proposal that the nature, extent and location of the tactile walkway materials must be known to the visually impaired group for whom it is intended, and the material must be reasonably easy to detect. The device is not useful or useable by all visually impaired people. It is most likely to be useful to those who use long canes, and those with partial vision, and the specific applications would have to be learned in the same way that other 'landmarks' have to be remembered. Some of these applications are discussed in Part 2 of this Volume.

Previous Studies

There has been little research to determine empirically the degree to which different variations in sidewalk surface texture can be detected by the visually handicapped. Certain types of textured surfaces have been used as guides for visually impaired pedestrians. Herms, Elias and Robins⁴ described the successful use of a tactile guide strip to help visually impaired pedestrians cross confusing and complex intersections in San Diego. The guide strip consisted of a two-inch (5.08 cm) wide strip of epoxy cement, with a $\frac{1}{4}$ " (.635 cm) thickness of pea gravel embedded in it. Visually impaired pedestrians touch the strip with their canes as they follow it across the intersection.

Special patterned pavement tiles (easily identified through soft-soled footwear) have been used in Japan to guide visually impaired pedestrians to touch posts at intersections. The touch posts provide information about the intersection through vibrations and audible signals. Raised studs have also been used as guides at the edges of crossings.

Several cities in the U.S. are using curb ramps that have been specially constructed using a textured surface. The texture is intended to inform visually impaired pedestrians of the presence of the ramp.

While various types of tactile guides have been used to aid visually handicapped

pedestrians, as described above, there appears to have been no attempt to determine experimentally what types of outdoor surface textures are most easily detected by the visually impaired. The purpose of this study is to test responses to a range of materials. Of special interest are textured surfaces which can be fitted to existing sidewalks and walkways.

Methodology

1. Course design. The test area illustrated in Figure 2 included eleven textured surfaces and a blank, or dummy section, consisting of a section of normal sidewalk and used as a control. The textured surfaces were all of a type that could be used (in conjunction with) concrete sidewalks. The types of textured surfaces used were as shown in Table 26.

Table 26 List of Textures for Evaluation

Texture No.	Texture Size	Texture Type	Description
1	Width 48" (121.9 cm) Length 72" (182.8 cm)	Thermoplastic Solid	Prismo Universal Corp. <u>Plastix S.D.</u> with no added reflective particals. 6" x 48" (15.24 x 121.9 cm) strips applied to entire 48" x 72" (121.9 x 182.9 cm) broom finish concrete to form a continuous surface. Epoxy Bond.
2	Width 48" (121.9 cm) Length 72" (182.8 cm)	Thermoplastic Strip	Prismo Universal Corp. <u>Plastix S.D.</u> with no added reflective particals. One 6" x 48" (15.24 x 121.9 cm) strip applied per each 12" (30.48 cm) of texture area length. Epoxy Bond.
3	Width 48" (121.9 cm) Length 72" (182.8 cm)	Ruled Concrete	$\frac{1}{2}$ " (1.27 cm) Depth cut every 2" (5.08 cm) of texture area length. (Grooves are at right angles to direction of subject movement).
4	Width 48" (121.9 cm) Length 72" (182.8 cm)	Exposed Aggregate	Coarse Pea Gravel set in concrete.

Table 26 List of Textures for Evaluation (continued)

Texture No.	Texture Size	Texture Type	Description
5	Width 48" (121.9 cm)	No Texture	Broom Finish Concrete.
6	Width 48" (121.9 cm) Length 72" (182.8 cm)	Kushionkote	Applied Chemical <u>Kushionkote</u> . Neutral color. <u>Exterior</u> Grade Application—3 coats applied with Squeegee onto broom finished concrete.
7	Width 48" (121.9 cm) Length 72" (182.8 cm)	Pliant Polymer	3M Pliant Polymer <u>Pave-</u> <u>ment Marking Film</u> .06" (.1524 cm) thickness. 6" x 48" (15.24 x 121.9 cm) strips applied to entire 48" x 72" (121.9 x 182.9 cm) broom finish concrete to form a continuous surface.
8	Width 48" (121.9 cm) Length 72" (182.8 cm)	Burlap Finish	Texture applied to con- crete using burlap during setting period.
9	Width 48" (121.9 cm) Length 72" (182.8 cm)	Paving Brick	<u>Paving</u> bricks set in con- crete. The 3/8" (.9525 cm) joints are recessed 1/8" (.3175 cm).
10	Width 48" (121.9 cm) Length 72" (182.8 cm)	Knicked (Washboard) Concrete	Texture applied to con- crete using edge of 2 x 4 wood member.
11	Width 48" (121.9 cm) Length 15' (4.575 m)	Concrete to Asphalt	Asphalt.
12	Width 48" (121.9 cm) Length 72" (182.8 cm)	Asphalt to Concrete	Broom finish concrete.

Each textured surface was 4 feet (1.22 m) wide and 6 feet (1.83 m) long (except #11, which was 15 feet (4.575 m) long) and was preceded by an approach path of light broom finish concrete. A six-foot (2.1 m) 'overrun' path of light broom finish concrete followed each textured surface. A dummy control section or path without a textured surface was included to test for errors of anticipation by subjects.

2. Subject selection. All the visually handicapped subjects were legally blind. The sample (see Table 2) included subjects with a range of visual defects. The categories of subjects used in texture detection were:

1. totally blind
2. high partial (subjects with a high degree of partial vision)
3. low partial (subjects with a low degree of partial vision)

A test where a subject attempted to detect white and yellow lines was used as a screening device to confirm the category to which the subject belonged.

3. Procedure. On arrival at the test site each subject was briefed on the experiment and asked to sign a statement of informed consent to the tests. The subject was then escorted around the course and shown the various textured surfaces. This allowed the subjects to familiarize themselves with the textured surfaces that they later attempted to detect.

After examining the textured surfaces, each subject was given the opportunity to detect the surfaces one at a time in random order. On each trial the subject was escorted to a starting point on the approach path. Starting points were randomly varied from 15 feet (4.575 m) to 22 feet (6.71 m) in front of the textured surface to prevent learning the texture location. The subjects were told to proceed along the approach path and stop immediately upon detecting any surface change. To prevent visually handicapped subjects from veering, guide ropes were placed at each side of the test sidewalks for their entire length (see Figure 6). The distance traveled over the textured surface was recorded as well as the time taken to travel the last 10 feet (3.05 m) of the approach path. If the textured surface was detected, the subject rated the ease of detectability on a four-point Likert scale (described in the section on scoring). In separate tests, the textured surfaces were also evaluated for their effects on the comfort, stability and balance of the handicapped subjects used in the ramp study discussed earlier. Subjects traveled over each textured surface and rated it using a four-point Likert scale.

The protocol used for testing subjects is presented in Appendix A.

4. Scoring. The subjects rated each textured surface using a four-point Likert scale. After detecting a surface texture, a subject was asked to rate how easy or difficult it was to detect the surface, using the following scale:

1. very easy

2. easy
3. somewhat difficult
4. very difficult

If a textured surface was successfully detected, the distance traveled across the surface before the subject indicated detection by stopping, was recorded. As an indicator, marks were placed every six inches (15.24 cm) along the edges of the surface. The tester recorded the distance in terms of the numbers of six-inch (15.24 cm) increments to the nearest index mark.

The physically handicapped but sighted subjects evaluated the various textured surfaces using a four-point Likert scale. After crossing over each surface, each subject was asked to rate how the surface affected comfort, stability and balance, using the following scale:

1. not at all
2. a little
3. moderately
4. very much



Figure 6 View of Texture Evaluation Test Site

Table 27 shows the success of the three groups of subjects in detecting the test materials. As one might suspect, the high partials had the greatest success: all of the materials were detected by 85% of this group of subjects. The low partials

Table 27 Material Detection Rankings

Number	Material	Totally Blind	Low Partial	High Partial
1	Thermoplastic Solid	B	A	A
2	Thermoplastic Strips	A	A	A
3	Ruled Concrete	B	B	A
4	Exposed Aggregate	A	A	A
5	No Texture	-	-	-
6	Kushionkote	A	B	A
7	Pliant Polymer	A	A	A
8	Burlap Bagged	C	C	A
9	Paving Brick	B	A	A
10	Knicked Concrete	A	B	A
11	Concrete to Asphalt	C	A	A
12	Asphalt to Concrete	C	A	A

A represents successful detection by 85% or more of the subjects.

B represents successful detection by more than 69% but less than 85% of the subjects.

C represents successful detection by 69% or less of the subjects.

were somewhat less successful: seven of the twelve materials were detected by 86% of the subjects, and another three materials were detected by 70% of the subjects. And the totally blind group were substantially less successful. Five of the twelve materials were detected by 86% of the subjects, and three additional materials were detected by 70% of the subjects.

From Table 27, three materials were successfully detected by 86% of the subjects from all three groups and therefore have an A-rating.

- thermoplastic strips
- exposed aggregate
- pliant polymer

And four other materials were slightly less satisfactory and received two A-ratings and one B-rating.

- thermoplastic (solid)
- Kushionkote
- paving brick
- knicked concrete

Table 28 shows ease of detection ratings for the seven materials found (from Table 27) to be detectable by most of the subjects. The figures in Table 28 show the mean of the subjects' ratings of each material in terms of ease of detection based on the four-point Likert scale (very easy, easy, difficult, very difficult). It can be seen from the table that, as might be anticipated, the totally blind found the materials to be more difficult to detect than those with partial vision.

If we eliminate from the table all those materials receiving a mean ranking of more than two, then the mean rankings of the remaining materials falls into the easy or very easy to detect range. Three materials satisfy this condition:

- thermoplastic strips
- exposed aggregate
- Kushionkote

And the exposed aggregate was clearly the easiest material to detect.

Table 29 indicates the distances (from the beginning of the materials) travelled by the subjects before they decided with confidence that they had detected a change of material and should stop walking. The table only lists the results for the seven

Table 28 Ease of Texture Detection—Mean Ratings

Number	Material	Totally Blind	Low Partial	High Partial
1	Thermoplastic Solid	2.15	1.20	1.14
2	Thermoplastic Strips	2.00	1.20	1.00
4	Exposed Aggregate	1.28	1.80	1.14
6	Kushinkote	1.88	1.25	1.86
7	Pliant Polymer	2.44	1.20	1.33
9	Paving Brick	2.07	1.80	1.71
10	Knicked Concrete	2.06	1.50	1.29

Table 29 Stopping Distance after Detection of Materials

Number	Material	Totally Blind	Low Partial	High Partial
1	Thermoplastic Solid	48" (121.9 cm)	12" (30.48 cm)	12" (30.48 cm)
2	Thermoplastic Strips	42" (106.68 cm)	12" (30.48 cm)	0"
4	Exposed Aggregate	48" (121.9 cm)	36" (86.4 cm)	0"
6	Kushionkote	42" (106.68 cm)	12" (30.48 cm)	18" (45.7 cm)
7	Pliant Polymer	30" (76.2 cm)	12" (30.48 cm)	24" (60.96 cm)
9	Paving Brick	42" (106.68 cm)	24" (60.96 cm)	18" (45.7 cm)
10	Knicked Concrete	42" (106.68 cm)	18" (45.7 cm)	0"

90 percent of the subjects stopped within the distances indicated

most detectable materials as indicated on Table 27.

The results indicate that the totally blind took longer to detect the materials than did the low partials, and the low partials (in general) took longer than the high partials. All of the seven materials were detected (by 90% of the totally blind subjects) within 48" (121.9 cm) of the beginning point of the materials. Several materials were detected within 42" (106.68 cm) and one within 30" (76.2 cm).

Table 30 shows the comfort rating of the materials by wheelchair users and by a group of ambulatory handicapped subjects. After traversing the materials the subjects rated the materials in terms of how they affected their comfort, stability and balance on a four-point scale:

1. not at all
2. little
3. moderately
4. very much

This table presents the average rating, and the percentage of subjects that rate the material as affecting them: 1 = not at all or 2 = a little.

The distance travelled over the materials is short, so the ratings may not be applicable to longer exposures to these materials.

Table 30 indicates that the exposed aggregate adversely affected those in wheelchairs. Twenty-five percent of the subjects rated the materials as affecting them 'moderately' or 'very much.' However, the material had a much worse rating from the remainder of the ambulatory handicapped subjects—more than half of them said it affected them 'moderately' or 'very much.'

With the exception of the exposed aggregate, all of the materials received mean ratings that fall in the 'not at all' or 'a little' range of affect on comfort, stability and balance.

Conclusions

1. Using the normal mobility techniques applicable to the outdoor environment (and this differs from that normally used indoors) most visually impaired subjects were able to detect certain types of the walkway surface material changes with little difficulty.
2. Further work will be necessary to determine standard specifications for detectable walkway surfaces, and to explore the range and types of materials that are practical for outdoor construction.
3. For the materials tested, a strip that is 30" (76.2 cm) to 48" (121.9 cm) long is

Table 30 Comfort Ratings of Surface Materials

Number	Material	Wheelchair Users (24 Subjects)		Ambulatory Handicapped (44 Subjects)	
		Average Rating	% Subjects Rating (1 or 2)	Average Rating	% Subjects Rating (1 or 2)
1	Thermoplastic Solid	1.04	100	1.24	95
2	Thermoplastic Strips	1.29	88	1.55	88
3	Ruled Concrete	1.16	100	1.19	95
4	Exposed Aggregate	2.13	75	2.35	49
5	No Texture	-----	---	-----	--
6	Kushionkote	1.04	100	1.14	98
7	Pliant Polymer	1.13	96	1.42	91
8	Burlap Bugged	1.04	100	1.17	98
9	Paving Brick	1.75	83	1.51	91
10	Knicked Concrete	2.00	75	1.86	79

needed for detection.

4. Pea gravel exposed aggregate was found to be easily detectable but in the form used (a rather rough surface), it was found to adversely affect the mobility of people travelling over it. Thermoplastic strips, 6" (15.24 cm) wide and spaced 6" (15.24 cm) apart and set across the normal direction of movement were found to be easily detectable and comfortable to move over. A tennis court covering material called Kushionkote, was found to be easy to detect and comfortable.

Recommendations.

1. In those locations where a detectable walkway surface can be used to provide warnings or general navigational information, at least a 48" (121.9 cm) length of the walkway should be treated.
2. Either a resilient material similar to Kushionkote should be considered or strips of thermoplastic 6" (15.24 cm) wide and set 6" (15.24 cm) apart across the walkway. If an exposed aggregate such as pea gravel is to be formed in the walkway, this should not be made so rough that it will affect adversely the passage of people in wheelchairs and other handicapped pedestrians.

a study of wheelchair dimensions

Background

The space needs of people in wheelchairs have become a major influence in the design of many elements of the built environment—for doorways, walks, and corridors; for access to water fountains, telephones, etc. And several studies have been directed at establishing these space needs. However, for the ramp study and for the preparation of The Implementation Package, "A Manual for Developing a Priority Accessible Network," it is necessary to reexamine current data not currently published.

In question is whether extant recommendations and standards are adequate for the very large range of chairs manufactured today. An examination of the data from a few manufacturers seems to suggest that some of the recommended dimensions (based on a 26" (66.04 cm) wide chair, for example) will be sufficient for an 'average' chair only. Obviously then, for people in chairs that are larger than this average, these recommendations will be unsuitable.

Recommendations for a wheelchair 'design vehicle' must be based on dynamic as well as static data. The space requirements for a wheelchair must take into account not only the measurements of the vehicle but also the tolerances that are needed between the chair (or the occupant) and the environment as the vehicle is maneuvered by the user. However, most tests of people in wheelchairs which have been directed at establishing these tolerances have been based on an 'average' wheelchair rather than the range of wheelchairs that are manufactured.

The study that follows does not deal with these dynamic tolerances, only the static

dimensions of the vehicle, and therefore the recommendations should be considered in this context. Dimensional data on most of the models that are currently manufactured is presented, and then recommendations based on these data are proposed.

The recommendations resulting from this study have a further limitation. The manufacturers' data does not include marketing information nor indications of how many of each model have been sold in the past. It is not possible therefore to conclude, for example, that 95% of all wheelchairs are (say) less than 28" (71.12 cm) wide. It is only possible to conclude, for example, that 95% of all wheelchair models currently manufactured are (say) less than 28" (71.12 cm) wide. However, this defect may not be fatal; the differences between these two conclusions may not be great. After all, it is probably reasonable to assume that if 20 models are made with a width of 20" (50.8 cm) and only one model with a width of 25" (63.5 cm), then if the manufacturers are responding directly to the demand, there are likely to be 20 times as many 26" (66.04 cm) wide wheelchairs as there are 25" (63.5 cm) wide chairs.

Methodology

To update available information on wheelchairs, both powered and unpowered, a survey of manufacturers in the United States was undertaken. Most manufacturers were prepared to cooperate in the study. Some manufacturers provided all information requested, others only a part of it. Some corporations listed as manufacturers turned out, on investigation, to be agents for others.

The following information was sought for each model:

- overall length
- overall width
- overall height
- wheel diameters, front and rear
- turning radius
- front and rear overhangs
- ground clearance
- weight
- wheel type
- propulsion system

The results of the survey are presented in five tables. Tables 31 to 34 list all the unpowered chairs manufactured by the respondent companies 'A', 'B', 'C' and 'D' respectively. Table 35 lists the powered wheelchairs manufactured by respondent companies 'A', 'C', 'E' and 'F'.

Table 31 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'A'

Model	Weight lbs	(L) Length inches	(W) Width inches	(H) Height inches	Wheel Diameter		Turning Radius inches	Overhang		Ground Clearance inches
					Front inches	Rear inches		Front inches	Rear inches	
1	44	43 3/4	25	36 3/8	5	24	30 1/4	16 1/2	0	4
2 Narrow	45	43 3/4	23	36 3/8	5	24	30 1/4	16 1/4	0	4
3	43	41 7/8	24 7/8	36 3/8	5	24	28 3/8	15 1/2	0	4
4	46	43	25	36 3/8	8	24	29 1/2	16 1/2	0	4
5 Narrow Adult	46	43	23	36 3/8	8	24	29 1/2	16 1/2	0	4
6	49	40 1/2	24 1/2	36 1/4	24	8	0	0	0	4
7	47	43	25	33 3/4	8	24	29 1/2	15 1/2	0	4
8	48	43	24 1/2	36 1/4	8	14	29 1/2	15	0	4
9	45	41 7/8	24 7/8	36 3/8	8	24	28 3/8	15 1/2	0	4
10	47	41 1/2	24 1/2	36 1/4	8	24	28	14 1/2	0	4
11 Narrow Adult	45	41 1/2	22 1/2	36 3/8	8	24	28 1/4	14 1/2	0	4
12	49	42 3/4	24 7/8	36 3/4	8	24	29 1/4	14	0	4
13		42 3/4	27 7/8	36 3/4	8	24	29 1/4	14	0	4
14		42 3/4	29 7/8	36 3/4	8	24	29 1/4	14	0	4
15		42 3/4	31 7/8	36 3/4	8	24	29 1/4	14	0	4
16	45	42 3/4	24 3/4	36 5/8	5	24	29 1/4	14	0	4
17	47	42 3/4	24 3/4	36 5/8	8	24	29 1/4	16 1/4	0	4
18	47	42 3/4	24 5/8	36 3/8	8	24	29 1/2	15 1/2	0	4
19	52	45	24 3/4	36 3/8	8	24	31 1/2	17	0	4
20	50	42	25	36 1/4	8	24	28 1/2	16	0	4
21	49	43 1/4	24 1/2	36 1/4	8	24	29 3/4	17	0	4
22	32	41	24 3/4	36 3/8	8	22	28 1/2	17	0	4
23	34	42	24 3/4	36 3/8	8	22	29 1/2	16 1/2	0	4
24	42	38 3/8	24 1/2	36 1/4	5	24	24 7/8	16 3/4	0	4
25	42	38 3/8	24 1/2	36 1/4	24	5 (reversible)	0	0	0	4

Table 31 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'A' (continued)

Model	Weight lbs	(L) Length inches	(W) Width inches	(H) Height inches	Wheel Diameter		Turning Radius inches	Overhang		Ground Clearance inches
					Front inches	Rear inches		Front inches	Rear inches	
26 Junior	47	38	19½	35 7/8	8	24	24½-30 3/4	4 3/4-9	0	4
27	47	38	20½	35 7/8	8	24	24½-30 3/4	4 3/4-9	0	4
28	46	38	21½	35 7/8	8	24	24½-30 3/4	4 3/4-9	0	4
29	46	44½	22½	35 7/8	8	24	24½-30 3/4	4 3/4-9	0	4
30	46	44½	23½	35 7/8	8	24	24½-30 3/4	4 3/4-9	0	4
31 Junior	37	34 3/4	22½	33½	5	20	23½	13½	0	4
32	38	33	21	37 7/8	5	20	21½	5 3/4	0	6
33 Wood chairs,* coil spring	82	NA**	25	50	24	8	NA**	0	0	NA**
34	84	NA	28½	50	24	8	NA	0	0	NA
35	54	NA	22	44½	24	8	NA	0	0	NA
36	84	NA	25	50	24	8	NA	0	0	NA
*Hospital Use		**Not Available								

Table 32 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'B'

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius
					Front	Rear	
1 Child	30	37½	21	34	8	24	24
2 Adult	37	43½	26	36½	8	24	30
3 Senior Adult	37	43½	24	34	8	24	30
4 Recliner	52	48½	26	49½	8	24	35

Table 33 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'C'

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance
					Front	Rear		Front	Rear	
1	40 $\frac{1}{2}$	44 3/8	24 $\frac{1}{2}$	36	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
2	45	45 7/8	24 $\frac{1}{2}$	51 5/8	8	24	36 3/4	16 $\frac{1}{2}$	0	4
3	50	49 3/8	24 $\frac{1}{2}$	53 1/8	8	24	40 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
4	45	44 3/8	26 $\frac{1}{4}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
5	32 $\frac{1}{2}$	44 3/8	24 $\frac{1}{2}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
6	31 $\frac{1}{2}$	44 3/8	26 $\frac{1}{4}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
7	38	42 7/8	24 $\frac{1}{2}$	33 $\frac{1}{4}$	8	22	32 5/8	16 $\frac{1}{2}$	0	4
8	43	42 7/8	26 $\frac{1}{4}$	33 $\frac{1}{4}$	8	22	33	13 7/8	0	4
9	33	42 7/8	24 $\frac{1}{2}$	33 $\frac{1}{4}$	8	22	31 3/8	14 $\frac{1}{2}$	0	4
10	51	45 5/8	26 $\frac{1}{4}$	51 5/8	8	24	36 3/4	16 $\frac{1}{2}$	0	4
11	37 $\frac{1}{2}$	42 7/8	26 $\frac{1}{4}$	33 $\frac{1}{4}$	8	22	33	13 7/8	0	4
12	55	49 3/8	26 $\frac{1}{4}$	53 1/8	8	24	30 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
13	39 $\frac{1}{2}$	44 3/8	24 $\frac{1}{2}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
14	45	45 5/8	24 $\frac{1}{2}$	51 5/8	8	24	36 3/4	16 $\frac{1}{2}$	0	4
15	50	49 3/8	24 $\frac{1}{2}$	53 1/8	8	24	40 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
16	44	44 3/8	26 $\frac{1}{4}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
17	51	45 5/8	26 $\frac{1}{4}$	51 3/8	8	24	36 3/4	16 $\frac{1}{2}$	0	4
18	53	49 3/8	26 $\frac{1}{4}$	53 1/8	8	24	40 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
19	31	44 3/8	24 $\frac{1}{2}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
20	35 $\frac{1}{2}$	44 3/8	26 $\frac{1}{4}$	38 3/4	8	24	35 $\frac{1}{2}$	16 $\frac{1}{2}$	0	4
21	40	44 3/8	24 $\frac{1}{2}$	38 3/4	8	24	33	14	0	4
22	45	45 5/8	24 $\frac{1}{2}$	51 5/8	8	24	32 $\frac{1}{4}$	14	0	4
23	45	44 3/8	26 $\frac{1}{4}$	36 $\frac{1}{4}$	8	24	31	14	0	4
24	44	43 1/8	24 $\frac{1}{2}$	38 3/4	5	24	29 3/4	14	0	4
25	39	34 $\frac{1}{2}$	24 $\frac{1}{2}$	38 3/4	8	24	21 $\frac{1}{4}$	13 $\frac{1}{4}$	0	4

Table 33 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'C' (continued)

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance
					Front	Rear		Front	Rear	
26	37	34 $\frac{1}{2}$	24 $\frac{1}{2}$	38 3/4	5	24	21 $\frac{1}{4}$	13 $\frac{1}{4}$	0	4
27	31	39 7/8	19 3/4	40 $\frac{1}{2}$	5	20	23	6 $\frac{1}{2}$	0	4
28	36	42 5/8	20 $\frac{1}{2}$	35	8	24	29 $\frac{1}{4}$	12	0	4
29	34	42 5/8	18 $\frac{1}{2}$	35	8	24	25	7 3/4	0	4
30	40	42 5/8	22 $\frac{1}{4}$	35	8	22	30 $\frac{1}{4}$	12	0	4
31	38	42 5/8	20 $\frac{1}{4}$	35	8	22	26	7 3/4	0	4
32	46	45 3/8	22 $\frac{1}{4}$	50 5/8	8	22	33	12	0	4
33	44	45 3/8	20 $\frac{1}{4}$	50 5/8	8	22	28 3/4	7 3/4	0	4
34	51	47 3/8	22 $\frac{1}{4}$	52 1/8	8	22	35	12	0	4
35	49	47 3/8	20 $\frac{1}{4}$	52 1/8	8	22	30 3/4	7 3/4	0	4
36	48	49 3/8	19	50 5/8	8	24	31	7	0	4
37	49	49 3/8	19	50 5/8	8	24	30 3/4	6 3/4	0	4
38	50	49 3/8	19	50 5/8	8	24	30 3/4	6 3/4	0	4
39	51	49 3/8	21	50 5/8	8	24	29 1/8	5 1/8	0	4
40	52	49 3/8	21	50 5/8	8	24	29	5	0	4
41	54	50 3/8	23	53	8	24	41 3/4	16 3/4	0	4
42	55 $\frac{1}{2}$	50 3/8	25	53	8	24	41 3/4	16 3/4	0	4

Table 34 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'D'

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
1	35	44 $\frac{1}{4}$	24 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Fixed Arms
2	34 $\frac{1}{2}$	44 $\frac{1}{4}$	22 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Fixed Arms
3	35	46	22 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Fixed Arms
4	34 $\frac{1}{2}$	46	22 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Fixed Arms
5	42	46	26 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Fixed Arms
6	41 $\frac{1}{2}$	46	24 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Fixed Arms
7	37	44 $\frac{1}{4}$	24 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Fixed Arms
8	36 $\frac{1}{2}$	44 $\frac{1}{4}$	22 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Space-Saver
9	39	44 $\frac{1}{4}$	24 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Space-saver arms
10	38 $\frac{1}{2}$	43 $\frac{1}{4}$	22 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Space-saver arms
11	37	46 $\frac{1}{4}$	24 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Space-saver arms
12	36 $\frac{1}{2}$	46	22 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Space-saver arms
13	44	46	26 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Space-saver arms
14	43 $\frac{1}{2}$	46	24 $\frac{1}{2}$	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Space-saver arms
15	36 $\frac{1}{2}$	44 $\frac{1}{4}$	26	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Space-saver arms
16	36	44 $\frac{1}{4}$	14	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Removable arms
17	38 $\frac{1}{2}$	44 $\frac{1}{4}$	26	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Removable arms
18	38	44 $\frac{1}{4}$	24	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	Removable arms
19	36 $\frac{1}{2}$	46	26	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Removable arms
20	36	46 $\frac{1}{4}$	24	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Removable arms
21	43 $\frac{1}{2}$	46	28	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Removable arms
22	43	46	26	35 3/4	8	24	33 $\frac{1}{4}$	12	0	4	Removable arms
23	54	45 $\frac{1}{4}$	25	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	4	Removable arms
24	53 $\frac{1}{2}$	45 $\frac{1}{4}$	23	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms
25	55 $\frac{1}{2}$	45 $\frac{1}{4}$	26 $\frac{1}{2}$	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms
26	55	45 $\frac{1}{4}$	24 $\frac{1}{2}$	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms
27	57 $\frac{1}{2}$	45 $\frac{1}{4}$	26 $\frac{1}{2}$	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms
28	57	45 $\frac{1}{4}$	24 $\frac{1}{2}$	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms
29	54	45 $\frac{1}{4}$	25	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	Av. 4	Removable arms

Table 34 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'D' (continued)

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
30	55½	45½	26½	49½	8	24	32½	12	0	Av. 4	Dual Drive. Operable from one side only. Two drive rims on one side, permitting steering from that side.
31	55	45½	24½	49½	8	24	32½	12	0	Av. 4	
32	57½	45½	26½	49½	8	24	32½	12	0	Av. 4	
33	57	45½	24½	49½	8	24	32½	12	0	Av. 4	
34	36	46	24½	35 3/4	8	24	33½	12	0	4	
35	36½	46	26	35 3/4	8	24	33½	12	0	4	
36	37	46	24½	35 3/4	8	24	33½	12	0	4	
37	42½	46	30	35 3/4	8	24	33½	12	0	4	
38	43½	46	28½	35 3/4	8	24	33½	12	0	4	
39	38	48	26	40	8	24	35½	12	0	4	
40	38½	48	24½	40	8	24	35½	12	0	4	
41	34	46	23	33	8	24	35½	12	0	4	
42	36½	46	24 3/4	33	8	24	35½	12	0	4	
43	37	46	23	33	8	24	35½	12	0	4	
44	42	46	26½	35 3/4	8	24	33½	12	0	4	
45	44½	46	28	35 3/4	8	24	33½	12	0	4	
46	45	46	26½	35 3/4	8	24	33½	12	0	4	
47	46½	46	32	35 3/4	8	24	33½	12	0	4	
48	47	46	30½	35 3/4	8	24	33½	12	0	4	
49	45	46	28	40	8	24	33½	12	0	4	
50	44½	46	26½	40	8	24	33½	12	0	4	
51	41	46	24 3/4	33½	8	24	33½	12	0	4	Hemiplegic: for primarily foot propulsion. Swing-away foot rest.
52	43½	46	26½	33½	8	24	33½	12	0	4	
53	44	46	24 3/4	33½	8	24	33½	12	0	4	
54	34	44½	25	33½	8	24	31½	12	0	4	
55	33½	44½	23	33½	8	24	31½	12	0	4	
56	36½	44½	27	33½	8	24	31½	12	0	4	
57	36	44½	25	33½	8	24	31½	12	0	4	
58	38½	44½	27	33½	8	24	31½	12	0	4	
59	37	44½	24	33½	8	24	31½	12	0	4	

Table 34 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'D' (continued)

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
60	36 $\frac{1}{2}$	44 $\frac{1}{4}$	23	33 $\frac{1}{4}$	8	24	31 $\frac{1}{2}$	12	0	4	Designed for large adults. Includes reclining chairs. Dimensions for upright travel position.
61	39	44 $\frac{1}{4}$	25	33 $\frac{1}{4}$	8	24	31 $\frac{1}{2}$	12	0	4	
62	42	44 $\frac{1}{4}$	30	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
63	43 $\frac{1}{2}$	44 $\frac{1}{4}$	28	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
64	44	44 $\frac{1}{4}$	30	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
65	43 $\frac{1}{2}$	44 $\frac{1}{4}$	29	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
66	42 $\frac{1}{2}$	44 $\frac{1}{4}$	28 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
67	42	44 $\frac{1}{4}$	26 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
68	44 $\frac{1}{2}$	44 $\frac{1}{4}$	28 $\frac{1}{2}$	35 3/4	8	24	31 $\frac{1}{2}$	12	0	4	
69	58 $\frac{1}{2}$	45 $\frac{1}{4}$	31	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	4	
70	60 $\frac{1}{2}$	45 $\frac{1}{4}$	31	49 $\frac{1}{2}$	8	24	32 $\frac{1}{2}$	12	0	4	junior models
71	38	44 $\frac{1}{4}$	26	40	8	24	31 $\frac{1}{2}$	12	0	4	
72	37 $\frac{1}{2}$	44 $\frac{1}{4}$	24	40	8	24	31 $\frac{1}{2}$	12	0	4	
73	38 $\frac{1}{2}$	44 $\frac{1}{4}$	24 $\frac{1}{2}$	40	8	24	31 $\frac{1}{2}$	12	0	4	
74	40 $\frac{1}{2}$	44 $\frac{1}{4}$	24 $\frac{1}{2}$	40	8	24	31 $\frac{1}{2}$	12	0	4	
75	40	44 $\frac{1}{4}$	22 $\frac{1}{2}$	40	8	24	31 $\frac{1}{2}$	12	0	4	
76	56 $\frac{1}{2}$	46 3/4	26 $\frac{1}{2}$	53 $\frac{1}{2}$			34	12	0	4	
77	56 $\frac{1}{2}$	46 3/4	26 $\frac{1}{2}$	53 $\frac{1}{2}$			34	12	0	4	
78	33 $\frac{1}{2}$	44 $\frac{1}{4}$	23	33 $\frac{1}{2}$	8	24	31 $\frac{1}{2}$	12	0	4	
79	33	43 3/4	23	33 $\frac{1}{2}$	8	24	31 $\frac{1}{2}$	12	0	4	
80	36	44 $\frac{1}{4}$	23	33 $\frac{1}{2}$	8	24	31 $\frac{1}{2}$	12	0	4	Reclining junior
81	35 $\frac{1}{2}$	43 3/4	23	33 $\frac{1}{2}$	8	24	31	12	0	4	
82	36 $\frac{1}{2}$	44 $\frac{1}{4}$	23	33 $\frac{1}{2}$	8	24	31 $\frac{1}{2}$	12	0	4	
83	38 $\frac{1}{2}$	44 $\frac{1}{4}$	23	33 $\frac{1}{2}$	8	24	31 $\frac{1}{2}$	12	0	4	
84	53	45 $\frac{1}{4}$	23 $\frac{1}{2}$	47	8	24	32 $\frac{1}{2}$	12	0	4	
85	57 $\frac{1}{2}$	45 $\frac{1}{4}$	25 $\frac{1}{2}$	47	8	24					
86	55	45 $\frac{1}{4}$	25 $\frac{1}{2}$	47	8	24					
87	53	45 $\frac{1}{4}$	23 $\frac{1}{2}$	47	8	24					
88	55 $\frac{1}{2}$	45 $\frac{1}{4}$	25 $\frac{1}{2}$	47	8	24					
89	41	43 3/4	24 $\frac{1}{2}$	35	8	24	31	12	0	4	

Table 34 Dimensional Data for Unpowered Wheelchairs: Manufacturer 'D' (continued)

Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
90	41	43 3/4	24 1/2	35	8	24	31	12	0	4	
91	45	43 3/4	26	35	8	24	31	12	0	4	
92	49	43 3/4	26 1/2	35			31	12	0	4	
93	46	43 3/4	24 1/2	35			31	12	0	4	
94	46	43 3/4	23	35			31	12	0	4	
95	50	43 3/4	26	35			31	12	0	4	
96	45	43 3/4	24 1/2	35			31	12	0	4	
97	45	53 3/4	24 3/4	37	8	24	41	12	0	4	
98	43	53 3/4	22 3/4	37	8	24	41	12	0	4	
99	73	53 3/4	34 3/4	48	8	24	41	12	0	4	
100	71	53 3/4	22 3/4	48	8	24	41	12	0	4	
101	41	53 3/4	26 3/4	37	8	24	41	12	0	4	
102	49	53 3/4	24 3/4	37	8	24	41	12	0	4	
103	76	53 3/4	26 3/4	48	8	24	41	12	0	4	
104	74	53 3/4	24 3/4	48	8	24	41	12	0	4	
105	58	53 3/4	25 1/2	37	8	24	41	12	0	4	
106	56	53 3/4	23 1/2	37	8	24	41	12	0	4	
107	78	53 3/4	25 1/2	48	8	24	41	12	0	4	
108	76	53 3/4	23 1/2	48	8	24	41	12	0	4	
109	41	43-3/4	24 1/2	36 1/2	8	24	31	12	0	4	
110	41	43 3/4	33 1/2	36 1/2			31	12	0	4	
111	29	38 3/4	19	33 1/2			26	12	0	4	
112	40	43 3/4	24 1/2	33 1/2			31	12	0	4	

Table 35 Dimensional Data for Powered Wheelchairs

Manufacturer & Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
A1	74 + batteries	42 3/4	26 1/4	37 3/8	8	24	39 1/4		0		Two 12-v. motors, joystick control, two-speed. F & R, 2 12-v. batteries, climbability 10%.
E2	78 + batteries	35	18*	36	6	4	36	4	0	3	One 12-v. motor, handlebar control, V. speed. F & R, up to 6 mph.
F3	300	39	28	39 1/2	8	12	39	0	2	6	Two 6-v. custom control. 4-speed F & R. Climbability 40%. Max. speed 12 mph or 18 mph.

Table 35 Dimensional Data for Powered Wheelchairs (continued)

Manufacturer & Model	Weight	(L) Length	(W) Width	(H) Height	Wheel Diameter		Turning Radius.	Overhang		Ground Clearance	Remarks
					Front	Rear		Front	Rear		
G4	300 (varies)	49 1/2 (varies)	26 1/2	41	10	8	NA				Two 12-v. custom control. 2-speed F & R. Max. speed 2 mph. Climbability 8%. 2 speed. 2 1/4 mph max. @ spd. Up to 4 3/4 mph.
	Note: Custom built for quadraplegics. Sophisticated prosthetic. Very few made.										
C5	80 + batteries	45 3/4	24 1/2	36	8	20	37 1/2	16 1/2	0	4	
C6	80 + batteries	45 3/4	24 1/2	36	8	20	37 1/2	16 1/2	0	4	
C7	80 + batteries	45 3/4	24 1/2	36	8	20	37 1/2	16 1/2	0	2 1/2	
*Outdoors uses stabilizer wheels which makes the width 24". Originally designed for MS Victims.											

Findings

There are two basic varieties of wheelchair--powered and manual. The typical manual wheelchair, despite its long production run and disadvantages, has been, and continues to be, manufactured in large numbers. There are a number of manufacturers in the field that produce custom-built chairs, particularly the powered variety, which are adapted to extreme disabilities. These are so diverse in size, and their production level is so low, that where dimensions are quoted for these manufacturers, they are "typical" dimensions or dimension ranges.

The powered chairs are further subdivided into "principally for indoor use" and "principally for outdoor use". The reason for this is that small, light, maneuverable devices are well adapted to use in office, supermarket and home, but because of the necessity of limiting weight and width, they are at a disadvantage where anything but flat or gentle inclines are encountered. The 'outdoor only' models make use of larger, balloon-tired wheels, making them capable of dealing with the terrain normally negotiable by a golf cart. This type of machine is supplied battery powered (as are all of the powered chairs surveyed) and with maximum speeds as high as 20 miles (12.428 km) per hour. Such a vehicle is less suitable for indoor use, and its overall dimensions are somewhat greater than the large traditional wheelchair.

Figures 7 to 10 show the frequency distributions for selected dimensions of wheelchairs--length, width, height and weight. For environmental design purposes, these are the data most frequently needed. From the distributions it can be concluded that:

● Length:

- 83% of all the models listed are less than 47" (119.38 cm) long.
- 55% of all the models listed are less than 46" (116.84 cm) long.
- 31% of all the models listed are less than 45" (114.3 cm) long.

The conventional wheelchair offers a wide range of leg and footrest adjustment. The front rigging is usually designed for two positions, for footrest and for legrest. In the footrest position the legs are bent at the knee in the usual sitting stance. In the legrest position, the leg is supported in a position approximately parallel to the ground. For the footrest and legrest position there is the opportunity for additional adjustments. For example, the 4" (10.16 cm) footrest ground clearance used in the tables is quoted by the manufacturers as a possible average for a traveling mode. This clearance can be reduced to almost nil, or increased to suit the needs of the user. The length (L) dimension in the tables is an overall length based on the maximum extension in the footrest position. For an 'average' extension (from information provided by the manufacturers), this overall length will be reduced by about 1½ inches (3.8 cm). (For a footrest rigging, the overall length will be increased by about 1½ inches (3.8 cm).

● Width:

- 89% of all the models listed are less than 28" (71.12 cm) wide.

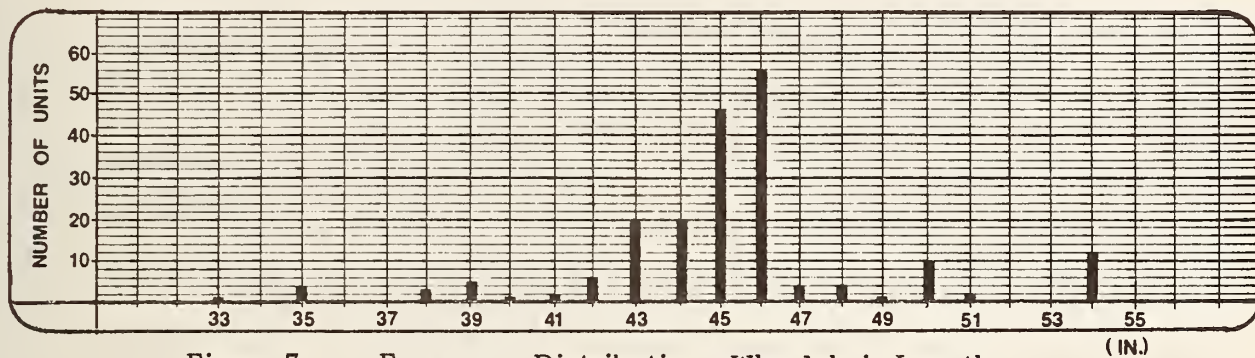


Figure 7 Frequency Distribution: Wheelchair Lengths

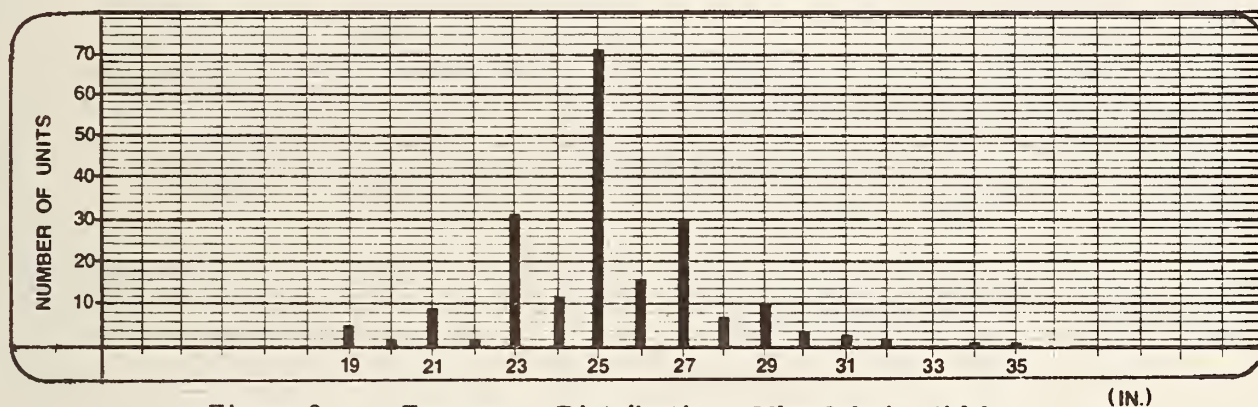


Figure 8 Frequency Distribution: Wheelchair Widths

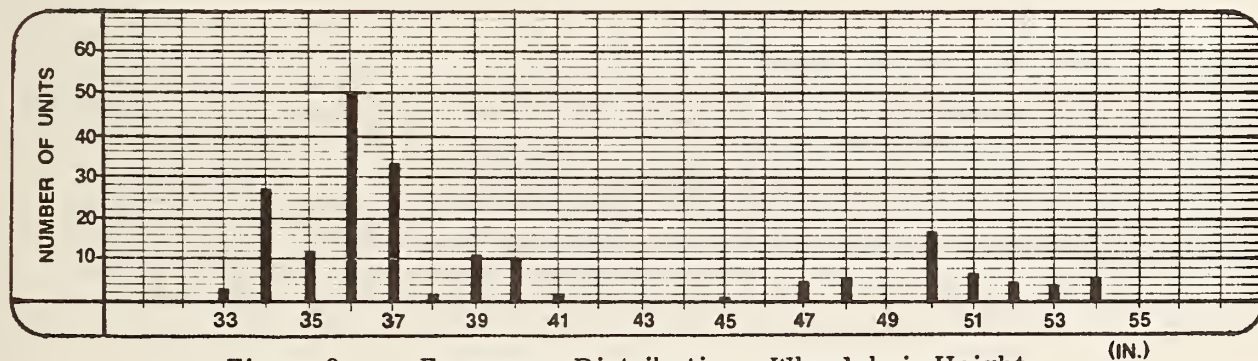


Figure 9 Frequency Distribution: Wheelchair Height

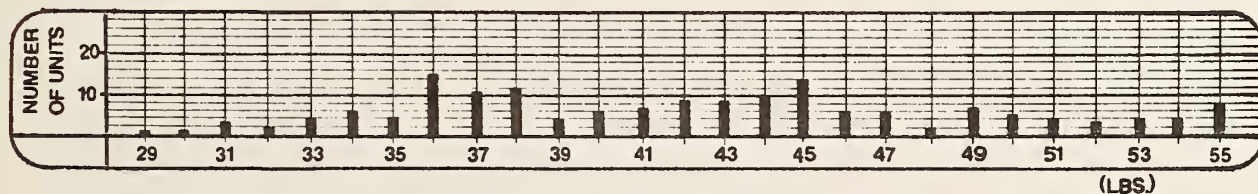


Figure 10 Frequency Distribution: Wheelchair Weight

- 76% of all the models listed are less than 27" (68.58 cm) wide.
- 66% of all the models listed are less than 26" (66.04 cm) wide.
- Weight:
 - 76% of all the models listed are less than 51 lbs. (37 kilos) in weight.
 - 62% of all the models listed are less than 46 lbs. (21 kilos) in weight.
 - 49% of all the models listed are less than 44 lbs. (20 kilos) in weight.

Conclusions and Recommendations

There are a very large number of wheelchair models manufactured; however, from the frequency distributions it can be seen that most of the available models have dimensions that are clustered into quite a small number of sizes. As it was not possible to obtain sales information on the various models (as discussed earlier), all we can do is to hypothesize that the bulk of the sales (and perhaps the models in use) are clustered in a similar manner.

The static dimensions of wheelchairs are an important influence on environmental design decisions. If the dimensions used as criteria are too small, then some wheelchairs will be unable to use the environment. If the recommended dimensions are too great, then the costs of providing for wheelchairs may be unnecessarily high. The selection of a design vehicle dimension that is only one inch greater than an alternative choice may make a substantial difference. For example, 55% of all models are less than 46" (116.8 cm) in length, but 83% are less than 47" (119.4 cm) in length. So the additional inch may affect 28% of the users.

A glance at Figure 7 suggests that to recommend a length of less than 46" (116.8 cm) would omit a large number of units, and to recommend more than 46" (116.8 cm) would not increase the number of units substantially. This suggests that an overall length of 46" (116.8 cm) should be used. This length, as discussed earlier, is based on the maximum extension in the footrest position. And it could be argued that an average extension might be more appropriate for design purposes--this would reduce the overall length by 1½" (3.81 cm). However, the chair users' footwear normally projects beyond the footrest, (by approximately 1½" (3.81 cm), and should therefore be included. For these reasons, an overall length of 46" is recommended.

The frequency distribution of wheelchair widths, Figure 8, also has a pronounced step characteristic at 27" (68.58 cm), and 89% of all models are 27" (68.58 cm) or less. Therefore, it is recommended that an overall width of 27" (68.58 cm) should be used.

The frequency distribution of wheelchair heights, Figure 9, is bi-modal with a major peak at 36" (91.44 cm) and a minor peak at 50" (127 cm). Nevertheless, 75% of all models are less than 42" (106.68 cm) high, so a design height of 42" (106.68 cm) is recommended. There are comparatively few circumstances where the height of a wheelchair without an occupant will be used (for storage, for example). The height with

an occupant will, or course, be substantially greater than this recommended static dimension.

There may be special circumstances where it is necessary to consider a maximum anticipated wheelchair weight, but to allow for 76% of all models a design weight of 51 lbs. (22.9 kilos) is recommended.

Summary of recommendations: (Fig. 11)

Length	46" (116.8 cm)
Width	27" (68.58 cm)
Height	42" (106.68 cm)
Weight	51 lbs. (22.9 kilos)

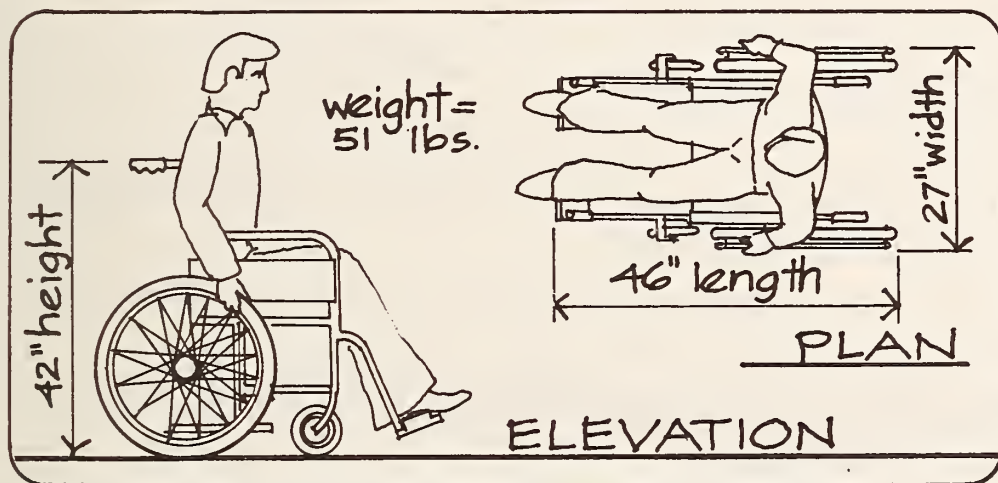


Figure 11 Recommended Static Wheelchair Dimensions

background

Volume 2 in this series, Hazards, Barriers, Problems, And The Law, described the identification of approximately 750 problems that restrict or prevent elderly and handicapped pedestrians from moving safely around the urban environment. This large list of problems was later reduced to 18 categories. And countermeasures were developed for some of these problem categories--those that were rated by a panel of elderly and handicapped mobility experts as being those for which solutions were most urgently needed.

The categories of problems for which solutions were sought, and the countermeasures developed are as follows:

Problem Categories

A. Safer and easier ways to cross streets. The specific problems for which solutions were sought are:

- i) Very wide streets are difficult or impossible to cross, for those without much stamina and agility, and pedestrians are exposed to moving and turning traffic for a substantial period of time.

For the problem, the following countermeasures have been developed for evaluation:

- Countermeasure No. 1, The Widened Sidewalk. In this solution, the road width is reduced at pedestrian crossings by widening the sidewalks out to the edge of the parking lane.
- Countermeasure No. 2, The Pedestrian Refuge Island. In this solution, the effective width of the street is reduced by introducing a pedestrian island which can be used to divide the crossing trip into two.

- ii) The drivers of vehicles often ignore midblock crosswalks and pedestrian rights, even if the crosswalk is well marked.

For this problem, the following countermeasure has been developed for evaluation:

- Countermeasure No. 3, Midblock Crossing, with painted markings that are intended to be more obvious to drivers.

- iii) Narrow crosswalks tend to be congested and increase the risks of people being jostled and losing their balance. And there is an increased risk of vehicles stopping within and blocking the crosswalk.

For this problem, the following countermeasure has been developed for evaluation:

- Countermeasure No. 4, Widened Crosswalks. In this solution, the crosswalks at intersections are increased in width.

- iv) Where right turns-on-red are permitted, and the driver's view to the left is obscured by a planting or buildings, etc., they tend to move their vehicles into the crosswalk to improve their sight lines. This causes increased inconvenience and hazard to pedestrians.

For this problem, the following countermeasure has been developed for evaluation:

- Countermeasure No. 15, Right-Turn-On-Red Prohibition. In this solution, where the view to the left is obscured, right-turn-on-red movements would be prohibited.

- v) At intersection crosswalks, parked vehicles, street furniture and plantings often obscure pedestrians, and particularly those in wheelchairs and those of small stature.

For this problem, the following countermeasure has been developed for evaluation:

- Countermeasure No. 1, The Widened Sidewalk. This countermeasure, which has already been described, puts the pedestrians who are preparing to cross, in a position where they will not be obscured by parked vehicles, etc.

B. Problems at curbs and curb ramps. The specific problems for which solutions were sought are:

- i) Curbs are a barrier for many elderly and handicapped pedestrians. Curb ramps are needed, but there is no single design that is universally applicable. Which of the frequently used designs work well (in the appropriate environment)?
- ii) Curb ramps may be hazardous to severely visually impaired pedestrians in that they may not notice them and wander off into the street. Which of the frequently used designs (and locations) are the safest and most detectable?
- iii) Some curb ramps must be located in sidewalks such that people in wheelchairs may have to travel across them in order to round a corner, etc. Which of these curb ramp designs are reasonably accessible from this point of view?

For these problems, the following curb ramp countermeasures have been developed for evaluation:

- Countermeasure No. 6, The Raised Box. In this solution, the whole street intersection is raised to the level of the sidewalks, eliminating the step at the curb. The pavement for vehicles is then ramped up to the intersection.
- Countermeasure No. 7, The Ramped Sidewalk. In this solution, for very narrow sidewalks, the sidewalk is ramped down to the corner quadrant.
- Countermeasure No. 8, The Single Conventional (Recessed) Curb Ramp. In this solution, a single curb ramp is set into the sidewalk at the corner, facing the center of the intersection.
- Countermeasure No. 9, The Double Conventional (Recessed) Curb Ramp. In this solution, two curb ramps are set into the sidewalk at the intersection at right angles (usually) to each other, and facing into the crosswalks.
- Countermeasure No. 10, The Ramped Quadrant Curb Ramp. In this solution, the whole outside of the sidewalk is ramped to form a continuous ramp that serves both crosswalks.
- Countermeasure No. 11, The Ramped Full Quadrant Curb Ramp. In this solution, the curb ramp occupies the entire sidewalk quadrant and serves both crosswalks.
- Countermeasure No. 12, The Single Add-On Curb Ramp. In this solution, the curb ramp is formed by making a ramp in the street beyond the curb, but following the curb line.
- Countermeasure No. 13, The Double Add-On Curb Ramp. In this solution, two curb ramps are added, each serving one crosswalk.

The location of street furniture. The specific problems for which solutions were sought are:

- i) Street furniture—mail boxes, trash containers, poles, parking meters, manhole covers, grating, etc., are located on sidewalks in a haphazard, unplanned manner. This furniture is useful to visually impaired people for locating themselves, but often the multiplicity of objects are confusing, and barriers to direct movement. And covers and gratings may cause slips and trips.

For this problem the following countermeasure has been developed for evaluation:

- Countermeasure No. 5, The Textured Strip. In this solution, the street furniture is separated from the main sidewalk path by means of a surface strip that is detectable by visually impaired pedestrians.
- Countermeasure No. 14, Painted Manhole Covers and Gratings. In this solution, manholes and gratings are made more obvious by painting them

a bright color that contrasts with the surrounding walkway.

D. Location and warning cues for the visually impaired. The specific problem for which a solution was sought is:

- i) That visually impaired people may fail to detect some potential hazard such as a stair or a ramp.

For this problem the following countermeasure has been developed for evaluation:

- Countermeasures No. 9 and No. 11, described earlier, where the curb ramps and their surroundings have been treated with a detectable surface.

E. The lack of places to rest in safety and comfort. The specific problem for which a solution was sought is:

- i) Few cities provide space on sidewalks for benches and for wheelchairs where people with limited stamina can rest.

For this problem, the following countermeasures have been developed for evaluation:

- Countermeasures No. 1, The Widened Sidewalk, and Countermeasure No. 2, Pedestrian Island Refuge. These solutions have been described earlier, and make possible additional space for these rest areas.

Site Selection and Preparation for Countermeasure Evaluation.

Sites that were suitable for the construction and evaluation of the countermeasures were chosen using the following steps.

- a. Establishment of criteria for selection of cities.
- b. Preparation and distribution of a brochure describing the countermeasures to be evaluated.
- c. Selection of cities from those that responded to the brochure and volunteered to participate.
- d. Selection of sites within those cities.
- e. Detailed design of the countermeasures.
- f. Countermeasure construction.
 - a. The criteria used in the selection of cities included:

- Size of cities: cities to be of sufficient size to have an active central business district with substantial pedestrian activity.
 - Climatic situation: the selected cities to be representative of various climatic conditions, including snow, high humidity, heavy rainfall, etc.
 - Elderly and handicapped: all selected cities to have a substantial number of elderly and handicapped persons within their total population.
 - Improvement programs: all cities to have on-going programs of providing improvements for the elderly and handicapped, and to be interested in participating in the research effort.
 - Consumer groups: all cities to have active consumer groups and organizations, and agencies that deal with the elderly and handicapped.
 - The cities to be willing to participate in the project and to construct the countermeasures.
- b. A brochure was prepared by the Pedestrian Research Laboratory, describing the countermeasures to be evaluated. The brochure was distributed by the Federal Highway Administration, through its regional offices, to cities throughout the United States. The intent of the brochure was to provide information about the project, the countermeasures to be evaluated, and the kinds of locations being sought, in order to give cities an opportunity to volunteer their participation in the research effort.

The contents of the brochure included the following, for each of the countermeasure designs:

- The problems intended to be solved by the countermeasure.
- Criteria to be used in selecting a site appropriate for testing the countermeasure.

Certain of these criteria differed for each countermeasure, for example:

- The ramped sidewalk is intended for a location which has a narrow sidewalk.
- The widened sidewalk is intended for a location where parking is permitted in the curb lane 24 hours per day.

Other criteria were common to all of the countermeasures, for example:

- A location with high pedestrian volumes.
- A location with substantial traffic volume.

- A location likely to be used by the elderly and handicapped.

Design features to be incorporated into the countermeasure design; for example, curb ramps, textured surfaces, handrails, benches, planters.

- A schematic design sketch of the countermeasure design. This sketch was intended as a suggestion of the physical design of the countermeasures. Final design was a result of a dialogue between The Pedestrian Research Laboratory, and engineers from the city which constructed the countermeasures.
- A time schedule, setting forth estimates for final design approval, construction, and evaluation.

c. City Responses to the Countermeasure Brochure.

More than 25 cities responded to the brochure. These cities, all of which indicated at least some degree of willingness to participate in the research project, were fairly evenly distributed geographically: five from the Northeast and Middle Atlantic Coast; eight from the Southeast; three from the Midwest; nine from the West and one from the Farwest.

The cities were asked to indicate which of the countermeasures they would be interested in constructing and having evaluated within their city. Some countermeasures were selected by many of the responding cities. This may have been because they were familiar designs, or were less costly to construct. Other countermeasures, specifically the Raised Intersection Box and The Ramped Sidewalk attracted no interest from the responding cities. This may be assumed to have been because they were perceived as being costly to install, and because they were unusual solutions.

The following countermeasures were not evaluated, because of cost, inability to find a city willing to install them, or other difficulties.

Countermeasure Number	Countermeasure Description
6	Raised Intersection Box
7	Ramped Sidewalk
10	Ramped Quadrant
13	Double Add-On Ramp
14	Paint Indications--Inlets and Covers

Six cities were finally selected for installation and evaluation of countermeasures: Atlanta, Georgia; Athens, Georgia; Baltimore, Maryland; Boulder, Colorado; Clearwater, Florida; and Sioux City, Iowa.

In these six cities, the following countermeasures were constructed and evaluated:

Countermeasure Number	City	Countermeasure Description	Number of Locations
1	Sioux City Boulder Athens	Widened Sidewalk at Intersection	3
2	Clearwater	Pedestrian Island	1
3	Atlanta	Midblock Crossing	1
4	Atlanta	Widened Crosswalk	3
5	Clearwater	Textured Furniture Strip	1
8	Sioux City	Single Conventional Curb Ramp	1
9	Baltimore	Double Conventional Curb Ramp	1
11	Baltimore Clearwater	Ramped Full Quadrant	3
12	Sioux City	Single Add-On Ramp	1
15	Atlanta	Right-Turn-On-Red Prohibition	1

d. Identification of Sites Within the Selected Cities.

The original communication with each of the volunteering cities was usually with a public official such as the mayor or city manager. Once the decision was made to proceed with the construction and evaluation of countermeasures in a particular city, it was important to establish a working relationship with the appropriate designing or implementing agency for that city. In most cases, this was the traffic engineering department of the city.

The following steps were then followed for selecting the specific sites (usually intersections) where the countermeasures would be constructed and the evaluations would take place:

- The cities were asked to suggest, or recommend, sites which were felt to be appropriate for countermeasure installation and which met the criteria for site selection as defined by The Pedestrian Research Laboratory. In most cases, the cities responded with a series of possible and alternative sites for the countermeasures.

- A member of The Pedestrian Research Laboratory team then visited the city, and reviewed each of the suggested sites, with respect to the criteria which had been established for site selection. In some cases, additional sites were suggested at this time. The final choice was made jointly by the city and the research laboratory. On this visit, control sites also were selected, as close as possible to the evaluation sites, and sharing similar physical and functional characteristics.

e. Detailed Design of Countermeasures.

The following steps were followed in the preparation of the detailed countermeasure designs:

- The Pedestrian Research Laboratory secured engineering drawings of the intersections describing their conditions: dimensions, elevations, materials, utilities and drainage, etc. Detailed design suggestions for the countermeasures were then prepared, adapting the designs to the specific site conditions. These were submitted to the appropriate city agency. They were asked to make whatever adaptations were necessary to conform to the engineering and construction practices followed by the city, while conforming to the basic intent and design concept for the countermeasure. An iterative process then ensued, which led to the designs which were finally constructed.

The results of this process varied substantially from place to place. In some cases, the final constructed measure was, to all intents and purposes, identical to the original schematic suggestion. In other cases, the process of detailing the design, including its development into a form acceptable to the implementing agencies in the city, or into a form responding to existing site conditions (such as storm inlets and utility pole locations) without undertaking major and costly changes, resulted in a final design which was substantially different from the original design and its intended function. This was a 'fact of life', resulting from the cities bearing the responsibility (and cost) of construction.

f. Construction of Countermeasures.

The cooperating cities had total responsibility for scheduling and constructing the countermeasures, including bearing the costs of engineering and construction.

Before construction started, it was necessary for 'before' data gathering and observations to be collected at each of the test and control locations. Once construction was completed, an acclimation period was set aside to permit persons using the area to become accustomed to the changed conditions. The length of this acclimation period was a minimum of 30 days.

The Development of Evaluation Criteria.

Criteria were developed to evaluate:

- The suitability of the countermeasure designs.
- The impact of the various countermeasures upon the normal users of the locations where they were installed.
- The degree of success with which the countermeasures solved the problems of the elderly and the handicapped for which they had been designed.
- Any other effects, both positive and negative, which the countermeasures might have generated.

The general procedures developed by The Pedestrian Research Laboratory involved on-site evaluations of two distinct types:

- Observations of use of the countermeasure locations by persons (including both pedestrians and drivers of vehicles) who normally use those locations.
- Observations of use of the same locations by specially arranged groups of handicapped persons.

Data on Use of the Test Locations By the General Population

These data involved measuring the use of the test locations by persons who normally travelled through those locations. This included measures of changes in the amount and type of use, by pedestrians and drivers; and examination of the behavior of pedestrians, with emphasis given to behavior which was unsafe, or which had the potential of leading to an accident.

- i) Changes in the amount and type of use. The objective, with these measures, was to determine whether there would be any change in pedestrian and vehicular activity which could be attributed to the design of the Countermeasure constructed at that location. For example, did the add-on ramp (Countermeasure No. 12), or the widened sidewalk at an intersection (Countermeasure No. 1) impede traffic flows, or adversely affect vehicular turning movements? Did pedestrians avoid the countermeasure location, or change direction intentionally in order to use it?

Three measures were selected to provide information about these kinds of effects:

- Vehicular traffic volumes
- Vehicular turning movements
- Pedestrian travel volumes

- ii) Pedestrian behavior. Two sets of measures were used for the evaluation of pedestrian behavior. The first examined the impact of the countermeasure upon pedestrian safety; the second was used to determine the impact upon mobility.

The frequency of pedestrian accidents is such that it is not feasible to station observers at locations where countermeasures have been constructed to watch for accidents to occur. Instead, it is necessary to identify potential accident causing behavior. The Pedestrian Research Laboratory based its measures of accident-causing behavior upon the work of Snyder and Knoblaugh,⁵ in particular, upon their identification of pedestrian accident types likely to occur at intersections, and upon their definition of the chain of pedestrian behavior events which can lead to, or prevent, an accident.

The accident types most likely to occur at, or near, intersections include the dart out, intersection dash, multiple threat, vehicle turn/merge with attention conflict, and bus stop related. The chain or sequence of events which describes activities performed by pedestrians, and which will result in an accident if not performed safely, includes course, search, detection, evaluation, decision and action. From these accident types, and from the event sequence, The Pedestrian Research Laboratory developed evaluation criteria that would identify and measure failures in the event sequence that could lead to these kinds of accidents. For example, during the evaluation observations were recorded of:

1. Pedestrians crossing the street in a location which is not within a marked or unmarked crosswalk.

This is a failure in pedestrian course (path of movement). The pedestrian is likely to appear in a location which has not been anticipated by drivers. If the selected course includes movement in the vicinity of possible visual obstructions, there is a potential for a dart out type accident, particularly, if the pedestrian is moving at faster than normal walking speed.

2. Pedestrians moving at faster than normal walking speed.

The pedestrian is visible for a shorter period of time, reducing his/her opportunity to search, detect, evaluate, and make a decision. In addition, since the pedestrian will appear more quickly, drivers will have reduced time for detection, evaluation, and taking any appropriate action. The result can be an increased potential for a dart-out or intersection dash type of accident.

3. Pedestrians moving at slower than normal walking speed.

This includes such specific actions as slowing while entering the intersection, or reducing walking speed while in the middle of an intersection. Such slowing may indicate that the pedestrian is having difficulty with search and detection, which may result from vision problems, obstructions to clear vision, and confusing or conflicting visual information. The slowing may suggest that problems in the design of the intersection are causing this search and detection difficulty--careful investigation of the location is necessary to determine whether this is the case. The fact that the pedestrian slows, may, by itself, be an action unanticipated by drivers. Also, the pedestrian is vulnerable to traffic for a longer period of time.

In the above two criteria, an increase or decrease from the pedestrian's own, normal, walking speed was used, instead of comparing walking speed to an average rate of travel of any kind. This was because individual walking speeds can deviate substantially from the average, and because changes in any individual's speed is likely to be more indicative of difficulties he/she is perceiving while travelling, or of unsafe actions on his/her part.

4. Pedestrians stop within the street while crossing.

This suggests a failure in the pedestrian's search, detection, or evaluation, and has required a change in actions, from an earlier decision that has become inappropriate.

5. Pedestrians step into the street as if to begin crossing, then return to the sidewalk.

This action, of stepping out, then returning, signifies a failure in the pedestrian's initial search or detection, requiring a re-evaluation of the traffic situation and the taking of a corrective or evasive action.

6. Pedestrians change course, or swerve, while crossing.

This action, taken to avoid a vehicle; or to step out of the path of a vehicle, suggests a failure in search or detection, which required a re-evaluation and taking corrective or evasive action. This new action can be one which is unanticipated or unexpected by drivers, and can result in the pedestrian's sudden appearance in a vulnerable situation, potentially causing a multiple-threat type of accident.

7. Pedestrians begin crossing the street against a "don't walk" or a "red" traffic signal.

This action suggests a failure in the pedestrian's search, or evaluation function, and can result in an intersection dash, dart out, or vehicle turn/merge with attention conflict type of accident, since the pedestrian is likely to suddenly appear in an unexpected location.

8. Pedestrians caught in the intersection by a signal change to red.

This signifies a failure in the pedestrian's evaluation of the situation. The pedestrian is suddenly in a vulnerable location, and may take an unexpected corrective or evasive action (such as dashing across the remaining traffic lanes, or unexpectedly changing course and returning to the point of departure into the intersection). These actions could precipitate one of several intersection accidents.

Conflicts and Potential Conflicts Between Pedestrians and Other Pedestrians.

These criteria are similar to those which were listed above as pedestrian-vehicular conflicts and potential conflicts. They involve the same pedestrian behavior event sequence: course, search, detection, evaluation, decision, and action; however, in this case these are related to the incidence of other pedestrians and to the potential for conflict with these pedestrians, rather than to other vehicles. The criteria include the following actions taken as a response to other pedestrians:

1. Slowing, while crossing the street.
2. Stopping, while within the intersection.
3. Changing direction, or veering, while in the intersection.

Data on Responses to Mobility Barriers and Problems. The countermeasures were designed primarily to resolve or reduce mobility barriers and problems. To determine whether the countermeasures were successful from this point of view, the following evaluation criteria were developed:

1. Difficulty in negotiating the level change at the curb--on entering the street, and on leaving the street. A four point difficulty scale was used, ranging from a difficulty factor of 1 (equivalent to walking on a level surface, with no perceptible change in walking speed or gait) to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all—including needing to grab onto a pole for support or balance, or nearly falling while negotiating the level change).
2. Length of time required to cross the street. The length of time required by each individual to cross the street was measured, as a means of establishing a mean crossing time, determined the deviation in walking time from the mean, and comparing these to the crossing period permitted by the traffic signal.
3. Pedestrians still within the street, in the act of crossing, at the time the traffic signal turns to red. This criterion represents an additional means of measuring to what extent a traffic signal might serve as a barrier to mobility. General practice is to time pedestrian crossing signals on the basis of the 50th percentile of walking speeds. As a result, half of all pedestrians find that the

signal change occurs before they are able to complete their crossing. A significantly higher proportion of elderly and handicapped are so affected.

4. Use of a ramp, or curb cut, to negotiate the level change at the curb; or use of the curb to negotiate the level change--on entering the street, and on leaving the street. A well designed curb ramp will eliminate a travel barrier for persons with certain disabilities: persons in wheelchairs, for example. However, the same ramp may be hazardous to others (for example, to those with disabilities which cause balance problems). This criterion measures which choice, curb or ramp, each pedestrian makes.
5. Change of direction in order to use the ramp, or the curb, to negotiate the level change; and change in direction to avoid the ramp, or the curb, to negotiate the level change on entering the street and on leaving the street. This measure indicates whether pedestrians choose to use, or avoid, the ramp (or the curb).
6. Maintaining a preferred, or safe, course; along the sidewalk or within the crosswalk.
7. Conflict with street furniture or other obstructions.
8. Effectiveness of tactile cues for assisting visually impaired pedestrians to establish and maintain a proper course, and avoid hazards.

Evaluation Data Collection Techniques.

Three complementary data collection techniques were used for the countermeasure evaluations.

1. Direct recording of specific events as defined by the evaluation criteria. In this approach, the observer recorded, in tabular form, only those events which had been identified by the criteria. The advantage of this technique was its simplicity and ease of use, in both the collection and processing of data. This technique was most appropriate for recording data on traffic volumes and turning movements, and pedestrian travel volumes, since these were specific events requiring no interpretation by the observer.
2. Time lapse photography. The actions and behavior of persons using the countermeasures were recorded on film. A time lapse interval was chosen which balanced the need to conserve film, for economic reasons, with the need for sufficient detail to enable the recording of specific events, such as when a person's foot first leaves the curb. Investigation indicated that a film speed of 6 frames per second was the optimum for the purposes of this work. Time lapse photography provides a permanent record which can be reviewed as often as desired, and it provides a record of all events occurring during the use of a countermeasure—including unexpected but potentially significant events, as

well as those which were set out in the evaluation criteria. Time lapse photography was used to record the responses of special groups of handicapped individuals.

3. Pedestrian 'tracking'. All relevant events were recorded on forms which were map representations of the intersections (or other locations) where the countermeasures were installed. With this technique, the precise paths followed by the pedestrians were recorded, as were all events which occurred as they proceeded along those paths.

Pedestrian 'tracking' was used to record pedestrian behavior, the impact of countermeasures on pedestrian safety, and to measure the reduction in barriers to mobility and increased accessibility for the elderly and handicapped.

Development of the Test Plan.

An important consideration in scheduling the data collection was the generation of a quantity of data sufficient to yield statistically useful results, without exceeding manpower or budgetary limitations. This was complicated by the fact that the evaluations were being performed in six cities scattered around the United States.

Approximately 100 observations of pedestrian behavior (factors relating to safety and mobility) were needed to yield useful data for each location. With the recording techniques being used (the recording of tracks or paths followed by the test subjects on a map representing the test location, plus a small amount of additional data about each subject) it was determined through pre-testing that $1\frac{1}{2}$ to 2 minutes per observation was required. This then, indicated a total required time of between 3 and $3\frac{1}{2}$ hours, at each location, to complete this portion of the evaluation.

Because each of the test locations had heavy traffic and pedestrian travel flows, significantly less time was required to record traffic flow data. Less than 1 hour each, for vehicular and pedestrian flows, was determined to be sufficient.

The scheduling of the evaluations, insofar as day of week and time of day, were also given careful consideration. Generally speaking, the selection of a typical, or average, day was considered the best way to proceed. Normally, this was a weekday, and in most cases, Tuesday through Thursday. However, of greater importance was the requirement that the selected day of the week have a substantial amount of pedestrian and vehicular activity; therefore, in some situations, such as when the evaluation site was located within a shopping district, for example, evaluation on a Saturday was considered advantageous.

As for time of day, it was felt that the time periods when testing was scheduled should be representative of the significant use or activity periods during the day: the morning peak, mid-morning, the noon peak, mid-afternoon, and the evening peak. Therefore, five test periods were scheduled for each location. Each of the types of data to be gathered was collected during each of the periods. A typical data collection schedule was as follows:

SCHEDULE OF EVALUATION PERIODS:

City: _____ Site: _____

Test Period	Test Location	Control Location
1	8:00-9:20	8:00-9:20
2	9:40-11:00	9:40-11:00
3	11:40-1:00	11:40-1:00
4	2:50-4:10	2:50-4:10
5	4:30-5:50	4:30-5:50

Within each evaluation period, the following schedule of data collection was followed:

- 10 minutes—vehicular traffic volumes and turning movements.
- 10 minutes—pedestrian volumes
- 10 minutes—vehicular behavior (including stopping location with respect to crosswalk and stop line, and striking or overrunning curb or extended ramps).
- 40 minutes—pedestrian behavior (including safety and mobility factors).

Evaluation of the Use of the Countermeasures by Special Groups of Handicapped Persons.

It was unreasonable to expect a significant number of handicapped persons to pass through the countermeasure locations during the evaluation of their use by the general population. Yet, it was important to have handicapped individuals participate in the countermeasure evaluations, for the following reasons.

- While most countermeasures are expected to benefit all persons, they were designed specifically to solve problems of the elderly and handicapped.
- The problems of the elderly and handicapped, particularly with respect to mobility, are more severe than are those of the general population. To have this group evaluate the countermeasures provides an important measure of their effectiveness.
- Certain of the improvements (for example, the textured, resilient furniture strip) are of benefit only to the handicapped. Only testing by appropriate groups of individuals can determine the effectiveness of these measures.

Members of two subgroups of handicapped persons were considered especially important to bring to the sites to evaluate the countermeasures. These were the visually handicapped, and persons using wheelchairs. It was felt that these groups

should participate in the evaluations because their needs differ from each other. Improvements which benefit one group may be harmful or dangerous to the other. A curb ramp installed for wheelchair users, for example, if not well designed or carefully located, may be hazardous to the blind.

The Pedestrian Research Laboratory contracted with organizations and agencies that work with the handicapped in each of the cities where evaluations were being undertaken. These organizations were asked to do the following:

1. Organize two groups of test subjects. The first group was made up of persons who were legally blind, but environmentally active. At least 80 percent of these subjects were totally blind, or able to perceive light only. The remaining members of the group had other vision conditions within the legal definition of blindness.

The second group was to be made up of persons using wheelchairs that were environmentally active. Several of the subjects used powered chairs, but the majority of the subjects used manually propelled chairs.

The groups were formed from between 15 and 25 test subjects.

2. Describe in a general way the nature of the evaluations to each of the test subjects. Include a description of the purpose the evaluations serve in the research efforts of the Pedestrian Research Laboratory. Ensure that the subjects had an understanding of why their help was being sought, and of the value of the research findings to improving safety and mobility for the elderly and handicapped.
3. Ensure that the test subjects were familiar with the test locations including the general design of the countermeasures. This was of particular importance with the blind subjects. Our interest was in observing how the subjects would use the various design features (curb ramps, surface textures, railings, benches, poles, etc.); we were not interested in determining how the subjects would adapt themselves to a new environment. The rationale was that the blind generally navigate by learning a particular route, and each of the events along that route.
4. Have the test subjects sign the informed consent form. Of concern to The Pedestrian Research Laboratory was the possibility of subjecting the members of the test groups to hazard or danger, since the test locations were in a real-life, heavily trafficked situation. It was felt that the countermeasures were, if anything, safer than the standard city designs they replaced. Yet there remained the consideration that the test subjects would be at those locations, at that specific time, only because of the evaluation program, and that an accident might occur.
5. Transport the test subjects to the evaluation location(s) on the date and at the time when the evaluations were to take place. The Pedestrian Research Laboratory performed the evaluations of these special groups and recorded the

data, as they had done with the evaluations by the general population. The test subjects, upon arrival at the evaluation location, were housed in a location (usually indoors) where they awaited their turn to use the countermeasures, and from which they could not observe the performance of other subjects during the test. This last consideration was intended to avoid the possibility of learning from the experiences of others, or, of competitiveness contaminating the results.

The subjects were asked to use the countermeasures individually. They were given a prescribed route to follow, which took them through the test location in several different ways, from several directions. A member of the research team remained a short distance from the test subject throughout the evaluation, not so close as to influence the use of the countermeasures, but close enough to intervene if a test subject embarked on a seriously dangerous path, or was otherwise in need of assistance.

Time lapse photography was used to record the evaluations. The cameras and operators were located in an unobtrusive location above the test location, such as on the roof of a building. Super-8 cameras were used, photographing at 6 frames-per-second, a speed which gave sufficient detail without using an excessive amount of film. Zoom lenses on the cameras permitted closing in on events during the evaluations, such as the effect of a $\frac{1}{2}$ " (1.27 cm) lip at the bottom of a curb ramp on the front wheels of a manually powered chair. Researchers also used the 'tracking' recording technique (described earlier) to supplement the time lapse photography.

The measures which were used in the evaluation of the countermeasures by the special handicapped groups were the same measures of pedestrian behavior as were used for the general population use evaluation. These included:

- The impact of the countermeasure on pedestrian safety.
 - Pedestrians crossing outside a marked or unmarked crosswalk.
 - Pedestrians moving at faster than their normal speed.
 - Pedestrians moving at slower than their normal speed.
 - Pedestrians stopping within the street while crossing.
 - Pedestrians stepping into the street as if to begin crossing, then returning to the sidewalk.
 - Pedestrians changing course, or swerving while crossing.
 - Pedestrians beginning to cross against a "don't walk" or a red traffic signal.
 - Pedestrians caught in the intersection by a signal change to red.

- The impact of the countermeasure upon mobility.
- Difficulty in negotiating the level change at the curb—or entering the street, and on leaving the street.
- Length of time required to cross the street.
- Pedestrians still within the street, in the act of crossing, at the time the traffic signal turns to red.
- Use of a ramp, or curb cut, to negotiate the level change at the curb; or use of the curb to negotiate the level change on entering the street, and on leaving the street.
- Change in direction in order to use the ramp or the curb; or change in direction to avoid the ramp or the curb; to negotiate the level change—on entering the street and on leaving the street.
- Maintaining a preferred or safe course; along the sidewalk or within a crosswalk.
- Conflict with street furniture or other obstructions.
- Effectiveness of tactile cues in establishing and maintaining a proper course, and in avoiding hazards.

Statistical Analysis.

Some of the measures observed in this experiment were continuous, quantitative variables. These measures were analyzed, using the analysis of variance. It was felt that the underlying assumptions of the analysis of variance were not substantially violated. The least significant difference method based on the T-test was used to determine which treatment combinations differ. Computations were performed using the Statistical Package for the Social Sciences program available at Georgia Institute of Technology.

However, most of the measures observed were "count" or attribute type data. For these data, it was necessary to test the equality of four proportions. The appropriate test procedure for these responses is a contingency table. Because of the relatively small sample sizes involved, a test procedure based on the arcsine of the observed proportion was employed. Multiple comparison procedures were used to determine which pairs of proportions differ. A computer program was written for this purpose.

Using both these two techniques, population parameters are declared to be significantly different if the test statistic for the hypothesis exceeds the 90th percentile of the distribution of the test statistic.

countermeasure 1

widening of sidewalk at intersection

Countermeasure No. 1 is a solution which is intended primarily to reduce pedestrian-vehicular exposure and confrontation by diminishing the time in which they must share the same space. It is applicable to locations where parking is permitted in the curb lane 24 hours per day. Its basic design feature is the widening of the sidewalks out to the edge of the parking lanes, as illustrated in Figure 12. This countermeasure is proposed for wide streets; and where the signal time available for crossing may be insufficient, or where elderly and handicapped pedestrians may experience difficulties with so long a crossing. This design can reduce the total width to be crossed by two lanes, or by about one third of a six lane street.

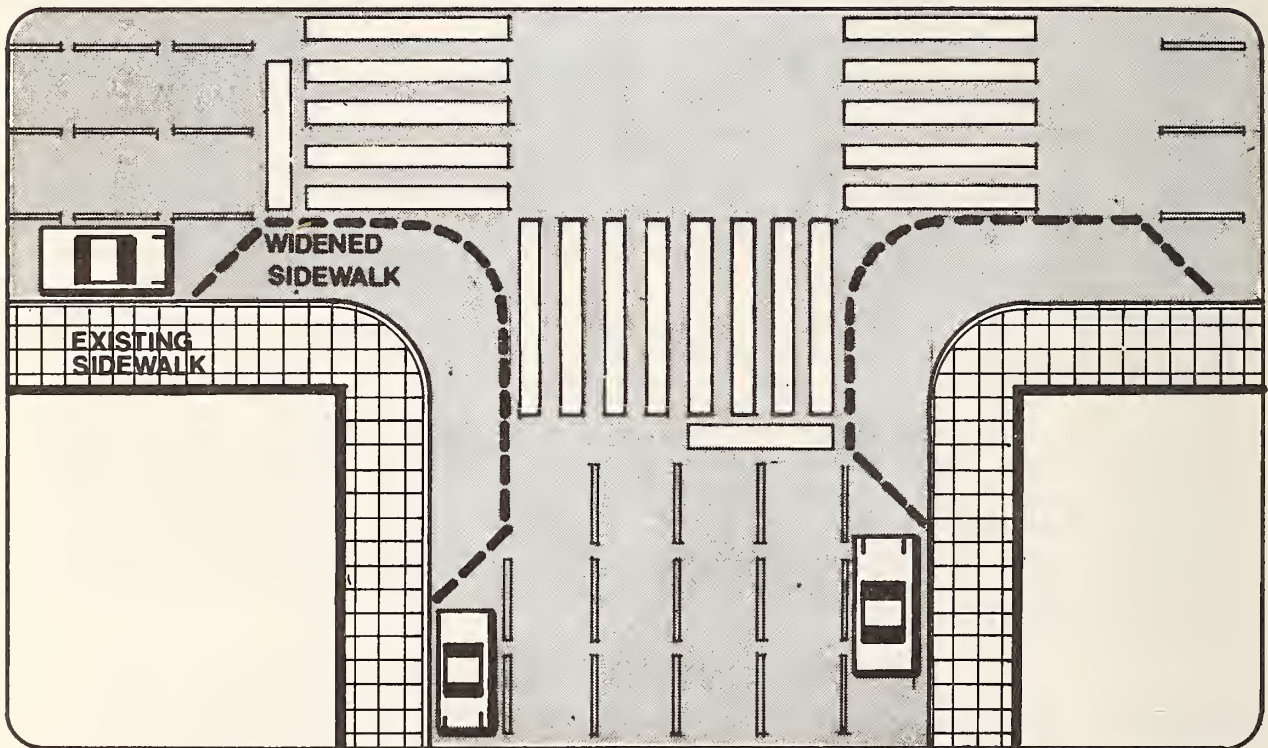


Figure 12 Widened Sidewalk Concept

As illustrated in Figure 13, there are other gains from this enlarged sidewalk. It enables pedestrians who are forming a "platoon", preparatory to crossing the road, to stand out of the stream of other pedestrian traffic. It provides additional space for a rest area, with benches, to permit those with little stamina to rest before or after the effort of crossing the street. It allows for space for locating curb ramps out of

the way of pedestrian traffic on the sidewalks. It can prevent vehicles from parking in the crosswalk. And finally, it permits street furniture to be located close to the corner and out of the flow of pedestrian traffic.

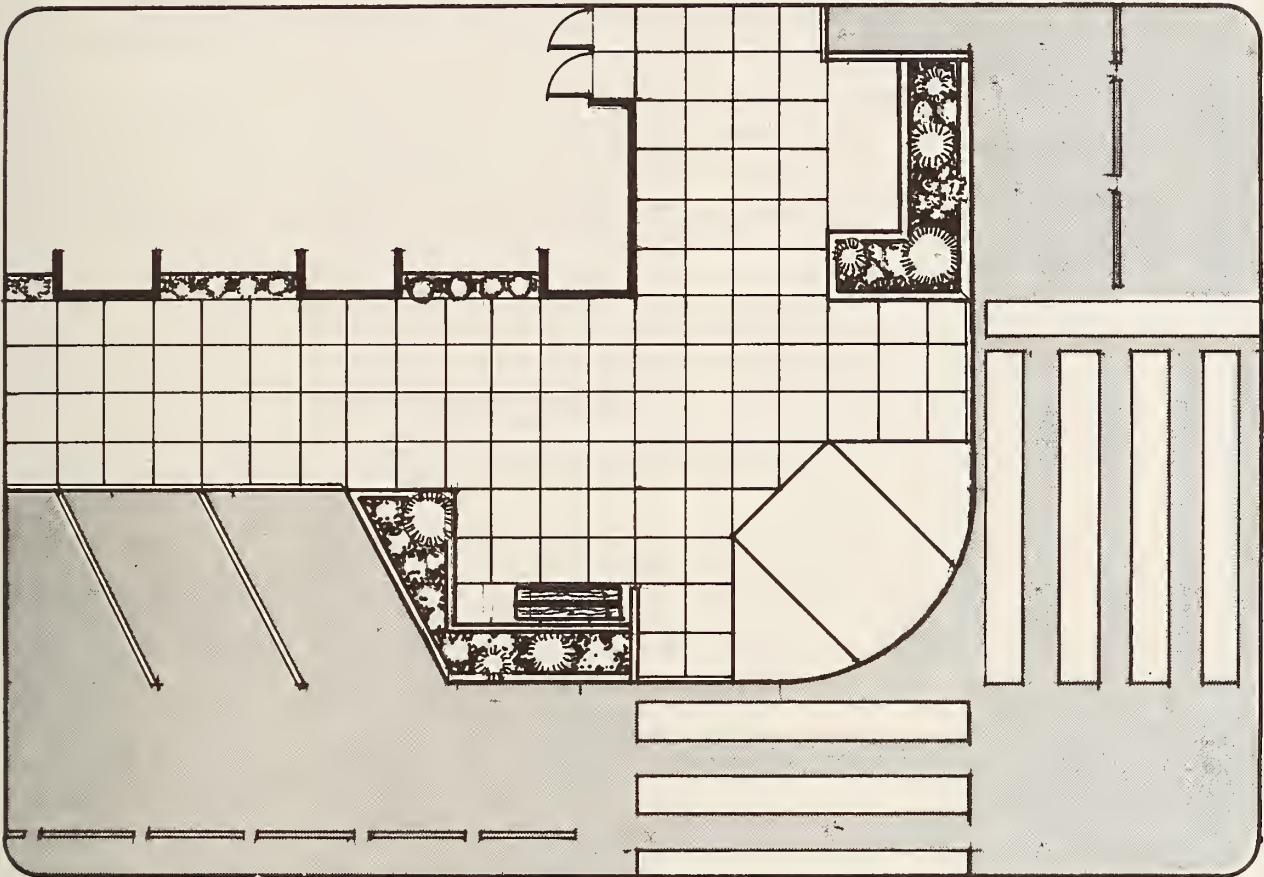


Figure 13 Benefits of Widened Sidewalk

Evaluation of Countermeasure No. 1.

Countermeasure No. 1 was evaluated based on its performance in solving the following problems:

- Long walking distances and exposure to traffic in crossing wide streets.
- Insufficient time for persons with slower than average walking speeds to cross a wide street.
- Pedestrians waiting to cross the street who are screened from view behind parked automobiles.
- Poor view of traffic by pedestrians wishing to cross a street.
- Insufficient space at a signalized intersection for platoons of pedestrians to assemble without blocking the sidewalk.

- Insufficient space on sidewalks for benches and other street furniture.
- Difficulty on the part of some elderly and handicapped pedestrians in negotiating the grade change at the curb.

In addition, as discussed earlier, evaluations were made of the general use of the locations where this countermeasure was installed, and of pedestrian behavior which might indicate problems in safety or mobility.

Evaluation Locations.

Countermeasure No. 1 was evaluated in three cities, Sioux City, Iowa; Boulder, Colorado; and Athens, Georgia.

I. LOCATION NO. 1: SIOUX CITY, IOWA

The city has an active downtown; the countermeasure was located within the principal shopping district, immediately adjacent to a major department store, within 1 block of a pedestrian mall, and within 2 blocks of several municipal parking structures (see Fig. 14).

Both streets which pass through the intersection are one-way routes. Both allow 3 lanes for moving traffic, plus two lanes in which parking is permitted 24 hours per day. Traffic volumes are fairly heavy, and are about equal on both streets, averaging about 750 vehicles per hour, or 500 vehicles/lane/hour of green on each street, (see Table 36).

TABLE 36. TRAFFIC VOLUMES - VEHICLES PER HOUR

SIOUX CITY: TEST SITE

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	500	535	550	700	815	600
Turning	60	110	125	170	175	130
Total	560	645	675	870	990	730
East-West Direction						
Thru	447	465	543	582	774	562
Turning	168	198	198	231	237	206
Total	615	663	741	813	1011	768

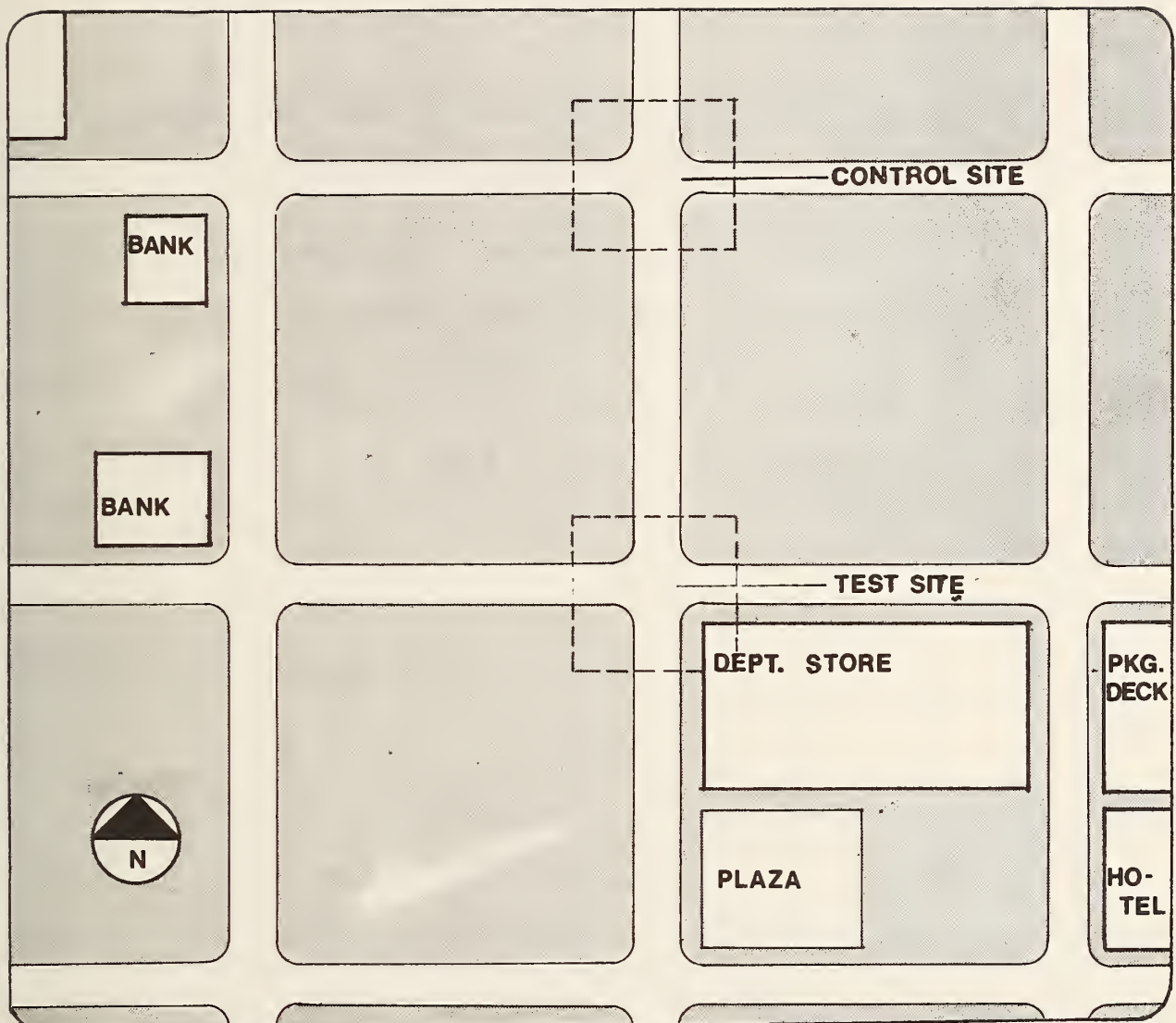


Figure 14 Sioux City: Test Location Map: Extended Sidewalk

The intersection is signalized, with right or left 'turn' arrows as applicable, and pedestrian 'walk'/'don't walk' indicators. Signal timing is set to allow a period for turning traffic at the beginning of the cycle before the pedestrian 'walk' period begins. The signal cycle is as follows:

	Elapsed time
Green, green 'turn' arrow	0 seconds
'Turn' arrow off, 'walk' begins	10
Flashing 'don't walk'	19
Yellow, continuous 'don't walk'	30
Red	33
Green	66

Note: signal time is equally distributed between the two directions.

All sidewalks are 14 feet (4.27 m) in width. Pedestrian flows were measured in the south and west crosswalks, and varried between average hourly flows of 150 and nearly 580 pedestrians per hours in both directions, (see Table 37).

TABLE 37. PEDESTRIAN FLOWS - PERSONS PER HOUR

SIOUX CITY: TEST SITE

West and South Crosswalks

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	70	110	190	265	140	155
East-West	80	125	370	315	200	220
TOTAL	150	235	560	580	340	375

The pedestrian crossing signal allowed 19 seconds of "walk" for crossing the 60 foot (18.3 m) wide street in those crosswalks which are not affected by turning traffic (the west and south crosswalks). However, because of the 10 second "turn" arrow cycle affecting the other two crosswalks (the north and east), only 9 seconds of "walk" was allowed for those crossings, requiring a walking speed of 6.7 feet (2.04 m) per second to complete crossing within the "walk" allotted time. This "walk" interval does not include the clearance time allowed for pedestrians to complete their crossing during the flashing "don't walk" phase.

Control Site: Sioux City.

The intersection used as a control site for the evaluation was located one block to the north of the test site. Its design and use were essentially the same as the test location's before the countermeasure was constructed. Traffic volumes at the control intersection, shown in Table 38, were somewhat higher in the north-south direction, but substantially higher in the east-west direction than at the test site. Pedestrian volumes were lower in both directions than at the test site, see Table 39.

TABLE 38. TRAFFIC VOLUMES - VEHICLES PER HOUR

SIOUX CITY: CONTROL SITE

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	315	425	485	640	780	530
Turning	335	320	280	415	310	330
Total	650	745	765	1055	1090	860
East-West Direction						
Thru	605	485	620	775	850	665
Turning	160	155	190	215	260	195
Total	765	640	810	990	1110	860

TABLE 39. PEDESTRIAN FLOWS - PERSONS PER HOUR

SIOUX CITY: CONTROL SITE

South and East Crosswalks

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	55	55	190	95	145	110
East-West	50	75	260	130	160	135
TOTAL	105	130	450	225	305	245

Description of Changes Made at Location No. 1—Sioux City, for Countermeasure No. 1, Extended Sidewalk.

Figures 15 and 16 illustrate the changes made at Location No. 1. Only the southwest quadrant of the intersection was modified (except for the addition of curb ramps at the other corners); and the sidewalk widening was made in only one direction, into the parking lane of the east-west street.

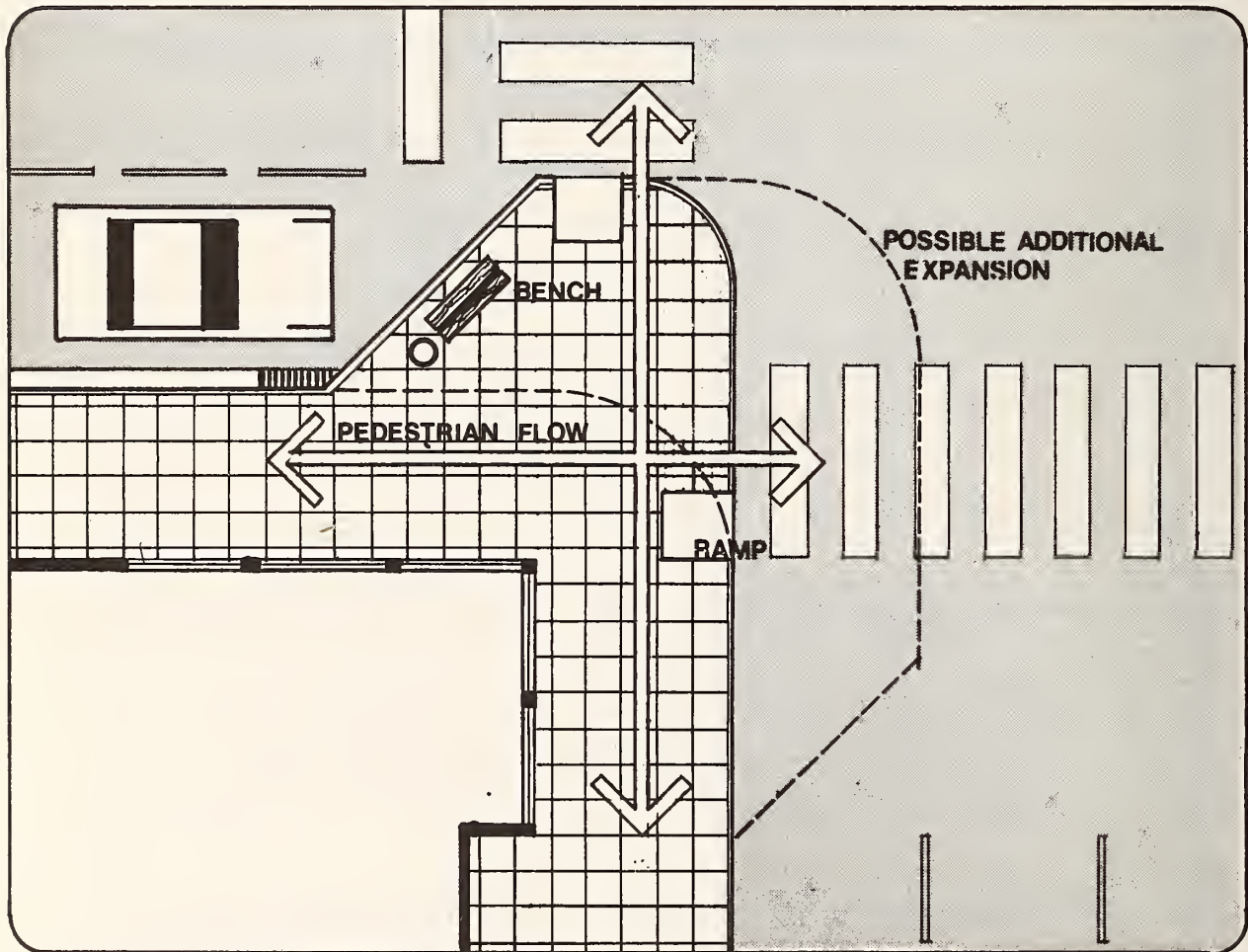


Figure 15 Sioux City: Test Location: Design Features

The widening added approximately 200 square feet (18.6 m^2) to the quadrant sidewalk area. Space was provided for two curb ramps, each 40 inches (101.6 cm) wide, with railings on either side of both ramps. Also provided was a wood bench, approximately 4 feet (1.22 m) in length, as well as additional space for pedestrians to gather, and form platoons, to await the crossing signal.

The decision to widen only the one quadrant, in the one direction, was based on the city's desire to minimize construction costs, as well as to avoid affecting turning movements at the test intersection.

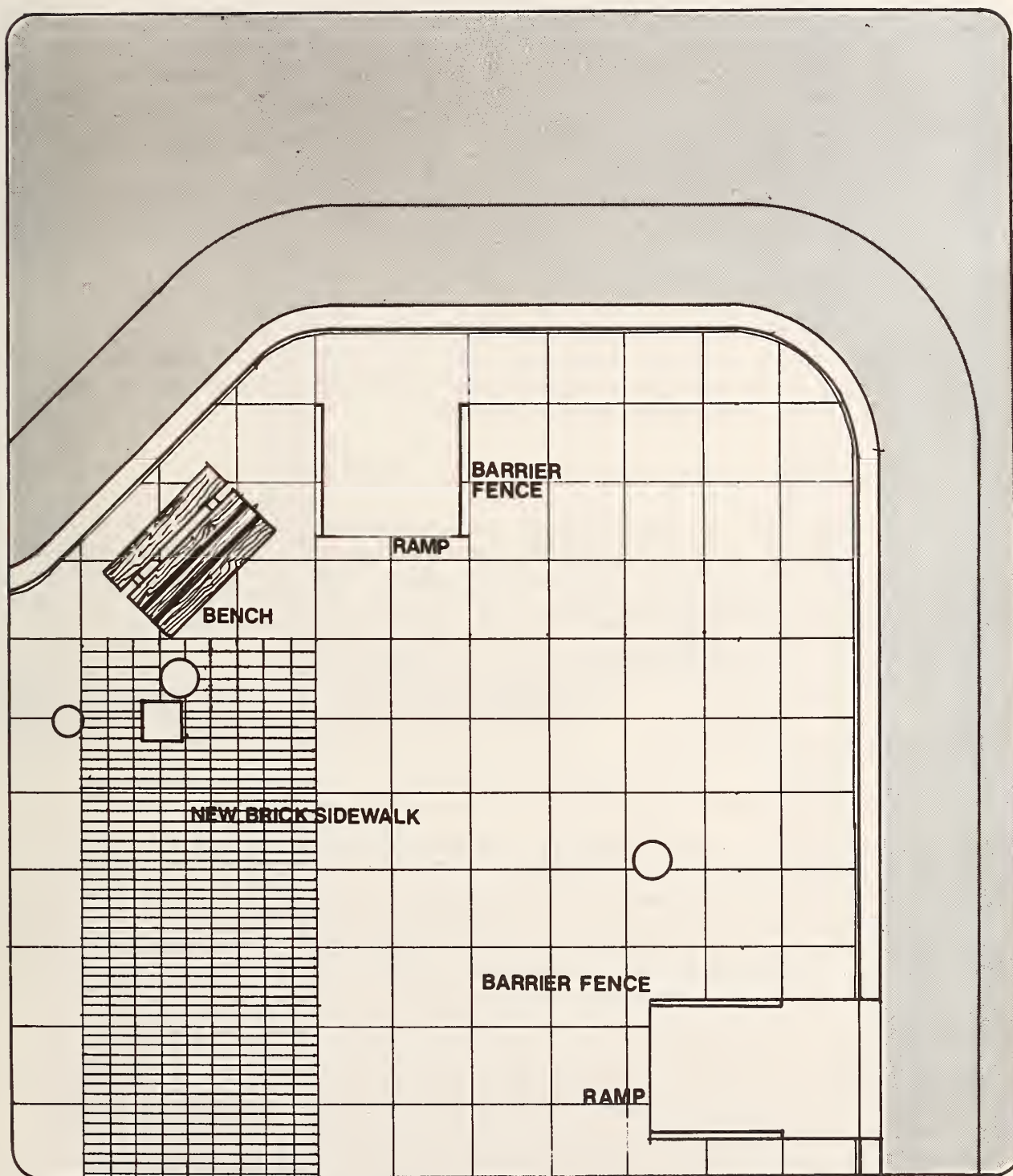


Figure 16 Sioux City: Test Location Map: Layout Test Site

The amount of widening was also restricted by the location of an existing storm inlet located less than 24 feet (7.32 m) from the corner on the west street. These restrictions resulted in the curb ramps being located in a less than optimum location. They were directly within the path of normal pedestrian flows instead of being offset from those flows, as recommended in The Implementation Package, "A Manual for Developing a Priority Accessible Network." As a result, the ramps and the railings were in a location not well suited to the needs of the visually impaired.

Other problems associated with the specific design of this countermeasure installation included the following:

- The handrails did not extend to the curb, but were stopped 21 inches (53.34 cm) short of the curb because it was feared they might interfere with snow removal. As a result, they may be too far removed from the street to be of assistance to someone having difficulty ascending from the street (either someone in a wheelchair, or for example, an elderly person with little strength, attempting to use the handrail while stepping onto the curb).
- The location of existing utility poles were left unchanged.

The final designs for the countermeasure were prepared in Sioux City based upon schematic proposals from the Pedestrian Research Laboratory. Many compromises were necessary but the design that was built was adequate for the tests. Of the three cities in which this countermeasure was evaluated, Sioux City was the only one in which the researchers participated in the design process. In the other two cities, the design work was already complete before the cities agreed to participate in the evaluations, and no changes were possible.

Evaluation Results--Location 1

1. Effect of Countermeasure Design on Traffic Flows.

This countermeasure was evaluated with respect to its possible effect upon the flow of traffic through the intersection in which it was located. The intent was to determine if the widening of the sidewalk at the intersection would impede the flow of traffic in any way. The widening was only to the north, into the curb parking lane of the east-west street. Since it did not extend as far as the moving traffic lanes, it was not anticipated that any restriction on flows would result. Whatever unanticipated effects might occur would be only in the east-west street, since no change in curb location occurred in the north-south street.

The results of the traffic observations (see Fig. 17) indicated an increase in vehicular flow on the east-west street. Flows, including turning movements, averaged about five percent higher after the countermeasure was installed than they had been before. Corresponding measurements at the control location showed an essentially unchanged flow (a decrease of less than one percent). As there was no evidence of a general increase in traffic flow in the area, these data support the assumption that the countermeasure had no adverse effect upon vehicular traffic.

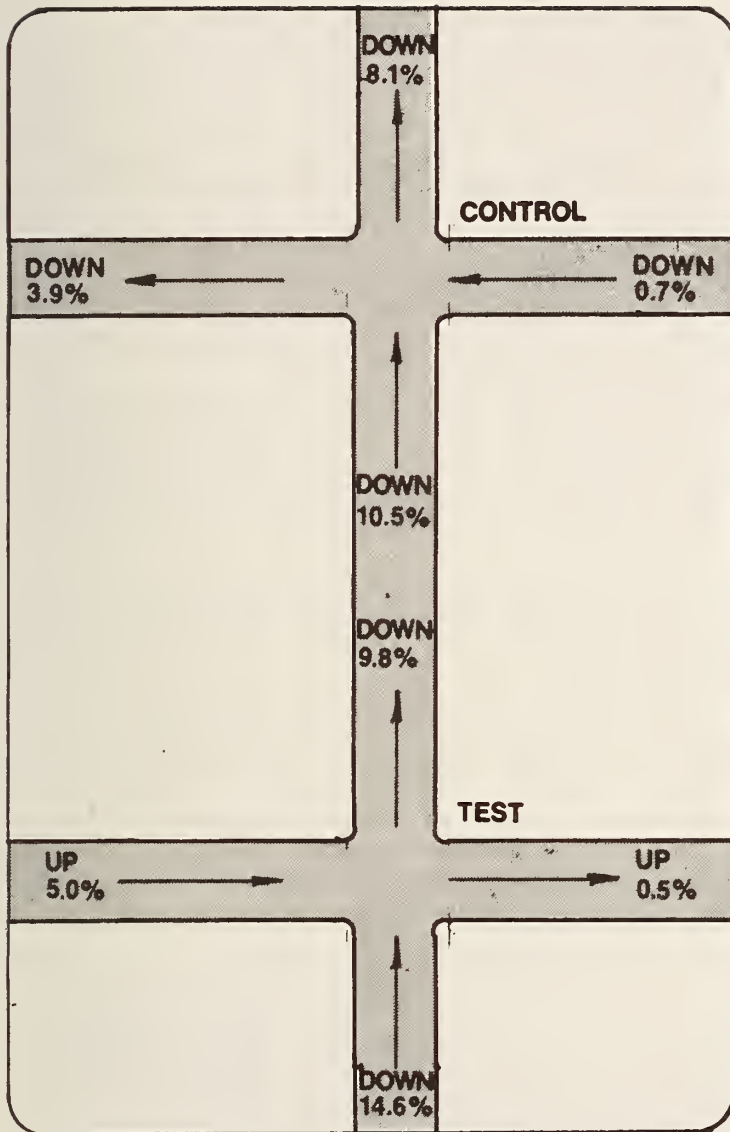


Figure 17 Sioux City: Test Site:
Vehicular Traffic Flow

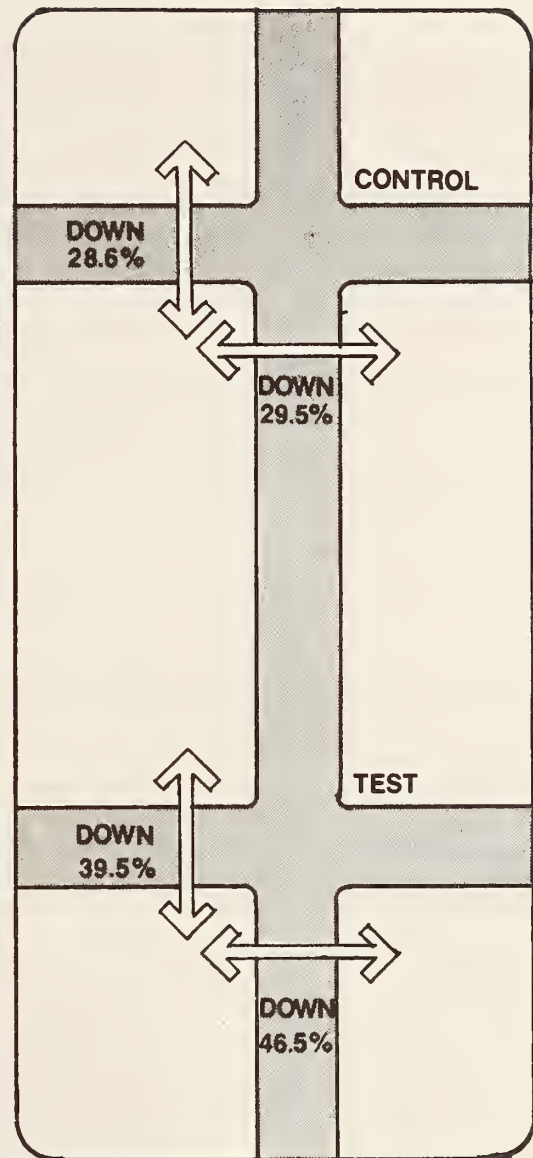


Figure 18 Sioux City: Test Site:
Pedestrian Traffic Flow

Traffic flows on the north-south street declined at both the test and control location. It is not likely that this resulted in any way from the installation of the countermeasure. (A building under construction one block south of the test location might have affected traffic flows during the after-tests).

2. Effect of Countermeasure Design Upon Pedestrian Flows.

The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians. The intent was to determine if the countermeasure design would have any effect in attracting or discouraging pedestrian use.

The results of the evaluations indicate a decrease in pedestrian flow through the test intersection (see Fig. 18), a decrease of about 43 percent in the two crossing directions observed. This compared with a corresponding 29 percent decrease at the control location.

Factors other than the design of the intersection might have affected pedestrian flows in such a way as to yield these results. The test location is closer than the control site to the center of the downtown shopping district of this city, and had substantially greater pedestrian activity. It is probable, therefore, that the test site would have been more heavily affected by an overall decrease in downtown shopping activity. There was evidence of reduced pedestrian activity at all intersections studied in this city (as discussed in the evaluation of other countermeasures).

In addition, the store at the south-west corner of the test location was vacant during the after-test period, which may have influenced the results.

The tests before the countermeasure was installed took place in July 1976, and the tests after the installation, in September, 1977.

3. Effect of the Countermeasure Design on the Drivers' View of Pedestrians; and Upon the Pedestrian's View of Approaching Vehicles.

Figures 19 and 20 illustrate the increased visibility of vehicle and pedestrian resulting from the countermeasure design. Moving the pedestrians approximately 9 feet (2.75 m) forward puts them in a position where the driver's view is no longer obstructed by a parked vehicle, and where the pedestrians have a clear view of approaching traffic. The photograph, taken from the first traffic lane about 100 feet (30.5 m) to the west of the crosswalk, illustrates the improved view of pedestrians standing on the widened sidewalk area.

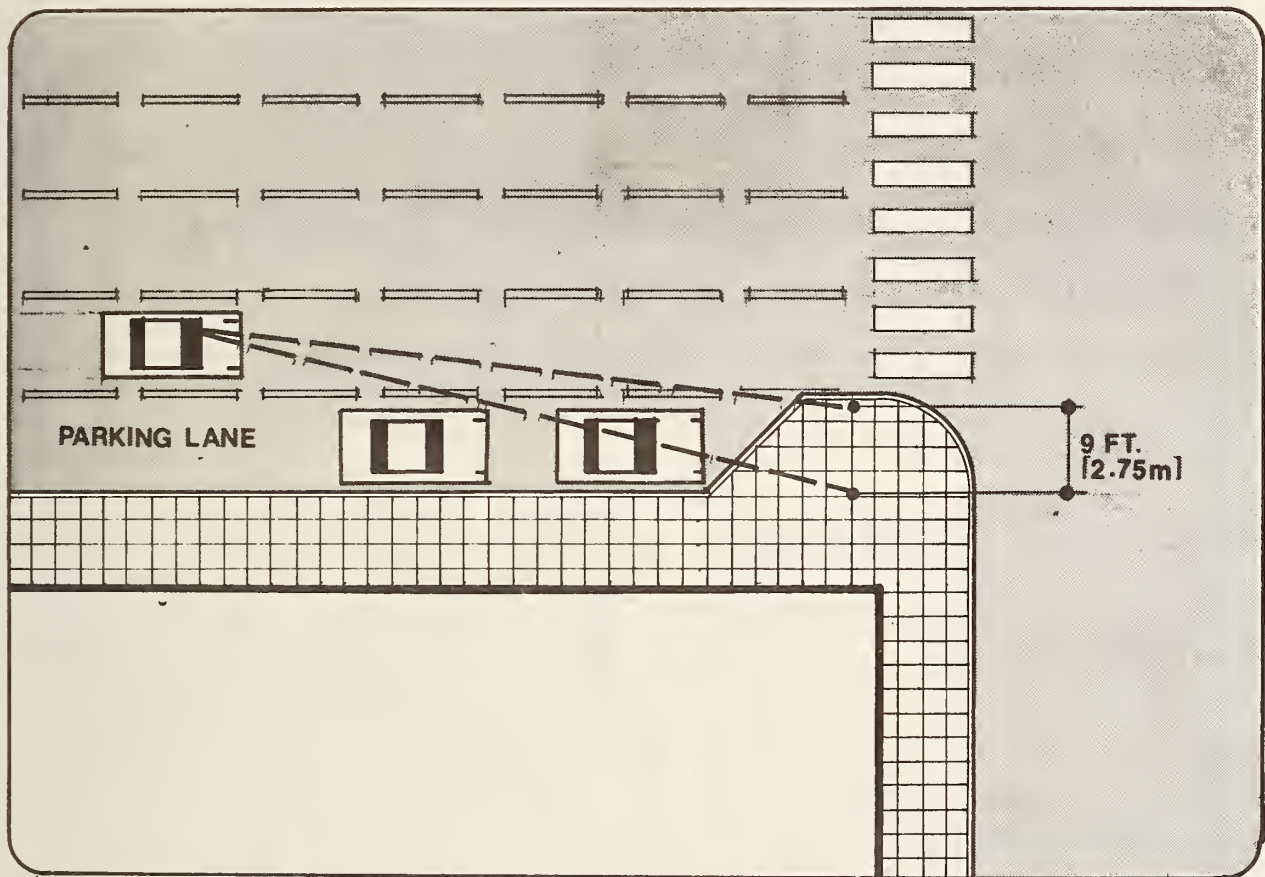


Figure 19 Sioux City: Test Location: Driver's View



Figure 20 Sioux City: Test Location: View of Test Site

4. Increase in Space on the Sidewalk for Platoons of Pedestrians to Form.

The drawings in Figure 21 illustrate the increase in platoon formation space resulting from the sidewalk extension. Before widening of the sidewalk, approximately 125 square feet (11.625 m^2) of space was available for platoons of pedestrians to form, much of this space obstructing normal flow on the sidewalk. With the sidewalk extension, approximately 285 square feet (26.5 m^2) of space became available for platoon formation. Of this total, 145 square feet (13.485 m^2) of space is on the sidewalk extension; platoons occupying this area would not obstruct the flow of persons using the sidewalk.

At an average of 5 square feet ($.465 \text{ m}^2$) per person, the size of the maximum platoon was increased from 25 to 57 persons; more important, a platoon of 20 persons could be accommodated without obstructing pedestrian flows.

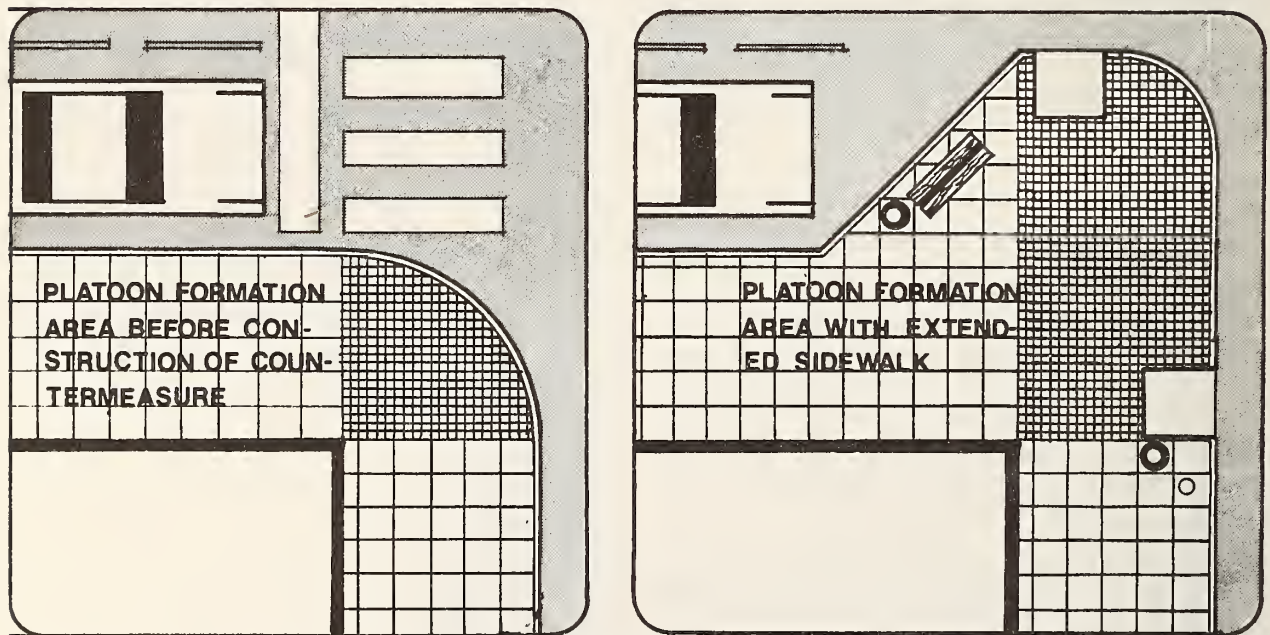


Figure 21 Sioux City: Test Location Map: Platoon Space

5. Space for Additional Design Features.

As Figures 15 and 16 illustrated, the sidewalk extension permitted the addition of a bench at this intersection. The design also permitted the location of the northernmost of the curb ramps out of the path of pedestrians travelling along the sidewalk in an east-west direction. If the extension had been larger, extending about four feet further to the west, the ramp would have also been completely out of the path of pedestrians travelling in the north-south direction. Similarly, extending the sidewalk into the parking lane of the street

to the east, as indicated on the drawing, would have permitted the location of the other ramp beyond normal pedestrian paths. Ramps located in this way are available for use by persons needing or desiring them, without being in the direct path of people with impaired vision.

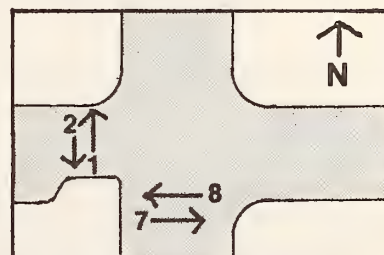
6. Crossing Time.

Countermeasure No. 1 was evaluated to determine whether it had any effect on the time required to cross the street. It was anticipated that the reduction in crossing distance, (from 52 feet to 44 feet) (15.85 m to 13.42 m), resulting from the extension of the sidewalk, would lead to a corresponding reduction in crossing time in the west crosswalk of the test intersection.

The results are shown in Table 40.

TABLE 40. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	13.5556	10.6400	13.4815	11.800	S	N
2	12.5789	9.8095	12.2000	11.7895	S	N
7	13.1923	11.9167	13.3077	11.4231	N	N
8	13.6667	12.6818	15.2857	12.9000	N	N

S = Significant

N = Not Significant

Conclusions.

There was a significant reduction in crossing time at the test site, and no significant reduction at the corresponding control site (nor at the east-west crossing at the test site where no changes had been made). The reduction in crossing time was about 22%, which approximately matches the reduction in distance to be crossed.

7. Effect of the Countermeasures Upon the Difficulty in Negotiating the Level Change at the Curb.

Curb ramps were installed as part of the countermeasure design at the test location. This evaluation was intended to determine the effect of these curb ramps on the general population using the test intersection. The evaluation was in two parts:

- (a) Determining whether the ramps were used;
- (b) Determining how the apparent difficulty of making the level change was affected by the installation of the ramps.

(a) Use of ramps. Table 41 illustrates the results of the evaluations of the use of the ramps.

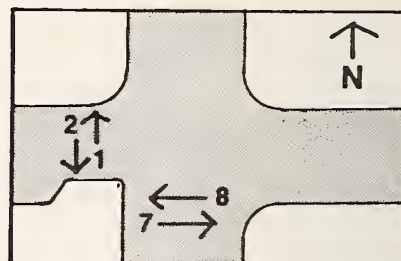


TABLE 41. PEDESTRIAN USE OF CURB RAMP

COUNTERMEASURE NO. 1; LOCATION NO. 1

Direction	Used ramp	Changed direction to use ramp	Changed direction to avoid ramp
1	14 / 30	0 / 30	0 / 30
7	17 / 30	0 / 30	0 / 30
2	11 / 23	0 / 23	0 / 23
8	4 / 24	0 / 24	2 / 24

From Table 41 we find that:

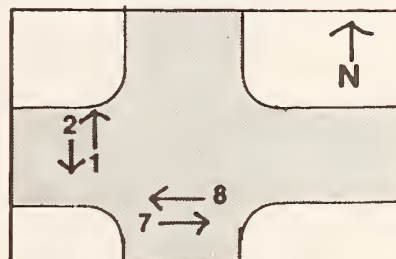
- approximately 50% of all the pedestrians used the ramps to enter the street;
- approximately 32% of all the pedestrians used the ramps to leave the street;

and, while no pedestrians appeared to obviously change direction to use the ramps, only two obviously changed direction to avoid using the ramps.

- (b) Apparent difficulty of level change. A four point scale was used to record observed difficulty of pedestrians in negotiating the level change at the curb. Separate evaluations were made for entering and leaving the street. The scale ranged from a difficulty factor of 1 (equivalent to walking on a level surface with no perceptible change in walking speed or gait), to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all—including needing to grab onto a pole for support or balance, or nearly falling). An average, able-bodied adult, mounting or descending a typical 8-inch (20.32 cm) curb, would score 2 on the scale.

Table 42 presents the results of this evaluation:

TABLE 42. APPARENT DIFFICULTY IN
MAKING LEVEL CHANGE AT CURBS: MEAN VALUES



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	2.1111	1.4800	2.0000	1.9500	S	N
7	2.1154	1.4583	2.0385	2.0000	S	N
2	2.0526	1.5714	2.0667	2.0000	S	N
8	2.1333	1.8182	2.0000	2.0000	N	N

S = Significant

N = Not Significant

It can be seen from Table 42 that there was no significant change in the difficulty that people had in negotiating the curbs at the control site. On the other hand, at the test site there was a significant improvement at the ramp on the extended sidewalk, and for people leaving the sidewalk to cross in the easterly direction. An improvement in the westerly

direction can also be seen, but this is not very significant.

8. Evaluation of the Countermeasures in Terms of Behavior Affecting Safety and Mobility.

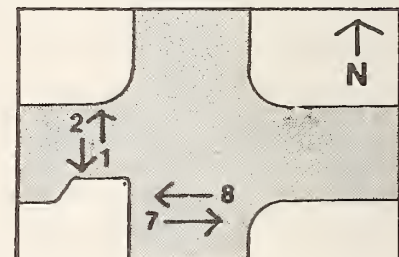
These included evaluations of:

- (a) Conflicts between crossing pedestrians and vehicles;
- (b) Conflicts between crossing pedestrians and other pedestrians;
- (c) Pedestrians waiting in the street for the crossing signal to change;
- (d) Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk;
- (e) Pedestrians crossing against the traffic signal.

- (a) Conflicts between crossing pedestrians and vehicles. As a measure of conflict, certain types of pedestrian behavior were measured—a sudden increase in speed, a reduction in crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

TABLE 43. CONFLICTS BETWEEN CROSSING PEDESTRIANS AND VEHICLES:

COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	4 / 22	1 / 30	5 / 31	8 / 27	N	N
2	1 / 21	2 / 24	4 / 17	5 / 28	N	N
7	1 / 33	4 / 30	4 / 29	2 / 27	N	N
8	1 / 16	1 / 24	3 / 18	3 / 21	N	N

S = Significant

N = Not Significant

From Table 43 it can be seen that there was no significant change in the number of pedestrian/vehicular conflicts as measured by this evaluation at the test and at the control site.

- (b) Conflicts between crossing pedestrians and other pedestrians. Pedestrian conflicts were recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

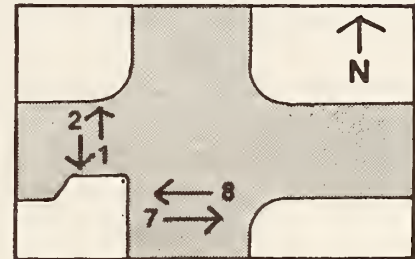


TABLE 44. CONFLICTS BETWEEN PEDESTRIANS AND VEHICLES:

COUNTERMEASURE NO. 1, LOCATION NO. 1

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	2 / 22	0 / 30	1 / 31	3 / 27	N	N
2	0 / 21	1 / 24	3 / 17	1 / 28	N	N
7	5 / 33	0 / 30	2 / 29	1 / 27	N	N
8	2 / 16	2 / 24	1 / 18	0 / 21	N	N

S = Significant

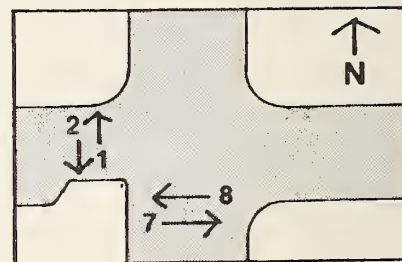
N = Not Significant

From Table 44 it can be seen that there was no significant change in the number of pedestrian conflicts.

- (c) Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing or decreasing the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, and were therefore vulnerable to traffic.

From Table 45, it can be seen that the only significant change was that there was an increase in people waiting in the street to cross in an easterly direction. The reason for this was not apparent.

TABLE 45. PEDESTRIANS WAITING IN THE STREET
FOR THE CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	0 / 22	0 / 30	0 / 31	1 / 27	N	N
2	0 / 21	0 / 24	0 / 17	0 / 28	N	N
7	0 / 33	6 / 30	0 / 29	2 / 27	N	S
8	0 / 16	0 / 24	0 / 18	0 / 21	N	N

S = Significant

N = Not Significant

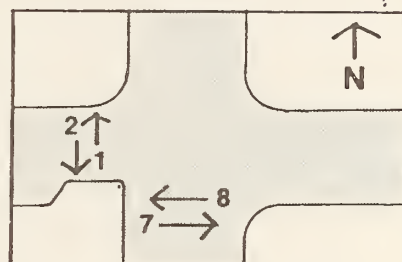
- (d) Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would walk and remain within the limits of the crosswalk.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

From Table 46 it can be seen that significantly more pedestrians stayed within the crosswalk when leaving the test site. There was no significant change at the control site, nor for pedestrians approaching the test site.

TABLE 46. PEDESTRIANS WALKING BEYOND LIMITS OF
CROSSWALK
COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	6 / 22	1 / 30	3 / 31	1 / 27	S	N
2	1 / 21	0 / 24	1 / 17	1 / 28	N	N
7	11 / 33	2 / 30	0 / 29	3 / 27	S	N
8	3 / 16	0 / 24	2 / 18	0 / 21	N	N

S = Significant

N = Not Significant

From Table 47 it can be seen that there was a significant increase in people leaving the crosswalk when leaving the test site in an easterly direction, and a similar pattern at the control site. There is no obvious explanation for these changes.

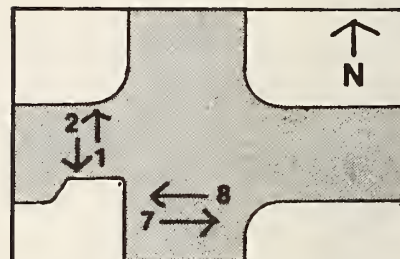
- (e) Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing or decreasing such behavior.

From Table 48 it can be seen that no significant change in the number of pedestrians crossing against the traffic signal occurred at the test or control sites.

TABLE 47. PEDESTRIANS LEAVING THE CROSSWALK

WHILE CROSSING THE STREET

COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	0 / 22	5 / 30	3 / 31	4 / 27	N	N
2	4 / 21	0 / 24	0 / 17	3 / 28	N	N
7	0 / 33	6 / 30	0 / 29	15 / 27	S	S
8	2 / 16	3 / 24	2 / 18	4 / 21	N	N

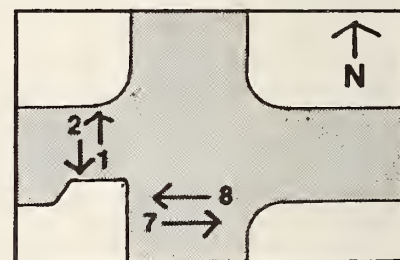
S = Significant

N = Not Significant

TABLE 48. PEDESTRIANS CROSSING AGAINST THE TRAFFIC

SIGNAL

COUNTERMEASURE NO. 1, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	2 / 22	3 / 30	2 / 31	2 / 27	N	N
2	4 / 21	0 / 24	0 / 17	3 / 28	N	N
7	6 / 33	1 / 30	2 / 29	3 / 27	N	N
8	2 / 16	3 / 24	2 / 18	4 / 21	N	N

S = Significant

N = Not Significant

Evaluation of Countermeasure No. 1 by Special Groups of Handicapped Persons

The procedures used in the special group evaluations were described earlier in this report, Methods. Two groups of handicapped persons were used for the evaluation of this countermeasure. These were the severely visually handicapped, and people in wheelchairs. The test subjects were brought to the site. They had been asked to familiarize themselves with the test locale. The test subjects used the test locations individually, following prescribed routes, while observations were made of their use of design features incorporated into those locations, and any apparent benefits or difficulties resulting from those design features. Time lapse photography (at six frames per second) was used to record the results of these evaluations. The results sought from this process were the same as sought in the observation of pedestrian behavior evaluations--the impact of the countermeasure design on pedestrian safety and on mobility. In addition, several specific questions relative to the needs and difficulties of these groups were addressed. These included the following:

- Visually Impaired Group:
 - Is the handrail useful; is it inconvenient;
 - Is the ramp detectable; can the visually impaired detect the points at which the ramp meets the street and the level sidewalk;
 - Is the ramp location and design satisfactory, or inconvenient.
- Wheelchair Group:
 - Is the handrail useful to persons in wheelchairs;
 - Is the ramp satisfactorily designed and located, or is it difficult to use.

Evaluation Results: Visually Impaired Group.

1. Usefulness of handrails to the visually impaired. Ten test subjects were involved in a total of 53 situations in which they entered or left the street at the quadrant on which the ramps with handrails were located. Of these 53 events, the subjects detected or made use of handrails for navigation purposes a total of 30 times, or on 57 percent of those occasions. Use of the handrails occurred more frequently in the north/south crossing, the direction in which the sidewalk widening was made. Also, there was more frequent use of the handrails on entering the street, presumably because of their location with respect to the normal path followed on sidewalks where the visually impaired make use of building frontages for navigation.

The following descriptions are typical of the use of the handrails by members of this group:

Entering street:

- The subject, moving along the sidewalk, approached the ramp. As he did so, he detected the railing with his cane and moved sideways slightly to align himself with the ramp. He then proceeded slowly down the ramp, and judging traffic conditions to warrant crossing, entered the street.
- The subject approached the ramp and detected the handrail with his cane. He slowed and stepped to the outside of the railing, so as to avoid the ramp. He proceeded to the curb, and waited for the traffic signal to change.
- The subject, walking north, reached the railing which he detected with his cane. He aligned himself with the ramp, and proceeded down towards the street. As his cane detected the slope change where the ramp met the street, he stopped, positioning himself at the ramp end.

Leaving street:

- The subject approached the ramp from the street. As he did so, his cane contacted the railing on the ramp's right side. The subject paused, felt with his cane, shifted position slightly towards the left, then proceeded forward and up the ramp.
- The subject approached the sidewalk on a line about two feet (.61 m) to the left of the ramp. His cane detected the curb, and as he stepped on the curb, detected the railing to his right. He then proceeded straight ahead onto the sidewalk.

It can be concluded that the handrails did provide useful locational information for the visually impaired, although nearly half the subjects did not make use of this information. It is possible that more people would use the handrails with increased familiarity of the general layout of the intersection.

2. Inconvenience caused by the handrails. One test subject inadvertently walked into one of the handrails during the 53 crossing events. The subject was walking north and, having strayed slightly off course, walked into the railing of the ramp leading to the east/west crosswalk. This incident occurred as follows:

- The subject, approaching the intersection from the south, walked into the railing for the east/west street crossing to his right. Then, after pausing and shifting slightly to his left, the subject continued forward with his right hand resting on the railing until he had passed it by. The subject proceeded very slowly, searching for the curb for some distance before reaching it. Finally, detecting the curb, he positioned himself in a location several feet to the right of the north/south ramp and waited for the proper time to cross.
- The railing the subject walked into was for the area of the quadrant which was not widened. Had it been extended in accordance with the original design recommendation, the event would not have occurred in this way.

3. Usefulness of the ramps to the visually impaired. Forty-five percent of the visually impaired subjects using the sidewalk extension to cross in a north/south direction used the ramp. Thirty five percent of those crossing in an east/west direction used the ramp to negotiate the level change.

Many subjects who used the curb rather than the ramp, were on a course which did not take them near enough to a ramp to use it. Therefore, it may be useful to identify which of those persons who detected a railing, and therefore were aware of the ramp location, chose to use the ramp, and which chose to use the curb.

Nearly three-fourths of those visually impaired subjects who detected the ramp handrail chose to use the ramp. Some of these persons detected the handrail while directly lined up with the ramp, and simply proceeded directly forward. It is also possible to separate out those subjects who detected the handrail while in a position where an equivalent choice of moving to one side or the other would lead them to the ramp or avoid the ramp, and those subjects who changed direction substantially; that is, went out of their way to use or avoid the ramp. Of these two groups of subjects, 60 percent chose to use the ramp, suggesting, at the least, that they considered the ramp useful, and not more difficult or hazardous to use than the curb.

4. Crossing time: visually impaired subjects. Table 49 compares the crossing time for the visually impaired group with the general population observations.

TABLE 49. CROSSING TIME

North/South Crossing	Blind Subgroup	General Population	% Difference
Mean	14.8 sec	10.2 sec	+ 45%
Min.	10.0	4.0	
Max.	23.0	20.0	
East/West Crossing			
Mean	17.4 sec	12.3 sec	+ 41%
Min.	13.0	9.0	
Max.	31.0	30.0	

Members of the visually impaired group required more than 40 percent additional time, on the average, to cross the same distance than members of the general population. Since the available time for crossing at this intersection was 19 seconds, one third of the subjects crossing in the east/west direction, the direction in which the sidewalk was not extended, were unable to complete their crossing in the available time; one fifth of the subjects crossing the north/south direction, the direction in which the sidewalk was extended, similarly could not complete their crossing in the allotted time.

5. Usefulness of ramps to wheelchair users. The ramps associated with the sidewalk extension appeared to be satisfactory to all wheelchair users. In all cases, the difficulty factor associated with use of the ramps to enter or leave the street was 1 (equivalent to travelling on a level surface with no perceptible change in travel speed) on a 4 point difficulty scale. The only apparent difficulty which affected the travel of members of this groups was pedestrians occupying space on the ramp or immediately in front of it, causing time delays to the wheelchair users. Several examples of this problem are illustrated below. On one occasion, pedestrians on the ramp caused a wheelchair user to use the sideslope on the outside of the railing to leave the street. The subject was willing to accept the small jolt of riding up the low half-curb, rather than wait for the ramp to clear.

Entering street:

- As the subject approached the street, he had to stop and wait short of the ramp because it was filled with four pedestrians. After the light changed to green, a bit more than two seconds of time elapsed for the subject to reach the street.

Leaving street:

- As the subject reached the midpoint of the street crossing, he had to slow to wait for pedestrians in front of him. The pedestrians finally did not use the ramp; the subject crossed behind them and ascended from the street.
- As the subject approached the ramp, a pedestrian was standing on the ramp waiting and holding onto the handrail. The subject came almost to a complete stop before the pedestrian recognized the problem and stepped back off the ramp onto the sidewalk so the subject could enter and ascend the ramp.
- The subject approached the ramp and saw that it was occupied by several pedestrians. Without slowing, the subject decided to pass to the outside of the ramp handrail and climb the half curb and ramp sideslope. He experienced some jarring from the curb lip.

6. Usefulness of the handrail. One of the wheelchair subjects made use of the handrail to assist with his ascent of the ramp:

- As the subject approached and entered the ramp he slowed considerably, then reached out with his left hand, grabbed onto the railing and helped pull himself up the ramp.

7. Crossing time: wheelchair subjects. Table 50 compares the crossing times for the wheelchair user group with the observations of members of the general population.

TABLE 50. CROSSING TIME

North/South Crossing	Wheelchair Subgroup	General Population	% Difference
Mean	13.0	10.2	27%
Min.	10.0	4.0	
Max.	17.0	20.0	
East/West Crossing			
Mean	15.7	12.3	27%
Min.	8.0	9.0	
Max.	24.0	30.0	

Members of the wheelchair group took 27% longer than the general population, on the average, to cross the same distance. The available crossing time at this intersection is 19 seconds. Three of the twenty east/west crossings, the direction in which the sidewalk was not extended, required more than the available time. All north/south crossings, the direction in which the sidewalk extension reduced the crossing distance, were completed in less than 19 seconds.

2. LOCATION NO. 2: BOULDER, COLORADO

The countermeasure was constructed on the fringe of the central business district of Boulder, Colorado, a western city of less than 80,000 population (see Fig. 22). The city has an active downtown, particularly in the vicinity of a new and successful pedestrian mall, located 2 blocks to the south of the test intersection. The intersection is less than one block from a large restored heavily used downtown hotel; however, the predominant uses in the immediate vicinity of the test location are residential—large, single family structures, some of which have been converted to multi-unit or institutional uses.

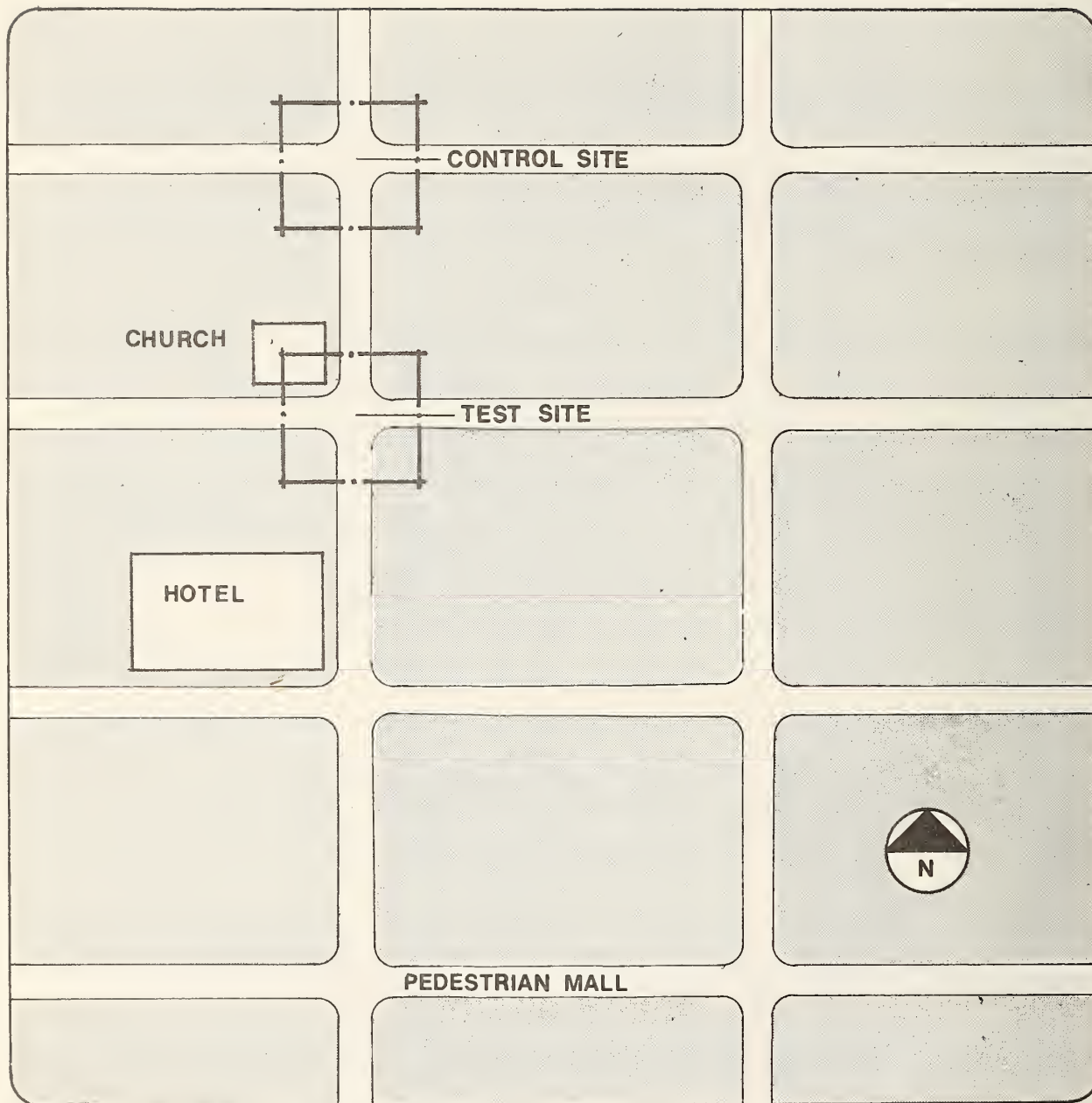


Figure 22 Boulder: Test Location Map

Both streets passing through the intersection allow one lane of moving traffic in each direction, plus two lanes in which parking is permitted 24 hours per day. Traffic volumes average 500 vehicles per hour in the east-west direction, and nearly 300 vehicles per hour in the north-south direction as shown in Table 51.

The intersection is not signalized. The east-west street is a through street, with its traffic having the right-of-way. Stop signs interrupt traffic flow in the north-south direction.

TABLE 51. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	70	35	55	75	170	80
Turning	115	185	250	215	265	205
Total	185	220	305	290	435	285
East-West Direction						
Thru	330	415	315	390	680	425
Turning	40	40	85	65	125	70
Total	370	455	400	455	805	495

Sidewalks are separated from the curb by a grass strip (see Fig. 23). Pedestrian flows were measured on all sidewalks in both directions. Pedestrian traffic was light at all times, varying from as low as 5 pedestrians per hour to as high as 125 per hour. Pedestrian flows are heaviest in the north-south direction (leading to and from the downtown business district), which is perpendicular to, and crosses, the heavier vehicular flows, see Table 52.

TABLE 52. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	50	50	125	40	80	70
East-West	5	20	20	15	10	15
TOTAL	55	70	145	55	90	85

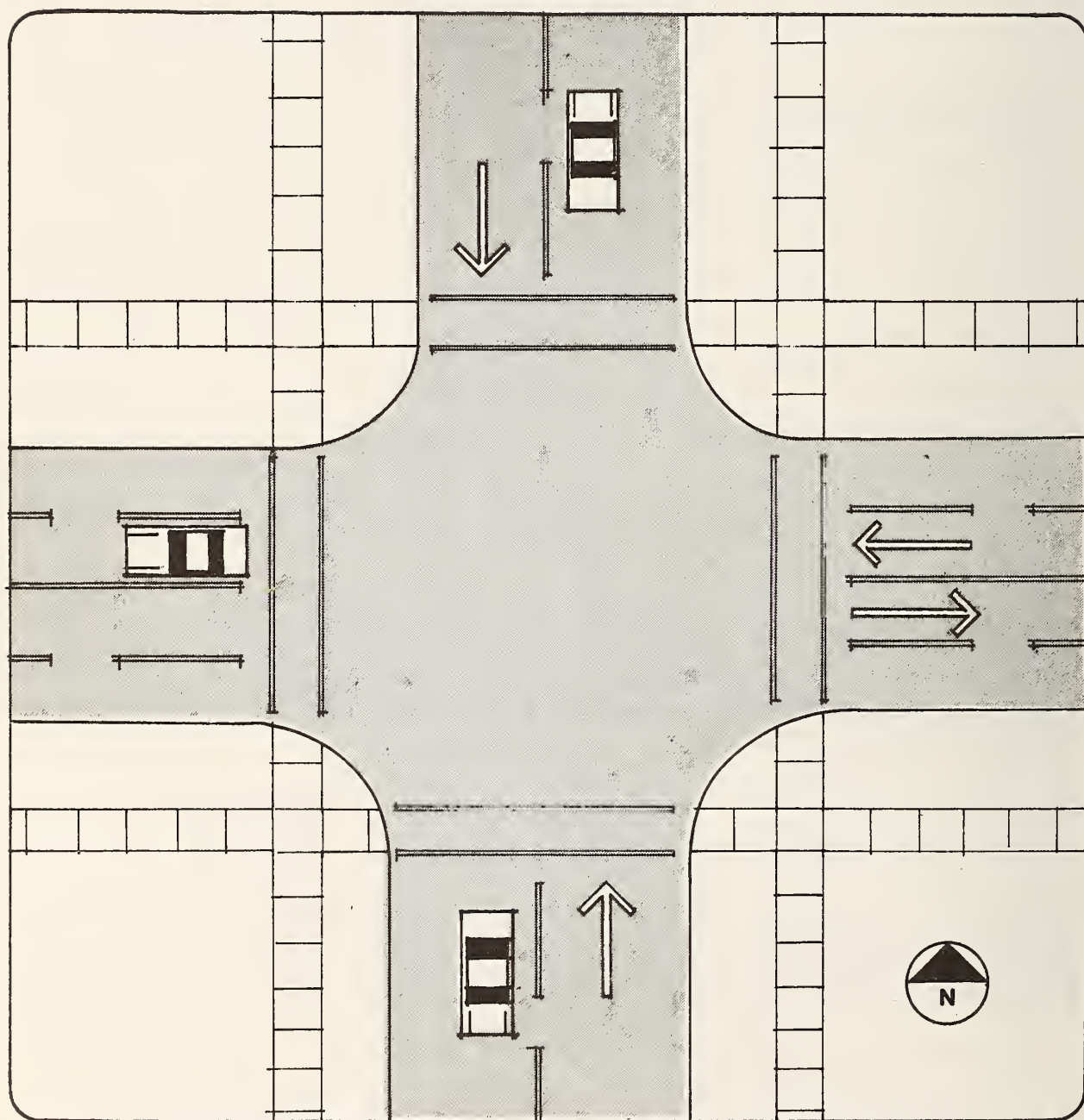


Figure 23 Boulder: Extended Sidewalk: Test Site Before Modification

Description of Changes Made at Location No. 2.

Figure 24 illustrates the changes made at Location No. 2. All four quadrants were widened into the area formerly occupied by parking lanes. The portions of the sidewalks located nearest to the intersections were repaved in brick. The crosswalks

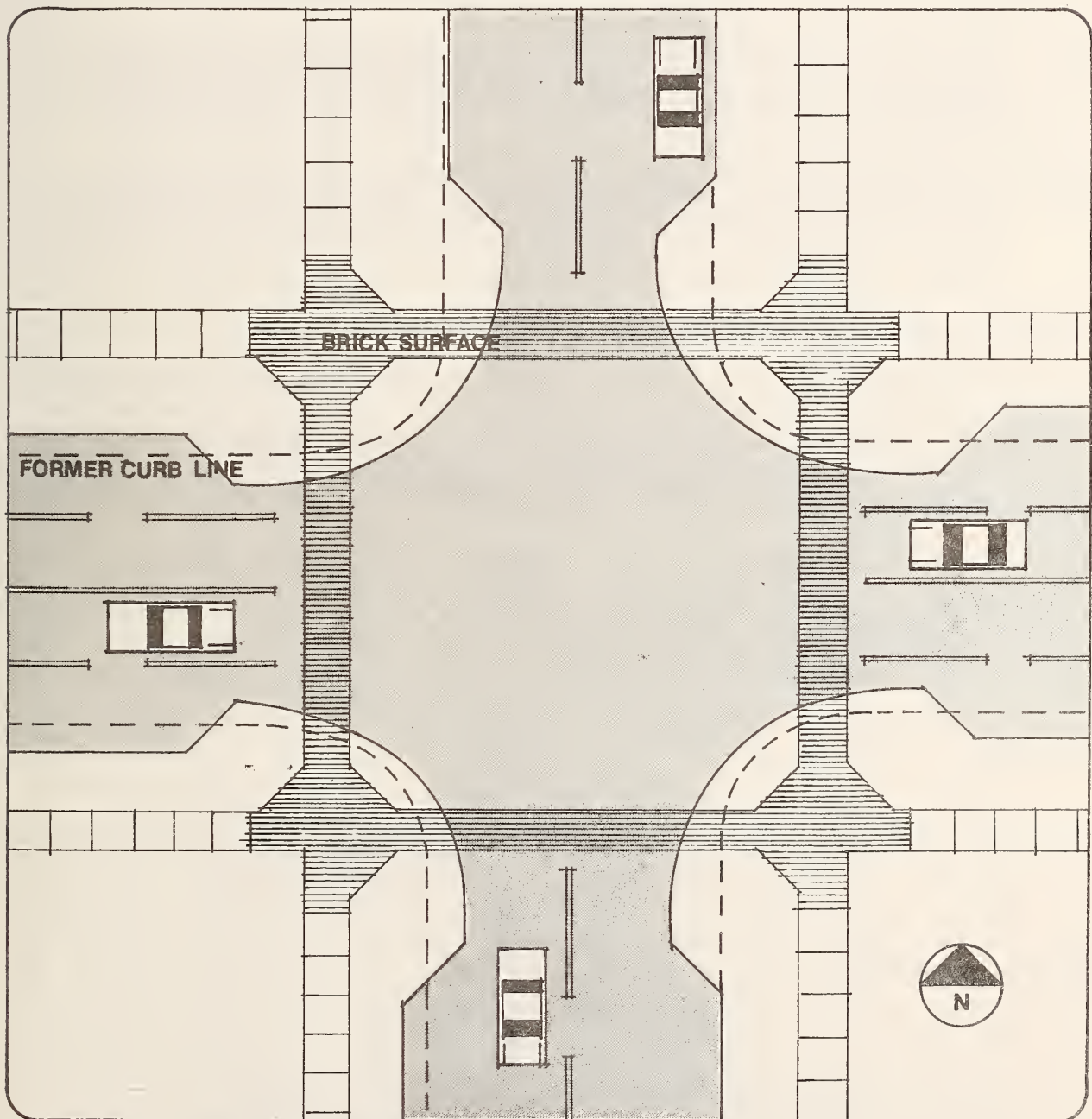


Figure 24 Boulder: Extended Sidewalk: New Layout

were repaved in brick as well. Areas between the sidewalks and the curbs were designated as planting areas.

The design of the street changes at Boulder had been completed before the Pedestrian Research Laboratory countermeasures had been developed. It was decided to evaluate the Boulder location because they were quite similar to the laboratory proposals and were about to be constructed. However, the Pedestrian Research Laboratory perceives several problems associated with the specific design adaptation of the widened sidewalk as constructed at Boulder. These include the following:

- The brick paving is seen as potentially uncomfortable for persons in wheelchairs, as well as others with walking difficulties. The brick may settle differentially, or may heave during freezing weather, becoming an obstacle to free movement.
- Lips, or construction tolerances, have been constructed at the bottoms of the ramps where they join the street and crosswalk. These lips vary in height from less than $\frac{1}{4}$ inch (.635 cm) to as much as $\frac{1}{2}$ inch (1.27 cm), and are seen as a barrier to mobility of persons in wheelchairs.

Control Location for Test Location No. 2.

The control intersection is located one block to the north of the test location. Its design and use are essentially the same as the test location's before the countermeasure was constructed. Traffic volumes were substantially lower in the east-west direction. Pedestrian volumes were somewhat lower in the north-south direction than at the test location as shown in Tables 53 and 54.

TABLE 53. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	145	165	245	245	365	235
Turning	20	30	55	50	65	65
TOTAL	165	195	300	295	430	300
East-West Direction						
Thru	80	20	25	55	65	50
Turning	25	20	20	20	55	25
TOTAL	105	40	45	75	120	75

TABLE 54. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	50	20	85	35	55	50
East-West	15	10	20	30	10	15
TOTAL	65	30	105	65	65	65

1. Effect of countermeasure on traffic flows. From Figure 25 it can be seen that there was a substantial decrease in north-south vehicular traffic after the countermeasure was installed. However, the decrease was substantially less at the test site than at the control site. In the east-west direction, traffic volumes at both the test and the control sites increased. On the basis of these data it is concluded that the countermeasure has not adversely affected traffic flow.

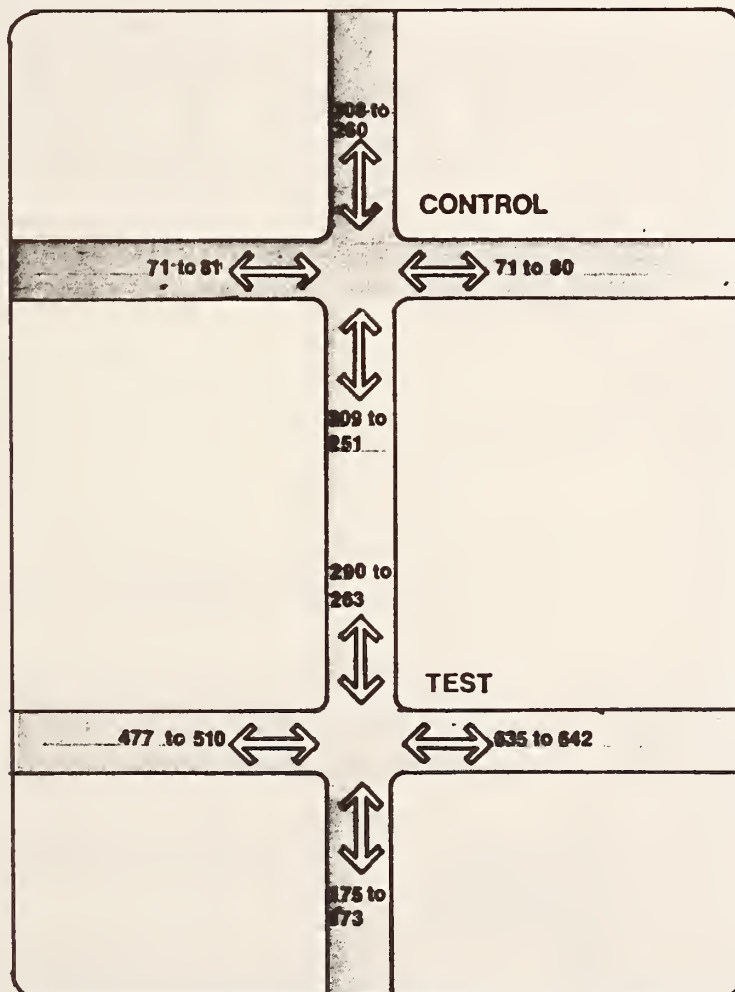


Figure 25 Boulder Extended Sidewalk: Traffic Flows

2. Effect on pedestrian flow volumes. From Figure 26, it can be seen that at both the test and the control sites, pedestrian activity declined substantially. There is no obvious explanation for this decline except that the numbers of pedestrians using the area are small so that an increase or decrease of only a few pedestrians makes a large difference.

From these data, it is concluded that the countermeasure has made no significant difference to pedestrian volumes.

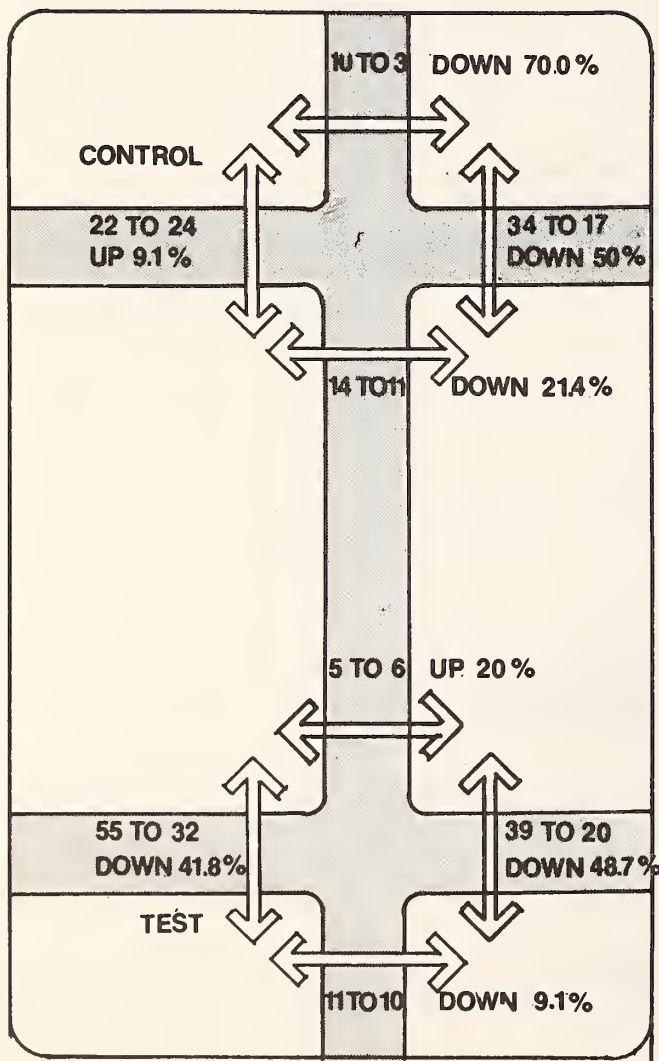


Figure 26 Boulder: Extended Sidewalk: Pedestrian Flows

3. Pedestrians' view of vehicles: drivers' view of pedestrians. From Figure 27, it can be seen that there is increased visibility resulting from the countermeasure design. The driver's view is no longer obstructed by a parked vehicle, and the pedestrians have a clear view of approaching traffic.

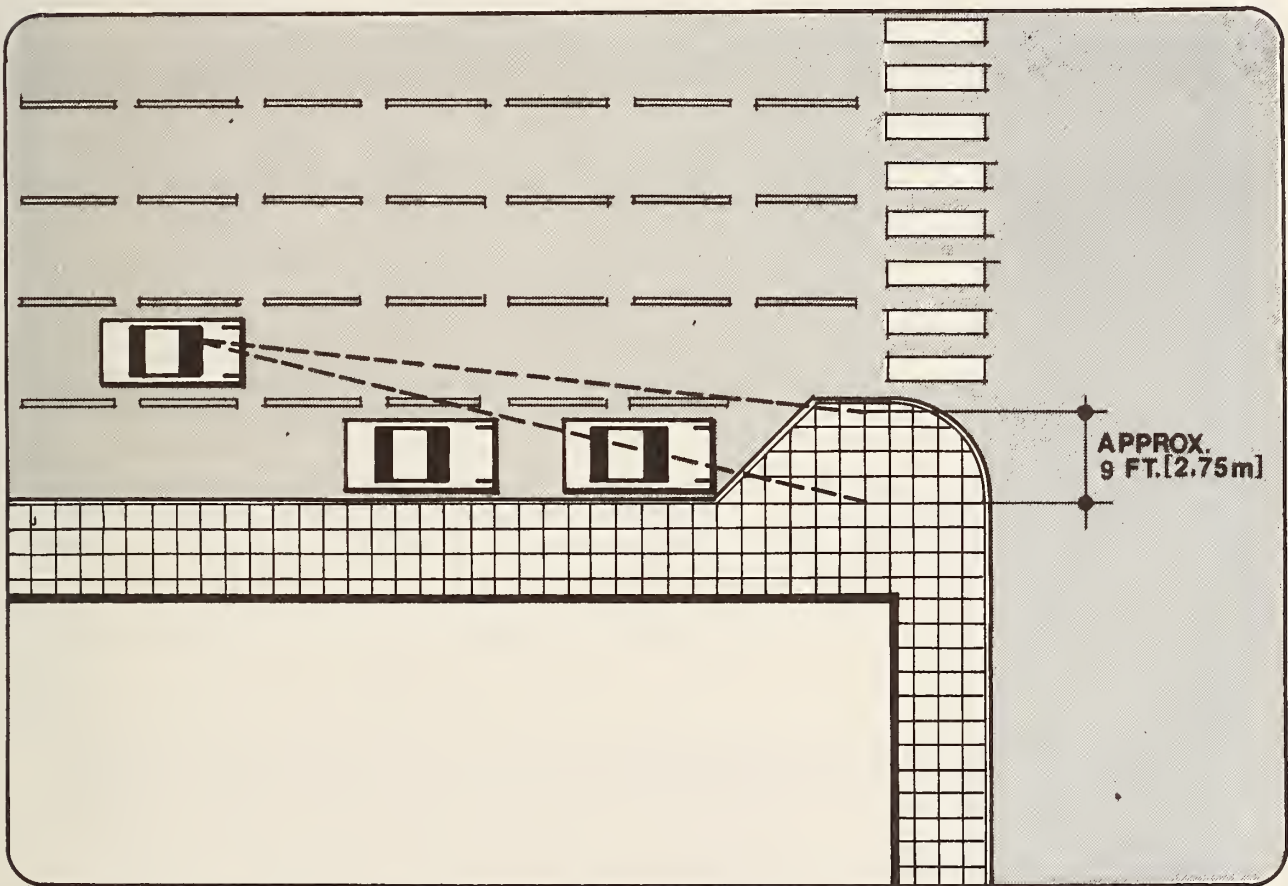


Figure 27 Boulder: Extended Sidewalk: Driver's View

4. Platoon formation space for pedestrians. The test site was not located at a signalized intersection, and the pedestrian flow volumes are low, so no special provisions for increased space have been provided. However, for the future, additional space on the quadrant could be paved.
5. Space for additional design features. Additional space for planting/landscaping was made possible, but not utilized.
6. Crossing time. Countermeasure No. 1 was evaluated to determine whether there was a change in the time taken by pedestrians in crossing the streets. It was anticipated that the reduction in crossing distance, resulting from the sidewalk extensions, would result in a reduced crossing/vehicular exposure time.

From Table 55 it can be seen that a reduction in crossing time occurred at both the test and the control sites. This reduction was significant at the test site. At the control site, this reduction was significant for traffic moving in an east-west direction but not significant in the north-south direction.

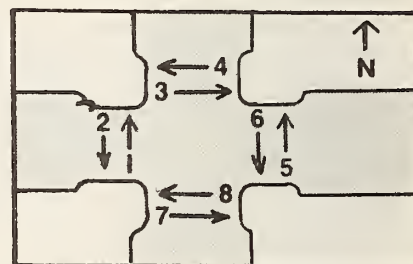


TABLE 55. CROSSING TIME

COUNTERMEASURE NO. 1; LOCATION NO. 2

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
1, 2, 5, 6	10.3030	7.7455	8.0882	7.2535	N	N
3, 4, 7, 8	11.4643	6.7273	11.4000	8.0625	S	N

S = Significant

N = Not Significant

7. Effect of the countermeasure upon the difficulty in negotiating the level change at the curb. Curb ramps were installed as part of the countermeasure design at the test location. This evaluation was intended to determine the effect of these curb ramps on the general population using the test intersection. An evaluation was carried out to determine whether the new ramps affected usage patterns. The apparent difficulty of making level changes is not included for these intersections because at both the new and the original conditions the sidewalk merged into the street pavement without an appreciable curb.

From Tables 56 and, 57, we find that:

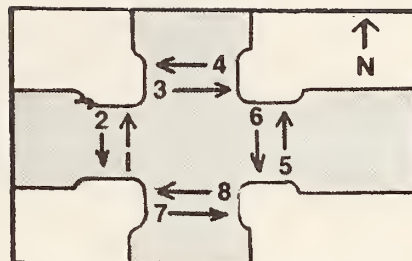
- 96% of all pedestrians used the ramps to enter the street.
- 95% of all pedestrians used the ramps to leave the street.

However, in these designs the only option available to those who did not wish to use the ramps was to travel on the planted areas of the sidewalk, rather than the concrete paved surfaces.

A similar reason may explain why no pedestrians deliberately sought to avoid using the ramp; and only one person appeared to modify his direction of travel in order to make use of a ramp.

TABLE 56. PEDESTRIANS USE OF CURB RAMP TO ENTER STREET

COUNTERMEASURE NO. 1; LOCATION NO. 2

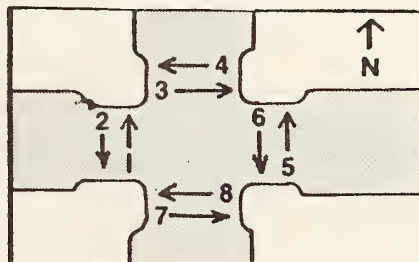


Direction	Used Ramp	Changed direction to use ramp	Changed direction to avoid ramp
1,2,5,6	97 / 101	0 / 101	0 / 101
3,4,7,8	33 / 34	0 / 34	0 / 34

TABLE 57. PEDESTRIANS USE OF CURB RAMP

LEAVING STREET

COUNTERMEASURE NO. 1; LOCATION NO. 2



Direction	Used Ramp	Changed direction to use ramp	Changed direction to avoid ramp
1,2,5,6	97 / 101	0 / 101	0 / 101
3,4,7,8	31 / 34	1 / 34	0 / 34

8. Evaluation of the countermeasure in terms of behavior affecting safety and mobility. These included evaluations of:

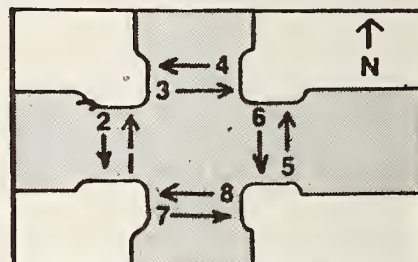
- Conflicts between crossing pedestrians and vehicles.
- Conflicts between crossing pedestrians and other pedestrians.
- Pedestrians waiting in the street for the crossing signal to change.
- Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- Pedestrians crossing against the traffic signal.

- a. Conflicts between crossing pedestrians and vehicles. As a measure of conflict, certain types of pedestrian behavior were measured--a sudden increase in crossing speed, a reduction in crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation are shown in Table 58.

TABLE 58. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND VEHICLES

COUNTERMEASURE NO. 1; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
1, 2, 5, 6	11 / 68	20 / 104	5 / 72	1 / 74	N	N
3, 4, 7, 8	5 / 29	1 / 35	0 / 19	5 / 19	N	S

S = Significant

N = Not Significant

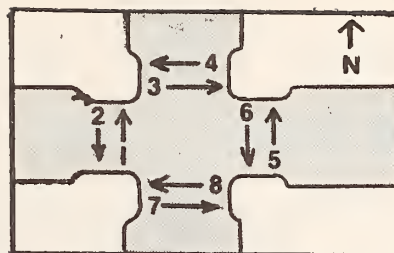
From Table 58, it can be seen that there was no significant change in the number of conflicts between pedestrians and vehicles at the test site.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Table 59.

TABLE 59. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND OTHER PEDESTRIANS

COUNTERMEASURE NO. 1; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
1, 2, 5, 6	0 / 68	0 / 104	0 / 72	0 / 74	N	N
3, 4, 7, 8	0 / 29	0 / 35	0 / 19	0 / 19	N	N

S = Significant

N = Not Significant

From Table 59, it can be seen that no pedestrian conflicts were observed before or after the installation of the countermeasure.

- c. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk and whether the countermeasure design would have any effect upon their doing so.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 60 and 61.

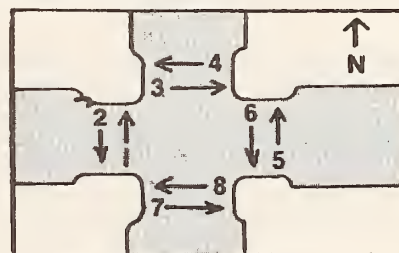


TABLE 60. PEDESTRIANS WALKING BEYOND THE
LIMITS OF CROSSWALK

COUNTERMEASURE NO. 1; LOCATION NO. 2

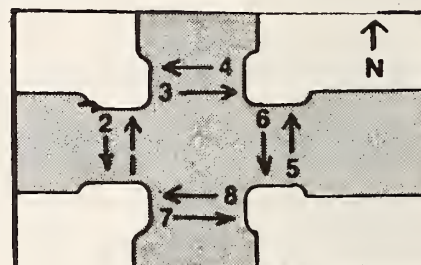
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
1, 2, 5, 6	7 / 68	5 / 104	20 / 72	5 / 74	N	S
3, 4, 7, 8	7 / 29	6 / 35	11 / 19	1 / 19	N	S

S = Significant

N = Not Significant

From Table, 60, there was no significant change in the number of pedestrians walking beyond the limits of the crosswalks. At the control site, there was a significant decrease in the number.

TABLE 61. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 1; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
1,2,5,6	0 / 68	6 / 104	8 / 72	3 / 74	N	N
3,4,7,8	0 / 29	3 / 35	0 / 19	2 / 19	N	N

S = Significant

N = Not Significant

From Table 61, it can be seen that there was an increase in the number of pedestrians leaving the crosswalk but this change was not significant.

Evaluation of Countermeasure No. 1: Location No. 2, By Special Groups of Handicapped Persons.

The procedures used in the special group evaluations were described earlier. The following tests were performed:

● By the visually handicapped.

1. Detectability of the ramp.
2. Usefulness of the ramp design and location.
3. Success in using the brick crosswalk texture as a location cue.

● By people in wheelchairs.

1. Ramp design.

Evaluation Results: Visually Handicapped Group.

1. Detectability of the ramp. Because of the gentleness of the ramp slope resulting from its length, evaluations were made to determine whether visually

impaired test subjects setting out to cross the street, could accurately determine where they should pause, or stop, to await safe crossing conditions. Their performance of this task was complicated by two factors: first, the fact that this was not a signalized intersection; therefore, they would have to be aware of traffic approaching the intersection from some distance; second, the paving material on the ramp and within the crosswalk was identical. Both were of brick without mortar joints. However, there was a concrete 'gutter' placed to separate the ramp from the crosswalk. This 'gutter' incorporated a nominal $\frac{1}{2}$ inch (1.27 cm) lip to denote the end of the ramp.

There were a total of 32 street crossing events in the evaluation. Of these, in 29 of the events, the test subject came to a complete stop on the ramp, short of the 'gutter' and the beginning of the crosswalk, suggesting a high factor of recognition of the ramp. The other 3 events included two in which the subject stepped into the street before stopping (on one of those events, the subject returned to the ramp upon realizing he had stepped beyond the end of the ramp). The third of the 3 events was a high partial subject who was able to determine that he could cross safely, and did so without stopping.

It can be concluded that as two of the 32 participants failed to detect the ramp end before entering the street the design is not satisfactory.

2. Usefulness of the ramp design and location. The ramps at this intersection were extensions of the sidewalk, and were designed to be used by all persons making a street crossing.

There were 32 events in which test subjects descended ramps to enter and cross the street; there were 25 events in which test subjects ascended ramps after crossing the street.

In none of these events did the ramps cause special difficulty to the visually impaired subjects, except as already discussed, despite the fact that the rate of slope is so gentle. The mean difficulty factors for the visually impaired test subjects were 1.77 for entering the street, and 1.68 for leaving the street. These values are close to the values of difficulty for the general population using these ramps.

3. Success in using the brick crosswalk texture as a cue. The crosswalks at this test installation were narrow--seven feet in width. The intersection design makes it important that persons crossing the street stay within the crosswalk. Straying out of the crosswalk directs one into areas that have been landscaped and planted. This evaluation was undertaken to determine whether members of the visually impaired group could detect the surface differences between the brick crosswalk and blacktop roadway.

There were a total of 32 crossing events; in 12 of these (38 percent) the test subject wandered beyond the limits of the crosswalk, and consequently did not reach the ramp on the far side of the street. Some of these 32 events involved

persons with high partial vision. If these are eliminated from consideration in this evaluation, 50 percent of the test events (12 of 24) involved test subjects straying beyond crosswalk limits. This suggests that the brick texturing is not sufficiently detectable to be useful to the blind.

The following examples illustrate these difficulties:

- The test subject stopped at the bottom of the ramp awaiting an opportunity to cross. When traffic conditions allowed, she set out across the street. Less than halfway across, she had wandered out of the crosswalk on the intersection side, thereby risking conflict with traffic passing through the intersection. She reached the far curb at the planter, stepped over the curb into the planter and continued until she found the sidewalk. (All four of this subject's crossings followed this same pattern).
- The test subject crossed, leaving the crosswalk while only one-third of the distance across the street. He reached the curb at the far side and followed it to his left until he detected the ramp. He then climbed the ramp to the sidewalk.

Evaluation Results: Wheelchair Group.

1. Design of ramps. A four point scale was used for this evaluation. The scale ranged from a difficulty factor of 1 (equivalent to maneuvering a level surface with no perceptible change in speed) to a difficulty factor of 4 (could negotiate only with the greatest difficulty, if at all). The ramps were so long and so gently sloped as to present little difficulty to persons using wheelchairs. However, the ramps all had 'lips', at the point where they joined the street, which varied from less than 1/8 inch (.3175 cm) to more than 1/2 inch (1.27 cm). These caused difficulties to a number of wheelchair users as can be seen in Table 62.

TABLE 62. DIFFICULTY FACTORS

ASCENT AND DESCENT OF CURB RAMPS - WHEELCHAIR USERS

	Total Events	Number of Events Scoring Factor of:				Mean Difficulty Factor
		1	2	3	4	
Entering Street	33	13	18	2	0	1.67
Leaving Street	33	17	4	6	6	2.03

The following narrative illustrates several of the problems caused by these ramp 'lips':

- The subject was in a chair being pushed by an attendant. As she reached the ramp to leave the street, the front wheels of her chair caught on the ramp 'lip', stopping the chair. The attendant had to lift the front wheels to enable him to push the chair onto the ramp.
- The subject was ready to cross the street, but a car was stopped completely blocking the crosswalk. The driver was attempting to see traffic coming on the intersecting street. After the car moved through the intersection, the subject crossed the street. As he reached the ramp, his front wheels caught on the ramp lip, jarring the chair. The subject lifted the wheels over the lip, but then the large rear wheels caught on the lip, again jarring the chair to a stop. An additional effort got the chair fully onto the ramp.
- The test subject was preparing to enter the street. As he began, he lifted his front wheels into the air to soften the jolt he anticipated from the ramp 'lip', while at the same time trying to accelerate. His chair nearly tipped over backwards, the front wheels rising well over 12 inches (30.48 cm) into the air before he recovered.

From this test, it can be concluded that the lip between ramp and street substantially increases the difficulty for wheelchair users, and the jolt that results may even be hazardous.

III. LOCATION NO. 3: ATHENS, GEORGIA

Athens, Georgia is a small university oriented city in the Southeast. While the downtown area of this city is relatively small, its main street is very active, with substantial vehicular and pedestrian flows. The countermeasure was evaluated at two adjacent intersections, both on the busiest part of the main street.

The plan, Figure 28, illustrates the intersection. The city's main street is a one-way street carrying traffic in a westbound direction. It has three lanes of moving traffic plus diagonal curb side parking on both sides, making a total curb to curb width of about 70 feet (21.35 m). Of the two illustrated intersecting streets, the westernmost carries two-way traffic and is four lanes in width plus two lanes of diagonal curb parking, for an overall curb to curb width of about 80 feet (24.4 m). The other intersecting street is a one-way street carrying three lanes of traffic in a southbound direction, with parallel curb parking on both sides and an overall curb to curb width of approximately 55 feet (16.775 m).

The changes representative of the installation of Countermeasure No. 1 were made by the city at several intersections within the downtown area including the two at which evaluations took place. At the westernmost of the two test intersections, all four quadrants were extended into the diagonal parking area of the street, as shown in Figure 28. This resulted in about 1300 square feet (120.9 m²) of additional space

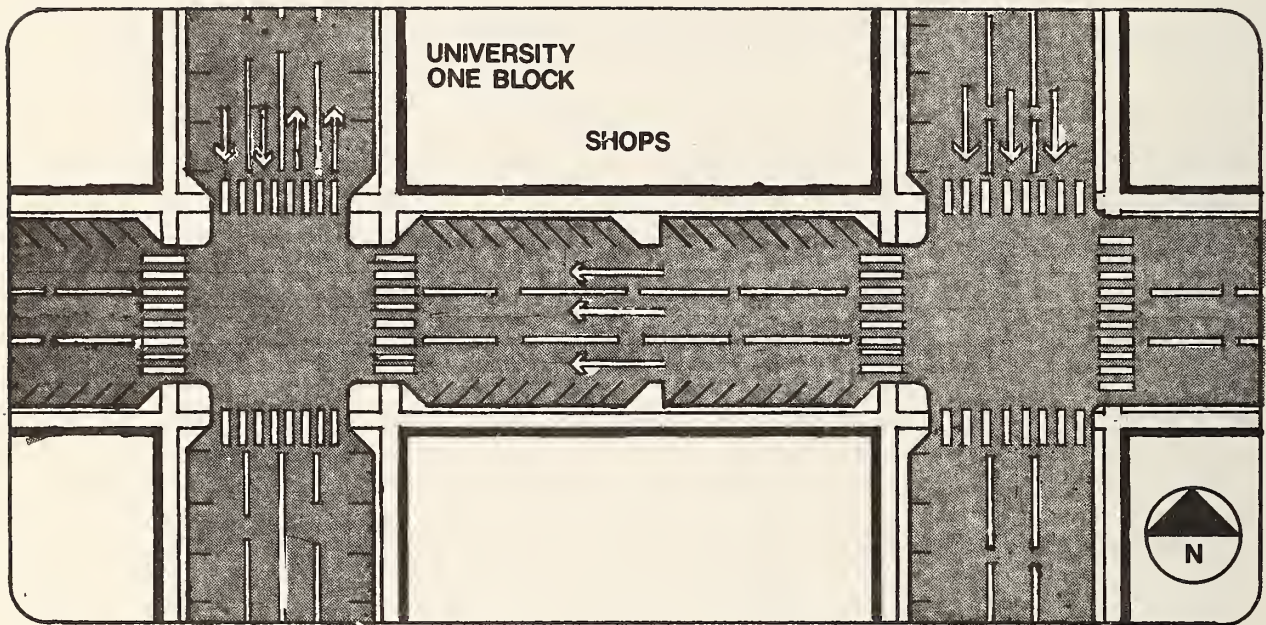


Figure 28 Athens: Test Location Map

being added to each of the four quadrants, and reduced the effective street crossing distance as follows:

- The north-south street: from 80 feet to 48 feet (24.4 m to 14.64 m) for a reduction of 40 percent.
- The east-west main street, from 70 feet to 36 feet (21.35 m to 10.98 m) for a reduction of nearly 50 percent.

At the easternmost part of the test intersection, only the two quadrants on the west side of the intersection were extended, as illustrated in Figure 29.

These extensions added about 1000 square feet (93 m^2) to each of the quadrants and reduced the crossing distances, at the intersection as follows:

- The north-south street: from 55 feet to 45 feet (16.775 m to 13.725 m), for a reduction of nearly 20 percent.
- The east-west main street, from 70 feet to 36 feet (21.35 m to 10.98 m), for a reduction of nearly 50 percent.

At both locations, the space gained by extending the sidewalk was used for several purposes:

- Additional sidewalk space was added, for groups or platoons of pedestrians awaiting a green crossing signal. At the westernmost intersection, for example,

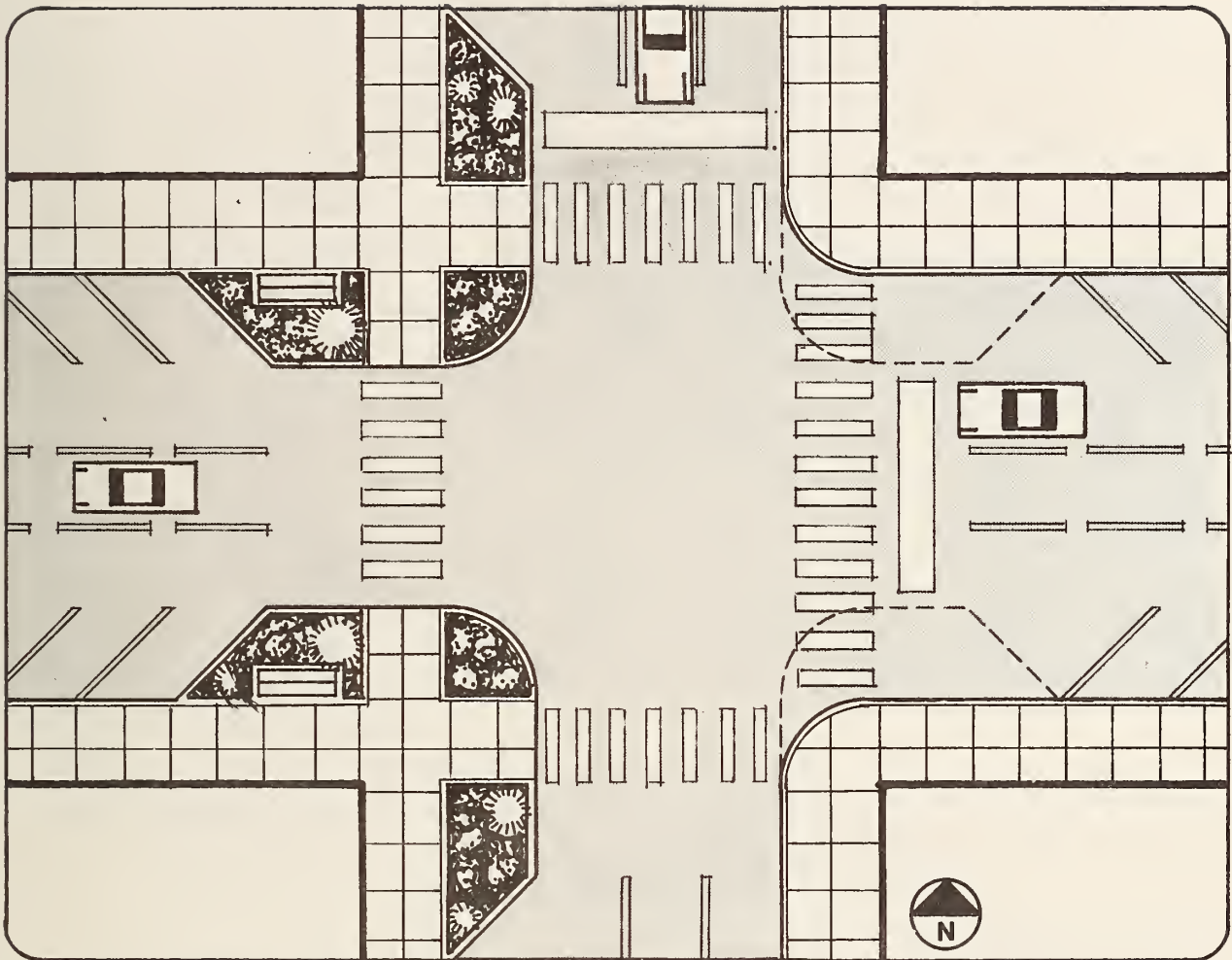


Figure 29 Athens: East Side Location Map

an area of about 225 square feet (20.95 m^2) was added as an approach to each of the crosswalks, as illustrated in Figure 30.

- This added space was used to ramp down from the sidewalk level to the street. Since the length is relatively long, a gentle ramp slope (of less than 6 per cent) was employed.
- Planters were added in the several illustrated locations. These were flush with the sidewalk surface, but were surrounded by an 8 inch (20.32 cm) concrete edge or curb. On the street side, the planters were set back about 12 inches (30.48 cm) from the street curb, and a brick paving surface was used to differentiate from the brushed concrete sidewalk surface.
- Benches were located, in some locations, within the space gained from the extension into the parking lanes.

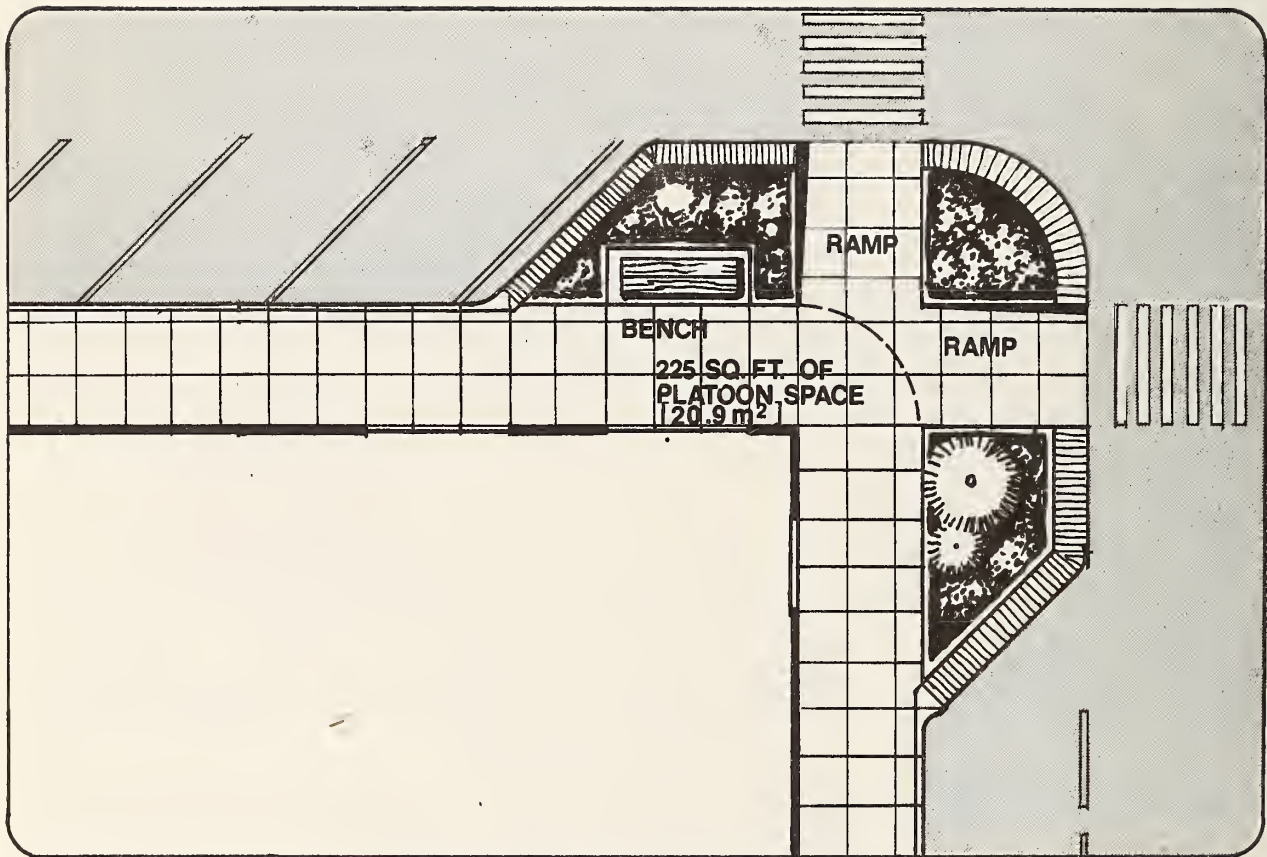


Figure 30 Athens: East Side Location Map: Detail

Evaluations at Location 3.

As the construction was already complete, the changes were evaluated by using special groups of handicapped persons, and using the same evaluative techniques employed at Sioux City and Boulder. In particular, the following questions were examined:

Visually Impaired Group.

- Is the ramp detectable; can the visually impaired detect the points at which the ramp meets the street.
- Is the ramp location and design useful or inconvenient.
- Are the 'curb' edges which define the planters useful to the visually impaired as navigation cues, or are they inconvenient.

Wheelchair Group.

- Are the ramps satisfactorily designed and located, or are they difficult to use or even hazardous.

Evaluation Results: Visually Impaired Group.

1. Detectability of the ramps. There appeared to be no difficulty in detecting the point at which the ramps joined the street. All test subjects were able to stop and wait for safe crossing conditions before entering the street. None wandered into the street inadvertently.
2. Usefulness of the ramps. The ramps because of their gentle slopes, appeared to cause no difficulties to any of the test subjects.
3. Usefulness of the 'curb' edges which define the planters. Of the test subjects who were either totally blind, or could only perceive light, all of them detected, or otherwise came into contact with, the planter curb edge. On nearly half of these occasions the planter edge caused some difficulty or confusion. The following examples describe typical problems caused by the 'curb' edge.
 - The test subject, on leaving the street, reached the ramp and began to ascend to the sidewalk, slowing somewhat from his normal walking speed. His course was slightly off-line, so that when about halfway up the ramp, his cane first detected the planter 'curb'. He came to a complete stop, obviously confused by the existence of a 'curb' at that location. After an explanation of the intersection design by an observer, the subject was able to proceed, and could use the planter edge for guidance without further difficulty.
 - While crossing the street, the subject had taken a course which took him beyond the edge of the crosswalk. As he reached the opposite side, his cane detected the street curb. He stepped onto it, and then walked into the planter. The subject then turned to his left and stepped down from the planter onto the ramp and proceeded without further difficulty.
 - The subject crossed the street. On reaching the far side, he was just outside the right side crosswalk line. His cane, as he used a sweeping motion, detected the planter edge which he mistook for a normal street curb. He stepped high, to step on the curb, but his foot came down on the ramp instead, giving him a jolt since it was about 8 inches (20.32 cm) lower than he expected. He then walked towards the left, diagonally across the ramp and soon reached the planter edge on the other side. At this point, the subject was confused and needed assistance from the observer.

On just over half of the occasions in which the visually impaired detected or came into contact with the planter 'curb' edge, no difficulty was involved; instead, the subjects were able to use the edge as a cue to navigate through the intersection quadrant. The following examples are indicative:

- The subject, on leaving the street, reached the ramp and detected, with his cane, the planter edge on the left side of the ramp. He proceeded slowly, using the planter as a guidance cue. He turned left, still following the

planter, and descended that ramp, positioning himself for the next street crossing.

- The subject, approaching the intersection to cross the street, detected the planter edge with his cane and followed it, using his cane tapping against the edge 'curb' as a means of maintaining the proper course. When the cane detected the end of the planter, the subject knew that he had reached the edge of the street and stopped, waiting for a safe time to cross.

There was evidence to support the contention that some of the difficulties caused by the planter 'curb' edge resulted from the test subjects not being sufficiently familiar with the intersection design. Most problems seemed to be associated with confusion, from finding what appeared to be a curb where a curb was not expected to be found.

Because the extended sidewalk design limited the entrances to the sidewalk from the street, it is necessary for all pedestrians to stay within the crosswalk. Because the crosswalk limits were not detectable, the visually impaired subjects had some difficulty in remaining within these limits. Therefore, for this specific design to be effective, detectable crosswalk lines are necessary.

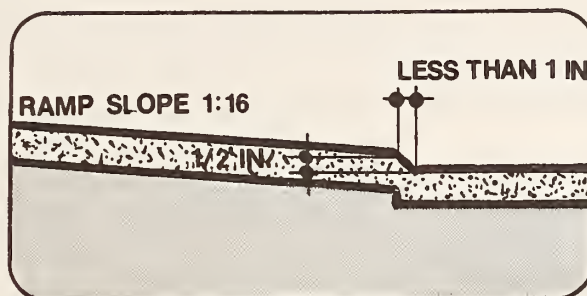
It was clear that, once having gone through the process of confusion and disorientation, followed by explanation of the intersection design, the subjects learned how to use the design features to their benefit in navigating the intersection. The following example illustrates the effect of this learning.

- On leaving the street, the test subject entered the ramp and walked up several steps. However, knowing he was to turn right 20 degrees and cross the intersecting street, he turned while only part way up the ramp and found the planter curb (with his cane) on the ramp's right side. This was confusing to him, and he stopped, unsure of how to proceed. The attending observer explained the design and directed the subject as to how to continue.
- On the subsequent crossing, again on leaving the street, the subject entered the ramp and found the planter edge 'curb' on his right. He slowed considerably, but continued to move as he followed the planter edge with his cane. Needing to turn 90 degrees to the right again, he did so when he detected the corner of the planter edge. He then followed the intersecting ramp leading to the next street crossing, and stopped when his cane detected the end of the ramp at the street.

Evaluation Results: Wheelchair Group.

1. Usefulness of the ramps. As described earlier, the ramps were very gently sloped (less than 6 percent) because of their length. However, the design condition at the point where the ramp joined the street was such that it caused difficulty to a large number of wheelchair users. At this point there was a short

section of very steep rise—approximately a 1:2 slope rising one-half inch (1.27 cm) or slightly more in less than 1 inch (2.54 cm) of horizontal distance.



2. Inconvenience caused by the ramps. The difficulty encountered by test subjects upon using a ramp to enter or leave the street was scored on the same four-point scale described earlier. The scale ranged from a score of "1", very easy, equivalent to travelling on a level surface with no apparent change of speed or effort required, to a score of "4", extremely difficult, very nearly impossible to negotiate. There were 56 test events in which a wheelchair user attempted to use a ramp to leave the street. The scores illustrating the difficulty associated with those events are shown in Table 63.

TABLE 63. DIFFICULTY SCORES - LEAVING STREET - WHEELCHAIR USERS

COUNTERMEASURE NO. 1 - TEST LOCATION NO. 3

Difficulty Scores	No. of Observations
1	29
2	10
3	10
4	7
Mean	1.91

The following examples are descriptive of some of the difficulties encountered by these subjects:

- The test subject reached the ramp, and his front wheels went over the 'lip'

without difficulty. However, as the rear wheels struck the 'lip', the chair was jarred considerably; the subject stopped briefly, then gave an extra push to climb the ramp.

- The subject's front wheels climbed over the lip without apparent difficulty, but the rear wheels did not. The chair was jarred to a stop, then rolled backwards about 18 inches (45.72 cm). The subject stopped the chair, gave an extra push and with some difficulty got up and over the ramp lip.
- The test subject was in a chair without a foot rest. As the rear wheels of his chair became hung up on the ramp lip, and he could not get over the lip using only his arms, he used one foot plus substantial arm effort to get up over the lip and onto the ramp.
- The subject began to ascend the ramp. His front chair wheels got up onto the ramp without much difficulty. When the rear wheels reached the lip, the chair was jolted to a stop. The subject tried a second and third time before managing to get the chair over the 'lip' and onto the ramp.

These data and examples confirm the substantial difficulty arising from the presence of the lip at the bottom of the ramps.

Summary of Findings: Countermeasure No. 1, Widened Sidewalk.

1. The countermeasure did not adversely affect vehicular flow at either of the two locations where this was evaluated.
2. At the Sioux City location there was a decrease in pedestrian flow, but this was almost certainly due to factors other than the countermeasure. At the Boulder location, there was a decrease in pedestrian flow at both the test and control sites, so the change cannot be attributed to the countermeasure.
3. The driver's view of pedestrians waiting to cross, and the pedestrian's view of vehicles, was greatly improved by the countermeasure.
4. The countermeasure made possible increased space for pedestrian platoon formation.
5. The countermeasure made possible additional space for benches, planters, curb ramps and street furniture.
6. There was a significant decrease in pedestrian crossing time, as a result of the countermeasure. Nevertheless, wheelchair users took an average of 25% longer to cross than the general population; visually impaired people took an average of 40% longer.
7. A substantial population of able bodied pedestrians chose to use the curb ramps,

rather than curbs, when the choice was available. A small minority of the able bodied pedestrians obviously chose to avoid the curb ramps.

Able bodied pedestrians demonstrated a significant decrease in difficulty in negotiating curbs after the ramps were installed.

8. There was no significant change in the number of pedestrian/vehicular conflicts.
9. There was no significant change in the number of pedestrian conflicts.
10. There was no significant change in the number of people waiting in the road to cross the street from the extended sidewalk (although there was an increase at the adjoining conventional curb ramp).
11. At the Sioux City test site, significantly fewer people crossed beyond the limits of the crosswalk. There was no significant change at the Boulder site.

There was no significant change in the number of people leaving the crosswalk after using the extended sidewalk.

12. There was no significant change in the number of pedestrians crossing against the traffic signal.
13. At Sioux City, and at Athens, the visually impaired subjects experienced no difficulty in detecting the curb ramps. At the Boulder site, the paving material on the ramp and the crosswalk is identical, and the gradient small. This design is not considered to be satisfactory because two of the thirty-two blind subjects failed to detect the curb ramp.
14. At Sioux City, where the curb ramps met the roadway without a step or lip, people in wheelchairs experienced no difficulty with the design. At Boulder and Athens, where lips of between $\frac{1}{4}$ " (.635 cm) and 1" (2.54 cm) were provided at the bottom of the ramps, wheelchair users experienced considerable difficulty (and even hazard) in using the ramps.
15. At Sioux City, where handrails were provided at the curb ramps, it was found that only one wheelchair traveller used the handrail. On the other hand, the majority of the visually impaired subjects used the handrails as landmarks. One visually impaired subject walked into a handrail after failing to detect its presence; however, this handrail was not well located.
16. At Boulder, where the crosswalk was formed from brick paving, many of the visually impaired subjects strayed beyond the limits of the crosswalk, as they failed to detect the difference between the brick and asphalt.
17. At Athens, where planters were used to define the edges of the crosswalk, the planters formed a barrier to access to the sidewalks from the roadway, for those visually impaired subjects who had strayed from the crosswalk.

Conclusions.

The countermeasure works well, with no obvious disadvantages, and many advantages. Specific detail design failures at test sites were apparent.

- Lips at the bottom of curb ramps should not be used.
- Crosswalks should be constructed from materials that are different from that of curb ramps (and sidewalks, if these are flush with the crosswalk). The materials should be detectably different for the visually impaired.
- Where planters, or other barriers (such as chains, etc.) to access to sidewalks from the street are used, and a crosswalk passes through the barrier, then the edges of the crosswalk should be constructed to be detectable to the visually impaired. These crosswalk definition lines should lead to the edges of the barrier, or, at least end at points within the opening in the barrier.

countermeasure 2 **pedestrian refuge island**

Countermeasure No. 2 is a solution which is designed for situations in which streets are wide, pedestrian crossing times long, and signal time available for crossing is insufficient for many members of the population; and, in particular, for the elderly and handicapped.

Countermeasure No. 2 is intended for those situations where it is not possible to shorten crossing distances by extending the sidewalk into the parking lane as in Countermeasure No. 1.

The primary benefit of the pedestrian refuge island is that it permits pedestrians to use two signal cycles to cross the street, requiring them to traverse only half of the distance on each cycle, and providing a safe haven at the mid-point of the crossing.

The basic design feature of Countermeasure No. 2 is the installation of a pedestrian refuge island in the median of a street, as shown in Figure 31. In those locations which do not already have a built up or painted median, space for such an island must be gained by borrowing space from the adjoining traffic lanes. The island must be at least six feet in width, to accommodate a wheelchair and attendant.

The island should have protected areas at each end, to separate it from the traffic lanes; however, the entire width of the crosswalk, passing through the island, should remain at street or crosswalk level, or, if it is raised above the street, include curb ramps, so as to pose no barriers to persons in wheelchairs. If the refuge area is flush with the roadway, the whole area should be formed from a material which is detectably different from that of the crosswalk so that the visually impaired can

find the refuge. The island should have signs to assure its visibility to drivers, and a railing.

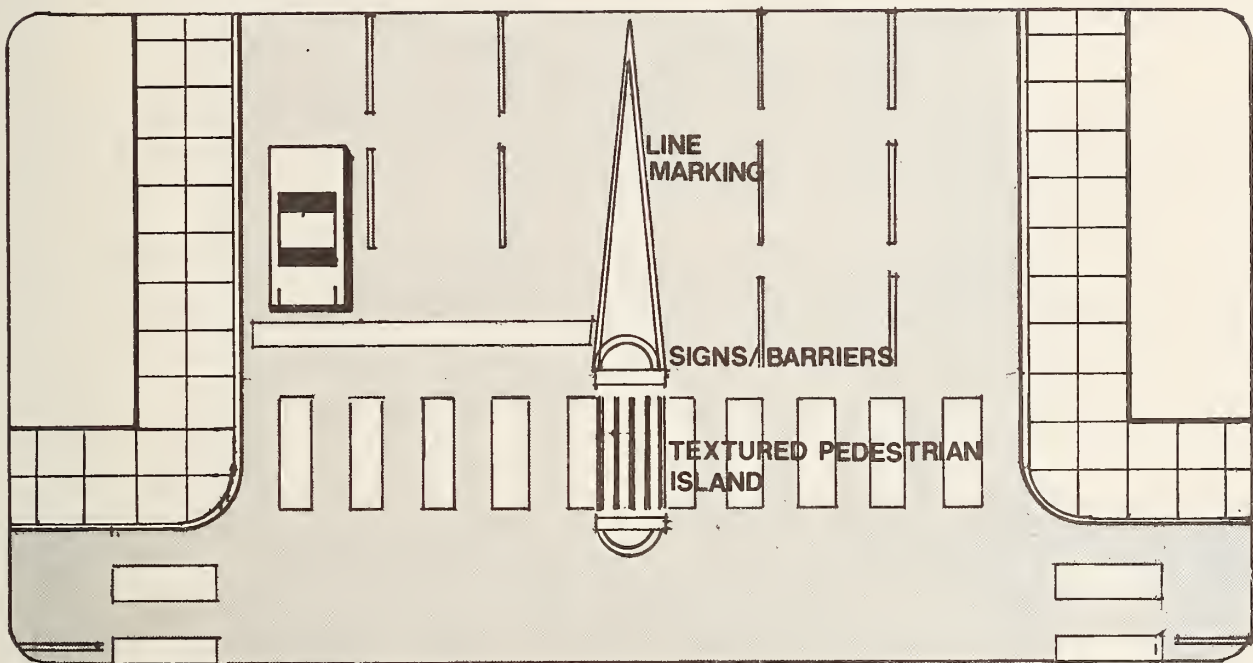


Figure 31 Conceptual Design: Refuge Island

Countermeasure No. 2 was evaluated, based on its performance in solving the following problems:

- Long walking distances in crossing wide streets.
- Insufficient time, for persons with slower than average walking speeds, to cross wide streets.
- Its design provisions as a pedestrian haven.

In addition, evaluations were made of the general use of the location where this countermeasure was installed, and of pedestrian behavior which might indicate problems affecting safety or mobility.

Evaluation Location.

Countermeasure No. 2, Pedestrian Refuge Island, was constructed and evaluated at a heavily travelled intersection in a business and recreation area of Clearwater, Florida, a mid-sized southeastern city. The location chosen was across a causeway, and about 1½ miles (2.4135 km) from that city's downtown business district.

The city, because of its climate and its consequent use as a retirement (as well as vacation) community, has a substantially higher than normal percentage of elderly persons in its overall population. For similar reasons, it has a high incidence of handicapped persons.

The location chosen for construction of this countermeasure is essentially a T-intersection. The roadway oriented north-south is about 62 feet (18.91 m) in width, carries four lanes of moving traffic (two in each direction) and provides for parallel parking along both curbs, 24 hours each day.

The intersecting street, coming from the east, is a one-way thoroughfare, carrying three lanes of moving traffic from the downtown business district across the causeway. Opposite this roadway is an entrance to a parking area serving the beach. See Figures 32 and 33.

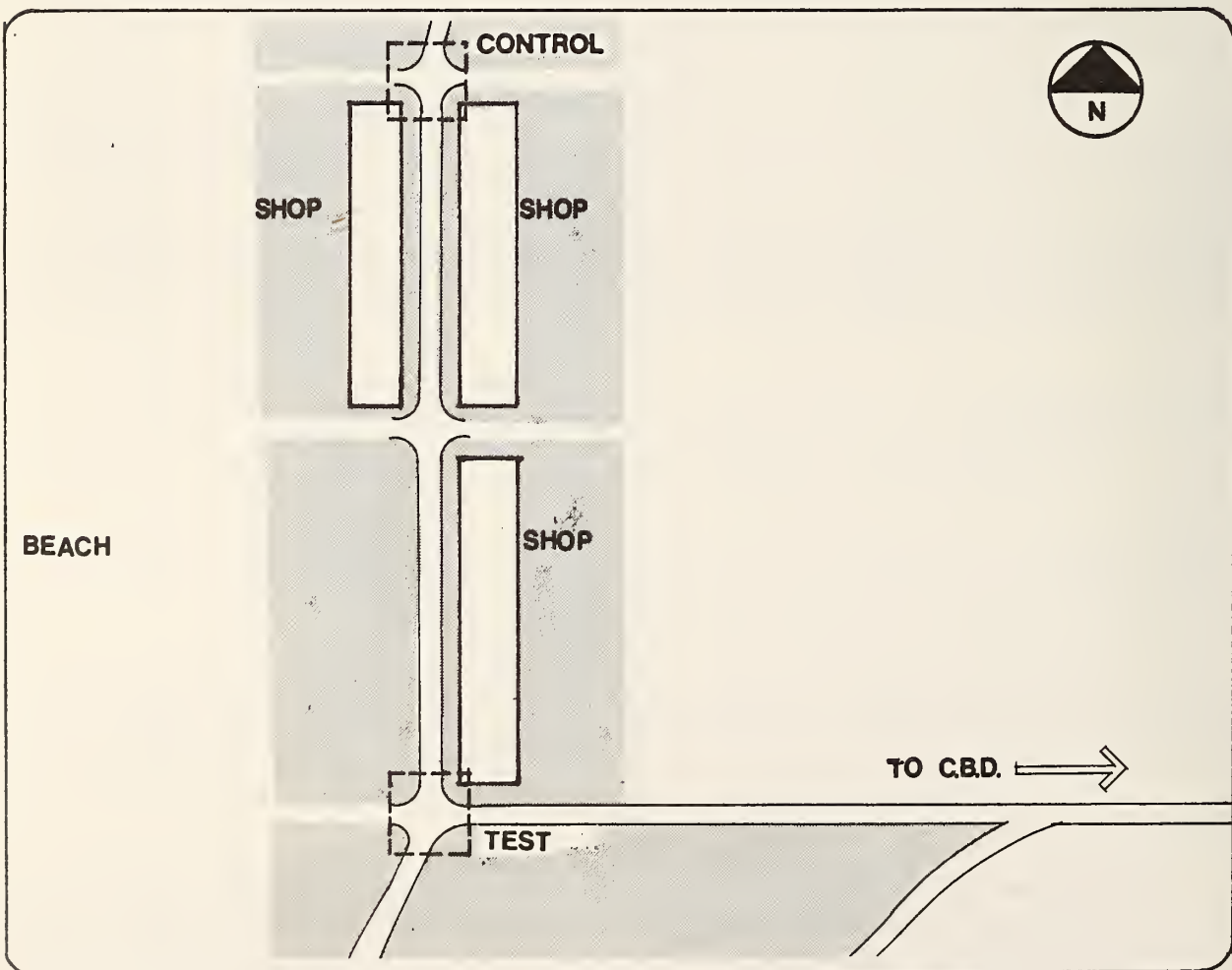


Figure 32 Clearwater: Test and Control Locations

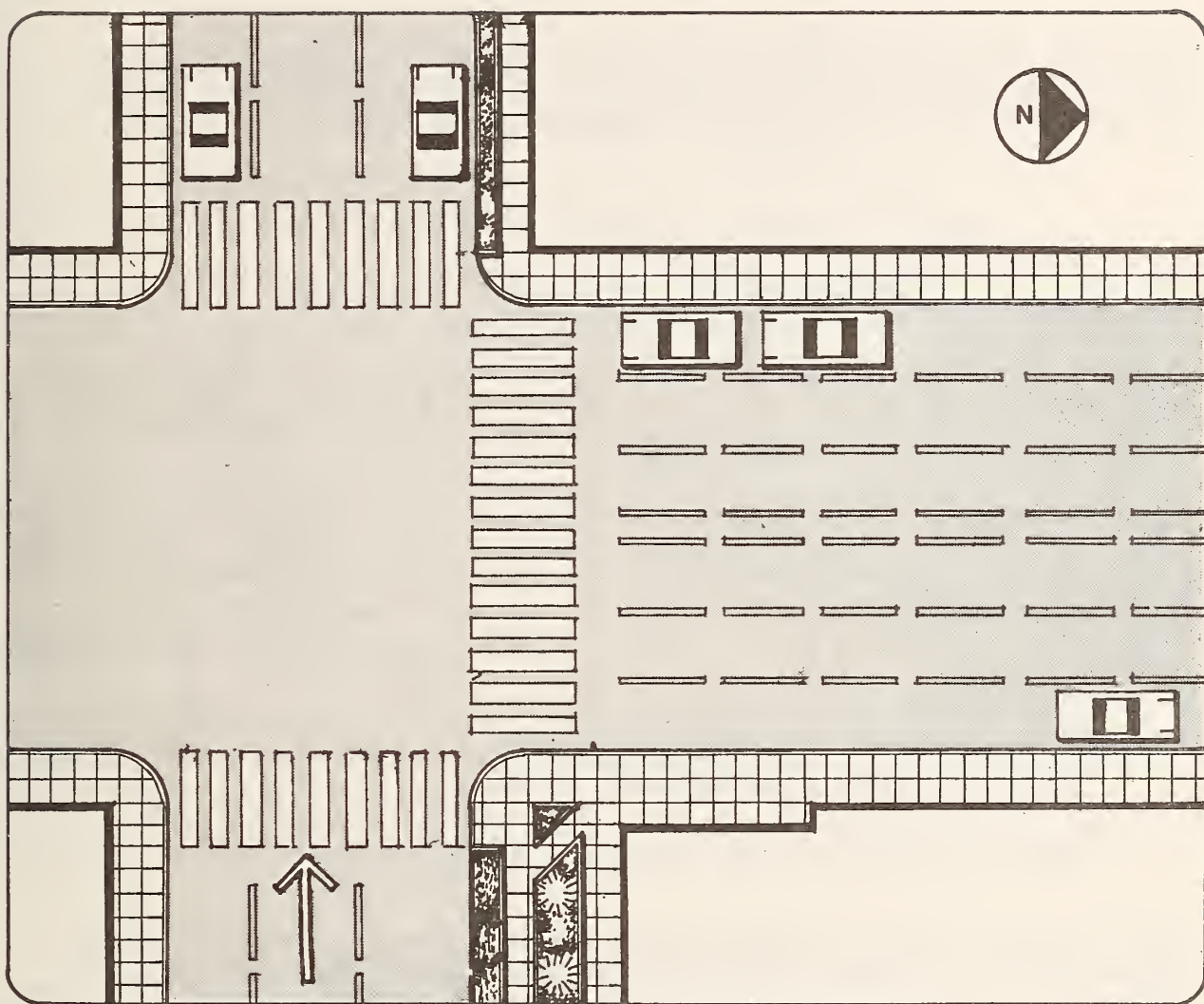


Figure 33 Test Site Before Changes

Beginning at the northeast quadrant of the test intersection, are retail stores (restaurants, speciality shops, gift and souvenir stores) which serve the leisure and recreation function of the immediate locale. There are important resulting pedestrian connections across the intersection, between this retail strip and the beach itself.

Traffic volumes on both streets are fairly heavy (see Table 64) especially on the one-way street providing access from the downtown across the causeway, reaching peak hour flows of over 475 vehicles per lane per hour of green time.

TABLE 64. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	290	555	560	625	470	510
Turning	10	40	25	30	35	30
Total	300	595	585	655	505	540
East-West Direction (Westbound Only)						
Thru	45	55	95	80	75	70
Turning	555	665	885	780	525	680
Total	600	720	980	860	600	750

The intersection is signalized, and has pedestrian "walk/don't walk" signals, as well. Signal timing is set to favor traffic on the westbound street. The signal cycle is as follows:

TRAFFIC SIGNAL TIMING - ELAPSED TIME IN SECONDS

	North-South Direction	Westbound Direction
Green; "Walk"	0	0
Flashing "Don't Walk"	7	7
Yellow; Solid "Don't Walk"	15	45
Red	20	50
Green	70	

Pedestrian flows were measured in the north crosswalk (see Table 65), and varied between hourly flows of 30 and 100 pedestrians per hour (both directions).

TABLE 65. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
Westbound	20	35	35	50	50	40
Eastbound	10	15	50	50	0	25
Total	30	50	85	100	50	65

The pedestrian crossing signal allowed 15 seconds, on the "walk" and "flashing don't walk" signal for crossing the street's six lanes (62 1/2 feet, 19.05 m) of width. This required a walking speed of nearly 4.2 feet (1.280 m) per second (2.864 miles, 4.608 km per hour) to complete the crossing in the allotted time.

Control Location for Countermeasure No. 2.

The control intersection is located two blocks (about 1500 feet, 457.5 m) to the north of the test intersection. Its design and use is essentially the same as at the test location before construction of the countermeasure (except that all the streets have two way traffic). (See Figure 34.) Traffic volumes on the north-south street were about 15 percent higher than at the test location; there was light traffic on the intersecting east-west street, since it did not connect to the mainland, see Table 66.

TABLE 66. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	465	650	665	795	770	670
Turning	35	75	50	75	40	55
Total	500	725	715	870	810	725
East-West Direction						
Thru	5	10	10	0	5	5
Turning	40	110	115	95	110	95
Total	45	120	125	95	115	100

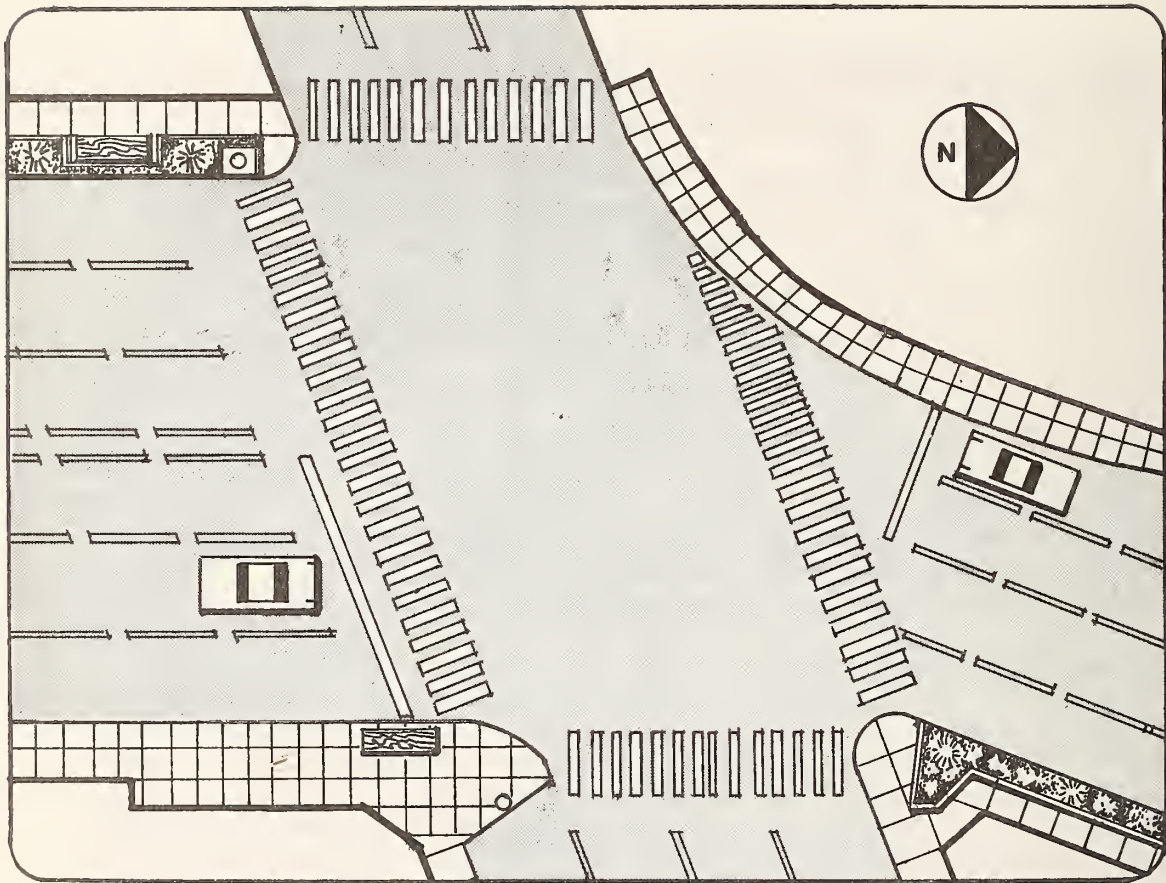


Figure 34 Control Site

Pedestrian volumes, crossing in the south crosswalk, were substantially lower than at the test location, see Table 67.

TABLE 67. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
Westbound	0	0	15	20	5	10
Eastbound	5	30	10	15	0	10
Average	5	30	25	35	5	20

Description of the Changes Made at the Test Location.

Figure 35 illustrates the design of Countermeasure No. 2, as it was constructed at the test location. The pedestrian refuge island was placed at the north crosswalk. The width of the island was six feet (1.83 m); its length was 18 feet (5.43 m),

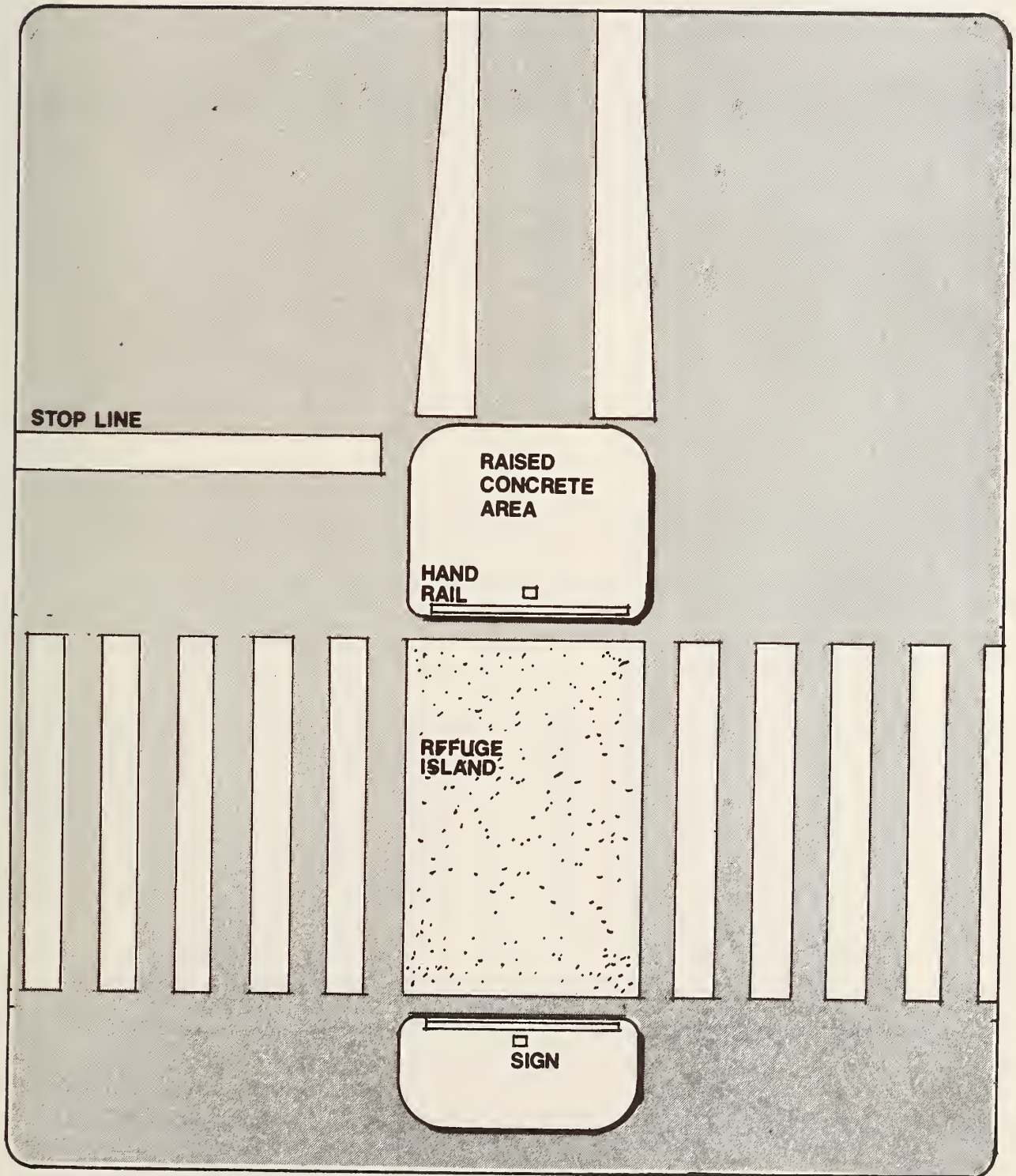


Figure 35 Detail of the Refuge Island as Constructed

including a ten foot (25.4 m) wide crosswalk. The area of the crosswalk, passing through the island, was flush with the street. Unfortunately, the detectable texture was not constructed. On both ends of the island, concrete surfaces were raised 6 inches (33.24 cm), to give definition to the island. Two railings were provided, for persons to lean upon while pausing on the island. Warning signs were provided for drivers.

Evaluation Results: Countermeasure No. 2.

The countermeasure was evaluated to determine whether it had any effect upon the flow of traffic in a north-south direction, on the street in which it was located. The intent of this evaluation was to determine whether traffic was impeded by the island in any way (see Fig. 36).

Clearwater is a vacation resort, with a population that responds to seasonal visitors, so the results of these measures are of limited value, particularly as the pretest was carried out close to the Fourth of July, while the post-test was carried out in August, a year later.

From Figure 36, it can be seen that, at the test site, vehicular traffic increased slightly while decreasing slightly at the control site; with the caveat mentioned earlier, it seems that vehicular traffic was not affected.

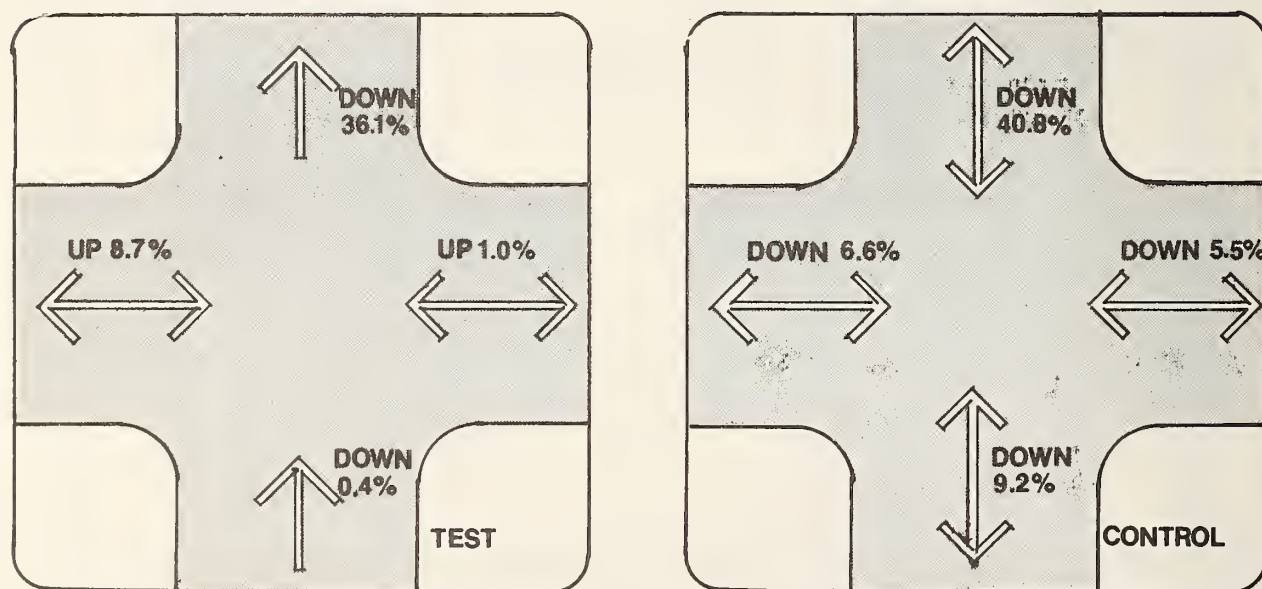


Figure 36 Vehicular Traffic Flow Changes: Test and Control Sites

From Figure 37, on the other hand, pedestrian traffic decreased very significantly at the test site, but only slightly at the control site. Other than the tourist seasonal differences, which are likely to affect the test site more than the control site, there is no obvious reason for this decline. The countermeasure is unlikely to have affected pedestrian volumes adversely.

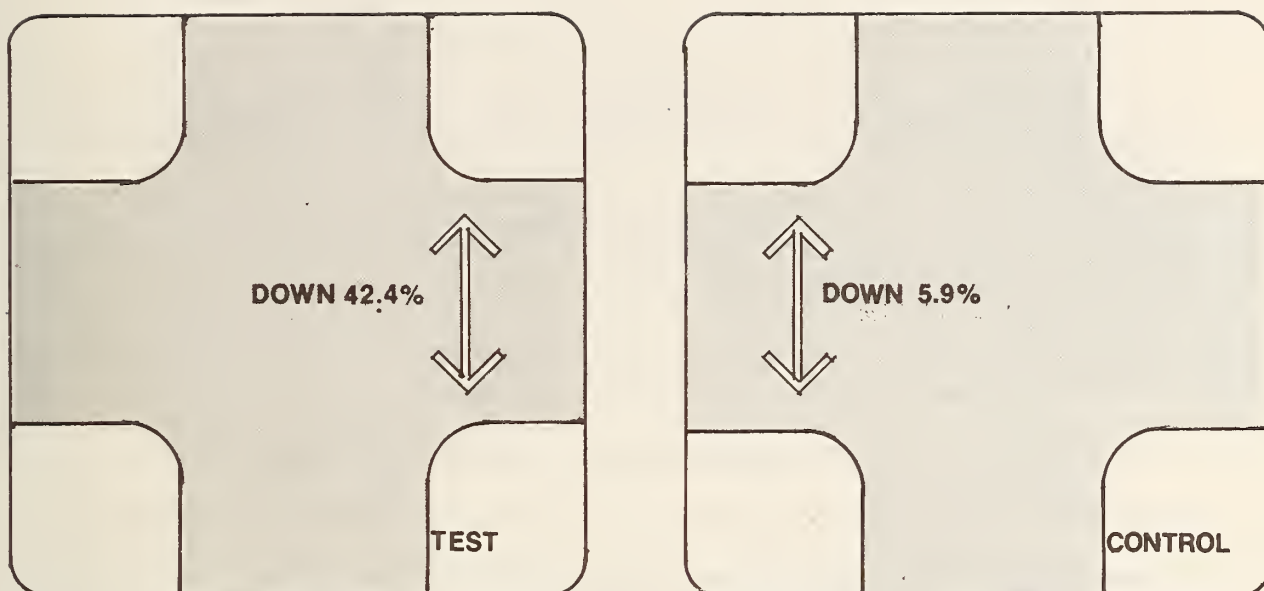


Figure 37 Pedestrian Traffic Flow Changes: Test and Control Sites

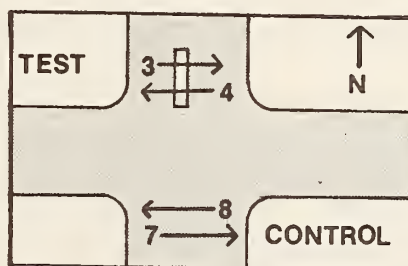
1. Crossing Time.

Countermeasure No. 2 was evaluated to determine whether the pedestrian island had any effect on the time required to cross the street. It was of interest to determine whether crossing time would have increased, as a result of pedestrians pausing or stopping within the pedestrian island. The results are shown in Table 68.

TABLE 68. MEAN CROSSING TIME IN

SECONDS

COUNTERMEASURE NO. 2; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	14.18	18.17	15.36	15.69	N	N
4, 8	16.00	17.09	16.46	16.00	N	N

S = Significant

N = Not Significant

From Table 68, it can be seen that crossing time increased substantially at the test site, but not at the control site, however, the increase was not found to be significant.

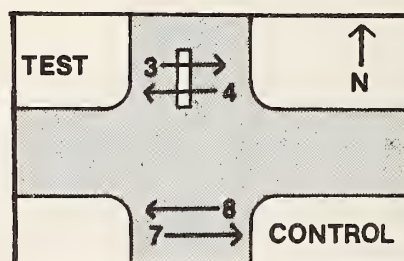
2. Evaluation of the Test and Control Intersections with Respect to Other Factors of Behavior Affecting Safety and Mobility.

These included evaluations of:

- a. Conflicts between crossing pedestrians and vehicles.
- b. Conflicts between crossing pedestrians and other pedestrians.
- c. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- d. Pedestrians crossing against the traffic signal.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in, or near, the intersection.

The results of this evaluation are shown in Table 69.

TABLE 69. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND VEHICLES
COUNTERMEASURE NO. 2; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	7 / 39	1 / 29	1 / 14	4 / 9	N	N
4, 8	1 / 30	2 / 32	2 / 14	0 / 6	N	N

S = Significant

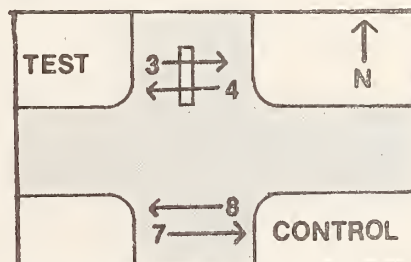
N = Not Significant

From Table 69, it can be seen that, for the eastbound pedestrian traffic at the test site, there was a substantial decrease in the number of conflicts between crossing pedestrians and vehicles; however, this is not quite significant. The other changes at the control site and the test site are not significant. An examination of the data shows that 5 of the eastbound pedestrians (17%), and 6 of the westbound pedestrians (19%) used the island to avoid conflicts with vehicles.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Table 70.

TABLE 70. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND OTHER PEDESTRIANS
COUNTERMEASURE NO. 2; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	0 / 39	0 / 29	0 / 14	0 / 9	N	N
4, 8	0 / 30	0 / 32	0 / 14	1 / 6	N	N

From Table 70, it can be seen that there was no significant increase in the number of pedestrian conflicts. S = Significant N = Not Significant

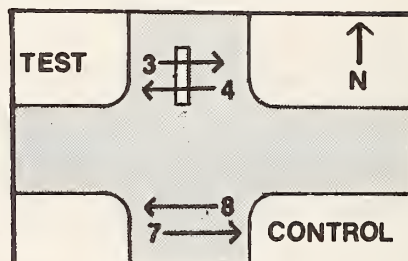
- c. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk, but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 71 and 72.

TABLE 71. PEDESTRIANS WALKING BEYOND THE
LIMITS OF THE CROSSWALK
COUNTERMEASURE NO. 2; LOCATION NO. 1



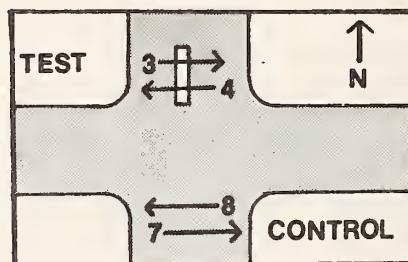
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	7 / 39	2 / 29	5 / 14	0 / 9	N	N
4, 8	3 / 30	5 / 32	1 / 14	0 / 6	N	N

S = Significant

N = Not Significant

From Table 71, it can be seen that there was no significant change in the number of people walking beyond the limits of the crosswalk.

TABLE 72. PEDESTRIANS LEAVING THE
CROSSWALK WHILE CROSSING THE STREET
COUNTERMEASURE NO. 2; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns - T test	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	7 / 39	9 / 29	5 / 14	5 / 9	N	N
6, 8	2 / 30	2 / 32	0 / 14	3 / 6	N	S

S = Significant

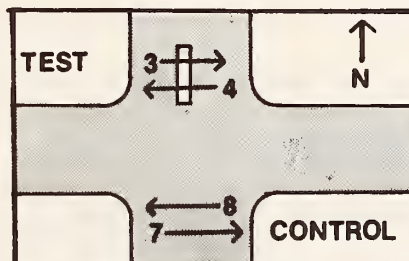
N = Not Significant

From Table 72, it can be seen that there was a significant increase in the number of pedestrians leaving the crosswalk while crossing the street, in a westerly direction, at the control site. But there was no significant change at the test site.

- d. Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing, or decreasing, such behavior.

The results of this evaluation are shown in Table 73.

TABLE 73. PEDESTRIANS CROSSING AGAINST
TRAFFIC SIGNAL
COUNTERMEASURE NO. 2; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1	Col. 2	Col. 3	Col. 4	Cols.	Cols.
	Before	After	Before	After	1&2	3&4
3, 7	6 / 39	8 / 29	0 / 14	2 / 9	N	N
4, 8	7 / 30	13 / 32	4 / 14	1 / 6	N	N

S = Significant

N = Not Significant

From Table 73, it can be seen that there was an increase in the number of pedestrians crossing against the traffic signal at both the test and the control site. But the increase is not significant.

Evaluation of Countermeasure No. 2 by Special Groups of Handicapped Persons.

The procedures used in the special group evaluations were described earlier. In brief, two groups of handicapped persons were considered to be sufficiently important, to the evaluation of this countermeasure, to warrant the use of special evaluation procedures. These were the visually impaired, and persons using wheelchairs. Approximately 15 persons, from each group, were brought to the site.

They were asked to have already become familiarized to the test locale. The test subjects used the test locations individually, following prescribed routes, while observations were made of their use of design features incorporated into those locations, and any apparent benefits, or difficulties, resulting from those design features. Time lapse photography was used to record the results of these evaluations. The results sought from this process were the same as sought in the observation of pedestrian behavior evaluations--the impact of the countermeasure design on pedestrian safety, and on mobility. In addition, several specific questions, relating to the needs and difficulties of these groups, were asked. These included the following:

- Visually Impaired Group.

- Was the pedestrian refuge island detectable.
- Was the island useful; or was it inconvenient to the visually impaired.

- Wheelchair Group.

- Was the island satisfactorily designed, and useful to persons in wheelchairs.

Evaluation Results: Visually Impaired.

1. Detectability of the Refuge Island.

Because the surface of the crosswalk, within the area of the island, was not given a textured surface to allow it to be distinguished from the street portion of the crossing, only those subjects who were near either side of the crosswalk, and whose canes detected the raised concrete areas of the island beyond the crosswalk limits, could detect the existence of the island. On four of the fourteen crossing events, such a detection of the island was made. The following narrative describes one of these events:

- The test subject was beyond the left crosswalk line by about one foot (.305 m). As he reached the island, his cane detected the raised concrete area. He reached out and touched the handrail with his left hand, and guided himself to the protected area of the crosswalk, waiting there about 7 seconds for traffic to clear, so that he could continue with his crossing of the street.

2. Usefulness or Potential Hazardousness of the Island.

Because there was no tactile indication of the edges of the crosswalk, the visually impaired faced great difficulty in maintaining a straight course, and staying within the limits of the crosswalk. Those who strayed only a short distance outside of the crosswalk, then faced the unexpected hazard of the raised concrete area of the pedestrian island. On nine of the fourteen crossing events, blind test subjects wandered beyond the limits of the crosswalk. The following narratives illustrate two of these events:

- The test subject was about 18 inches (45.72 cm) beyond the limits of the crosswalk, on the left. He was walking at a fast pace, and reached the raised concrete part of the island rather abruptly, striking it with a foot, just after his cane had detected it, but before he could react. He stopped for a moment, then, using his cane for detection, proceeded into the crosswalk, across the island, and completed his crossing.
- The test subject was beyond the crosswalk line on the left. He reached the raised concrete portion of the island, detecting it with his cane; paused briefly, then elected to pass to the outside of the island, on the street side, to complete his street crossing. He was vulnerable to traffic for the remainder of his trip across the street.

Evaluation Results: Wheelchair Group.

1. Satisfactory Design and Usefulness of the Island.

There were a total of 26 crossing events in the evaluation of the Pedestrian Refuge Island by the special group of persons using wheelchairs. This evaluation determined whether these subjects found the island a useful location for slowing, pausing, or stopping to survey traffic conditions before completing their crossing of the street. Following are the results of this evaluation:

- Subjects who came to a complete stop 4
- Subjects who paused briefly 4
- Subjects who slowed while within the island 5

These results illustrate that, in 13 of the 26 crossing events, the island was used as location for surveying traffic, and that in 8 of the 26 events (30 percent), the subjects briefly paused, or came to a full stop, within the island. The following narratives are illustrative of these events:

- The subject crossed to the island. When, within the protected area of the island, he came to a complete stop, waiting about 5 seconds while surveying the traffic situation. He then continued, and completed his crossing of the street.
- The wheelchair subject was propelled by an attendant. As they neared the island, they slowed somewhat; then, while within the island, they came to a nearly full stop; to see whether an auto, approaching the intersection, would stop. As the car stopped, they continued, and reached the ramp on the far side of the street.
- As the subject approached the island, he stopped, propelling the wheelchair; and coasted, with his hands lightly touching the wheel as he checked traffic.

Then, seeing all was clear, he continued across the second half of the street. The same subject, on his return across the street, came to a brief, but full stop within the island, before completing his crossing.

- The subject reached the island and slowed to a nearly full stop, while awaiting turning traffic to clear the crosswalk. He then continued with his crossing.

Summary of Findings: Countermeasure No. 2, Pedestrian Refuge Island

1. There was no significant change in the vehicular traffic flow.
2. Pedestrian traffic decreased at the test and control sites, but this change does not appear to be related to the countermeasure.
3. A significant increase in mean crossing time was noted for pedestrian crossing in the one direction. But no significant increase occurred in the other direction, nor at the control site. This change indicated that the island was being used as a stopping place.
4. There was a decrease in the number of conflicts between pedestrians, crossing in an easterly direction, and vehicles; but this was not significant, and there was no significant change for the other direction, nor at the control site. However, it was also found that 17% of the eastbound pedestrian, and 19% of the westbound pedestrians used the island to avoid conflicts with vehicles.
5. There was no significant change in the number of pedestrian conflicts.
6. There was no significant change in the number of people walking beyond the limits of the crosswalk, nor of pedestrians leaving the crosswalk while crossing the street.
7. There was no significant change in the number of pedestrians crossing against the traffic signal.
8. The refuge island was not detectable to the visually impaired, because it had no textural or grade change.
9. Because the crosswalk markings were not detectable, some visually impaired people strayed from the crosswalk, and the refuge island became an unexpected event.
10. In one half of all crossings, the subjects in wheelchairs paused or stopped on the island, using it as a refuge.

Conclusions.

The countermeasure was effective in principle, but certain details of the design, as constructed, proved to be inappropriate for visually handicapped people.

The refuge island had no adverse affects on the flow of pedestrian and vehicular traffic.

The countermeasure was obviously useful for people using wheelchairs. However, for visually impaired people, two changes are necessary.

- a. The floor surface of the island must be detectable either by an appropriate change of texture, or by raising the surface by means of a curb with curb ramps.
- b. If the refuge island forms a physical barrier to crossing the street at any point (as was the case in the test design), then the boundaries of the crosswalk should be detectable to the visually impaired, so that they can be directed through the island.

For refuge islands to be successful, of course, they must be designed to protect all the pedestrians that use them, and to be highly visible to vehicle drivers.

countermeasure 3

midblock crosswalk markings

Countermeasure No. 3 is a design intended to increase the safety of pedestrians using marked midblock crosswalks, by drawing greater driver attention to the crosswalk, so that the drivers will either reduce speed or search more effectively for pedestrians attempting to cross.

Most attempts to increase driver awareness of midblock crosswalks have been shown by previous research to have not been successful. These attempts have included painted treatment of the crosswalk itself, warning signs, painted markings on the roadway in advance of the crosswalk, and flashing warning lights. Only the flashing lights have been shown to have had some effect on driver speed.

Countermeasure No. 3 is an attempt to design a roadway marking for the approach to a midblock crosswalk which will be more effective than those which have been used in the past. Those previous measures have generally been designs painted on the roadway in a direction parallel to the crosswalk; that is, perpendicular to the movement of the vehicle. The result has often been that drivers come upon such markings suddenly, too quickly to take any action. The purpose of Countermeasure No. 3 is to determine whether markings on the roadway which are parallel to the line of vehicular movement, extending farther in advance of the crosswalk, will be more effective in increasing the safety of midblock crossings.

Evaluation of Countermeasure No. 3.

Countermeasure No. 3 was evaluated based on its performance in solving the following problem:

- Drivers do not notice or react to the crosswalk which is at a midblock location.

In addition, evaluations were made of the general use of the location where this countermeasure was installed, and of pedestrian behavior which might indicate problems affecting pedestrian safety or mobility.

Evaluation Location.

Countermeasure No. 3 was evaluated in the downtown area of Atlanta, Georgia, a southeastern city of over 1 million metropolitan population. Two midblock crossings, on the same street but one block apart, were found which were very similar and could serve as the test and control locations (see Figure 38). The street was in the midst of a dense and active retail shopping district which included a major department store, a large federal post office facility, and two-story retail structures. The street was 48 feet (14.64 m) in width, with two lanes provided for traffic flow in each direction.

Traffic volumes on the street for the test location and the control location are shown in Tables 74 and 75.

TABLE 74. TRAFFIC VOLUMES - VEHICLES PER HOUR

TEST LOCATION

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South	505	455	595	560	550	555

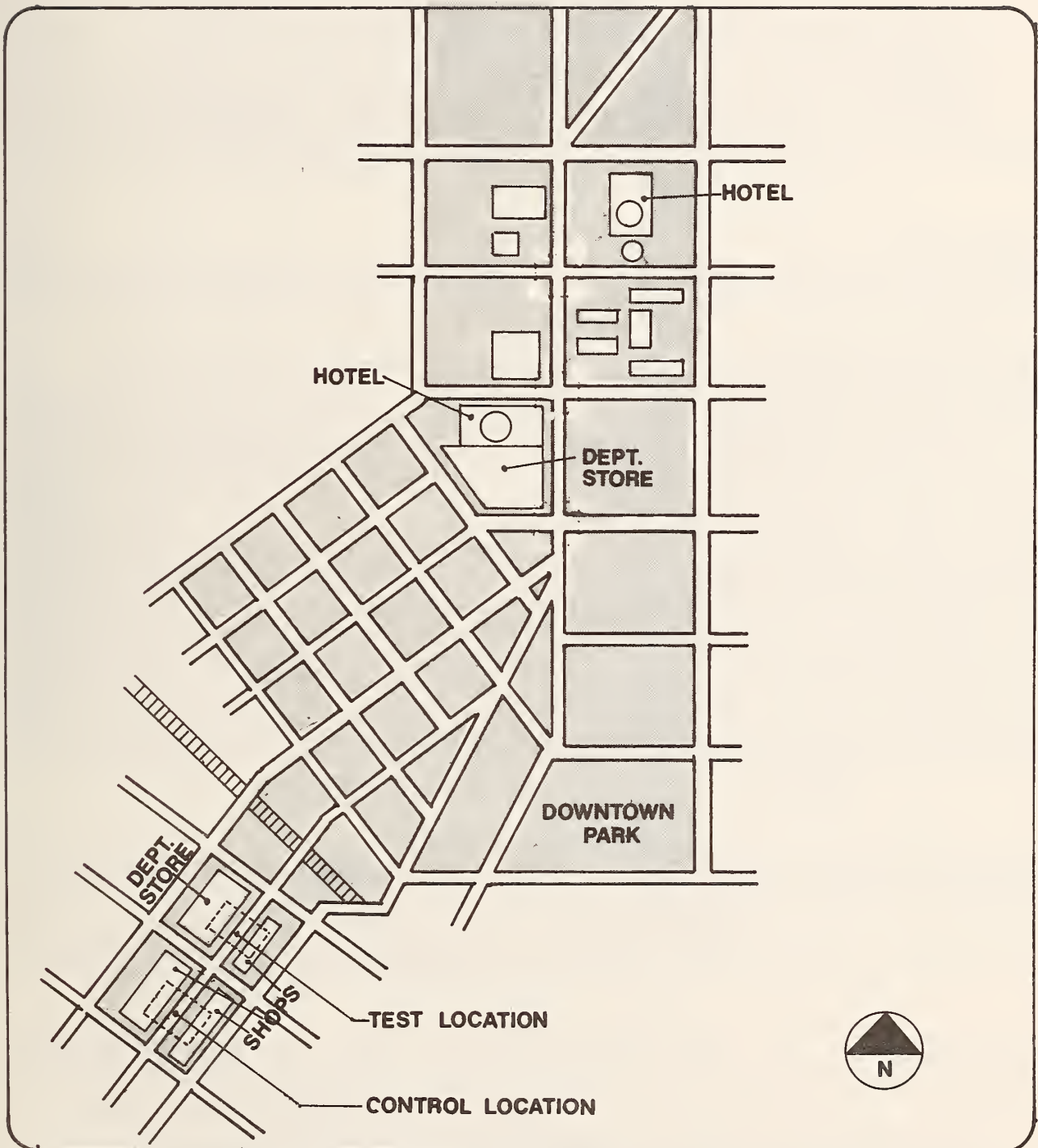


Figure 38 Atlanta: Test and Control Location Map

TABLE 75. TRAFFIC VOLUMES - VEHICLES PER HOUR

CONTROL LOCATION

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Thru	335	440	420	410	455	410

Pedestrian flows, for persons crossing at the midblock crossings, are shown for the test and control locations in Tables 76 and 77.

TABLE 76. PEDESTRIAN FLOWS - PERSONS PER HOUR

Test Location

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
East-West	295	405	565	545	510	465

TABLE 77. PEDESTRIAN FLOWS - PERSONS PER HOUR

Control Location

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
East-West	45	95	70	90	70	75

Description of the Changes Made at the Test Location.

Figure 39 illustrates the painted indication developed as Countermeasure No. 3, to serve as a midblock crossing warning. Painted lines were used in such a manner as to give the appearance of the traffic lanes becoming narrower as they approached the midblock crosswalk. The assumption to be evaluated was that the apparent narrowing of the traffic lanes would result in drivers reducing their speed.

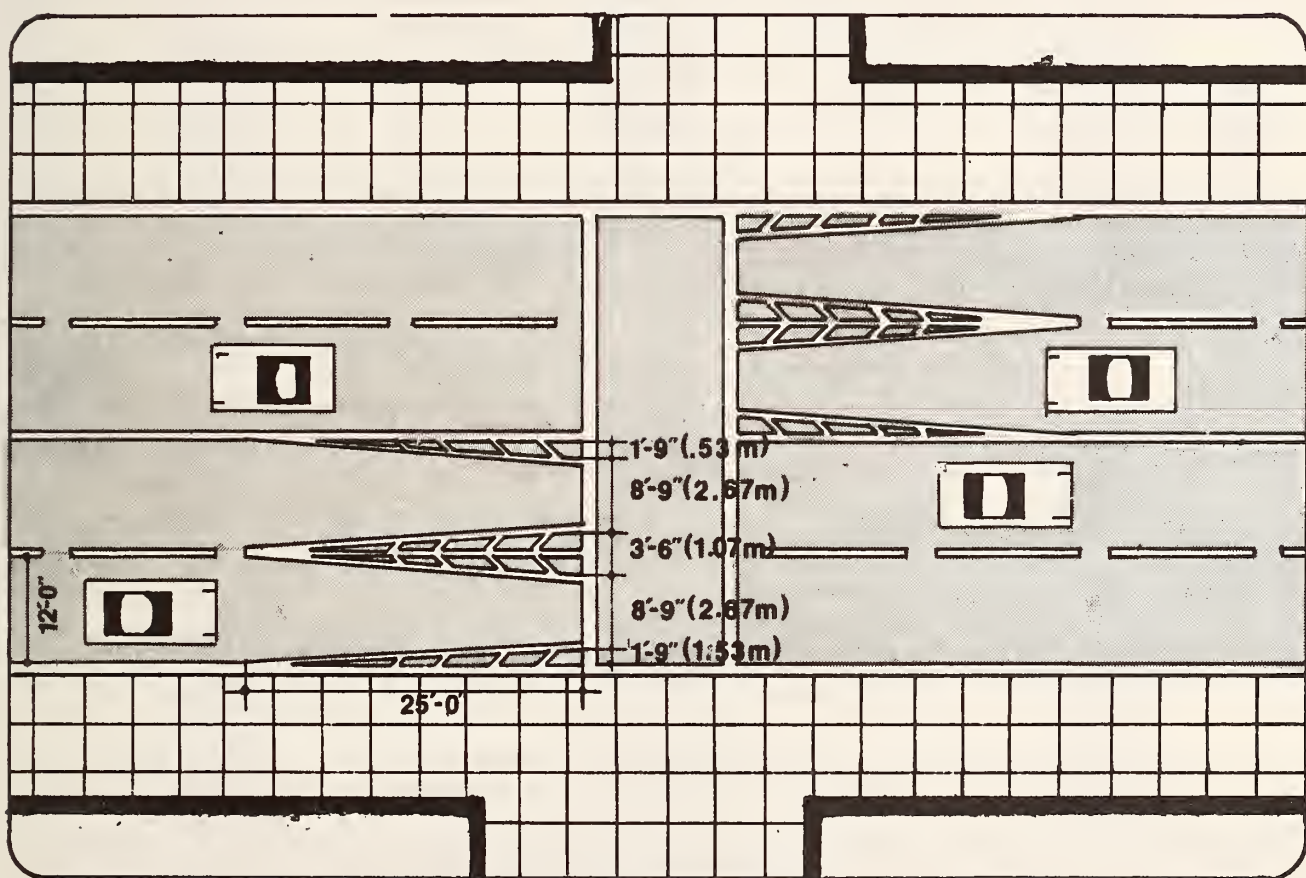


Figure 39 Midblock Crossing Design

Evaluation Results.

1. Effect on traffic volumes.

From Figure 40, it can be seen that there was a small decrease in traffic volumes at the test site and a small increase at the control site, but these changes are not significant.

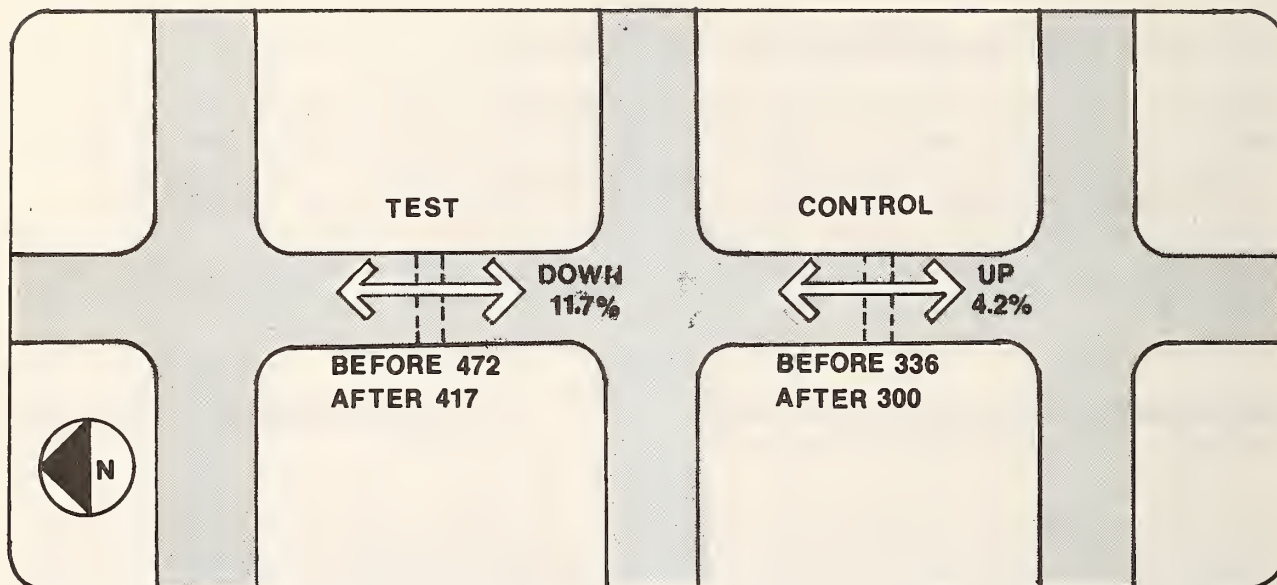


Figure 40 Vehicular Traffic Volume Changes

2. Effect of the countermeasure upon vehicular speed. An additional evaluation was performed to determine whether the painted crosswalk warning designation had any effect upon the speed of vehicles approaching the midblock crossing.

The results of this evaluation are shown in Figure 41.

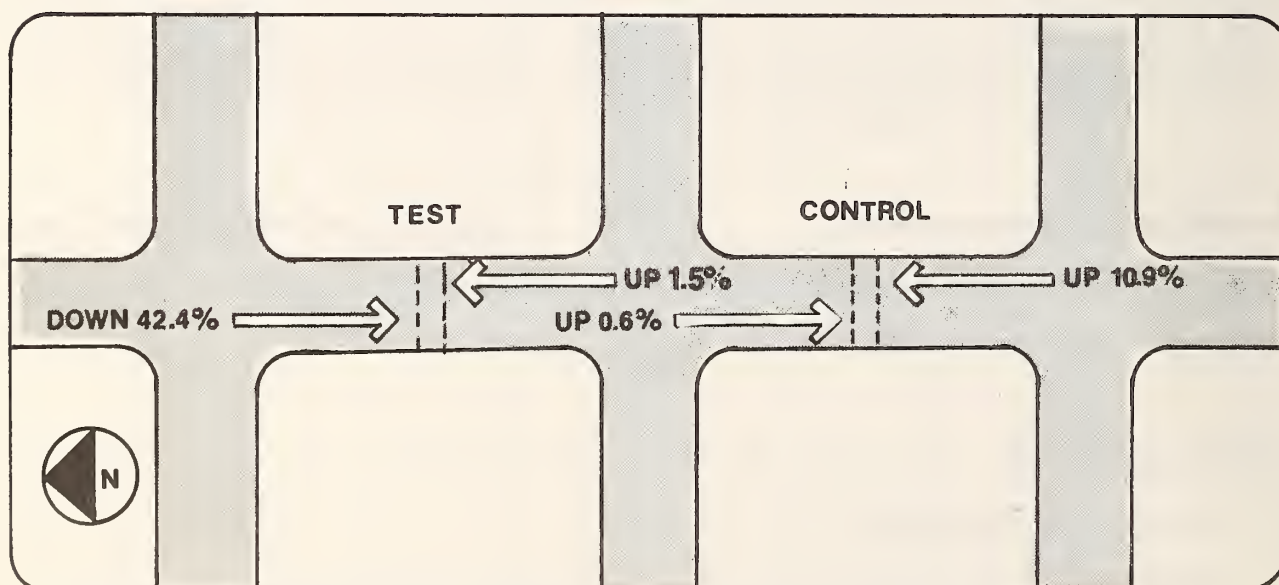


Figure 41 Vehicular Speed Changes

From Figure 41, it can be seen that there was a significant decrease in vehicular speed of southbound traffic at the control site, but no significant change in northbound traffic, nor in traffic using the control site. The results are therefore inconclusive.

3. Effect upon pedestrian volumes. Figure 42 shows the changes in pedestrian volumes.

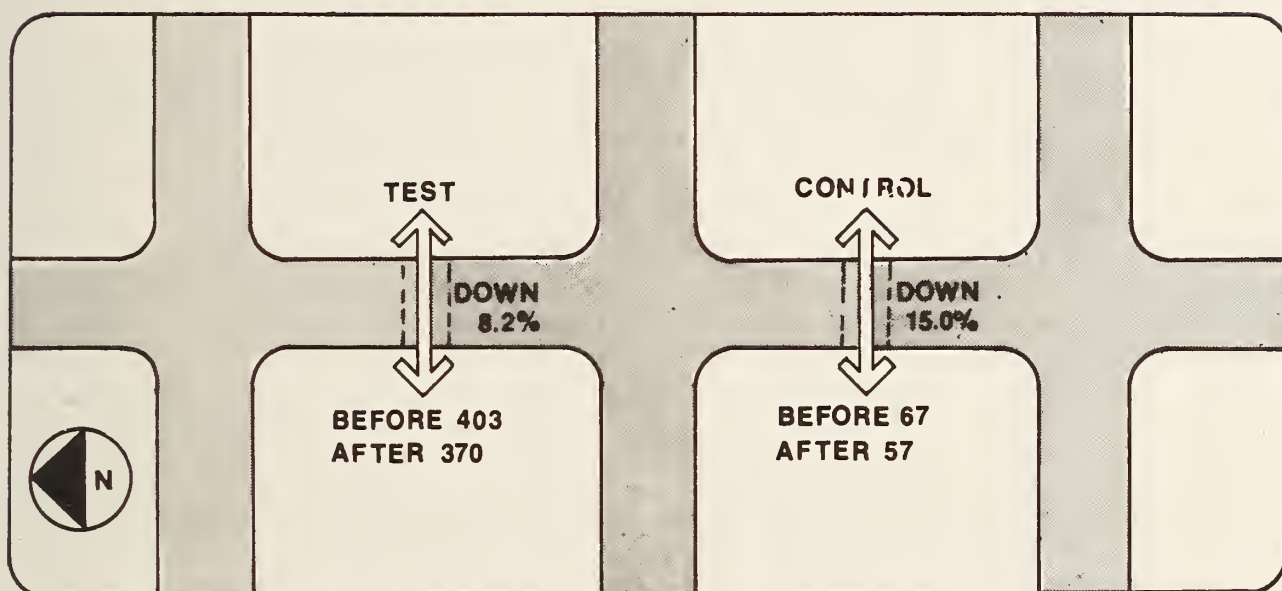


Figure 42 Pedestrian Traffic Volume Changes

From Figure 42, it can be seen that there was a decrease in pedestrian volumes at both the test and the control sites. From these results it is concluded that the countermeasure did not adversely affect pedestrian flow.

4. Crossing Time. Countermeasure No. 3 was evaluated to determine whether it had any effect upon the time required to cross the street.

The results are shown in Table 78.

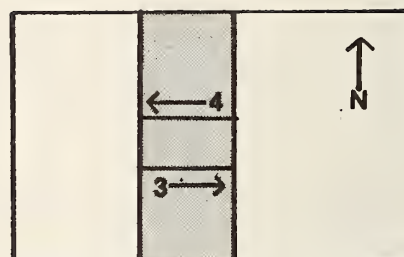
From Table 78, it can be seen that at both test and control sites, there was a decrease in crossing time. However, the change was only significant for the westbound pedestrian. As this significant decrease occurred at both the test and the control locations, it is concluded that the countermeasure was not the cause of this change.

5. Evaluation of the Test and Control Intersections With Respect to Other Factors of Behavior Affecting Safety and Mobility.

These included evaluations of:

TABLE 78. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 3; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	10.684	10.257	13.194	12.811	N	N
4	10.800	10.026	14.022	12.600	N	N

S = Significant

N = Not Significant

- a. Conflicts between crossing pedestrians and vehicles.
 - b. Conflicts between crossing pedestrians and other pedestrians.
 - c. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation are shown in Table 79.

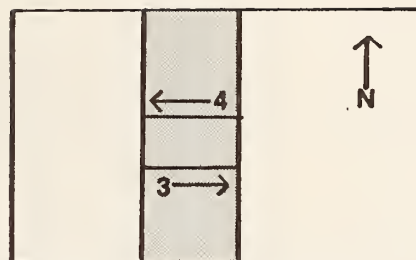
From Table 79, it can be seen that there was a significant increase in conflicts between crossing pedestrians and vehicles. However, a similar change occurred at the control site. Therefore, it is not possible to conclude that the change results from the countermeasure.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Table 80.

TABLE 79. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND VEHICLES

COUNTERMEASURE NO. 3; LOCATION NO. 1



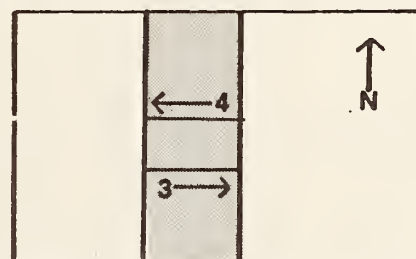
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	8 / 40	25 / 35	6 / 31	21 / 38	S	S
4	15 / 45	26 / 39	8 / 46	32 / 40	S	S

S = Significant

N = Not Significant

TABLE 80. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND OTHER PEDESTRIANS

COUNTERMEASURE NO. 3; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	1 / 40	1 / 38	0 / 31	0 / 38	N	N
4	1 / 45	0 / 39	0 / 46	0 / 40	N	N

S = Significant

N = Not Significant

From Table 80, it can be seen that there was no significant change in the number of conflicts between pedestrians.

c. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.

This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

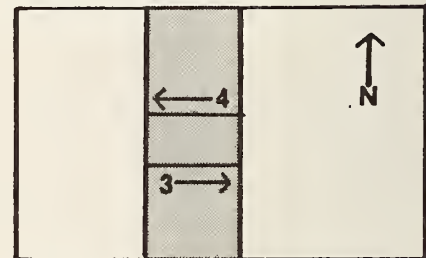
The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 81 and 82.

TABLE 81. PEDESTRIANS WALKING BEYOND LIMITS OF
CROSSWALK

COUNTERMEASURE NO. 3; LOCATION NO. 1

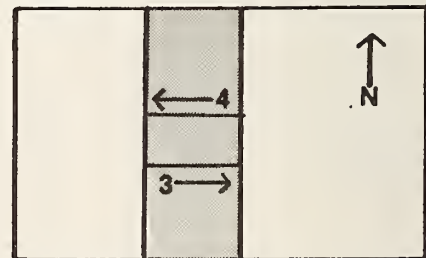


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	10 / 40	2 / 35	18 / 31	17 / 38	N	N
4	13 / 45	2 / 39	17 / 46	16 / 40	S	N

S = Significant N = Not Significant

TABLE 82. PEDESTRIAN LEAVING THE CROSSWALK
WHILE CROSSING THE STREET

COUNTERMEASURE NO. 3; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	0 / 40	3 / 35	0 / 31	8 / 38	N	S
4	0 / 45	11 / 39	12 / 46	8 / 40	S	N

S = Significant N = Not Significant

From Table 81, it can be seen that the number of pedestrians walking beyond the limits of the crosswalk decreased significantly in the one direction, and almost as significantly in the other.

From Table 82, it can be seen that the number of pedestrians leaving the crosswalk while crossing the street increased significantly in the one direction, and the same pattern (in the opposite direction) is apparent at the control site.

Summary of Findings: Countermeasure No. 3, Midblock Refuge Island

1. There was no significant change in the traffic volumes.
2. Although there was a significant decrease in vehicular speed of southbound traffic at the crosswalk, there was no significant change of speed by northbound traffic. So the results are inconclusive.
3. A decrease in pedestrian volumes occurred at both the test and control sites. It appears that the countermeasures did not adversely affect pedestrian traffic flow.
4. A significant decrease in pedestrian crossing time occurred at both the test and control sites, but only for the westbound traffic. It cannot be concluded that the countermeasure effected the change.
5. Although a significant increase in the number of conflicts between pedestrians and vehicles occurred at the test site, the same change occurred at the control site. Therefore, it cannot be concluded that the countermeasure has effected a change.
6. There was no significant change in the number of conflicts between pedestrians.
7. The number of pedestrians walking beyond the limits of the crosswalk decreased significantly in the one direction and almost significantly in the other.
8. The number of pedestrians leaving the crosswalk while crossing the street increased significantly for the one direction only. At the control site, the same pattern occurred for the other direction only.

Conclusions.

The results of the test are inconclusive but suggest that further evaluations of this countermeasure are worth consideration. There is some evidence that vehicle drivers respond more to the countermeasure than to the conventional crosswalk. And there is good evidence that pedestrians are less likely to cross away from the crosswalk. And, the countermeasure does not seem to have engendered adverse effects.

countermeasure 4

widened crosswalk

Countermeasure No. 4 is intended for congested crosswalks where that congestion makes crossing difficult for people with balance problems and where it restricts the flow of pedestrians, reduces their crossing speed (increasing crossing time and their exposure to traffic) and results in increased conflicts between pedestrians.

In addition, Countermeasure No. 4 may result in increasing the driver's view of crossing pedestrians by moving back the stop line as well as the crosswalk line and further separating vehicles from most of those pedestrians. This is illustrated in Figure 43.

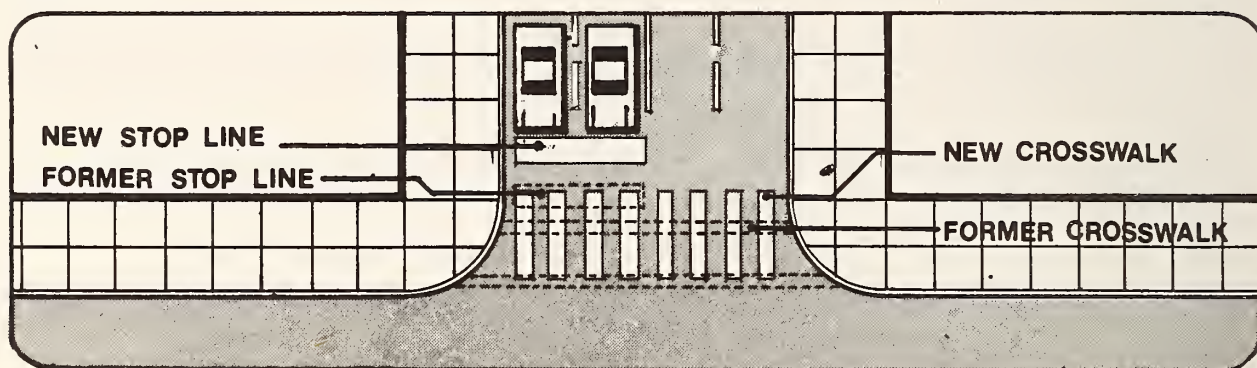


Figure 43 Widened Crosswalk Concept

Countermeasure No. 4 was evaluated based on its performance in solving the following problems:

- Crosswalk congestion.
- Stopped or parked cars obstructing the driver's view of crossing pedestrians.

In addition, evaluations were made of the general use of the locations where this countermeasure was installed, and of pedestrian behavior which might indicate problems affecting safety or mobility.

Evaluation Locations.

Countermeasure No. 4 was evaluated at three adjacent intersections on the main downtown street of Atlanta, Georgia (see Figure 44), a large southeastern city of about 1 3/4 million people (metropolitan area). The street is in the midst of the downtown commercial retail, office, and entertainment district. It is a location that has been developed intensively with hotels, high rise office structures and other facilities, all of which generate considerable pedestrian traffic.

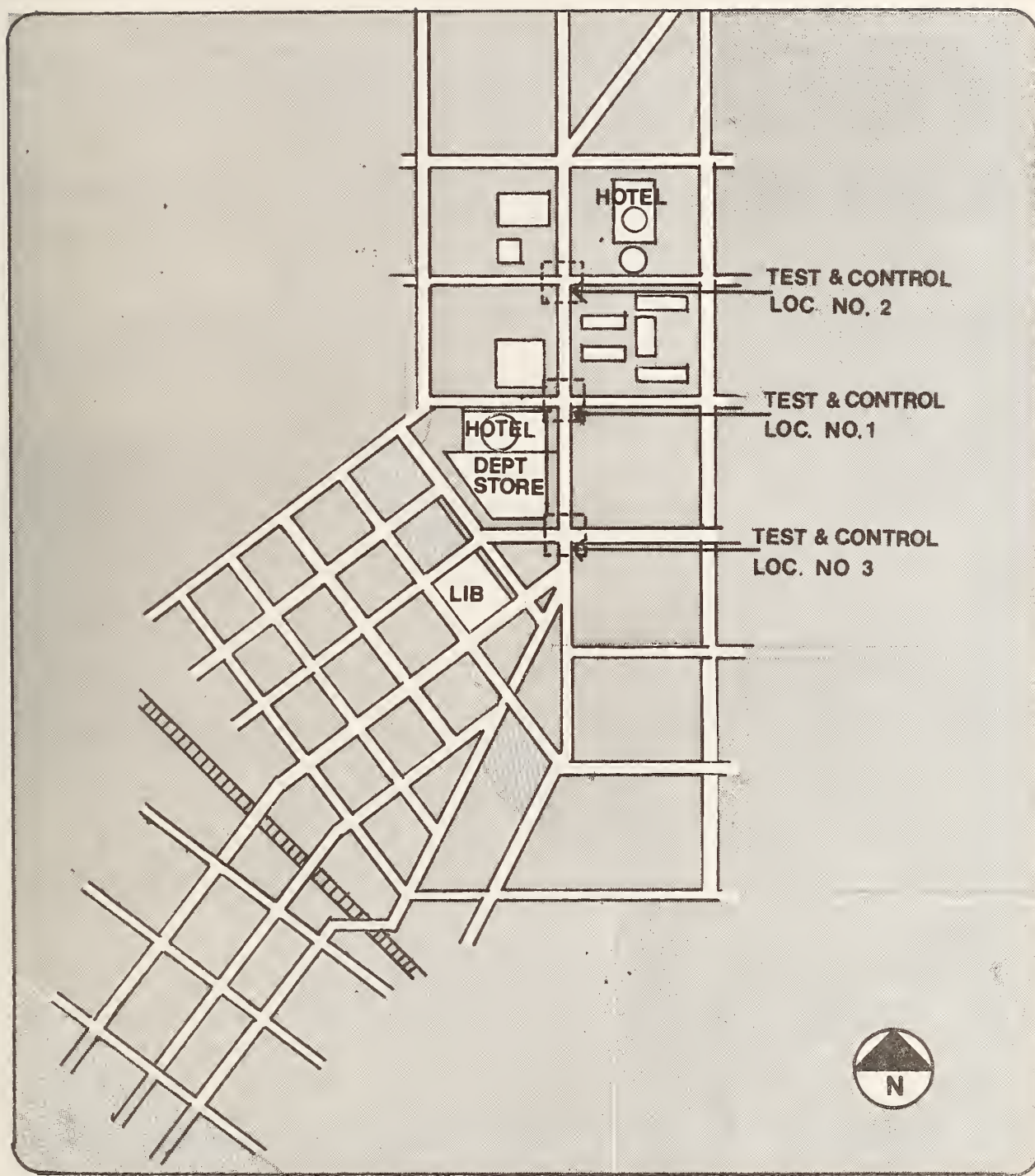


Figure 44 Atlanta: Widened Crosswalk: Test and Control Sites Location

The main north-south street is 48 feet (14.64 m) in width and provides two lanes of traffic flow in each direction. The three intersecting streets are each about 40 feet (12.2 m) in width. Each has provision for service parking or loading areas along one curb, and has two lanes of moving traffic.

At all three locations, the countermeasure was installed at the west crosswalk; the east crosswalk served as the control location.

Location No. 1.

Location No. 1 is located between the other two intersections (see Figure 45). The intersecting street is one way westbound, carrying two lanes of moving traffic. Before the change, the test and control crosswalks were 12'6" (3.81 m) and 12'9" (3.89 m) in width respectively.

Traffic volumes on both streets were considerable, averaging 325 vehicles/lane/hour on the north-south street and 290 vehicles/lane/hour on the westbound intersecting street. Traffic flows remained consistently heavy throughout the day (see Table 83).

TABLE 83. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	1165	1060	1360	1180	1295	1210
Turning	135	95	105	95	95	105
Total	1300	1155	1465	1275	1390	1315
East-West Direction						
Thru	465	385	410	370	510	430
Turning	175	105	140	145	170	145
Total	640	490	555	515	680	575

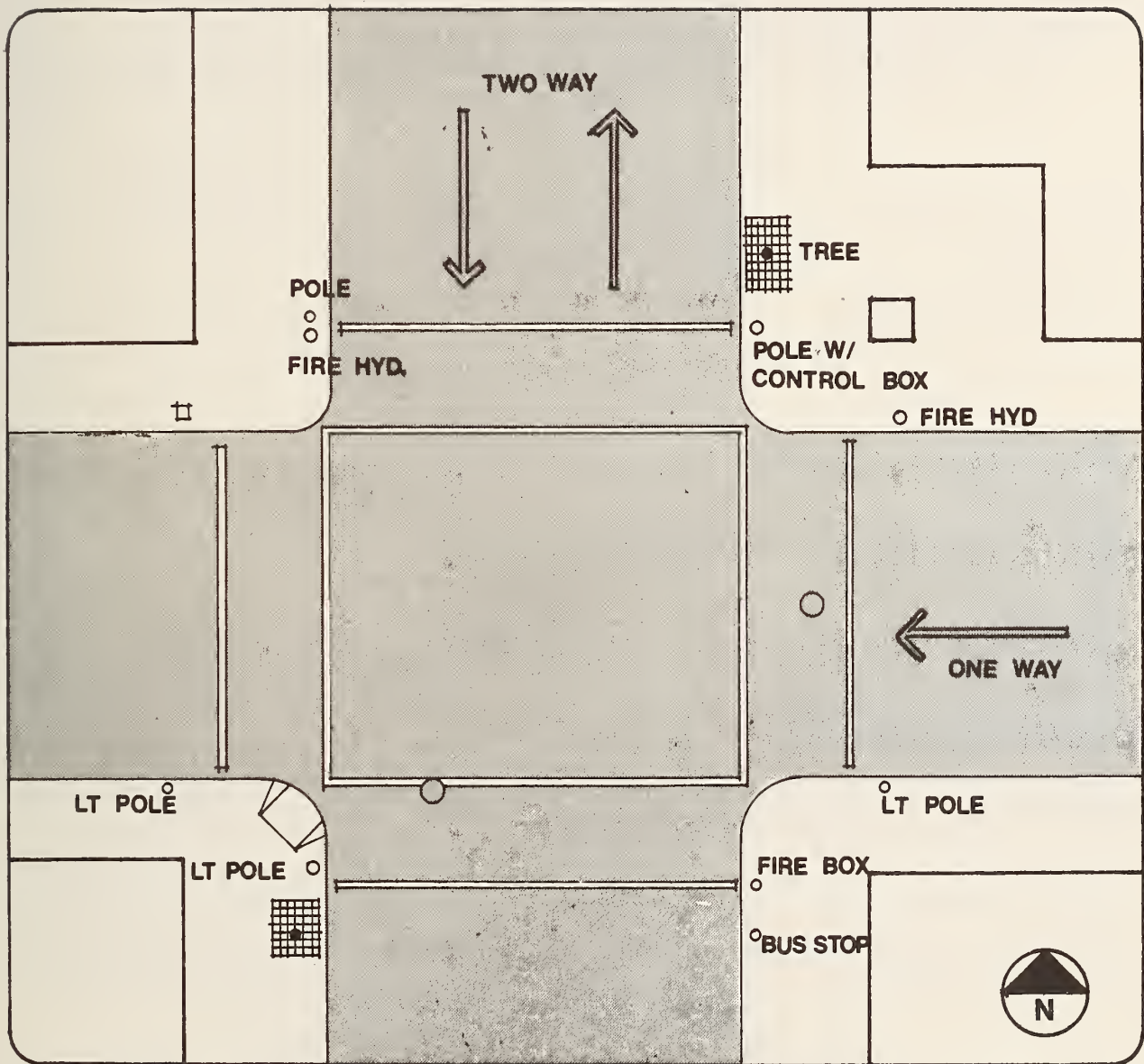


Figure 45 Atlanta: Widened Crosswalk: Location 1 Before Alteration

The intersection was signalized, and had pedestrian "walk"/"don't walk" signals as well. Signal timing was as follows:

North-South Street	Elapsed time in seconds	East-West Street	Elapsed time in seconds
Green/"Walk"	0	Red/"Don't Walk"	0
Flashing "Don't Walk"	34	Green/Walk	46
Yellow/Continuous "Don't Walk"	42	Flashing "Don't "Walk"	60
		Yellow/Continuous "Don't Walk"	66
Red	46		
Green	70	Red	70

Pedestrian flows were measured in the west and east crosswalks and were heavy throughout the day, reaching their peak during the noon hour as shown in Table 84.

TABLE 84. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
West side	638	785	1533	765	1010	945
East side	525	640	1310	585	805	775

Description of Changes Made at Location No. 1.

The west crosswalk was widened, on the side away from the intersection, from 12 feet 6 inches (3.82 m) to 16 feet (4.88 m) in width. The east crosswalk was left unchanged to serve as the control location. Figure 46 illustrates the changes made.

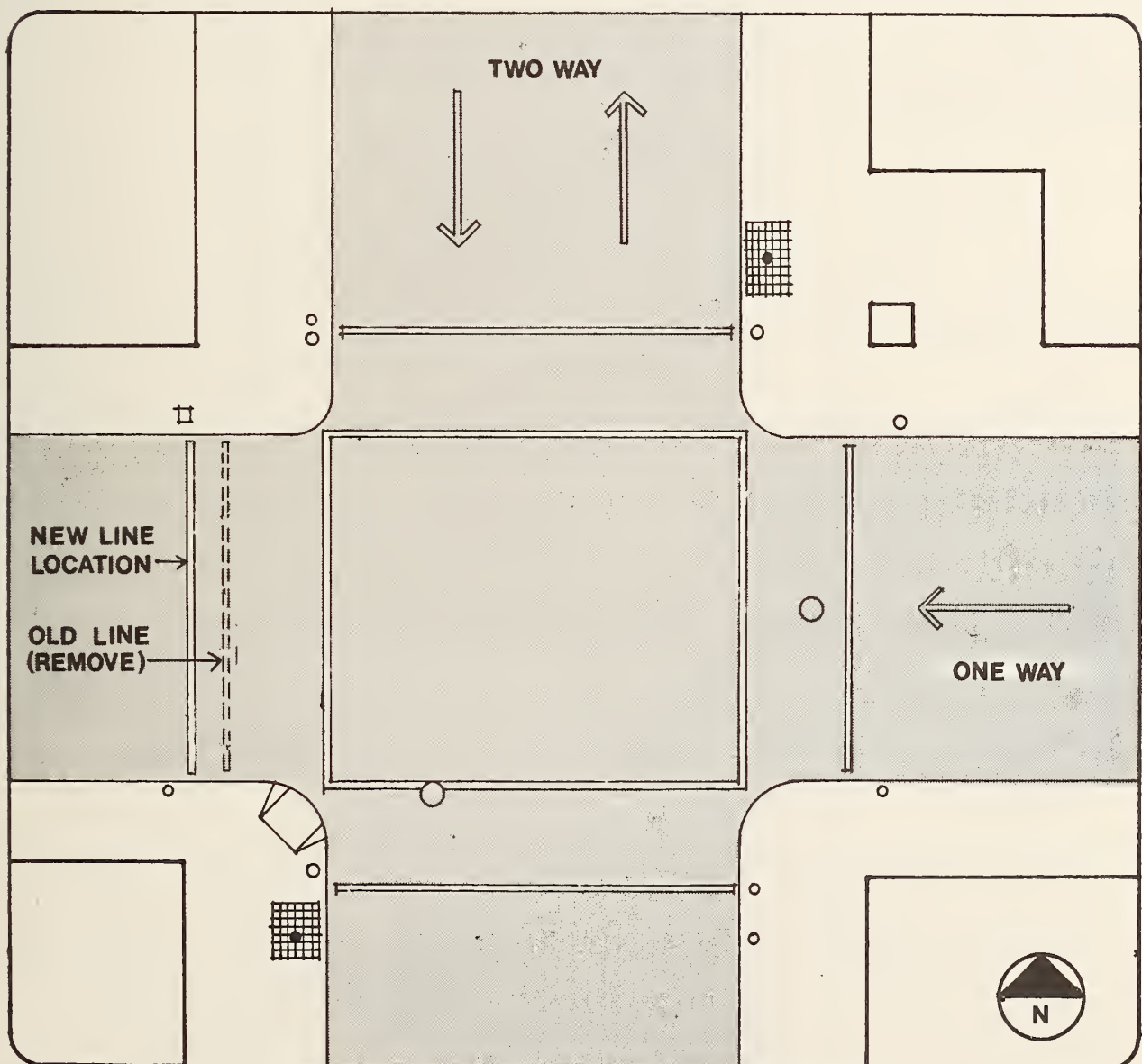


Figure 46 Atlanta: Widened Crosswalk: Location 1 After Modification

Location No. 2.

Location No. 2 was the northernmost of the three intersections. The intersecting street flows one way, eastbound. Traffic volumes on both streets are heavy, averaging 300 vehicles/lane/hour on the north-south street, and 250 vehicles/lane/hour on the intersecting east-west street (see Table 85).

TABLE 85. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	1030	990	1150	1240	1175	1115
Turning	105	90	100	100	120	105
Total	1135	1080	1250	1340	1295	1220
East-West Direction						
Thru	260	245	310	275	295	275
Turning	205	200	240	235	230	225
Total	465	445	550	510	525	500

The intersection was signalized, and had pedestrian "walk"/"don't walk" signals as well. Signal timing was as follows:

North-South Street	Elapsed time in seconds	East-West Street	Elapsed time in seconds
Green/"Walk"	0	Red/"Don't Walk"	0
Flashing "Don't Walk"	30	Green/"Walk"	45
Yellow/Continuous "Don't Walk"	40	Flashing "Don't Walk"	55
		Yellow/Continuous "Don't Walk"	65
Red	45		
Green	70	Red	70

Description of Changes Made at Location No. 2.

The west crosswalk was widened, on the side away from the intersection, from 12 feet 8 inches (3.86 m) to 18 feet (5.49 m) in width. The east crosswalk was left unchanged to serve as the control location. Figures 47 and 48 show the widened crosswalk before and after modification.

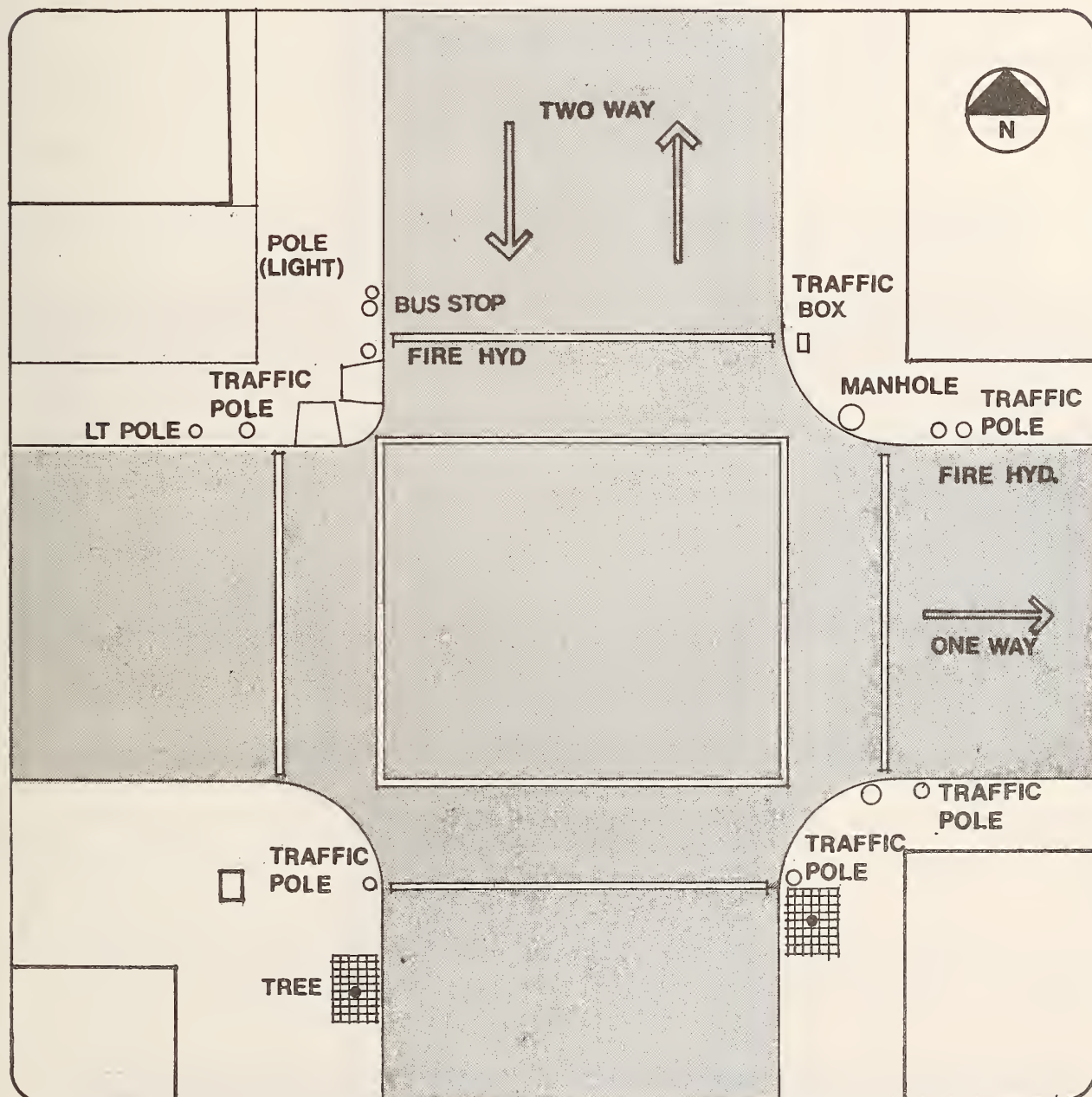


Figure 47 Atlanta: Widened Crosswalk: Location 2 Before Alteration

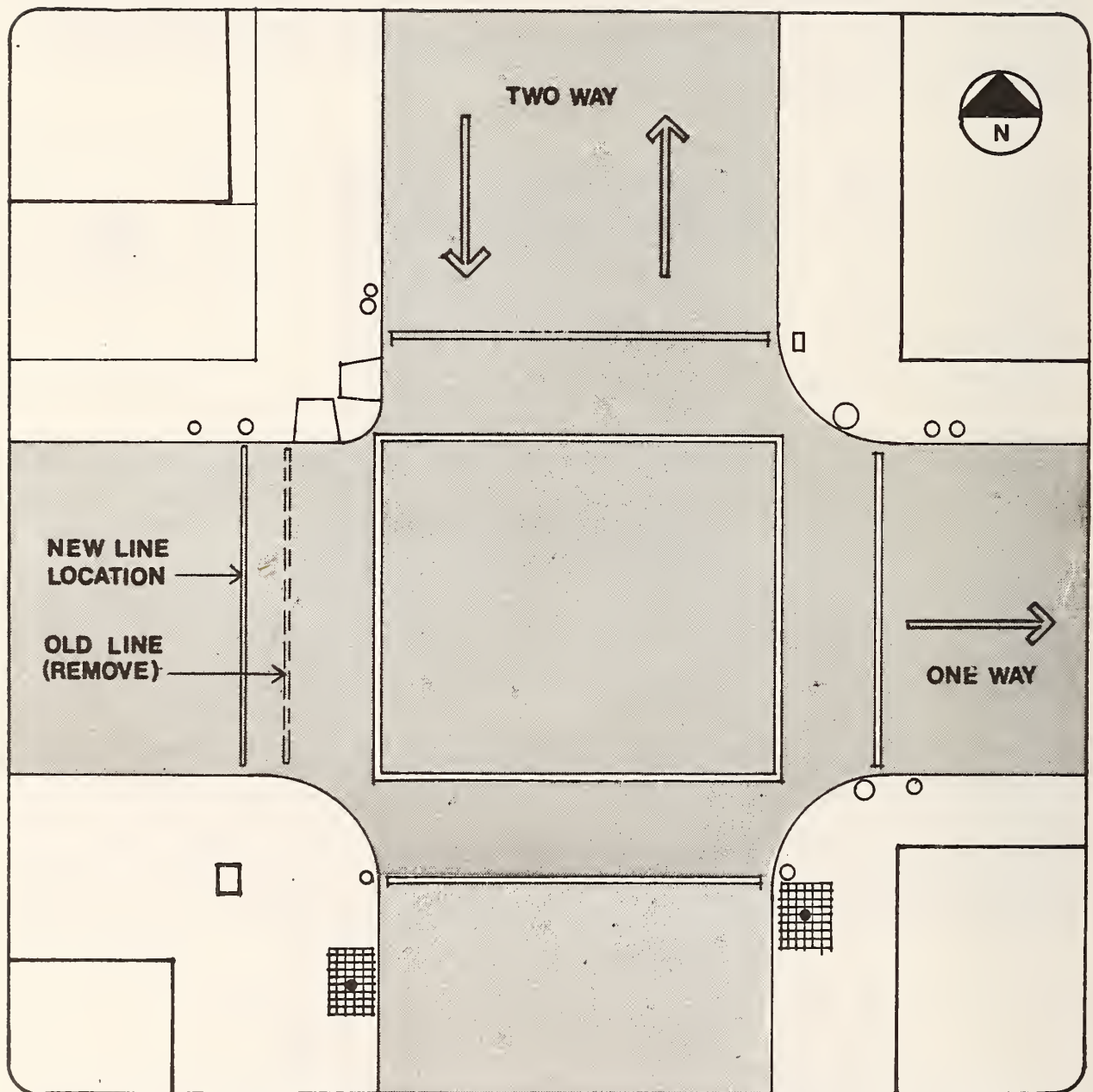


Figure 48 Atlanta: Widened Crosswalk: Location 2 After Modification

Pedestrian flows were measured in the west and east crosswalks and were heavy throughout the day, reaching their peak during the noon hour (see Table 86).

TABLE 86. PEDESTRIAN FLOWS - PERSONS PER HOUR

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
West side	360	370	1070	505	545	570
East side	375	385	845	650	615	575

Location No. 3.

Location No. 3 (see Figure 49) was the southernmost of the three intersections involved in this evaluation. The intersecting street is one-way eastbound, providing for two lanes of moving traffic. Traffic volumes on both streets are heavy, although flows on the eastbound street are the lowest of the three intersecting the north-south main street which are involved in the evaluation of Countermeasure No. 4. Traffic volumes average 340 vehicles/lane/hour on the north-south street, and 180 vehicles/lane/hour on the intersecting eastbound street (see Table 87).

TABLE 87. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	1250	1295	1270	1105	1295	1245
Turning	100	70	95	70	110	90
Total	1350	1365	1365	1175	1405	1335
East-West Direction						
Thru	270	320	320	280	385	315
Turning	40	25	50	35	50	40
Total	310	345	370	325	435	355

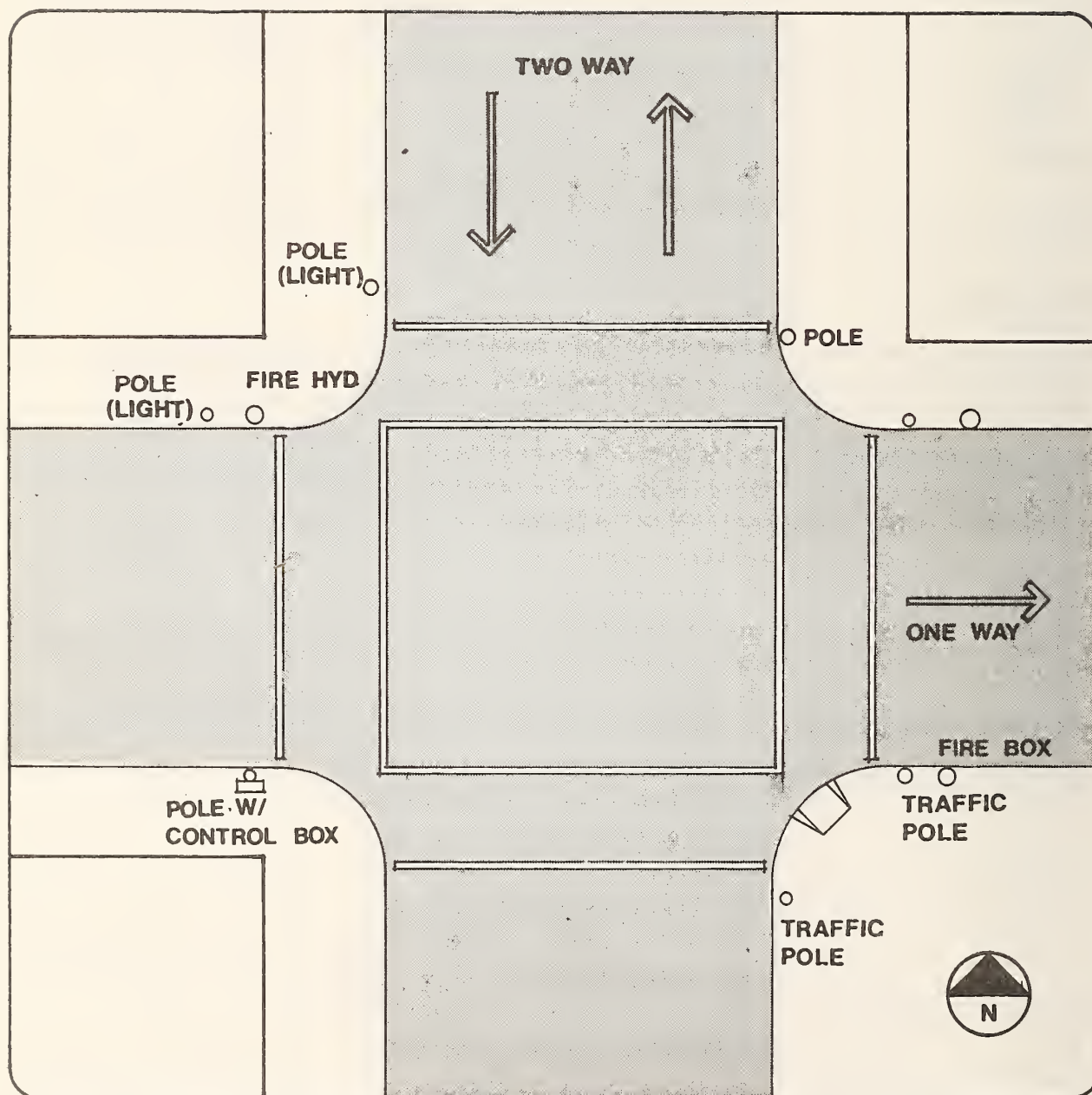


Figure 49 Atlanta: Widened Crosswalk: Location 3 Before Alteration

Description of Changes Made at Location No. 3

The west crosswalk was widened on the side away from the intersection, from 12 feet 7 inches (3.84 m) to 14 feet 6 inches (4.41 m). The east crosswalk was left unchanged to serve as the control location. Figure 50 illustrates the changes made.

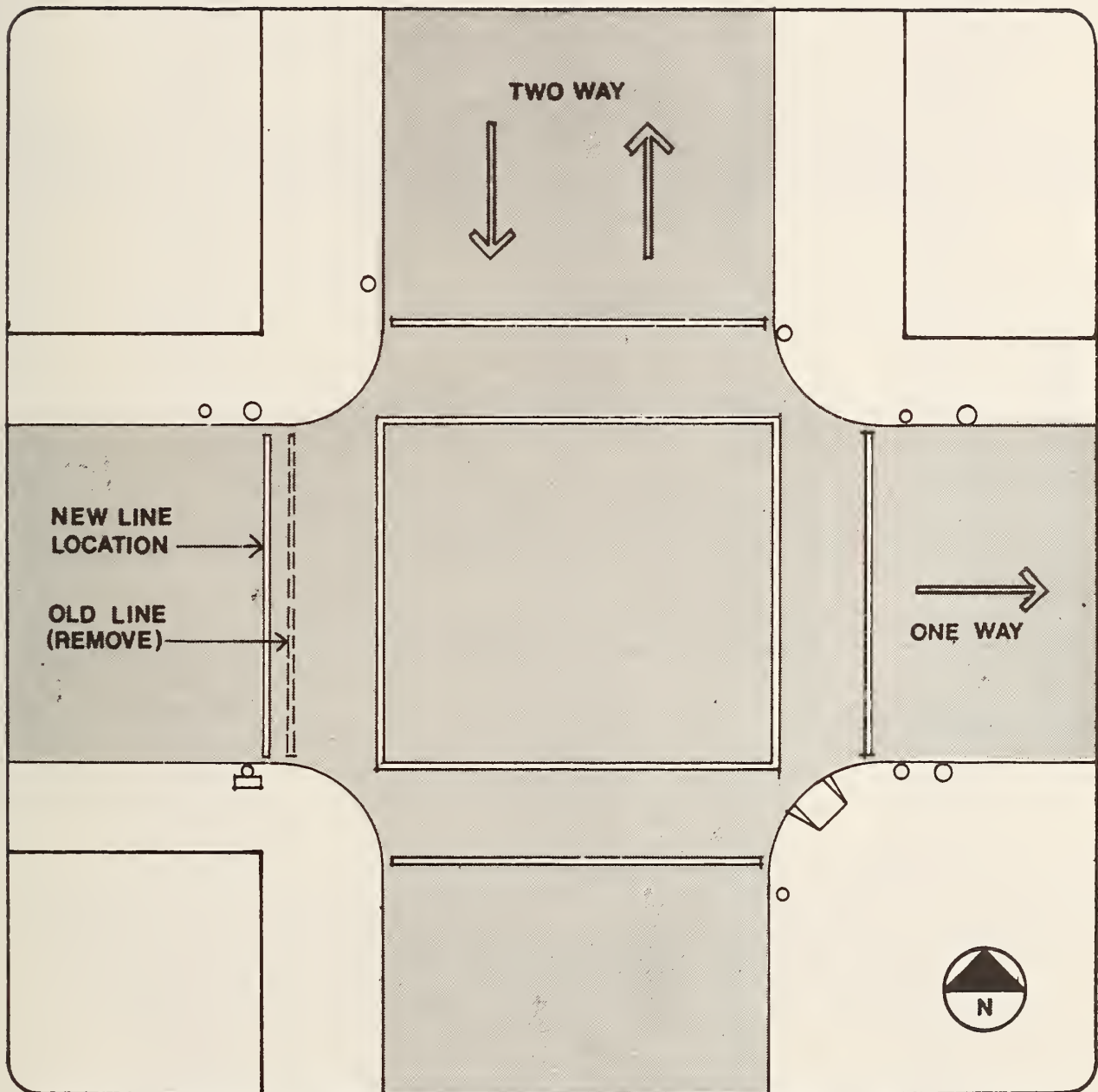


Figure 50 Atlanta: Widened Crosswalk: Location 3 After Modification

The intersection was signalized, and had pedestrian "walk"/"don't walk" signals as well. Signal timing was as follows:

North-South Street	Elapsed time in seconds	East-West Street	Elapsed time in seconds
Green/"Walk"	0	Red/"Don't Walk"	0
Flashing "Don't Walk"	30	Green/"Walk"	55
Yellow/Continuous "Don't Walk"	35	Flashing "Don't Walk"	65
		Yellow/Continuous "Don't Walk"	69
Red	40		
Green	70	Red	70

Pedestrian flows were measured in the west and east crosswalks and were heavy throughout the day, reaching their peak during the noon hour (see Table 88).

TABLE 88. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
West crosswalk	485	635	1005	775	985	775
East crosswalk	465	645	1235	685	975	800

Evaluation Results: Countermeasure No. 4.

1. Effect of countermeasure design on traffic flows. The countermeasure was evaluated with respect to its possible effect upon the flow of traffic through the intersections in which they were located. The intent was to determine whether widening the crosswalks and moving the stoplines, impeded traffic in any way.

The results of this evaluation were as shown in Figure 51.

The results are inconclusive. At Location No. 3, the traffic flow increased by 51% while the control site increased by about 12%. This increase can not be explained in terms of the countermeasure. At Location No. 1, the traffic flow decreased by 12%, but this is less than the 27% decrease at the control site. At Location No. 3, the traffic flow decreased by 21%, while the control flow increased by 6%. However, in view of the inexplicable flow changes at the other sites, it would be unrealistic to attribute the decreased traffic flow at Location No. 2 to the countermeasures.

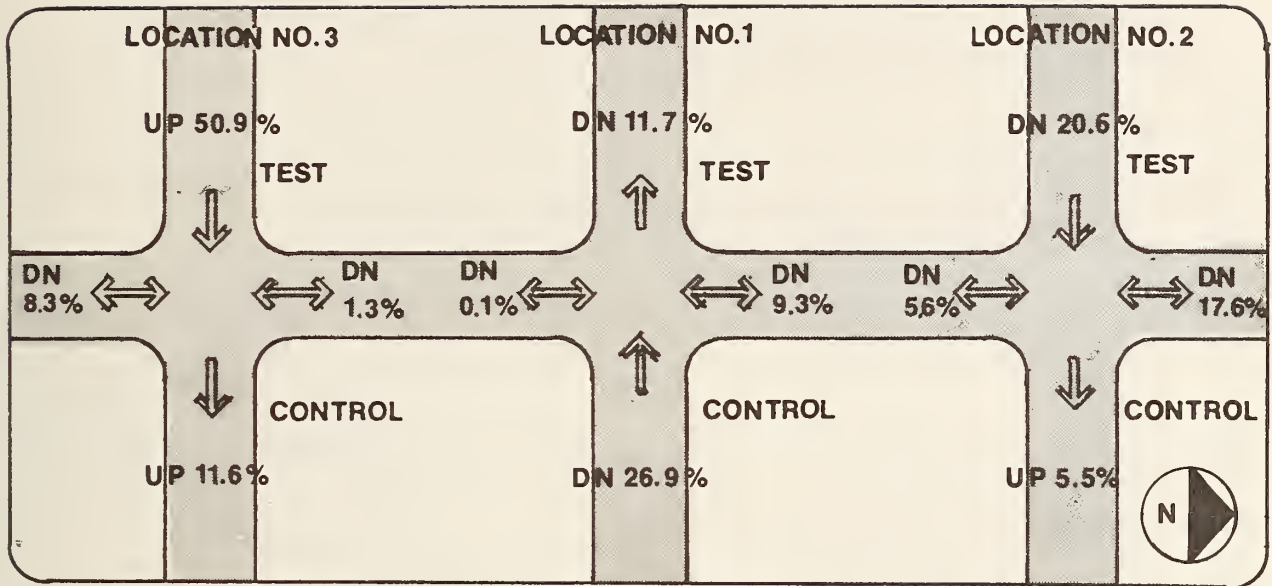


Figure 51 Changes in Vehicular Traffic Flows

2. Effect of countermeasure design upon pedestrian flows. The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians. The intent was to determine whether the widening of the crosswalks would have any effect on pedestrian use.

The results of this evaluation are shown in Figure 52.

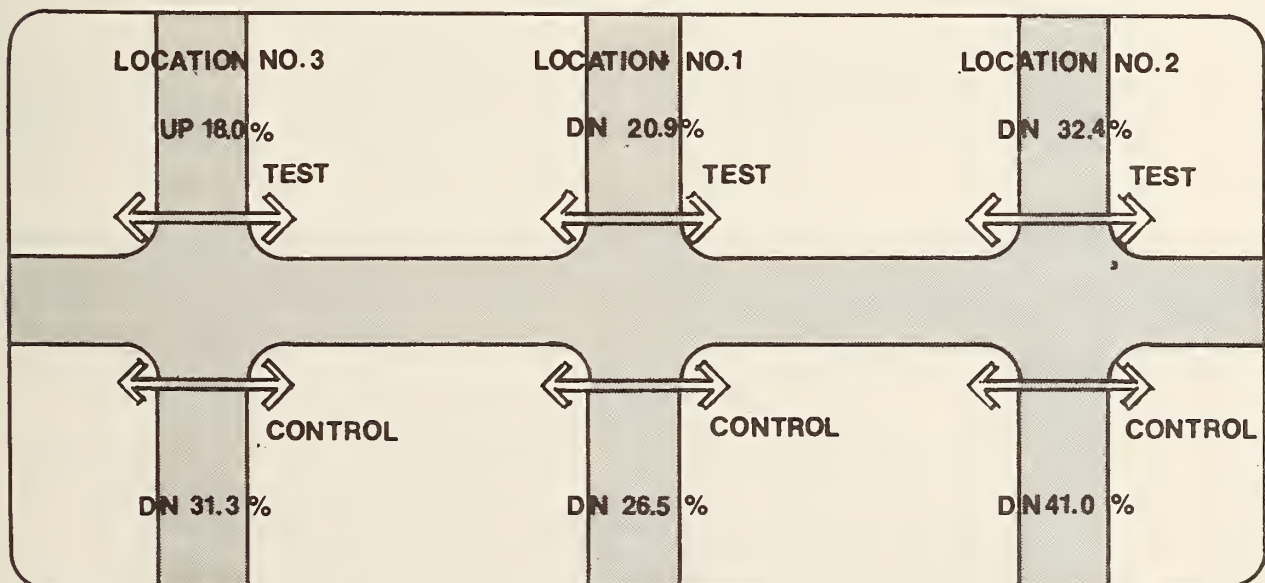


Figure 52 Changes in Pedestrian Traffic Flows

Pedestrian traffic at Location No. 1 was down by 21%, which was less than the 27% at the control site. At Location No. 2, pedestrian traffic was down by 32% compared to the 41% decrease at the control site. At Location No. 3, the pedestrian traffic increased by 18% compared to the 31% decrease at the

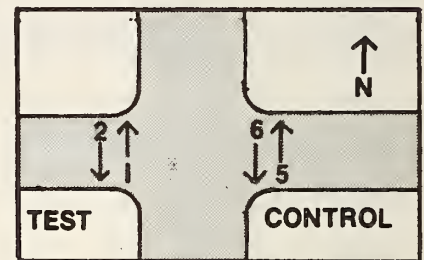
control site. From these results, it can be concluded that the countermeasure did not decrease pedestrian traffic flow.

Crossing Time.

Countermeasure No. 4 was evaluated in terms of the time required to cross the street. Evaluation results are shown below in Tables 89, 90 and 91.

TABLE 89. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 4; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	10.071	9.720	9.200	9.800	N	N
2,6	9.778	10.158	8.684	10.071	N	N

S = Significant

N = Not Significant

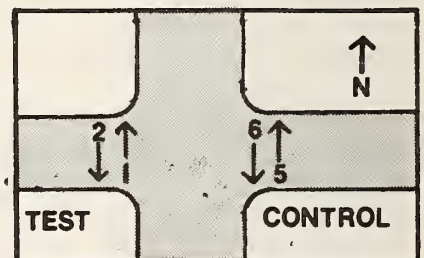


TABLE 90. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 4; LOCATION NO. 2

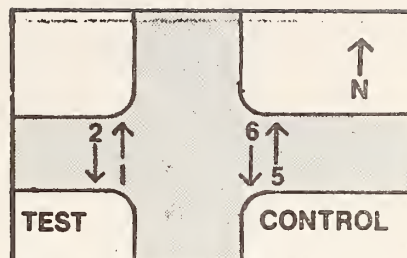
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	6.083	10.200	6.000	10.000	N	N
2,6	6.240	10.539	5.8750	10.3125	N	N

S = Significant

N = Not Significant

TABLE 91. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 4; LOCATION NO. 3



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	9.320	10.238	9.583	11.000	N	N
2,6	9.348	10.750	10.2381	9.636	N	N

S = Significant

N = Not Significant

From Tables 89, 90, and 91, it can be seen that at all the test and control sites there was no significant decrease in crossing times; in fact, in most cases, there was a significant increase in crossing time at both test and control sites and there is no obvious explanation for these increases. So it must be concluded that the countermeasure did not significantly decrease crossing time.

Evaluation of the Test and Control Intersections With Respect to Other Factors of Behavior Affecting Safety and Mobility.

These included evaluations of:

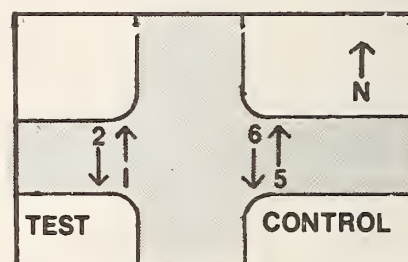
- Conflicts between crossing pedestrians and vehicles.
 - Conflicts between crossing pedestrians and other pedestrians.
 - Pedestrians waiting in the street for the crossing signal to change.
 - Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or

reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation are shown in Tables 92, 93, and 94.

TABLE 92. CONFLICTS BETWEEN CROSSING PEDESTRIANS
& VEHICLES

COUNTERMEASURE NO. 4; LOCATION NO. 1



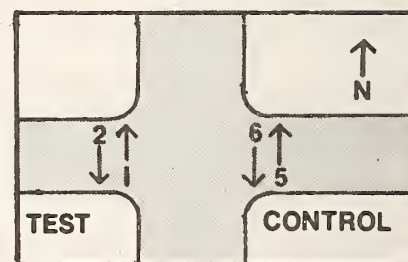
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	5/28	8/27	5/21	5/15	N	N
2,6	4/27	7/19	1/19	6/14	N	S

S = Significant

N = Not Significant

TABLE 93. CONFLICTS BETWEEN CROSSING PEDESTRIANS
& VEHICLES

COUNTERMEASURE NO. 4; LOCATION NO. 2

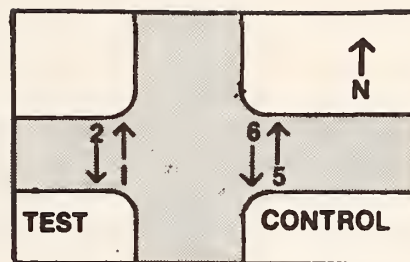


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	7/26	11/22	1/11	1/11	N	N
2,6	5/25	10/26	3/16	1/16	N	N

S = Significant

N = Not Significant

TABLE 94. CONFLICTS BETWEEN CROSSING PEDESTRIANS
& VEHICLES



COUNTERMEASURE NO. 4; LOCATION NO. 3

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	2/25	6/22	4/24	2/19	N	N
2,6	4/23	6/21	6/21	1/11	N	N

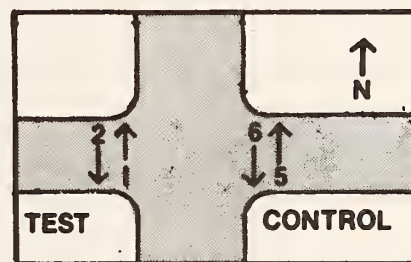
S = Significant N = Not Significant

From Tables 92, 93, and 94, it can be seen that although there was an increase in conflicts between pedestrians and vehicles at the three test sites, it was not significant.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Tables 95, 96, and 97.

TABLE 95. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND OTHER PEDESTRIANS



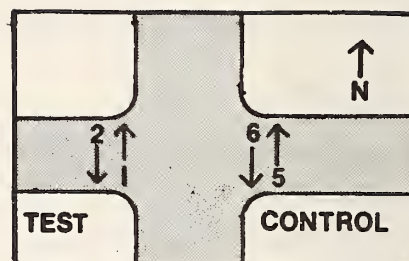
COUNTERMEASURE NO. 4; LOCATION NO. 1

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	7/28	5/27	4/21	4/15	N	N
2,6	10/27	2/19	4/19	2/14	N	N

S = Significant N = Not Significant

TABLE 96. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND OTHER PEDESTRIANS

COUNTERMEASURE NO. 4; LOCATION NO. 2



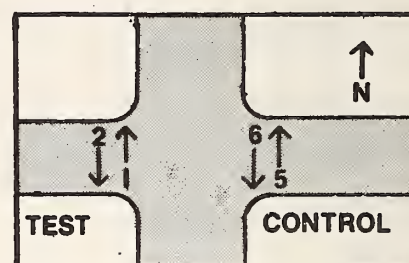
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	2/26	4/22	8/31	0/11	N	N
2,6	3/25	0/26	2/16	2/16	N	N

S = Significant

N = Not Significant

TABLE 97. CONFLICTS BETWEEN CROSSING PEDESTRIANS
AND OTHER PEDESTRIANS

COUNTERMEASURE NO. 4; LOCATION NO. 3



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	7/25	6/22	2/24	3/19	N	N
2,6	2/23	1/21	1/21	4/11	N	N

S = Significant

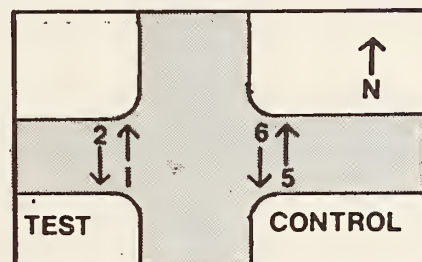
N = Not Significant

From Tables 95, 96, and 97, it can be seen that there was no significant change in the number of conflicts between crossing pedestrians and other pedestrians at the three test sites.

- c. Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing or decreasing the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, and were therefore vulnerable to traffic.

The results of this evaluation are shown in Tables 98, 99, and 100.

TABLE 98. PEDESTRIANS WAITING IN THE STREET
FOR THE CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 4; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	3/28	4/27	0/21	0/15	N	N
2,6	1/27	2/19	4/19	4/14	N	N

S = Significant

N = Not Significant

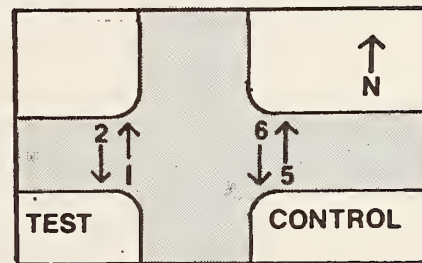


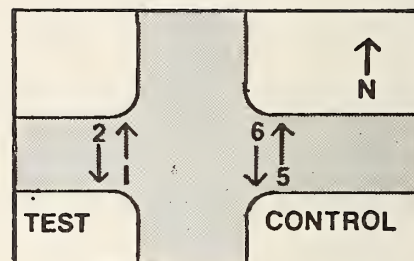
TABLE 99. PEDESTRIANS WAITING IN THE STREET
FOR THE CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 4; LOCATION NO. 2

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0/26	0/22	0/31	0/11	N	N
2,6	0/25	1/26	3/16	0/16	N	N

S = Significant

N = Not Significant

TABLE 100. PEDESTRIANS WAITING IN THE STREET
FOR THE CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 4; LOCATION NO. 3



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0/25	1/22	2/24	1/19	N	N
2,6	7/23	5/21	0/21	4/11	N	S

S = Significant

N = Not Significant

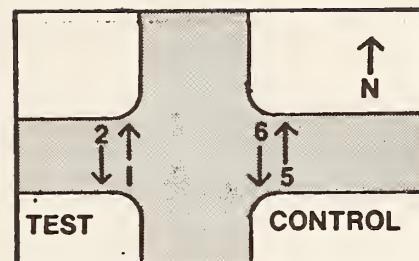
From Tables 98, 99, and 100, it can be seen that there was no significant change at any of the test sites in the number of pedestrians waiting in the street for the crossing signal to change.

- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.
1. The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The results of these evaluations are shown in Tables 101, 102, and 103.

TABLE 101. PEDESTRIANS WALKING BEYOND LIMITS
OF CROSSWALK

COUNTERMEASURE NO. 4; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	7/28	2/27	4/21	2/15	N	N
2,6	10/27	3/19	2/19	1/14	N	N

S = Significant

N = Not Significant

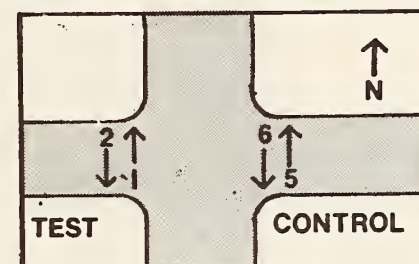


TABLE 102. PEDESTRIANS WALKING BEYOND LIMITS
OF CROSSWALK

COUNTERMEASURE NO. 4; LOCATION NO. 2

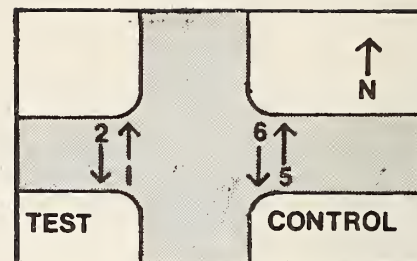
Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	5/26	1/22	7/31	1/11	N	N
2,6	2/25	1/26	2/16	2/16	N	N

S = Significant

N = Not Significant

TABLE 103. PEDESTRIANS WALKING BEYOND LIMITS
OF CROSSWALK

COUNTERMEASURE NO. 4; LOCATION NO. 3



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	2/25	2/22	6/24	7/19	N	N
2,6	5/23	3/21	8/21	2/11	N	N

S = Significant

N = Not Significant

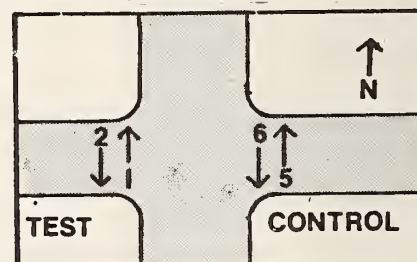
From Tables 101, 102, and 103, it can be seen that there was no significant change at the three test locations in the number of pedestrians walking beyond the limits of the crosswalks.

2. Pedestrians leaving the crosswalk while crossing the street.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 104, 105 and 106.

TABLE 104. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 4; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0/28	2/27	0/21	3/15	N	N
2,6	0/27	1/19	0/19	1/14	N	N

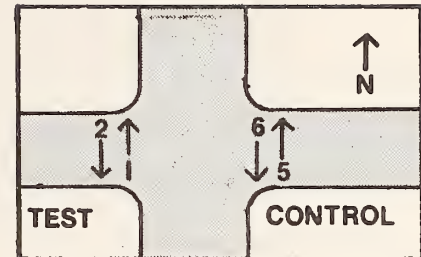
S = Significant

N = Not Significant

TABLE 105. PEDESTRIANS LEAVING THE CROSSWALK

WHILE CROSSING THE STREET

COUNTERMEASURE NO. 4; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0/26	0/22	0/31	0/11	N	N
2,6	0/25	0/26	0/16	0/16	N	N

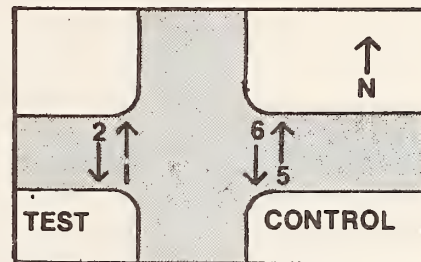
S = Significant

N = Not Significant

TABLE 106. PEDESTRIANS LEAVING THE CROSSWALK

WHILE CROSSING THE STREET

COUNTERMEASURE NO. 4; LOCATION NO. 3



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0/25	1/22	0/24	1/19	N	N
2,6	2/23	0/21	0/21	0/11	N	N

S = Significant

N = Not Significant

From Tables 104, 105, and 106, it can be seen that there was no significant change at the three test locations in the number of pedestrians leaving the crosswalk while crossing the street.

Summary of Findings and Conclusions: Countermeasure No. 4, Widened Crosswalks.

- The effect of the countermeasures on traffic flow was inconclusive.
- Widening the crosswalks did not significantly alter the pedestrian traffic flow.
- Widening the crosswalks did not decrease crossing time.
- There was no significant change in conflict between pedestrians and vehicles.
- There was no significant change in conflicts between pedestrians and other pedestrians.
- There was no significant change in the number of pedestrians waiting in the street for the crossing signal to change.
- There was no significant change in the numbers of pedestrians walking beyond the limits of the crosswalks, nor leaving the crosswalk while crossing the street.

From these findings, it can be concluded that for the level of pedestrians and vehicular activity present at the test sites, widening the crosswalks did not improve the flow characteristics nor the safety of the pedestrians.

countermeasure 5

textured furniture strip

The location selected for the installation and evaluation of Countermeasure No. 5, Textured Furniture Strip, was on a sidewalk of the main downtown business street of Clearwater, Florida, a mid-size southeastern city.

It is a city that, because of its climate and its consequent use as a retirement (as well as vacation) community, has a substantially higher than normal percentage of elderly persons within its general population. It has a greater than normal share of handicapped individuals as well.

The selected sidewalk test location is fronted upon by typical intensive mixed downtown uses: office buildings, banks, retail and food service. The location is also a short distance from the major downtown department store. As a result, pedestrian flows along this sidewalk are quite substantial.

The sidewalk is 14½ feet (4.42 m) in width. The area closest to the curb, approximately the first 6 feet of sidewalk width, is covered with a dense and varied mix of street furniture, including poles, planters, newspaper sales boxes, hydrants, etc.

The purpose of Countermeasure No. 5 was to evaluate the usefulness of a special surface applied to a certain portion of the sidewalk as a directional cue for the visually impaired. A number of problems faced by the visually impaired, in maintaining a proper and safe course while moving through the pedestrian environment, have been identified and described in this research (see Volume 2: Hazards, Barriers, Problems, and the Law). In addition, several possible countermeasures were originally given consideration as potential solutions to these problems. Countermeasure No. 5 was selected from these potential solutions for evaluation.

Countermeasure No. 5 defines an area of the sidewalk for street furniture as separate and distinct from the pedestrian traffic area of that sidewalk. It does this in one of two ways:

- by using a textured surface to identify the furniture area, or
- by providing a textured strip, dividing the two parts of the sidewalk.

In both cases, the textured surfacing establishes an edge to the furniture strip which can be used by the visually impaired for navigation. Figures 53(a) and 53(b) illustrate the two possible forms this countermeasure can take.

The material used for the test had been verified as detectable by the visually impaired, as discussed earlier. The purpose of this evaluation was to determine if this material, applied to a heavily used real-world situation, would assist the navigational needs of the visually impaired.

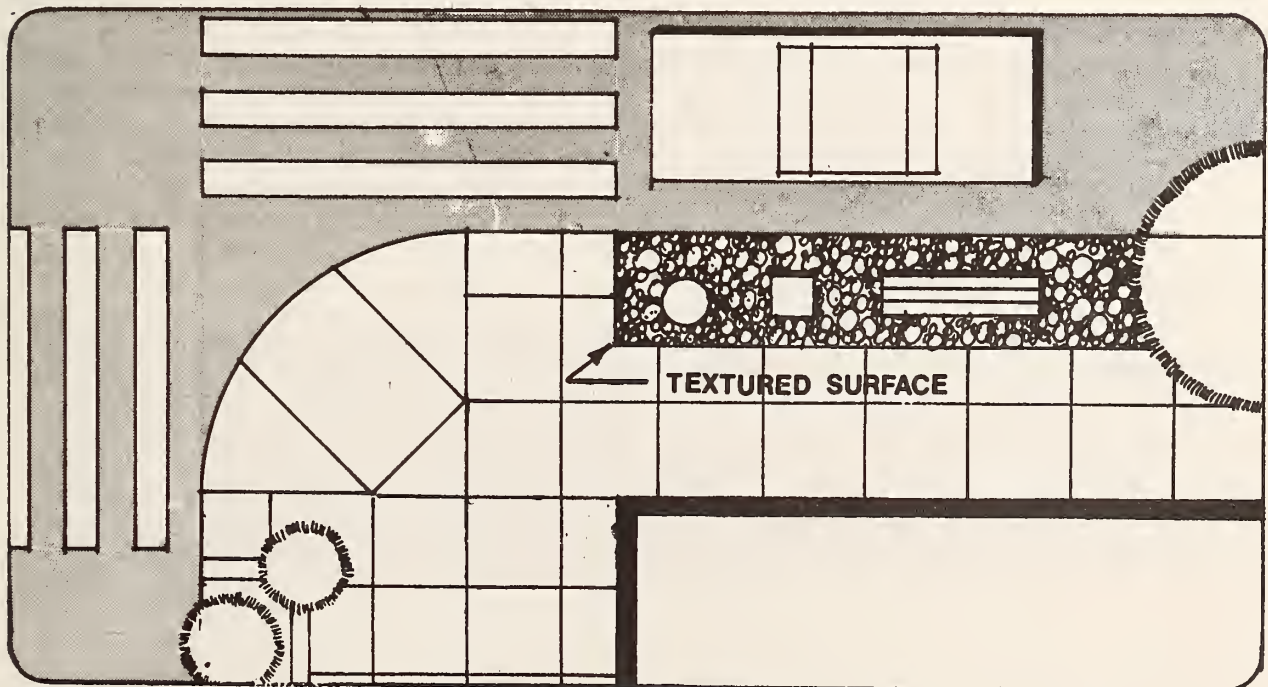


Figure 53 (a) Furniture Separation by Surface Change

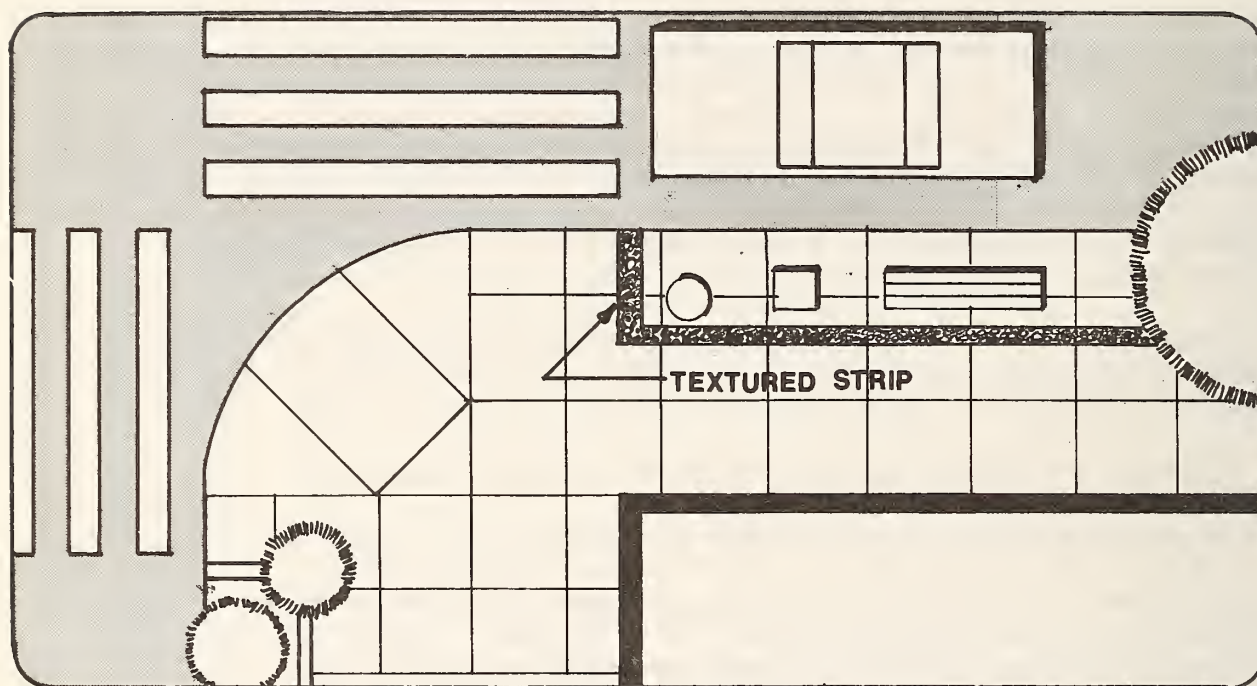


Figure 53 (b) Furniture Separation by Textured Strip

Description of the Countermeasure as Designed and Constructed.

Figure 54 illustrates the countermeasure as it was designed and constructed. A 12 inch (30.48 cm) strip of a thick, resilient paint, similar to that used to surface outdoor tennis courts, was applied for a distance of approximately 100 feet (30.5 m) along the sidewalk, as a separation between the area used for street furniture and the clear travel area of the sidewalk. The next 100 feet (30.5 m) of sidewalk was not treated in any way to serve as the control location for this evaluation.

Evaluation of Countermeasure No. 5.

This countermeasure was evaluated by a special group of visually impaired individuals. Seven visually impaired test subjects were brought to the site. They had been asked to familiarize themselves with the test location. During the evaluation, the test subjects went through the test location individually, following a prescribed route, while observations were made of their use of the design feature incorporated into that location, as well as any apparent benefits or difficulties resulting from the design features. Time lapse photography was used to record the results of the evaluations. The results sought were answers to specific questions.

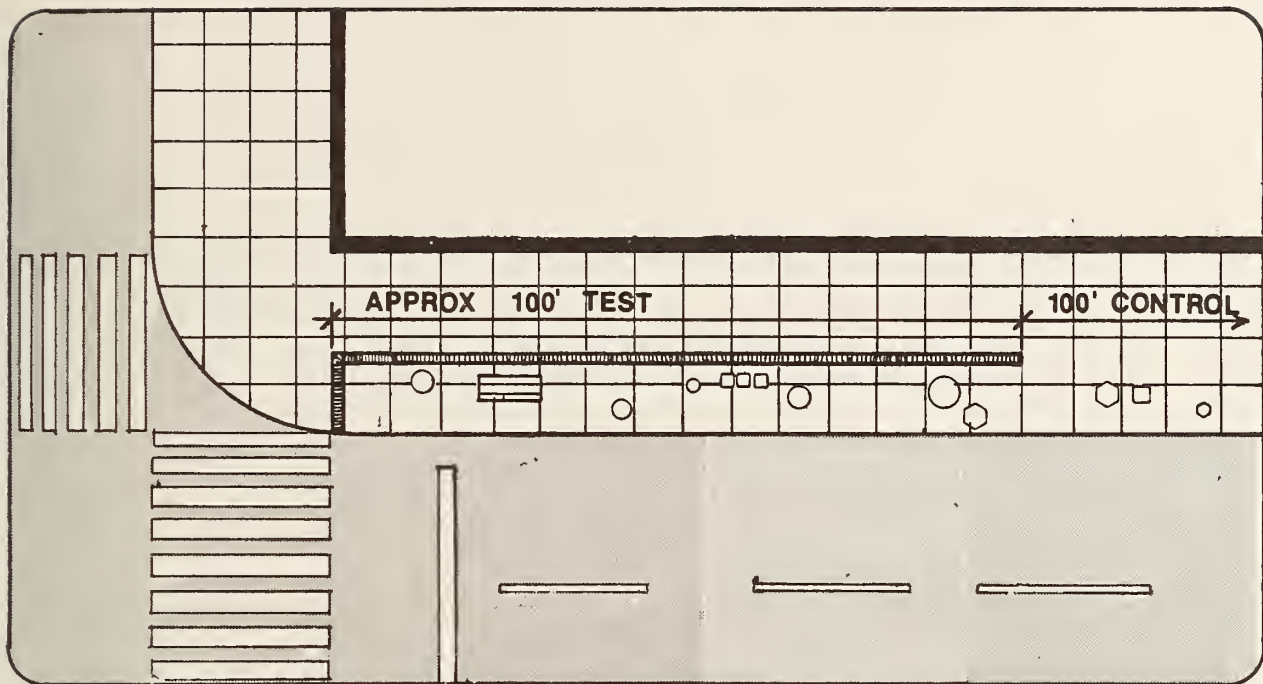


Figure 54 Textured Strip Test and Control Site

- Was the textured strip useful to the visually impaired as a cue;
- Were they able to maintain a direct course, without encountering street furniture.

Evaluation Results.

The test subjects followed a route along the portion of the sidewalk, approximately 100 feet (30.5 m) in length, on which the textured furniture strip was installed. then continued along the sidewalk for a distance of equal length; that distance to serve as the control evaluation. The subjects then were asked to turn and retrace their steps, repeating first the control and then the test segment of the course. Table 107 illustrates the results of the evaluation.

The test subjects were able, without exception, to follow the textured furniture strip and use it to maintain a safe and direct course, without encountering street furniture. At the control area, however, all but one of the subjects wandered off from a direct course, and had to change direction after encountering street furniture. The test results indicate that, on the average, each subject had two incidents of walking into street furniture; or, having missed the street furniture, wandering over to the curb and the edge of the street. Fewer instances of walking into the building frontage were recorded, since only those cases where subjects struck the buildings in an abrupt and potentially harmful way were considered events of this type. Other subjects were able to move over to the building frontage and use it as a shoreline.

TABLE 107. BEHAVIORAL RESPONSES BY BLIND SUBJECTS: TEXTURED FURNITURE STRIP

		Test Subjects							
		1	2	3	4	5	6	7	Total
Test Area	. Deviated to street side - struck furniture or curb	0	0	0	0	0	0	0	0
	. Deviated to building side - struck building face	0	0	0	0	0	0	0	0
Control Area	. Deviated to street side - struck furniture or curb	3	1	0	2	2	3	3	14
	. Deviated to building side - struck building face	3	0	0	0	0	0	0	3

The following test description serves as an example of the test results:

- The test subject, an elderly man of about 70 years of age, had no difficulty using the resilient painted stripe to maintain a straight course. He walked to the left of the line, about 18 inches (45.72 cm) away, and using a sweeping cane technique, maintained a perfectly straight course. Once past the end of the test furniture stripe, he maintained a straight course for about 25 feet (7.625 m), then began to wander to his left, until he walked into the building facade. From that point, he used the building facade as a reference point to maintain proper direction. On his return, the subject began wandering to his left, and walked into a 4 foot (1.22 m) high concrete planter tub. He corrected his course, but wandered off line to the right and into several newspaper sales boxes. He took a few seconds to determine on which side of the boxes he should pass, proceeded on a slight diagonal to his left, found the beginning of the test strip, and maintained a straight and true course for the remainder of his 'walk'.

Conclusion

The textured strip is considered to be an effective and successful countermeasure.

countermeasure 8

single conventional ramp

Countermeasure No. 8, Single Conventional Ramp, was constructed and evaluated in the downtown area of Sioux City, Iowa (see Figures 55 and 56). The countermeasure was located within the principal shopping district, immediately adjacent to a major department store, within one block of a pedestrian mall and two blocks from several municipal parking structures.

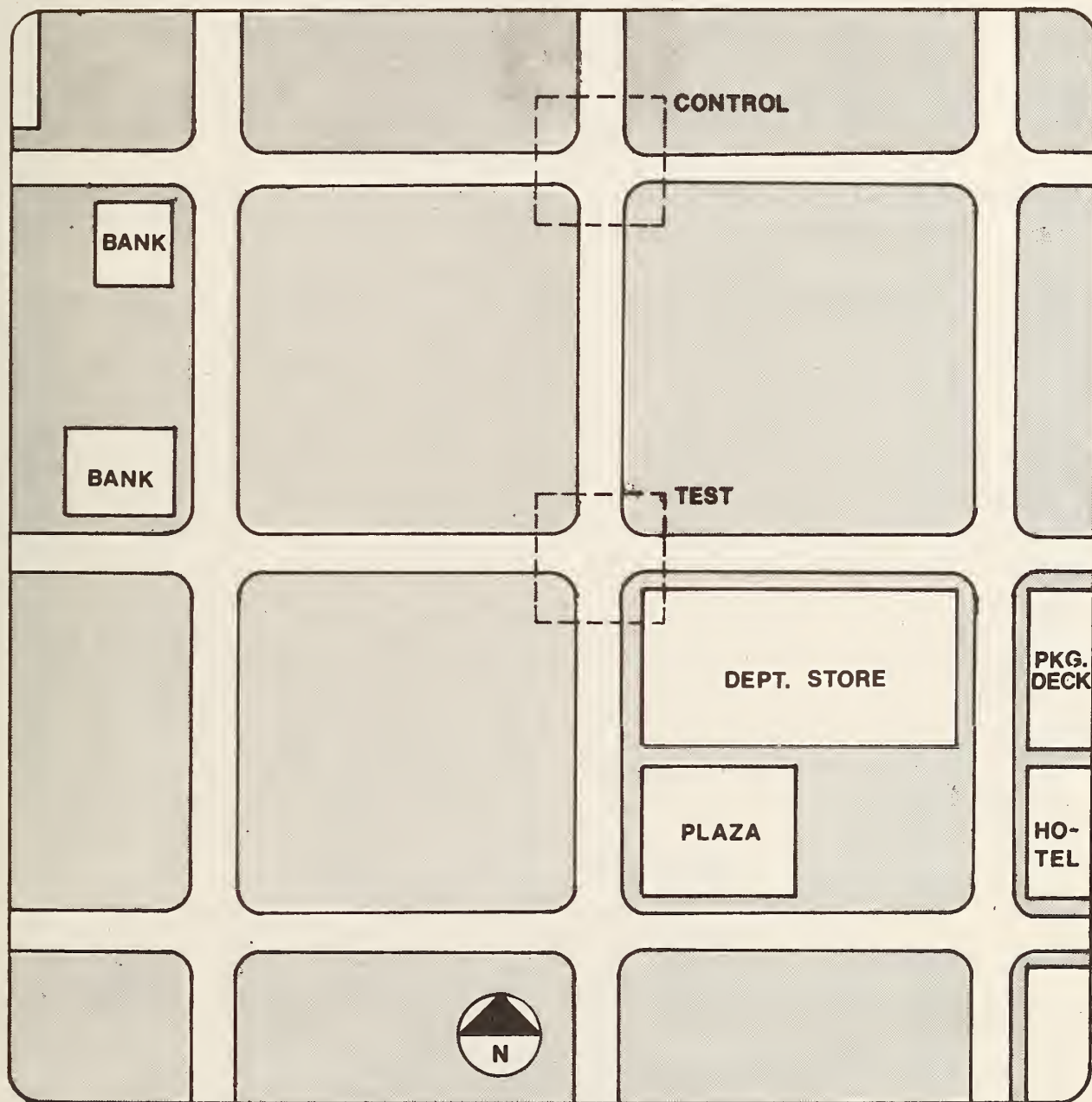


Figure 55 Sioux City: Single Conventional Ramp: Test and Control Site Locations

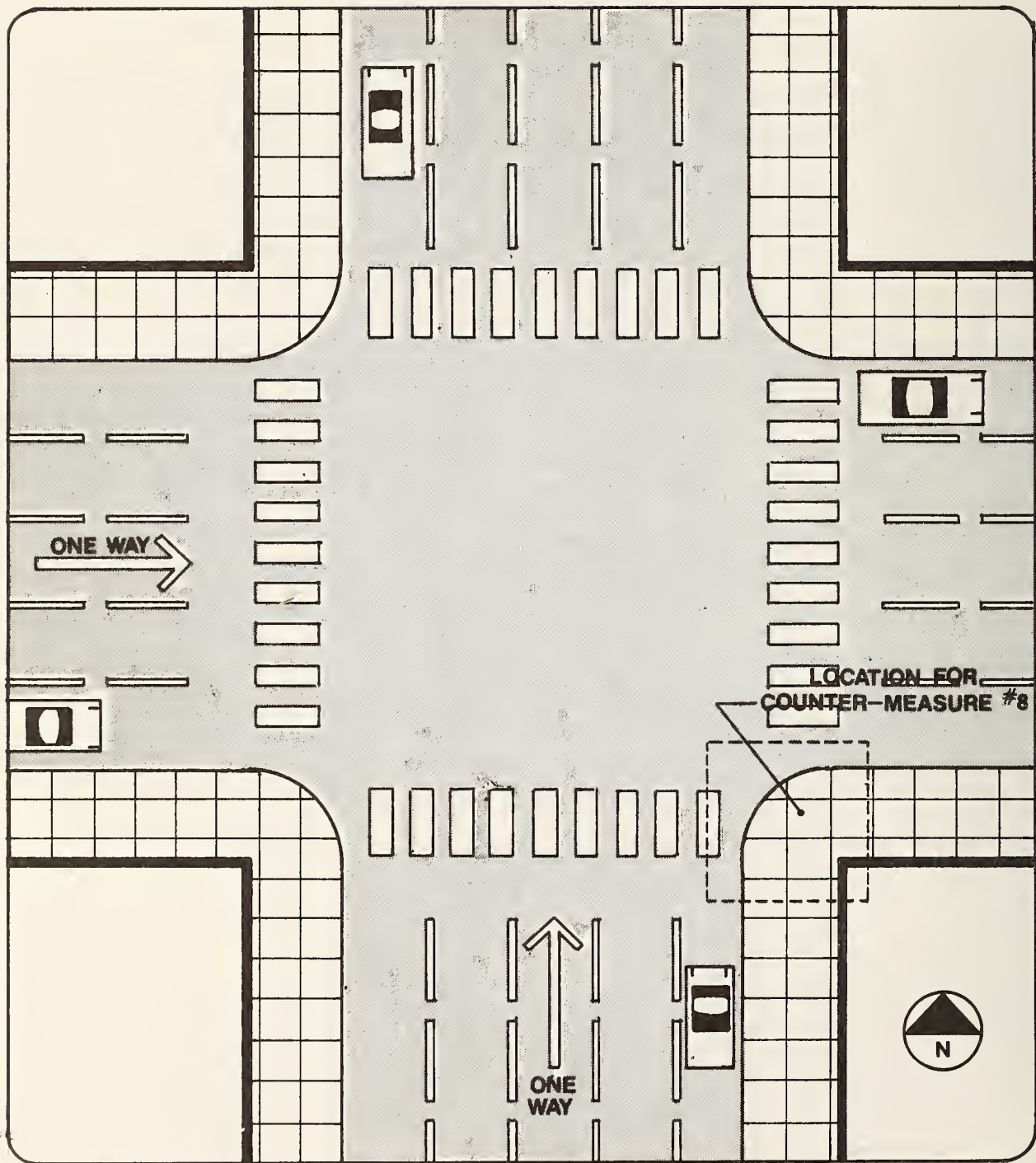


Figure 56 Sioux City: Single Conventional Ramp: Site Before Modification

Both streets which pass through the intersection in which the countermeasure was installed, are one-way routes. Both have provisions for 3 lanes of moving traffic, and parallel parking along the curb on both sides of the street. Traffic volumes are fairly heavy, averaging about 750 vehicles per hour on the east-west street (see Table 108).

TABLE 108. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	500	535	550	700	815	600
Turning	60	110	125	170	175	130
Total	560	645	675	870	990	730
East-West Direction						
Thru	447	465	543	582	774	562
Turning	168	198	198	231	237	206
Total	615	663	741	813	1011	769

The intersection is signalized with right or left turn arrows, as applicable, and pedestrian 'walk', 'don't walk' indicators. Signal timing is set to allow a short period for turning traffic at the beginning of the cycle before the pedestrian 'walk' period begins. The signal cycle is as follows:

	Elapsed time
Green, green 'turn' arrow	0 seconds
'Turn' arrow off, 'walk' begins	10
Flashing 'don't walk'	19
Yellow, continuous 'don't walk'	30
Red	33
Green	66

Note: Signal time is equally distributed between the two directions.

All sidewalks are 14 feet (4.27 m) in width. Pedestrian flows were measured in the south and west crosswalks, and varied between average hourly flows of 150 and nearly 580 pedestrians per hour in both directions (see Table 109).

TABLE 109. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	70	110	190	265	140	155
East-West	80	125	370	315	200	220
Total	150	235	560	580	340	375

The pedestrian crossing signal allowed 19 seconds of "walk" for crossing the 60 foot (18.3 m) wide street in those crosswalks which are not affected by turning traffic (the west and south crosswalks). However, because of the 10 second "turn" arrow cycle affecting the other two crosswalks (the north and east), only 9 seconds of "walk" was allowed for those crossings, requiring a walking speed of 6.7 feet (2.04 m) per second to complete crossing within the "walk" allotted time. This "walk" interval does not include the clearance time allowed for pedestrians to complete their crossing during the flashing "don't walk" phase.

Control Location for Test Location No. 1.

The control intersection was located one block to the north of the test site. Its design and use and traffic signal phasing and timing were essentially the same as the test location's before the countermeasure was constructed (see Figure 57). Traffic volumes at the control intersection were somewhat higher in both the north-south and the east-west directions than at the test site. Pedestrian volumes were lower in both directions than at the test site (see Tables 110 and 111).

Description of Single Conventional Ramp Installation.

The Single Conventional Ramp, constructed at the test location, was an adaptation of the schematic ramp design by the test city's engineers to meet local conditions and local design standards. The design, as constructed, featured a ramp at least 4 feet (1.22 m) in width, with a 1:12 slope (8.33 percent), located diagonally within the intersection quadrant, so as to provide equal access to both crosswalks. Ramp sideslopes were adjusted to fit local grade conditions, but were generally equal to the ramp itself in a gradient. The ramp was constructed of concrete, with a broom finished surface to be similar to adjacent sidewalk areas. The design incorporated a flush junction with the street; however, construction variation at this, and other similar installations in the city, often resulted in a ramp lip of as much as $\frac{1}{4}$ inch (.635 cm). Figures 58 and 59 illustrate the design, as constructed.

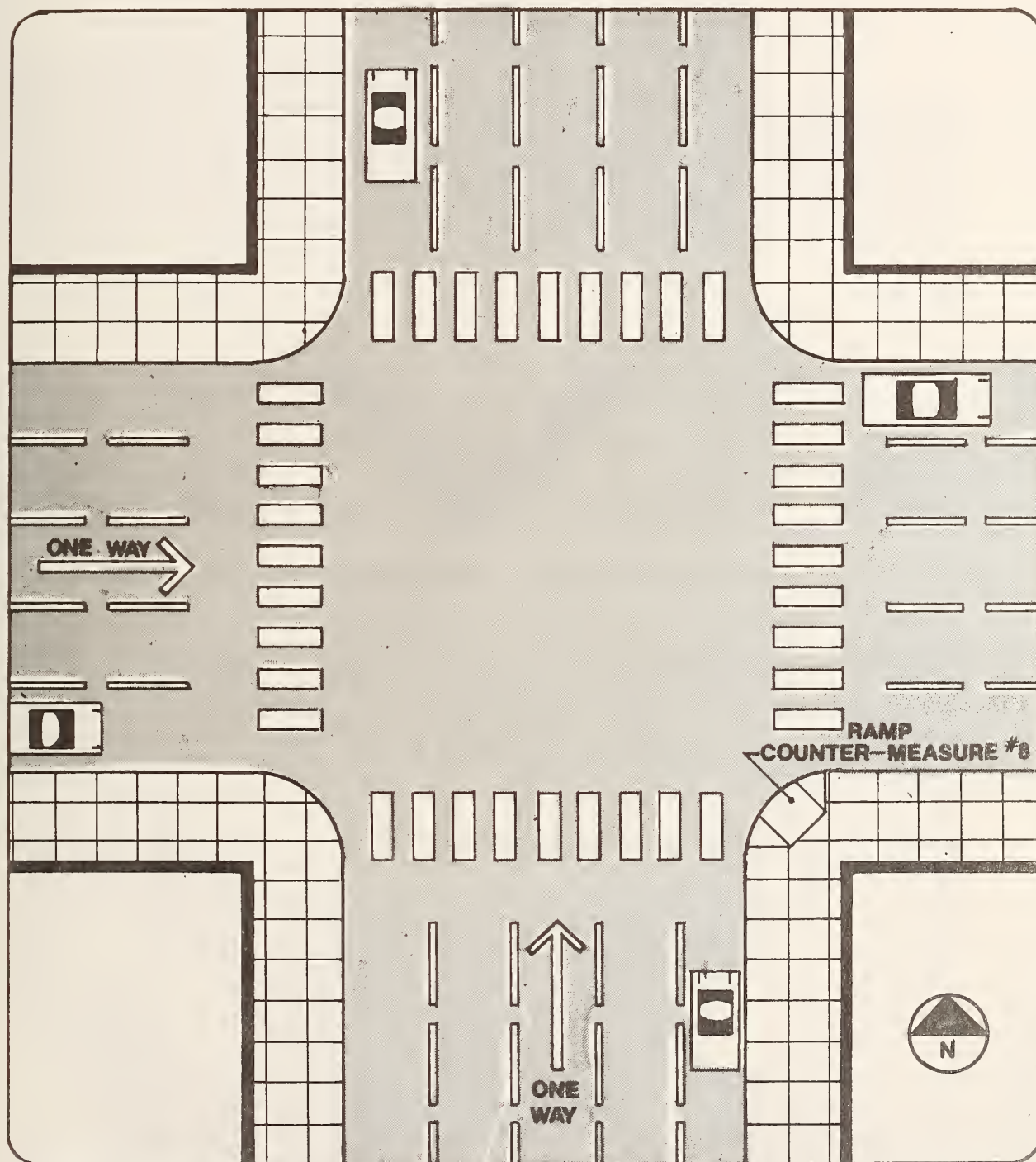


Figure 57 Sioux City: Single Conventional Ramp: Site After Modification

TABLE 110. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	315	425	485	640	780	530
Turning	335	320	280	415	310	330
Total	650	745	765	1055	1090	860
East-West Direction						
Thru	605	485	620	775	850	665
Turning	160	155	190	215	260	195
Total	765	640	810	990	1110	860

TABLE 111. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	55	55	190	95	145	110
East-West	50	75	260	130	160	135
Total	105	130	450	225	305	245

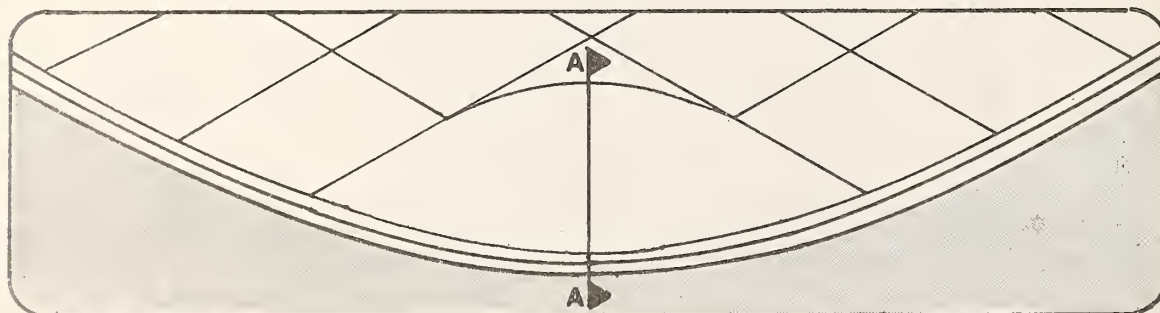


Figure 58(a) Details of Single Conventional Curb Ramp

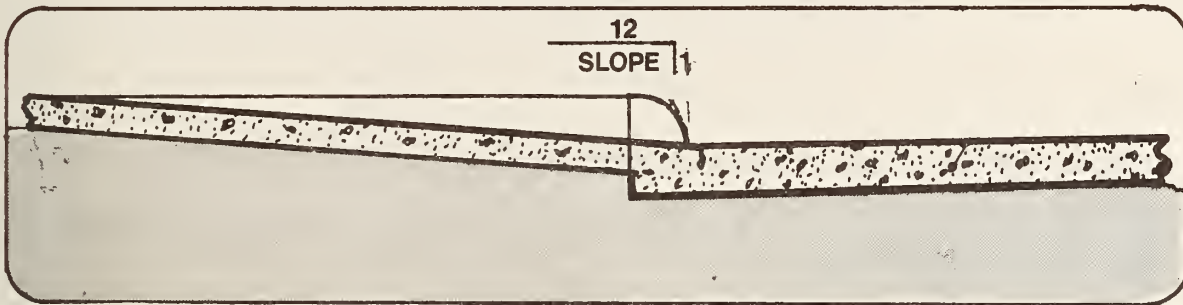


Figure 58(b) Details of Single Conventional Curb Ramp

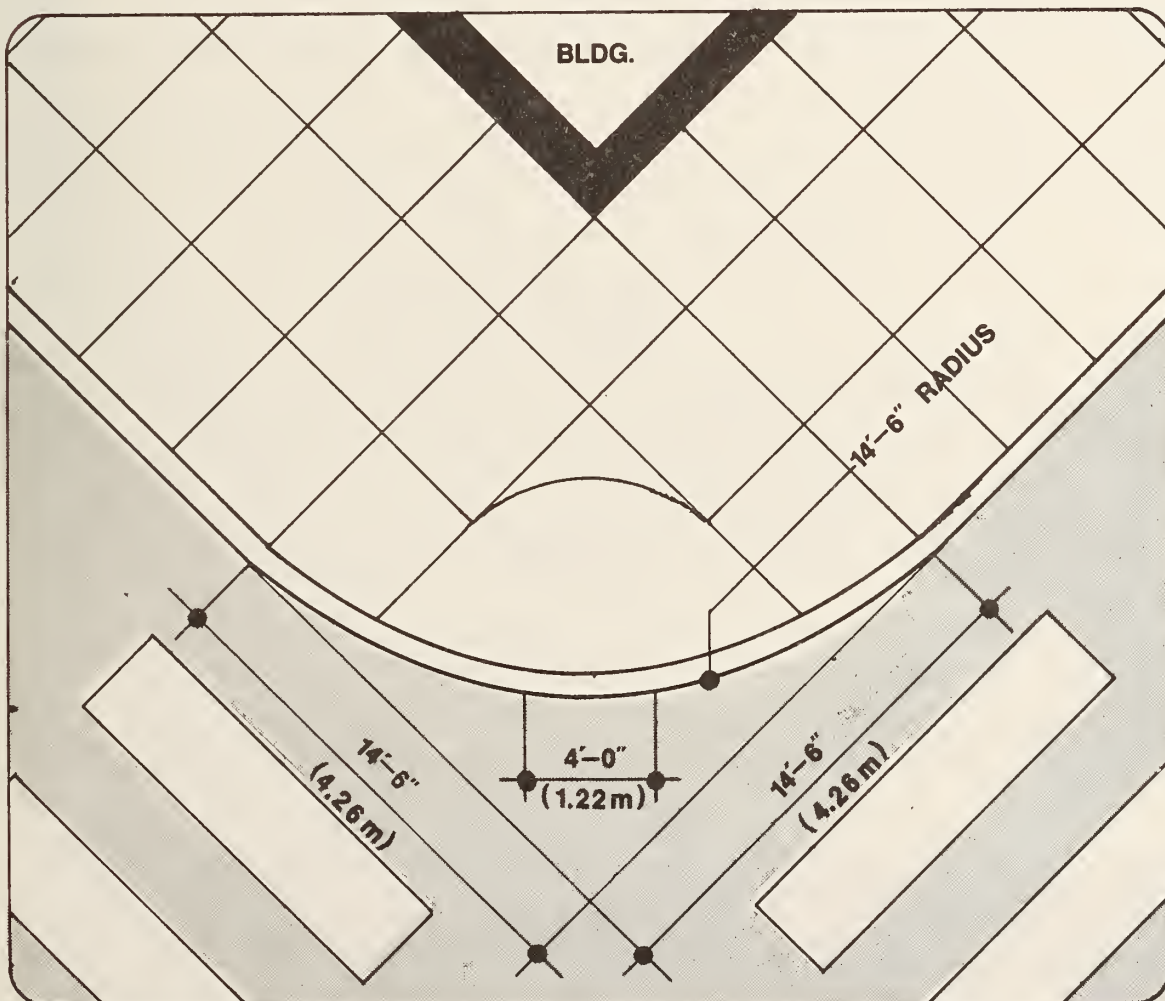


Figure 58(c) Locational Details of Single Conventional Curb Ramp

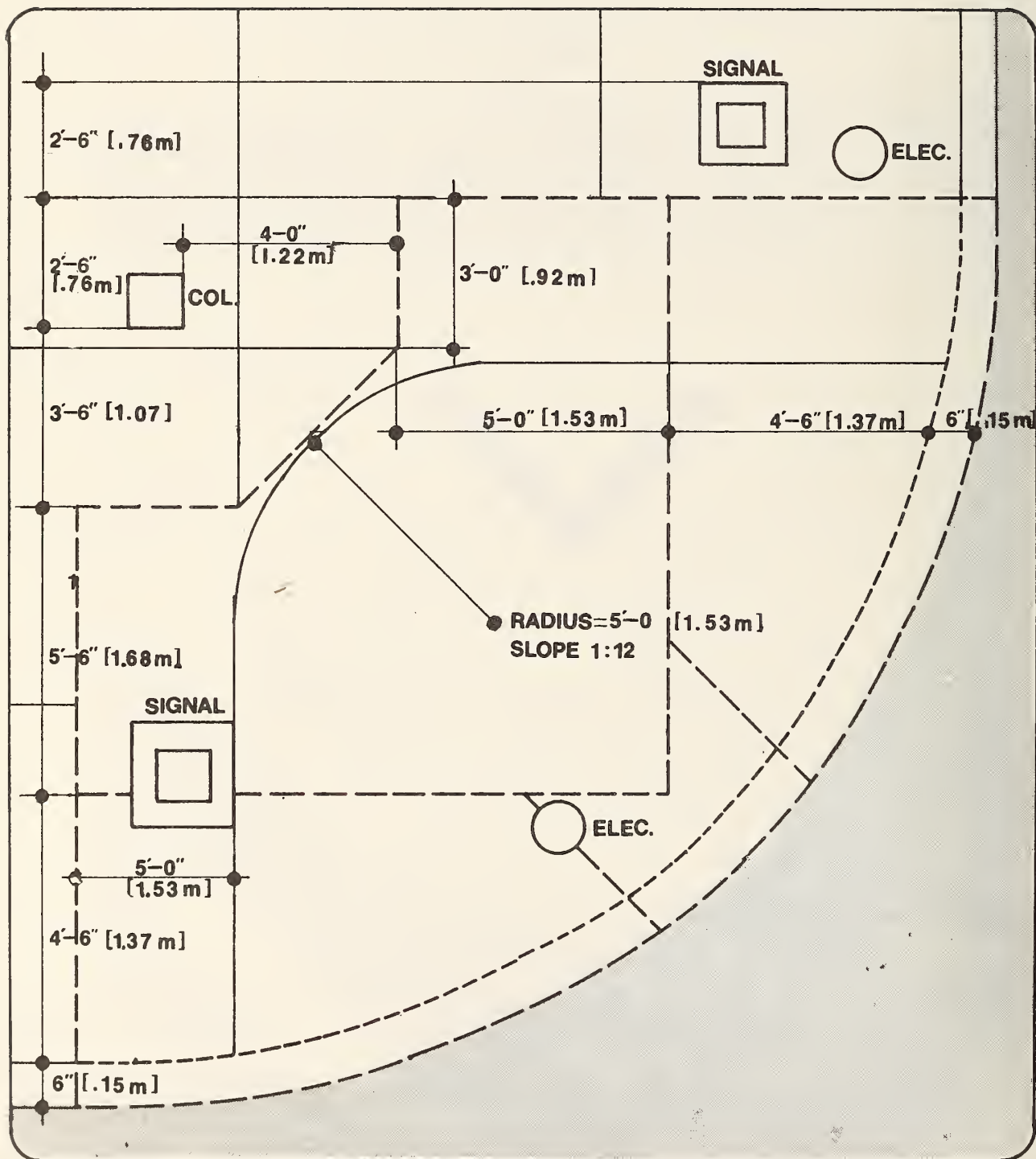


Figure 59 Single Conventional Curb Ramp as Constructed

Evaluation of Countermeasure No. 8: Simple Conventional Ramp.

Countermeasure No. 8 was evaluated only by special groups of handicapped persons.

Evaluation of Countermeasure No. 8 By Special Groups of Handicapped Persons.

The procedures used in the special group evaluations were described in detail earlier in this report. The purpose of the evaluation was to find out, for the:

- Visually Impaired Group:
 - Is the ramp detectable; can the visually impaired detect the point at which the ramp meets the sidewalk and the street;
 - Is the ramp location and design useful, or inconvenient for the visually impaired.
- Wheelchair Group:
 - Is the ramp satisfactorily located and designed;
 - Is it useful, or inconvenient for persons in wheelchairs.

Evaluation Results: Visually Impaired Group.

1. Detectability of the ramp. To determine whether the members of the visually impaired group could detect the ramp, and specifically, the point at which the ramp joins the street, the test subjects were observed as they used the ramp to enter the street. The evaluation was based on the location at which the visually impaired test subjects stopped while awaiting a safe opportunity to cross. Consistency in stopping location would be the basis for an assumption that the ramp was readily discernible. Following are the results of this evaluation:
 - There were fifteen test events in which the visually impaired used the single conventional ramp to enter the street. In fourteen of those events, the test subjects stopped at the bottom of the ramp, within one or two feet (.305 or .610 m) of the street. On the other occasion, the test subject stopped at the midpoint of the ramp. These results suggest that the visually impaired had little difficulty detecting the ramp.
2. Usefulness of the ramp design and location. The single conventional ramp, located, as it is, on the corner at the intersection, facing diagonally, gives persons a choice of using the ramp, or the curb, to enter the street. An indication of the usefulness of a ramp is the choice persons make of whether to use or avoid it. The visually impaired test subjects were observed, on both entering and leaving the street, to determine their preference for using the ramp or curb. Following are the results of this evaluation:

- There were ten crossing events in which visually impaired subjects entered the street.
 - 3 test subjects used the curb
 - 1 test subject used the sideslope, or half curb
 - 6 test subjects used the ramp
- There were ten crossing events in which visually impaired test subjects left the street.
 - 5 test subjects used the curb
 - 5 test subjects used the ramp
- In all cases, entering or leaving the street, the test subjects used whichever (ramp or curb) they first came in contact with. None of the subjects, having detected either a ramp or a curb, chose to change direction to avoid it and seek the other.

The following descriptions illustrate these results:

- The test subject, a 36 year old male, slowed as his cane detected the beginning of the ramp. He then stepped onto the ramp, resumed his normal speed, and continued on up to the sidewalk.
- The test subject, an 81 year old male, slowed considerably as he detected the ramp from the sidewalk. He descended the ramp slowly, and as his cane detected the edge of the ramp at the street, he stopped and awaited an opportunity to cross safely.
- The test subject, a 76 year old male, detected the curb as he was completing his crossing of the street. He slowed nearly to a stop, then stepped over the curb onto the sidewalk.
- An 83 year old female test subject, completing her crossing of the street, detected the beginning of the ramp and turned so as to climb the ramp in a direction perpendicular to its principal direction and slope. She climbed slowly, to assure balance on the sloped surface; then at the top, turned again to travel down the sidewalk, resuming her normal walking speed.

Evaluation Results: Wheelchair Group.

1. Usefulness of the ramp location and design. The single conventional ramp was evaluated with respect to its design and its orientation. The ramp had a slope of approximately 1 in 12 (8.33 percent) and had, as well, a lip of about $\frac{1}{4}$ inch

(.635 cm) where it joined the street. The lip was a by product of construction methods and tolerances, rather than an intentional part of the ramp design.

Eleven test subjects used the ramp to enter and leave the street. None appeared to have difficulty with the slope of the ramp, in either direction. The ramp lip caused difficulty to three of the 11 subjects. One subject chose not to use the ramp at all. The following narratives describe several of these problems:

- The test subject, a 29 year old male, was jarred to a stop by the ramp lip as he was about to use the ramp to leave the street. It was necessary for him to lean back and lift his front wheels off the roadway and over the lip, in order to get onto the ramp and climb to the sidewalk.
- A 42 year old male test subject, as he approached the ramp from the street, turned at the last moment and stopped in the street at the base of the ramp, to await the signal change that would permit him to cross the intersecting street.

The ramp was oriented so that it faced diagonally, towards the center of the intersection. Wheelchair users were observed, to determine whether this ramp location and orientation would force them out of the crosswalk so as to be vulnerable to moving traffic, as illustrated in Figure 60. There were 20 crossing events, in which test subjects in wheelchairs used the ramp to leave the street. On 18 of those events, the subjects remained within the limits of the crosswalk. On two occasions they travelled beyond the crosswalk boundary.

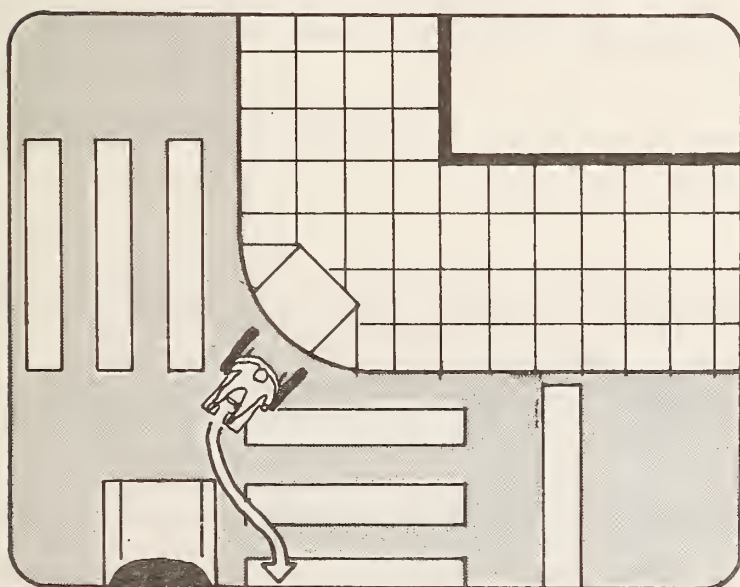


Figure 60 Wheelchair Movement: Conventional Curb Ramp

There were 20 crossing events in which wheelchair test subjects used the ramp to enter the street. In 14 of those events, the path used by the test subjects took them beyond the limits of the crosswalk. This number, substantially higher than for leaving the street, was probably the result of the speed generated by the descent of the ramp, and the resultant difficulty in quickly slowing and turning the chair, a particularly difficult maneuver. The following narratives describe several of these events:

- A 29 year old male test subject descended the ramp. His path took him about 2 feet (.61 m) beyond the limits of the crosswalk. Because of approaching pedestrians in the crosswalk, he remained outside the walkway line until reaching the midpoint of the street he was crossing.
- A 56 year old male test subject attempted to turn very quickly, once having entered the street, but was unable to do so before his momentum carried him about 12 inches (.305 m) beyond the limits of the crosswalk.
- A 26 year old male test subject, while crossing the street, swung about 18 inches (.457 m) beyond the crosswalk line, to align himself with the ramp he was about to ascend.

Summary of Findings: Countermeasure No. 8, The Conventional (Recessed) Curb Ramp.

1. The visually impaired subjects all detected the presence of the ramp, and the majority of them used the ramp, rather than the adjoining curb.
2. All of the wheelchair users were able to use the ramp; however, a small lip at the bottom of the ramp caused considerable difficulty to about 25% of the subjects.
3. The corner location of the curb ramp caused 70% of the wheelchair users to go out beyond the limits of the crosswalk.

Conclusions

Where the curb radius is small, the distance from the bottom of this type of ramp to the boundary of the crosswalk may also be very little (or nothing). Under these conditions, as was found at the test site, wheelchairs may be forced to travel into the path of moving vehicles. This is potentially hazardous, particularly for those of limited strength and agility. For this reason, the countermeasure is considered to be unsuitable, unless there is at least 4 feet (1.22 m) between the bottom of the curb ramp and the nearest crosswalk boundary line. The tests confirmed earlier findings, that any lip at the bottom of curb ramps causes considerable difficulty to some people in wheelchairs.

countermeasure 9

double conventional ramp

Countermeasure No. 9, Double Conventional Ramp, was constructed and evaluated in Baltimore, Maryland, a large northeastern city, having a population of about 900,000 (see Figures 61 and 62). The specific location chosen was on a heavily travelled artery nearly 2/3 of a mile (2.68 km) north of the city's center. It is a location that receives heavy use by the elderly; there is a large, newly constructed community and recreation center for the elderly immediately adjacent to the test location. Also, because of the substantially higher than normal elderly proportion of the population using the site, there is a greater number of handicapped persons present.

Most users of the recreation center come by public transportation. Those coming from the north, from predominately outlying areas of the city, arrive at a bus stop in front of the center. Persons coming from downtown, and those using bus connections that have taken them through the downtown, arrive on a northbound street one block to the east. The majority of persons appear to come from this direction.

Consequently, the north crosswalk (east-west direction) of the intersection selected for evaluation received pedestrian flows substantially greater than found in the other crosswalks. These are summarized in Table 112.

TABLE 112. PEDESTRIAN FLOWS - PERSONS PER HOUR

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
East Crosswalk	10	10	20	15	10	15
West Crosswalk	35	25	50	25	35	35
East-West Direction						
North Crosswalk	40	45	35	35	85	45

The main street passing through the intersection is a one-way south-bound street carrying three lanes of moving traffic. Parallel parking is permitted on the east side of the street only. This street serves as a major arterial feeding the city's downtown, and has heavy vehicular flows throughout the day. The problem caused by the heavy flows is compounded by the speed at which the vehicles operate—speeds of 40 to 50 miles (64 to 80 km) per hour not being uncommon; also the vehicles round a bend in the road just to the north of the recreation center so that



Figure 61 Baltimore: Test and Control Sites: Double Conventional Curb Ramps

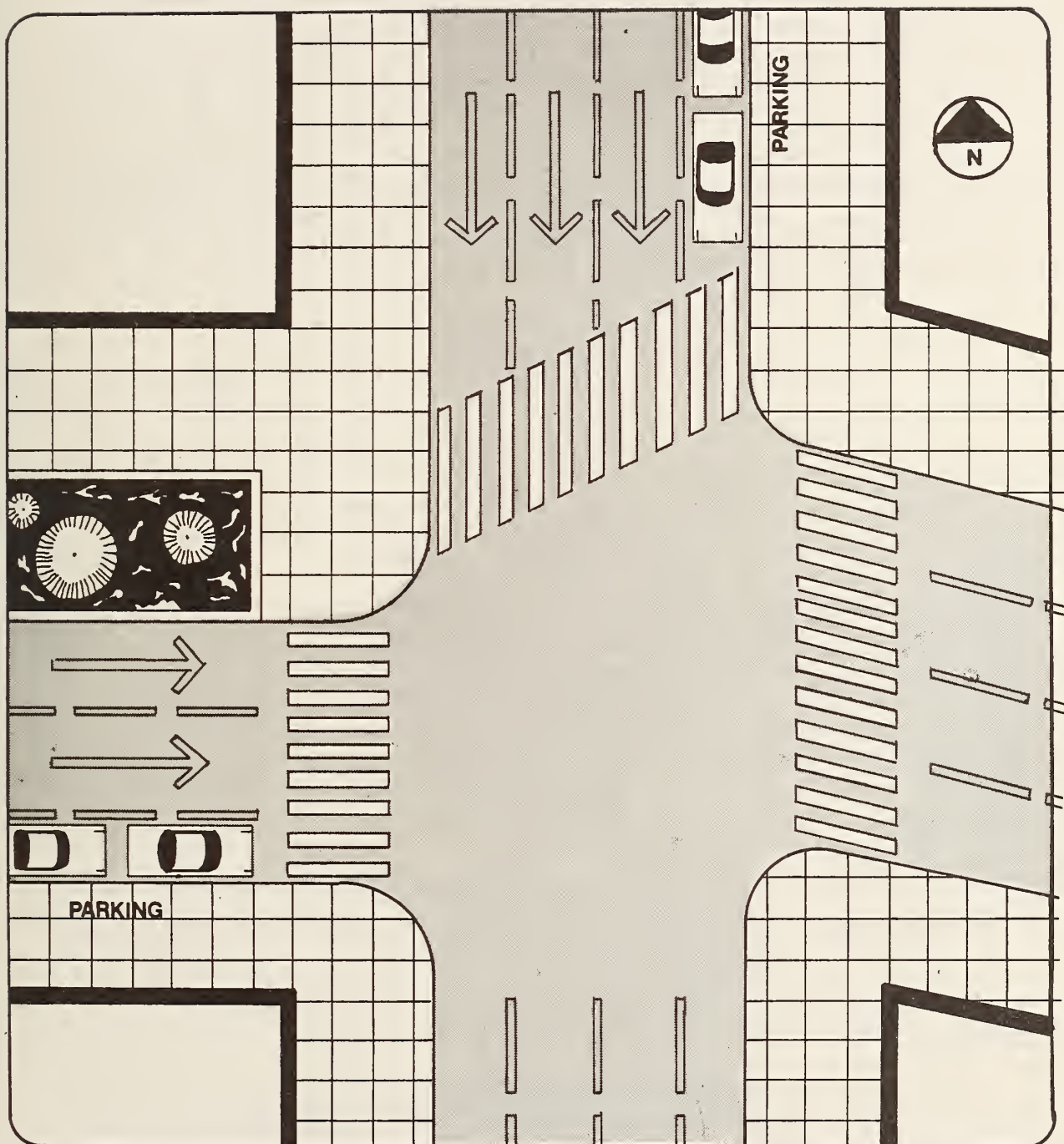


Figure 62 Baltimore: Double Conventional Curb Ramps: Test Site Before Modification

there is a short time period in which pedestrians crossing the street are visible, or drivers visible to them. The distance from the curve to the crosswalk is about 450 feet (137.25 m). Drivers travelling at 40 miles (64.36 km) per hour around the curve have less than $7\frac{1}{2}$ seconds to spot pedestrians crossing the street and take any necessary action.

The physical characteristics of the intersection add to this problem. The intersection is somewhat offset. As a result, the most heavily used crosswalk runs at an oblique angle. Therefore, a large proportion of pedestrians crossing at that point walk beyond the limits of the crosswalk, increasing their vulnerability to the rapidly approaching traffic. This is illustrated in Figure 63.

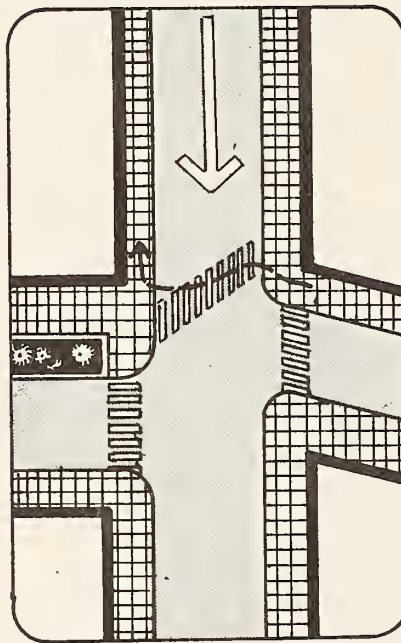


Figure 63 Pedestrian Route Through Oblique Angle Crosswalk

The intersecting street is one way eastbound, and carries very little traffic, moving at slow speed.

Signalization.

The test intersection is signalized. The timing of the traffic signals are as follows:

Eastbound Street	Elapsed Time
Green; "Walk" Signal	0 seconds
Flashing "Don't Walk"	28
Yellow; Continuous "Don't Walk"	32
Red	36
Green	60

Southbound Street

Red	0 seconds
Green; "Walk" Signal	36
Flashing "Don't Walk"	52
Yellow; Continuous "Don't Walk"	56
Red	60

The length of the "walk" signal, for crossing the intersection in the north crosswalk (which has the heaviest pedestrian crossing flow) is 16 seconds. This is equivalent to requiring a walking speed, for persons beginning to cross immediately at the start of the walk signal, of about 3 feet (.915 m) per second, or 2 miles (3.218 km) per hour. This "walk" interval does not include the clearance time allowed for pedestrians to complete their crossing during the flashing "don't walk" phase.

Control Location.

The control location for this countermeasure (see Figure 64) is located one block south of the test site. Its design is not precisely the same as the test intersection. The intersecting street is oriented more on a diagonal, and not quite so much offset. However, traffic flows were nearly the same. Pedestrian flows through the control intersection were substantially higher, averaging nearly twice the flow rates observed at the test intersection. The flow rates for the control intersection are shown in Table 113.

Description of Changes Made at Countermeasure No. 9 Test Location.

The research laboratory design for the intersection proposed the installation of double conventional curb ramps on the northwest quadrant of the test intersection. The design, illustrated in Figure 65, was for ramps having a slope of 1:12 (8.33 percent) and sideslopes of 1:10 (10 percent). Also proposed was an exposed aggregate surface treatment of sidewalk areas immediately adjacent to the ramp sideslopes, to serve as a warning of the existence of the sideslopes, and as a

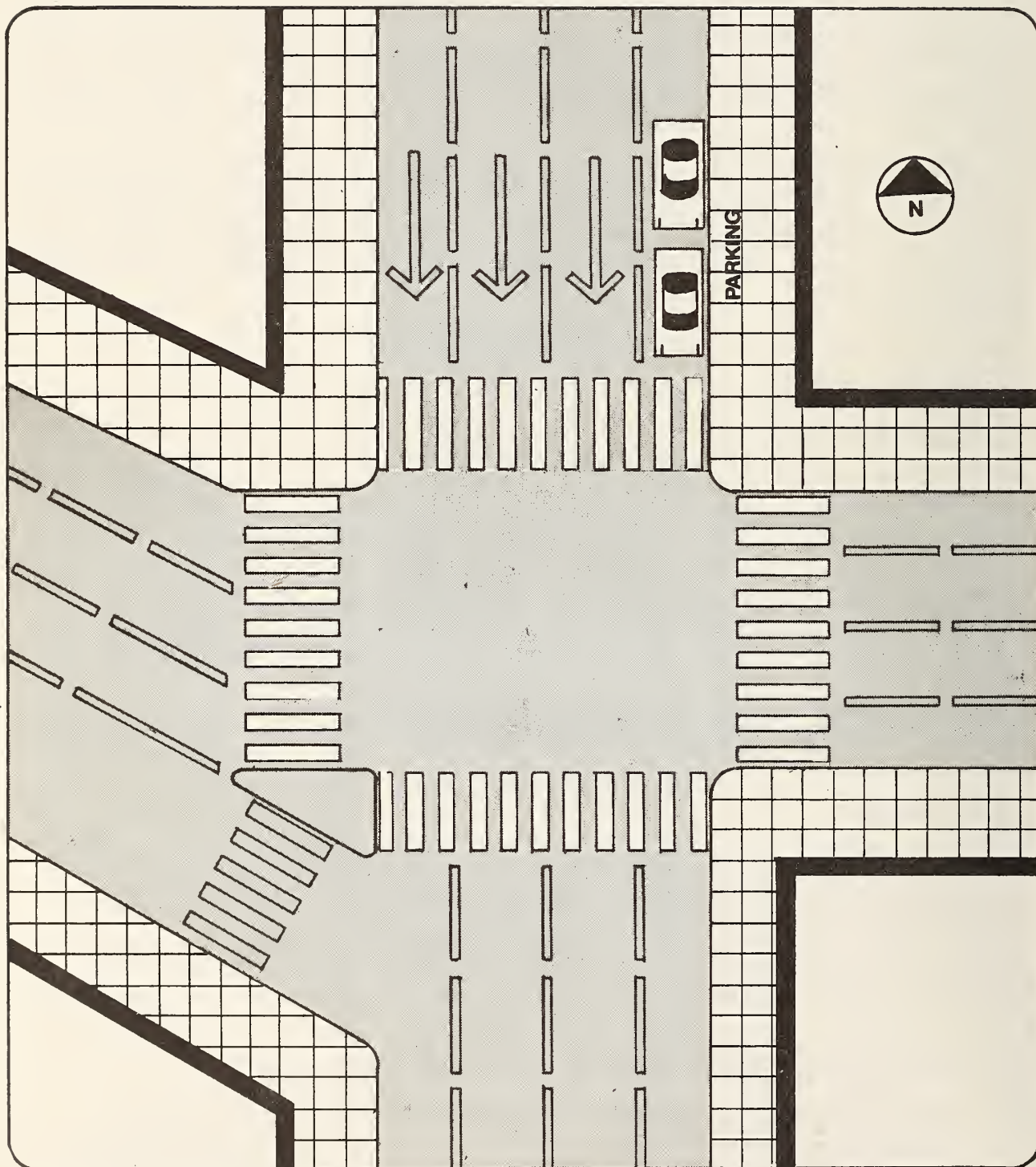


Figure 64 Baltimore: Double Conventional Curb Ramp: Control Site

TABLE 113. PEDESTRIAN FLOWS - PERSONS PER HOUR

Direction	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
East Crosswalk	50	40	40	30	35	40
West Crosswalk	85	55	75	35	30	55
East-West Direction						
North Crosswalk	65	55	100	80	70	75

locational cue for visually impaired pedestrians. The ramps were located, as much as possible, out of the direct path of persons moving along the sidewalk, so that their location would not be in the direct route of the visually impaired, and so that they need be used only by persons choosing, or requiring, to use them. Crosswalks were widened so that they were accessible by persons choosing to use either the ramps or the curb. An important feature for this specific intersection was changing the northern boundary line of the north crosswalk to run perpendicular to the curbs. This recognized the normal crossing movements made by many pedestrians at oblique angle crosswalks, and moved the vehicular stop line well back clear of these paths.

Countermeasure Design as Constructed.

There were two significant changes from the research laboratory's design in the adaptation constructed by the test locality (see Figure 66). The first of these involved the use of an exposed aggregate surface on the ramps and sideslopes as well as on areas of the sidewalk adjacent to the sideslopes. It has been common practice in the test city to surface ramps in this way; it is their opinion that this is necessary to make the ramps detectable to the visually impaired. The second change was in the design of the north crosswalk. The test city's traffic engineering department was not willing to accept the modified crosswalk lines; consequently, the crosswalk remained diagonal in orientation as it was before countermeasure construction.

Evaluation Results: Countermeasure No. 9.

Countermeasure No. 9 was evaluated, by observing its use by members of the general population, on the basis of the following criteria:

- Criteria determining its effect upon pedestrian safety:
 - Conflict between pedestrians and vehicles for persons entering the street from the countermeasure location.
 - Conflict between pedestrians and other pedestrians for persons entering the street from the countermeasure location.

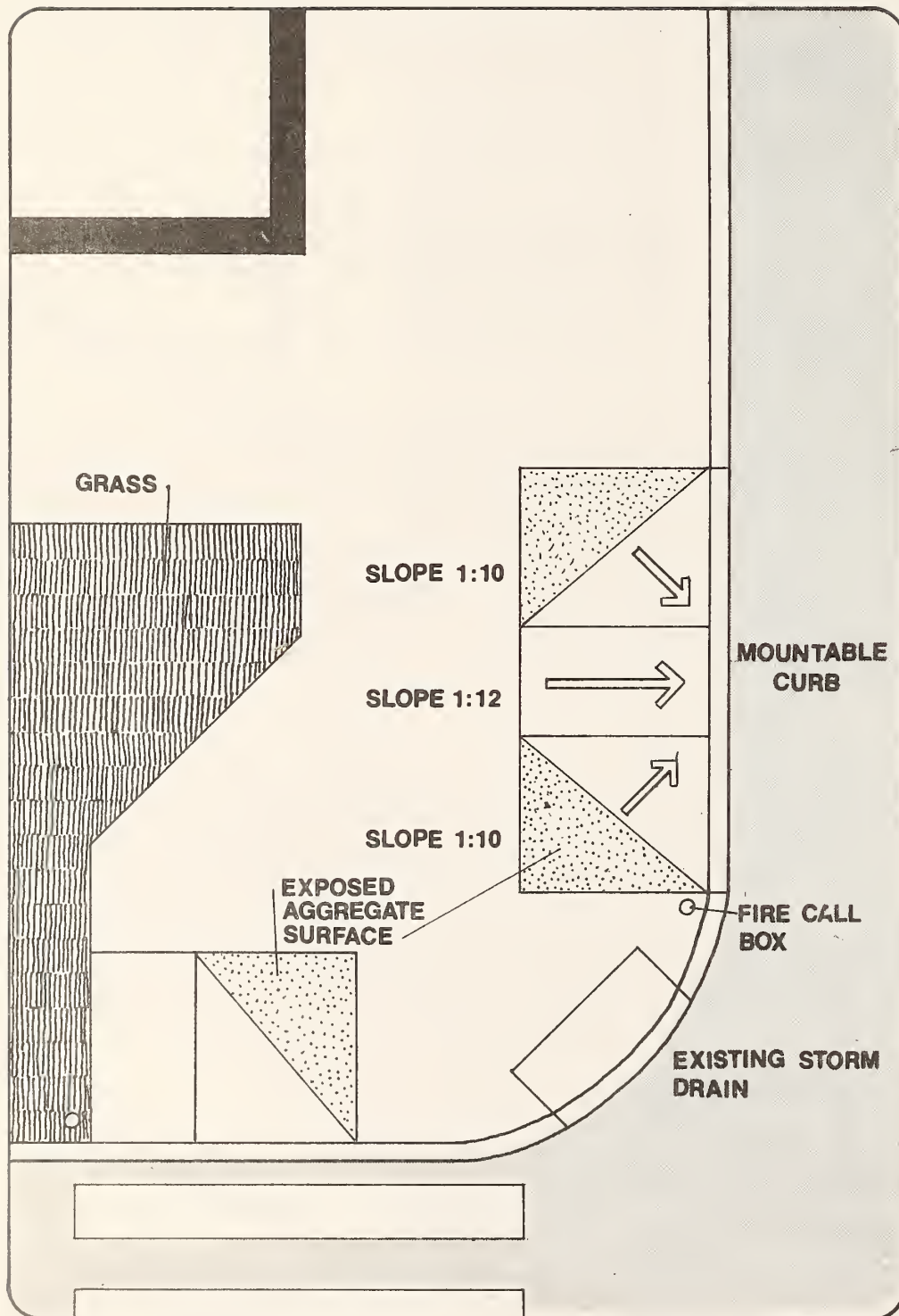


Figure 65 Double Conventional Curb Ramp: Conceptual Design

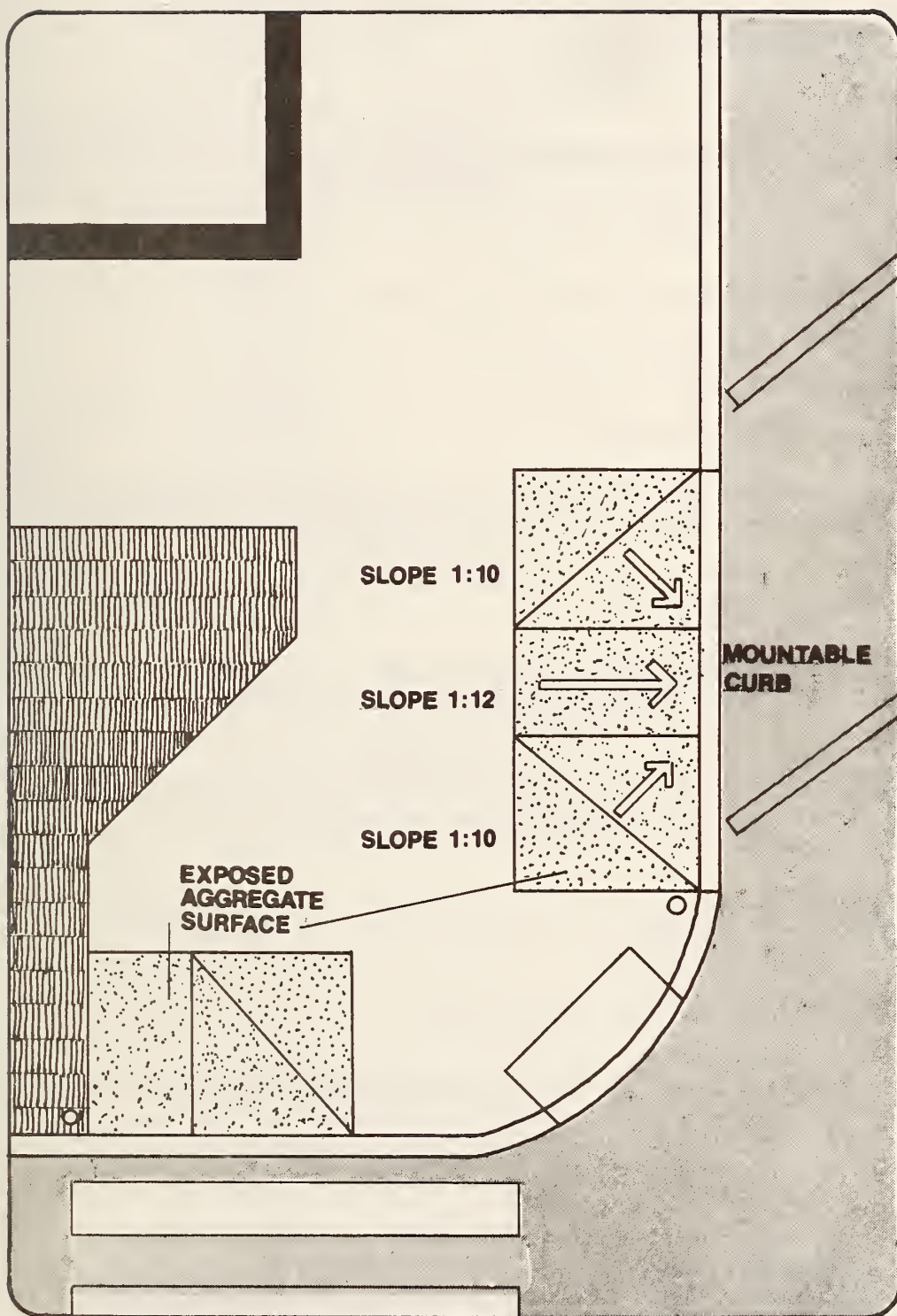


Figure 66 Double Conventional Curb Ramp: Design as Constructed

- Persons walking beyond the limits of the crosswalk for persons entering the street from the countermeasure location.
- Persons waiting in the street for the traffic signal to change.
- Criteria determining its effect upon pedestrian mobility:
 - Persons using the ramp; using the curb to enter the street and to leave the street.
 - Persons changing direction, or veering, to use or avoid the ramp to enter or leave the street.
 - Difficulty factor in entering the street.
 - Difficulty factor in leaving the street.

In addition, the countermeasure was evaluated by special groups of handicapped persons. These evaluations are described following the evaluation descriptions of the general population.

Evaluation Results: Countermeasure No. 9.

1. Effect of countermeasure design upon pedestrian flows. The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians.

The results of the evaluation were as shown in Figure 67.

From Figure 67, it can be seen that pedestrian activity increased substantially at the test site, while at the control site there was a marked decrease.

2. Crossing Time.

Countermeasure No. 9 was evaluated to determine whether the time required to cross the street increased or decreased.

The results of the evaluation are shown in Table 114.

From Table 114, it can be seen that there was no significant change in the crossing time.

3. Use of Ramps.

Curb ramps were installed as part of the countermeasure design at the test location (see descriptions of improvements made). This evaluation was intended to determine the usefulness of these curb ramps to the general population, using the test intersection. The evaluation was in two parts: one, determining whether the ramps were used; two, determining how the apparent difficulty of making the level change was affected by the installation of the ramps.

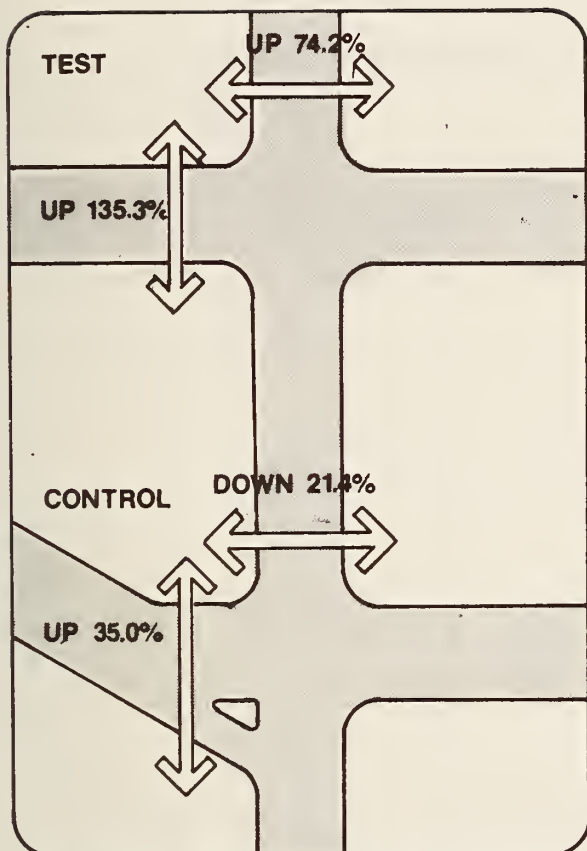
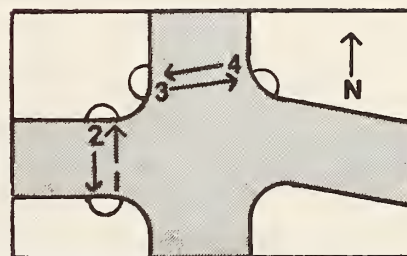


Figure 67 Double Conventional Curb Ramp:
Changes in Pedestrian Flow

TABLE 114. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	9.500	7.889	10.231	10.333	N	N
2	10.222	8.64	13.8182	10.1429	N	N
3	10.563	12.500	9.7059	9.7692	N	N
4	12.120	11.050	9.760	11.5000	N	N

S = Significant

N = Not Significant

Table 115 illustrates the results of the evaluation of the use of the ramps.

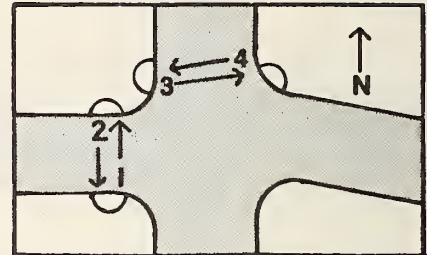


TABLE 115. PEDESTRIAN USE OF CURB RAMP

COUNTERMEASURE NO. 9, LOCATION NO. 1

Direction	Used ramp	Changed direction to use ramp	Changed direction to avoid ramp
1	2 / 9	0 / 9	0 / 9
2	18 / 25	1 / 25	1 / 25
3	10 / 15	0 / 15	0 / 15
4	4 / 33	0 / 24	0 / 24

From Table 115, it can be seen that a high percentage of people (70%) chose to use the test ramps to enter the streets. One person changed direction in order to use the ramp, and one person to avoid using them.

Only 12% of those leaving the street chose to use the ramps but this figure is clearly affected by the pattern of crosswalk use as discussed later.

Apparent Difficulty of Level Change.

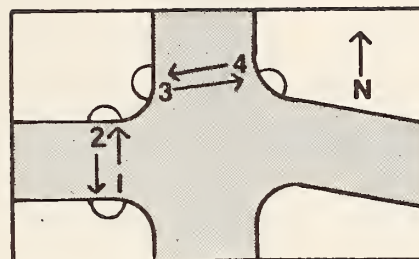
A four point scale was used to record observed difficulty of pedestrians in negotiating the level change at the curb. Separate evaluations were made for entering and leaving the street. The scale ranged from a difficulty factor of 1 (equivalent to walking on a level surface with no perceptible change in walking speed or gait) to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all—including needing to grab onto a pole for support or balance, or nearly falling). An average, able bodied adult, mounting or descending a typical 8 inch (20.32 cm) curb, would score 2 on the scale.

Table 116 presents the results of this evaluation:

From Table 116, it can be seen that there was a decrease in difficulty in making level changes at the curbs, but the decrease is not significant.

4. Evaluation of the Test and Control Intersections With Respect to Other Factors of Behavior Affecting Safety and Mobility.

TABLE 116. APPARENT DIFFICULTY IN MAKING LEVEL
CHANGE AT CURBS:
MEAN VALUES
COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	2.333	1.779	2.3077	2.0000	N	N
2	2.222	1.7143	2.364	2.143	N	N
3	2.125	1.600	2.059	2.154	N	N
4	2.320	2.000	2.040	2.300	N	N

S = Significant

N = Not Significant

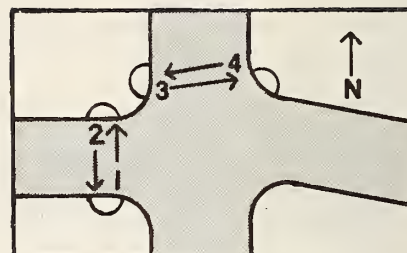
These included evaluations of:

- a. Conflicts between crossing pedestrians and vehicles.
 - b. Conflicts between crossing pedestrians and other pedestrians.
 - c. Pedestrians waiting in the street for the crossing signal to change.
 - d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
 - e. Pedestrians crossing against the traffic signal.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation are shown in Table 117.

TABLE 117. CONFLICTS BETWEEN CROSSING
PEDESTRIANS & VEHICLES

COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
2	1 / 9	0 / 25	2 / 12	2 / 16	N	N
3	4 / 20	0 / 16	0 / 17	0 / 14	N	N
1	1 / 6	0 / 9	1 / 16	3 / 17	N	N
4	3 / 27	0 / 24	1 / 28	3 / 16	N	N

S = Significant

N = Not Significant

From Table 117, it can be seen that there was a decrease in conflicts between crossing pedestrians and vehicles at the test site, and an increase at the control site, but these are not significant.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

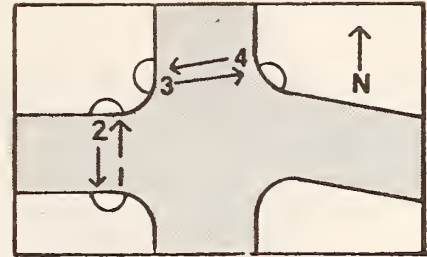
The results of this evaluation are shown in Table 118.

From Table 118, it can be seen that there was no significant change in the number of conflicts between crossing pedestrians and other pedestrians.

- c. Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing or decreasing the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, and were therefore vulnerable to traffic.

The results of this evaluation are shown in Table 119.

TABLE 118. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND OTHER PEDESTRIANS
COUNTERMEASURE NO. 9, LOCATION NO. 1

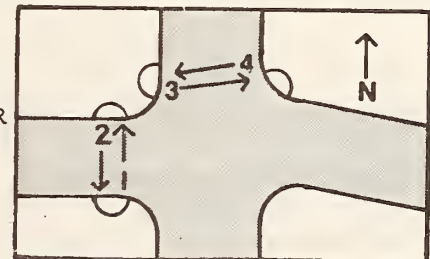


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
2	0 / 9	0 / 25	0 / 12	0 / 16	N	N
3	0 / 20	0 / 16	0 / 17	1 / 14	N	N
1	0 / 6	0 / 9	0 / 16	0 / 17	N	N
4	0 / 27	1 / 24	1 / 28	1 / 16	N	N

S = Significant

N = Not Significant

TABLE 119. PEDESTRIANS WAITING IN THE STREET FOR
THE CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
2	0 / 9	2 / 25	0 / 12	1 / 16	N	N
3	4 / 20	1 / 16	0 / 17	1 / 14	N	N
1	0 / 6	0 / 9	0 / 16	0 / 17	N	N
4	0 / 27	0 / 24	0 / 28	0 / 26	N	N

S = Significant

N = Not Significant

From Table 119, it can be seen that there was no significant change in the number of pedestrians waiting in the street for the crossing signal to change.

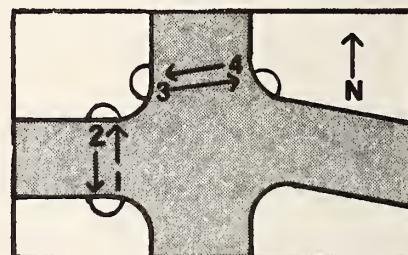
- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk and whether the countermeasure design would have any effect upon their doing so.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 120 and 121.

TABLE 120. PEDESTRIANS WALKING BEYOND THE
LIMITS OF THE CROSSWALK
COUNTERMEASURE NO. 9, LOCATION NO. 1

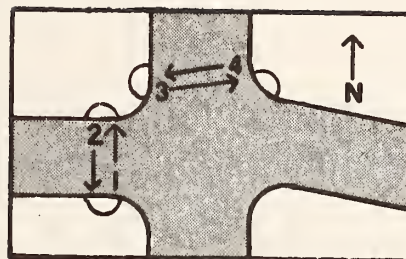


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
2	1 / 9	6 / 25	4 / 12	3 / 16	N	N
3	10 / 20	11 / 16	4 / 17	2 / 14	N	N
1	0 / 6	0 / 9	1 / 16	7 / 17	N	S
4	13 / 27	18 / 24	2 / 28	6 / 16	N	S

S = Significant

N = Not Significant

TABLE 121. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1	0 / 6	0 / 9	1 / 16	1 / 17	N	N
4	6 / 27	0 / 24	1 / 28	0 / 16	N	N
2	0 / 9	3 / 25	2 / 12	0 / 16	N	N
3	0 / 20	2 / 26	4 / 17	0 / 14	N	N

S = Significant

N = Not Significant

From Tables 120 and 121, it can be seen that at the test site there is no significant change in the numbers of pedestrians walking beyond the limits of the crosswalk and leaving the crosswalk while crossing the street. However, the results confirm that at oblique angle crosswalks, many pedestrians will tend to take the shortest route directly across the road and beyond the limits of the crosswalk. At the test site oblique crosswalk, about half of the pedestrians behaved in this way.

- e. Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing or decreasing such behavior.

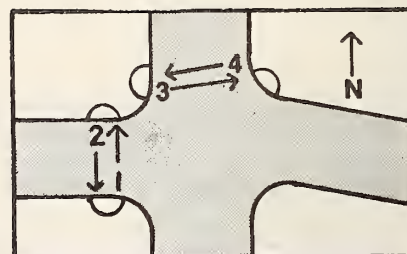
The results of this evaluation are shown in Table 122.

From Table 122, it can be seen that in direction one, there was a significant increase in the number of pedestrians crossing against the traffic signal. However, as there was no significant increase in any of the other directions, it does not seem likely that this is attributable to the countermeasure.

Evaluation of Countermeasure No. 9 by Special Groups of Handicapped Persons.

The procedures used in the special group evaluations were described in detail earlier

TABLE 122. PEDESTRIANS CROSSING AGAINST
TRAFFIC SIGNAL
COUNTERMEASURE NO. 9, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
2	2 / 9	6 / 25	0 / 12	4 / 16	N	N
3	6 / 20	7 / 16	5 / 17	6 / 14	N	N
1	0 / 6	6 / 9	3 / 16	5 / 17	S	N
4	7 / 27	9 / 24	8 / 28	6 / 16	N	N

S = Significant N = Not Significant

in this report. In brief, two groups of handicapped persons were considered to be sufficiently important to the evaluation of this countermeasure to warrant the use of special evaluation procedures. These were the visually impaired, and persons using wheelchairs. Approximately 15 persons from each group were brought to the site. They had been asked to familiarize themselves with the test locale. For the test, the subjects used the test locations individually, following prescribed routes, while observations were made of their use of design features incorporated into those locations and any apparent benefits or difficulties resulting from those design features. Time lapse photography was used to record the results of these evaluations. The results sought from this process were the same as sought in the observation of pedestrian behavior evaluations: the impact of the countermeasure design on pedestrian safety and on mobility. In addition, several specific questions relating to the needs and difficulties of these groups were considered. These included the following:

Visually Impaired Group.

- Were the ramps obviously detectable by the visually impaired; could they detect the points at which the ramps meet the level sidewalk and the street.
- Was the exposed aggregate ramp surfacing useful for the visually impaired.
- Were the ramps, in their location and design, useful or inconvenient to the visually impaired.

Wheelchair Group.

- Was the exposed aggregate ramp surfacing useful or inconvenient for persons in wheelchairs.
- Were the ramps satisfactorily designed and located, or were they inconvenient.

Evaluation Results: Severely Visually Impaired.

1. Detectability of ramps. The ramps were sloped, at a 1:12 ratio (8.33 percent). As described earlier, the ramps, sideslopes and warning area 3 feet (.915 m) in width were all given an exposed pebble aggregate finish. This evaluation was intended to determine if visually impaired test subjects were able to identify the sloped ramp surface itself, and in particular, the point at which it met the street. The method used for this evaluation was to observe whether, and precisely where, visually impaired subjects who used the ramps to enter the street stopped and waited for a safe time to cross.

There were nine crossing events involving subjects using the ramps to enter the street. For 7 of these events, the subjects stopped at the bottom of the ramp, just short of the street, and waited for a safe crossing opportunity. On the other two occasions, both involving the crossing of the very lightly travelled eastbound street, the subjects intentionally entered the street and crossed without stopping. These results indicate that the visually handicapped subjects were able to successfully identify the point at which the ramp joins the street.

2. Usefulness of the exposed aggregate ramp surfacing. The purpose of this evaluation was to determine whether the exposed pebble aggregate surfacing of the warning area, ramp sideslopes, and ramps was of use to the visually handicapped as a cue to assist them in negotiating their way through the pedestrian environment. The visually handicapped test subjects were observed as they approached the textured area for indications that they detected the pebble surface. Eight of the eleven visually handicapped test subjects reacted very quickly to the textured surface upon reaching it, the primary reactions being an abrupt change in direction, or change in walking speed. Two other subjects also changed their direction or speed; however, it was not as clear, in their cases, that the texturing of the surface was the reason for those responses. The following narratives are examples of the visually handicapped making use of the textured surfaces:

- The subject approached and reached the pebble surfaced warning area adjacent to the ramp leading to the southbound street. He detected the surface first with his cane and then with his feet. Immediately upon stepping on the warning surface, he changed direction, turning to his left about 120 degrees. He moved quickly to the utility pole adjacent to the curb, found it with his cane and his free hand, and positioned himself at the curb, waiting for a safe time to cross.

- The test subject was walking along the sidewalk and reached the pebble surfaced upper edge of the ramp and warning areas for the crossing of the southbound street. Not wishing to use that ramp, the subject used his cane to follow that edge for nearly ten feet (3.05 m), then turned 45 degrees to his right and found the textured surfacing of the ramp leading to the crossing of the eastbound street.
- The subject, walking south on the sidewalk, reached the pebble textured surface; turned immediately to his left, descended the ramp and waited at the bottom of the ramp for an opportunity to cross.

The surfacing of the ramp and sideslope, as well as the warning areas, resulted in very large continuous areas being treated with the pebble surfacing; for example, the ramp leading to the southbound street was within an area of more than 160 square feet (14.88 m²) being so treated. Such a large area being given a special surface treatment was confusing to at least one test subject who became disoriented, as follows:

- The test subject reached the warning area and sideslope of the ramp leading to the southbound street. He walked down the sideslope in a direction perpendicular to that of the ramp, crossed the ramp and continued up the side slope on the other side. He apparently felt that he had descended the ramp in the proper direction, for once he reached the brushed concrete sidewalk surface beyond the ramp he increased his walking speed as if believing he was now in the street and crossing it. Within a few steps he bumped into a fire call box.

3. Usefulness of the ramps. The ramps were located such that pedestrians would have a choice between using the ramp or curb to negotiate the level change between the sidewalk and the street. The ramp leading to the southbound street was located in a position that resulted in persons approaching the intersection from the sidewalk to the west would miss the ramp entirely if they continued to walk straight, and reach the curb instead. This evaluation was intended to determine how many visually impaired test subjects chose to use the curb, the ramp, or the half curb at the sideslope to enter or leave the street. In addition, factors of difficulty were measured for all test subjects to determine the relative ease or difficulty of using the ramp, curb, or sideslope. The results are shown in Table 123.

The results shown in Table 123, indicate that the visually impaired group did not all have the same preferences; some chose to use the ramp, some the flares, and some the curbs.

Evaluation Results: Wheelchair Group.

1. Location and design of ramps. The two ramps were located to provide direct access to both crosswalks. The purpose of this evaluation was to determine whether the ramp slope or surface resulted in any difficulty to wheelchair users. The results are shown in Table 124.

TABLE 123. ASCENT AND DESCENT OF RAMPS - BLIND USERS

	Number of Events	Mean Difficulty Factor
Entered Street		
Used Curb	7	2.43
Used Sideslope	11	2.00
Used Ramp	6	1.83
Left Street		
Used Curb	8	2.63
Used Sideslope	6	2.50
Used Ramp	9	2.00

TABLE 124. ASCENT & DESCENT OF RAMPS - WHEELCHAIR USERS

		Wheelchair Users													Mean Difficulty Factor
		1	2	3	4	5	6	7	8	9	10	11	12	13	
		Difficulty Factors													
North Ramp .	Enter Street	2	2	-	2	2	3	3	2	2	2	2	2	2	2.17
	. Leave Street	2	2	-	2	-	3	3	3	4	2	2	2	2	2.45
South Ramp .	Enter Street	2	2	2	3	-	2	2	2	2	2	2	2	2	2.08
	. Leave Street	2	3	3	4	3	2	4	3	3	4	2	2	2	2.84

The ramp slopes themselves appeared to give no difficulty to the wheelchair users. However, the pebble textured surface did cause substantial problems, because of its roughness and because of some loose material on the ramp surface.

- The test subject reached the ramp. Immediately upon beginning to ascend, the front wheels of her chair caught on the pebble surface and were turned to the side, forcing the chair to an abrupt stop. The subject attempted to

move the chair several times and was unable to do so. Finally, assistance was required for her to complete her ascent of the ramp.

- The test subject, in a motorized chair, came to a stop on the ramp when the front wheels of his chair became jammed on the rough ramp surface. The subject was able to move ahead a few inches (centimeters), then the wheels would jam again and the chair would roll back. This was repeated several times before the ascent was finally completed.
- The test subject was stopped at the midpoint of his ascent of the ramp when his chair wheels became caught in the rough pebble surface. As he attempted to move forward, his rear wheels spun loose, there not being sufficient traction on the ramp surface.

Summary of Findings: Countermeasure No. 9, Double Conventional (Recessed) Curb Ramps.

1. The countermeasure did not adversely affect pedestrian flow (in fact, pedestrian activity increased substantially, while at the control site activity decreased).
2. There were changes in the crossing time but none that were significant.
3. Seventy percent of the people entering the street chose to use the ramps rather than the curb. Only 12% chose to use the ramps when leaving the street, but about half of those crossing the oblique crosswalk left the crosswalk in order to take a more direct route at right angles to the curb and therefore reached the opposite side of the road some distance from the ramp.
4. There was a significant decrease in difficulty in making level changes at the test site.
5. There was an insignificant decrease in conflicts between crossing pedestrians and vehicles.
6. There was no significant change in the number of conflicts between crossing pedestrians.
7. There was no significant change in the number of pedestrians waiting in the street for the crossing signal to change.
8. There was no significant change in the numbers of pedestrians walking beyond the limits of the crosswalk or leaving the crosswalk.
9. At the oblique angle crosswalk about half of the pedestrians left the crosswalk to take the shortest and more direct route. This confirms the unsatisfactory nature of oblique crosswalks.
10. Although in one direction at the test site there was a significant increase in the number of pedestrians crossing against the traffic signal, there seems to be no reason for attributing this change to the countermeasure.

11. The severely visually handicapped were able to detect the ramps, and in particular, the bottom of the ramps.
12. The severely visually handicapped were able to detect the exposed aggregate textured surfaces and to use the texture as an environmental one.
13. The value of the textured surface was decreased because the treated area was too great and some of the visually handicapped users had to search around for an unnecessarily long time within this large space in order to find the edges and other landmarks.
14. Some of the visually impaired chose to use the ramp rather than the curbs.
15. The wheelchair users had no difficulty in using the ramps except for the surface finish.
16. The pebbles in the exposed aggregate were not adequately bonded into the concrete and were loose. This made passage for the wheelchairs very difficult.

Conclusions.

- The countermeasure worked well with no obvious disadvantages except that the exposed aggregate surface was improperly constructed and tended to come loose causing considerable difficulty to wheelchair users.
- Able bodied people tend to ignore the marked lines in an oblique crosswalk, and take the shortest route. Visually impaired people have greatly increased difficulty in orienting themselves and crossing oblique angled crosswalks.

countermeasure 11

ramped full quadrant

Countermeasure No. 11, Ramped Full Quadrant, was constructed and evaluated at the following two locations:

- Location No. 1, Baltimore, Maryland, a large northeastern city.
- Location No. 2, Clearwater, Florida, a southeastern city.

1. LOCATION NO. 1: BALTIMORE, MARYLAND

The location selected for the construction and evaluation of this countermeasure was on a heavily travelled artery, nearly 2/3 of a mile (1.073 km) north of the city's center (see Figures 68 and 69). The specific location chosen was the northeastern quadrant of the same intersection used for the evaluation of Countermeasure No. 9. The location receives heavy use by the elderly; there is a large, recently constructed, community and recreation center for the elderly immediately adjacent to the test location. Also, because of the substantially higher than normal

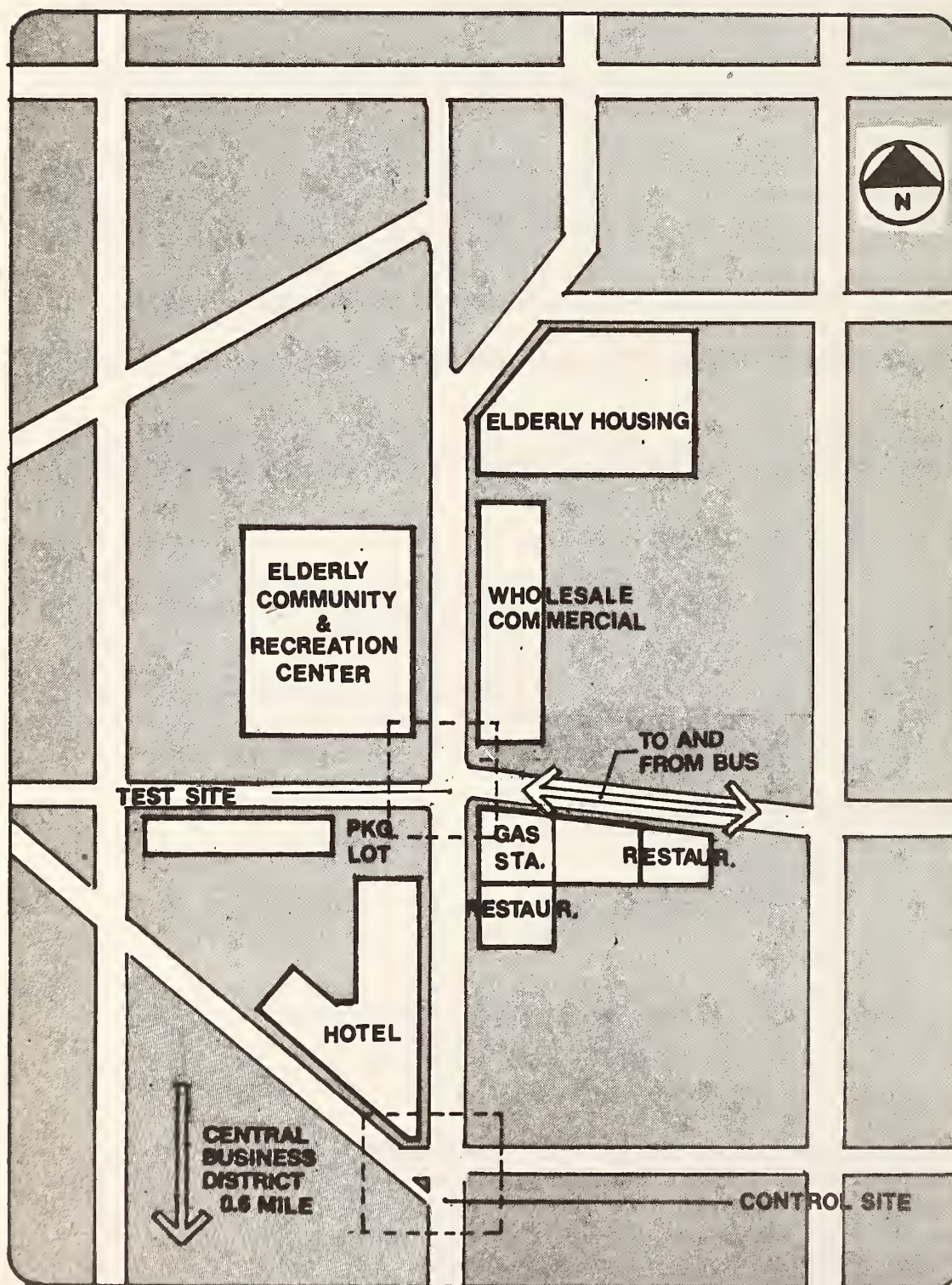


Figure 68 Baltimore: Ramped Full Quadrant: Test and Control Site Locations

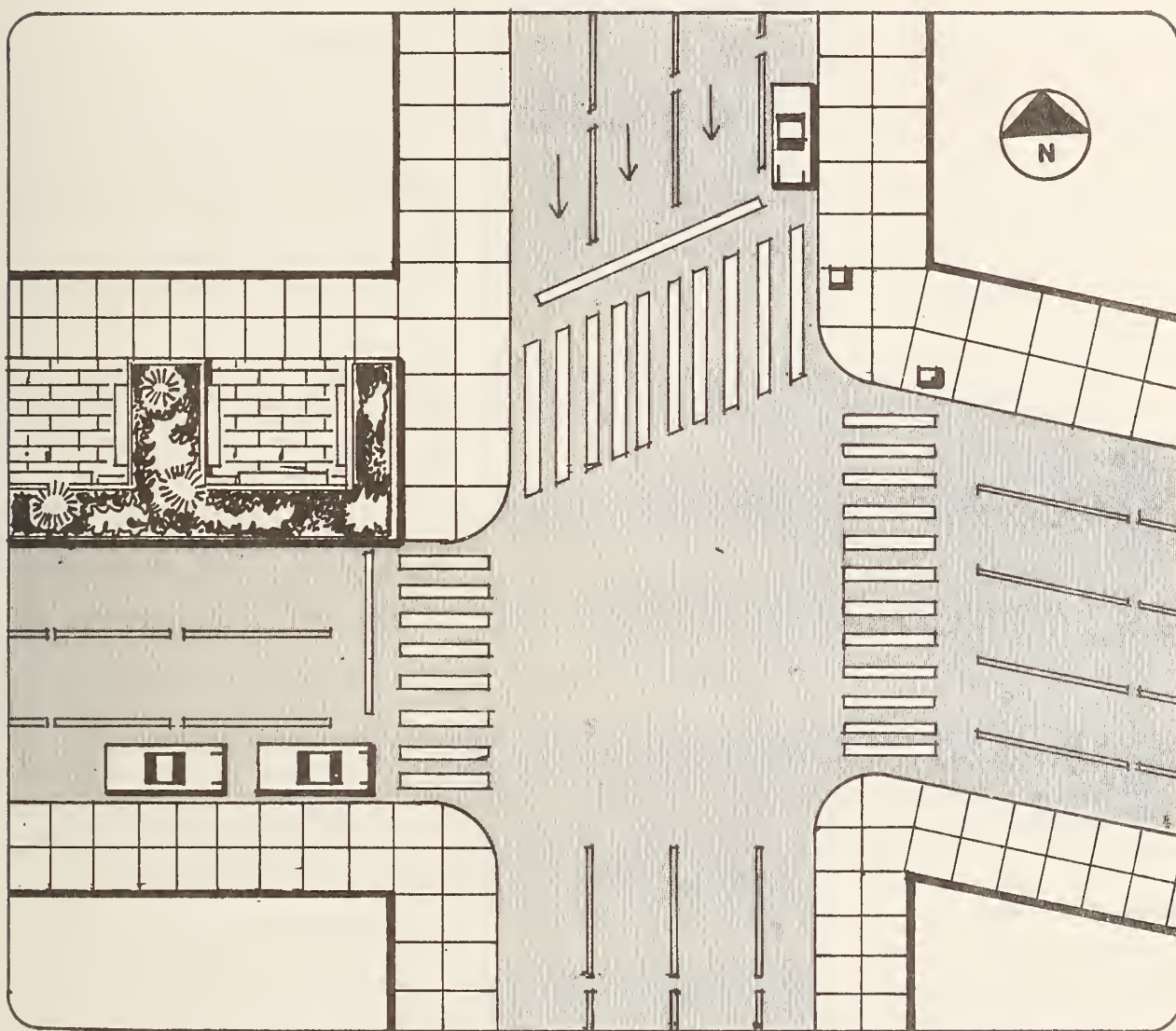


Figure 69 Baltimore: Ramped Full Quadrant: Test Site Before Modification

proportion of the elderly population using the site, there is a greater number of handicapped persons present, as well.

Most users of the recreation center come by public transportation. Those coming from the north, from predominately outlying areas of the city, arrive at a bus stop in front of the center. Persons coming from downtown, or via bus connections that have taken them through the downtown area, arrive on a northbound street, one block to the east. The majority of persons appear to come from this direction.

Consequently, the north crosswalk of the intersection selected for evaluation, received pedestrian volume flows substantially greater than found in the other crosswalks. These are summarized in Table 125.

TABLE 125. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
East Crosswalk	10	10	20	15	10	15
West Crosswalk	35	25	50	25	35	35
East-West Direction						
North Crosswalk	40	45	35	35	85	45

The main street passing through the intersection is a one-way south-bound street, carrying three lanes of moving traffic. Parallel parking is permitted on the east side of the street only. This street serves as a major arterial, feeding the city's downtown, and has heavy vehicular flows throughout the day. The problem caused by the heavy flows is compounded by the speed at which the vehicles operate—speeds of 40 to 50 miles (64.36 to 80.45 km) per hour not being uncommon; also, vehicles round a bend in the road just to the north of the recreation center, so that there is a short time period in which pedestrians crossing the street, are visible, or drivers visible to them. The distance from the curve to the crosswalk is about 450 feet (137.25 m). Drivers travelling at 40 miles (64.36 km) per hour around the curve have less than $7\frac{1}{2}$ seconds to spot pedestrians crossing the street, and take any necessary action.

The intersecting street is one-way eastbound, and carries very little traffic; all of which is local, or access traffic, moving at slow speed.

Signalization.

The test intersection is signalized. The timing of the traffic signal is as follows:

Eastbound Street	Elapsed Time
Green; "Walk" Signal	0 seconds
Flashing "Don't Walk"	28
Yellow; Continuous "Don't Walk"	32
Red	36
Green	60

Southbound Street

Red	0 seconds
Green; "Walk" Signal	36
Flashing "Don't Walk"	52
Yellow; Continuous "Don't Walk"	56
Red	60

The length of the "walk" signal, for crossing the intersection in the north crosswalk (which has the heaviest pedestrian crossing flow) is 16 seconds. This is equivalent to requiring a walking speed, for persons beginning to cross immediately at the start of the walk signal, of about 3 feet (.915 m) per second, or 2 miles (3.218 km) per hour. This "walk" interval does not include the clearance time allowed for pedestrians to complete their crossing during the flashing "don't walk" phase.

Control Location

The control location for this countermeasure (see Figure 70), is located one block south of the test site. Its design is not precisely the same as the test intersection; the intersecting street being oriented more on a diagonal, and not being quite so offset at the intersection. However, traffic flows were very nearly the same. Pedestrian flows through the control intersection were substantially higher, averaging nearly twice the flow rates observed at the test intersection. The flow rates for the control intersection are shown in Table 126.

TABLE 126. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
East Crosswalk	50	40	40	30	35	40
West Crosswalk	85	55	75	35	30	55
East-West Direction						
North Crosswalk	65	55	100	80	70	75

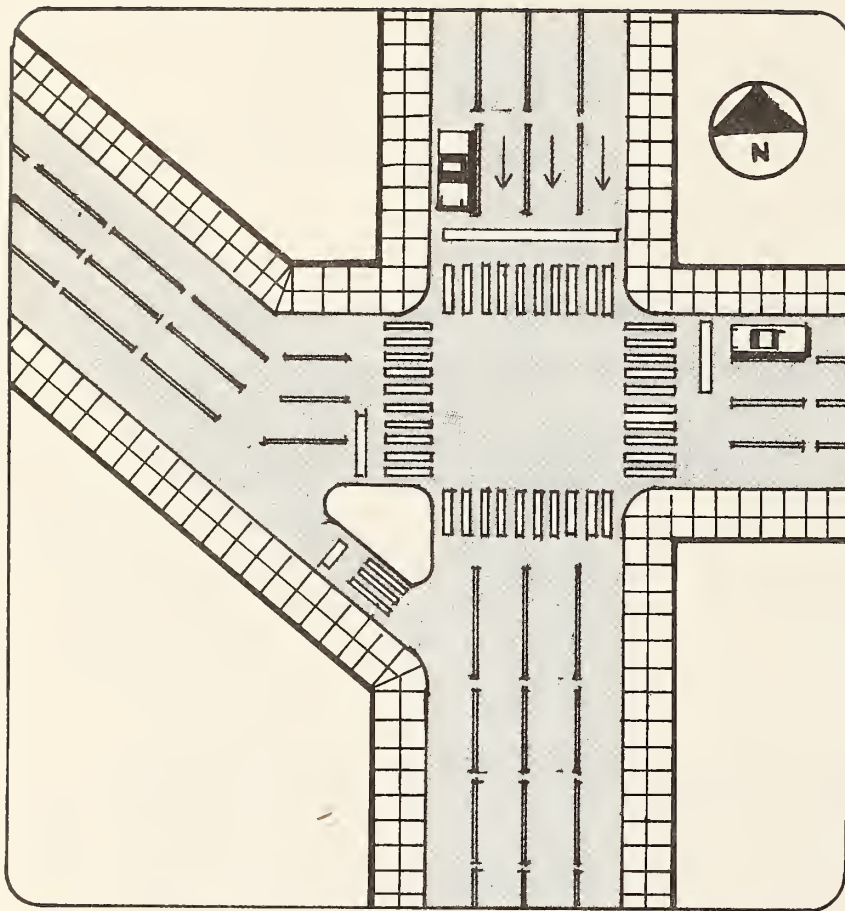


Figure 70 Baltimore: Ramped Full Quadrant: Control Site

Description of Changes Made at Location No. 1.

The research laboratory design for the intersection proposed the installation of the ramped full quadrant design on the northeast quadrant of the test intersection. The design, illustrated in Figure 71, included a gently sloped quadrant, extending from the curb to the corner building, and providing access to both crosswalks. A 1:12 (8.33 percent) sideslope was provided for transition to the sidewalk elevation. The curb was to be chamfered (or sliced at an angle as shown in Figure 71), rather than rounded, to give visually impaired persons orientation points (specifically where the curb angle changes) to align themselves properly with the crosswalks.

Countermeasure Design as Constructed.

The countermeasure design, as adapted and constructed by the test locality, was essentially as designed by the research laboratory, (see figure 72), differing in three respects:

- The width of the ramped quadrant was reduced somewhat, to place the sideslopes in the areas occupied by the existing utility poles.

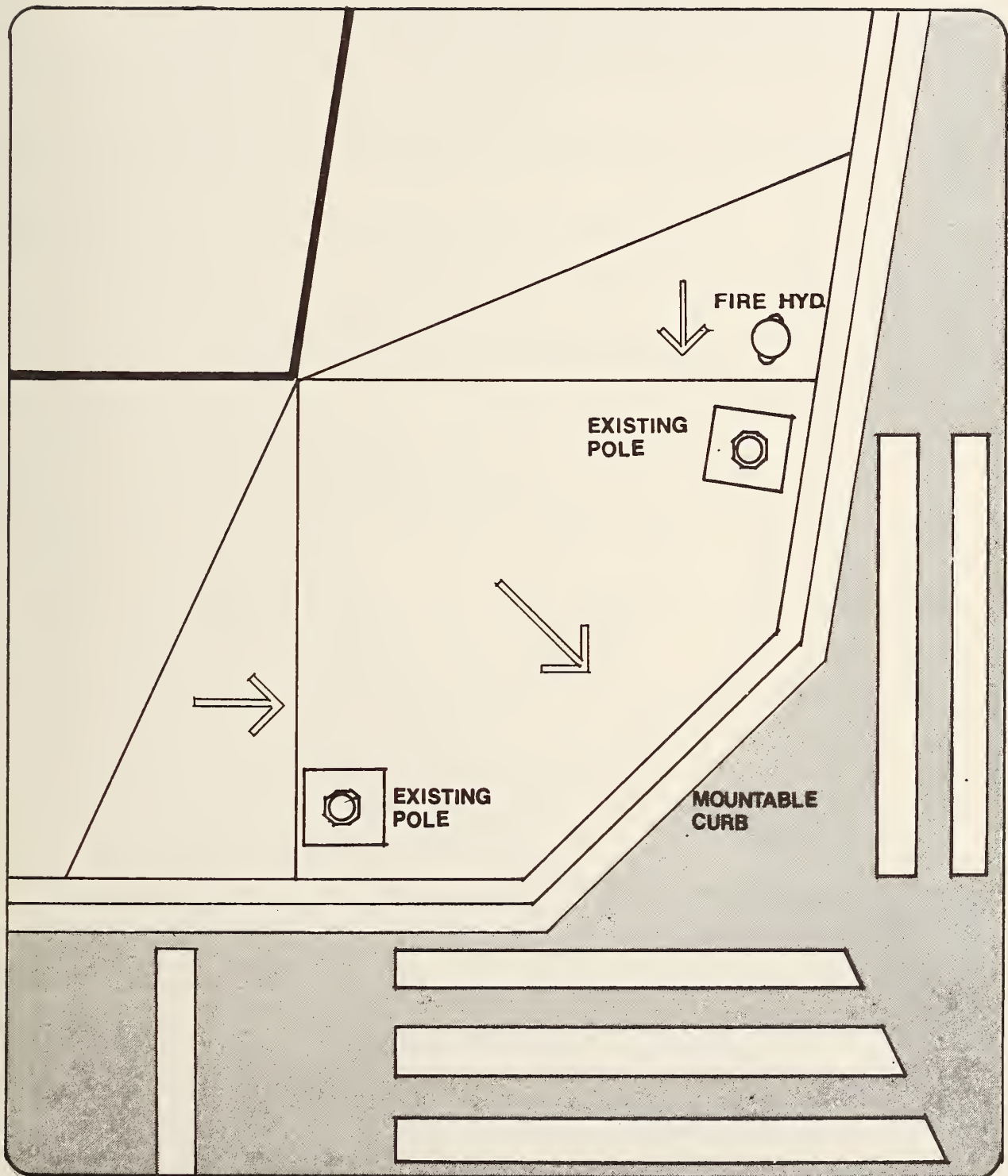


Figure 71 Baltimore: Ramped Full Quadrant: Original Design

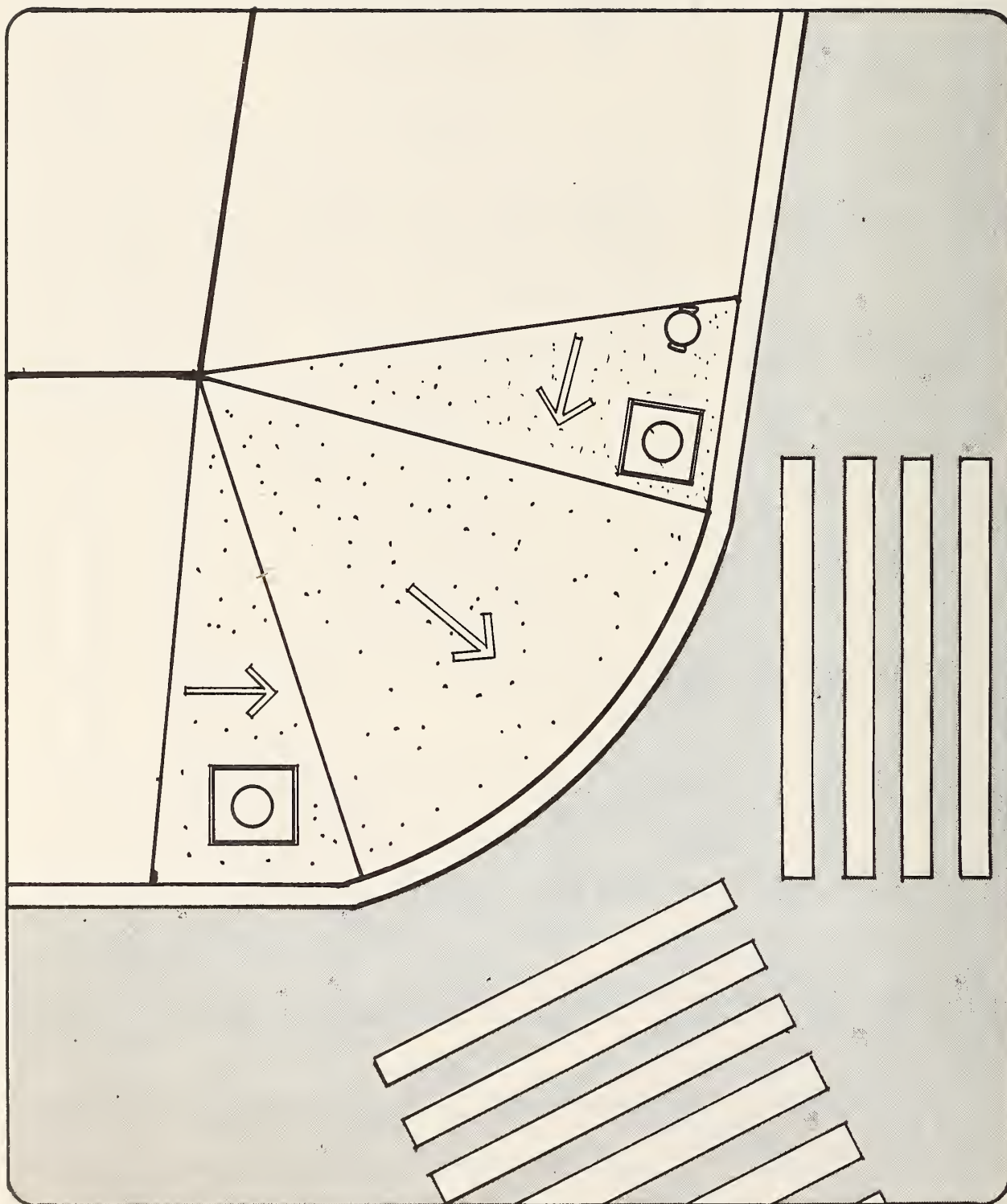


Figure 72 Baltimore: Ramped Full Quadrant: Design as Constructed

- The entire surface of the ramped quadrant, and its sideslopes, were given an exposed aggregate surface. It has been common practice in the test city to surface ramps in this way; it is their opinion that this is necessary to make the ramps detectable to the visually impaired.
- The chamfered (angled) curb design was not used; instead, the traditional curved curb line was retained.

Evaluation Results: Countermeasure No. 11, Test Location No. 1.

Countermeasure No. 11 was evaluated, by observing its use by members of the general population, on the basis of the following criteria:

- Criteria determining its effect upon pedestrian mobility.
 - Pedestrian flow
 - Pedestrian crossing time
 - Persons using the ramp; using the curb—to enter the street, and to leave the street.
 - Persons changing direction, or veering, to use, or avoid the ramp—to enter, or leave, the street.
 - Difficulty factor in entering the street.
 - Difficulty factor in leaving the street.
- Criteria determining its effect upon pedestrian safety.
 - Conflict between pedestrians and vehicles, for persons entering the street from the countermeasure location.
 - Conflict between pedestrians and other pedestrians, for persons entering the street from the countermeasure location.
 - Persons walking beyond the limits of the crosswalk, for persons entering the street from the countermeasure location.
 - Persons waiting in the street for the traffic signal to change.

In addition, the countermeasure was evaluated by special groups of handicapped persons. These evaluations are described following the evaluation descriptions of the general population.

Evaluation Results: Countermeasure No. 11.

1. Effect of Countermeasure Design Upon Pedestrian Flows.

The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians. The intent was to determine if the countermeasure design would discourage pedestrian use.

From Figure 73, it can be seen that pedestrian volumes at the test site increased significantly, while those at the control site decreased generally. From this it can be concluded that the countermeasure did not appear to discourage pedestrian use.

2. Crossing Time.

Countermeasure No. 11 was evaluated to determine whether it had any effect upon the time required to cross the street. The results of this evaluation are shown in Table 127.

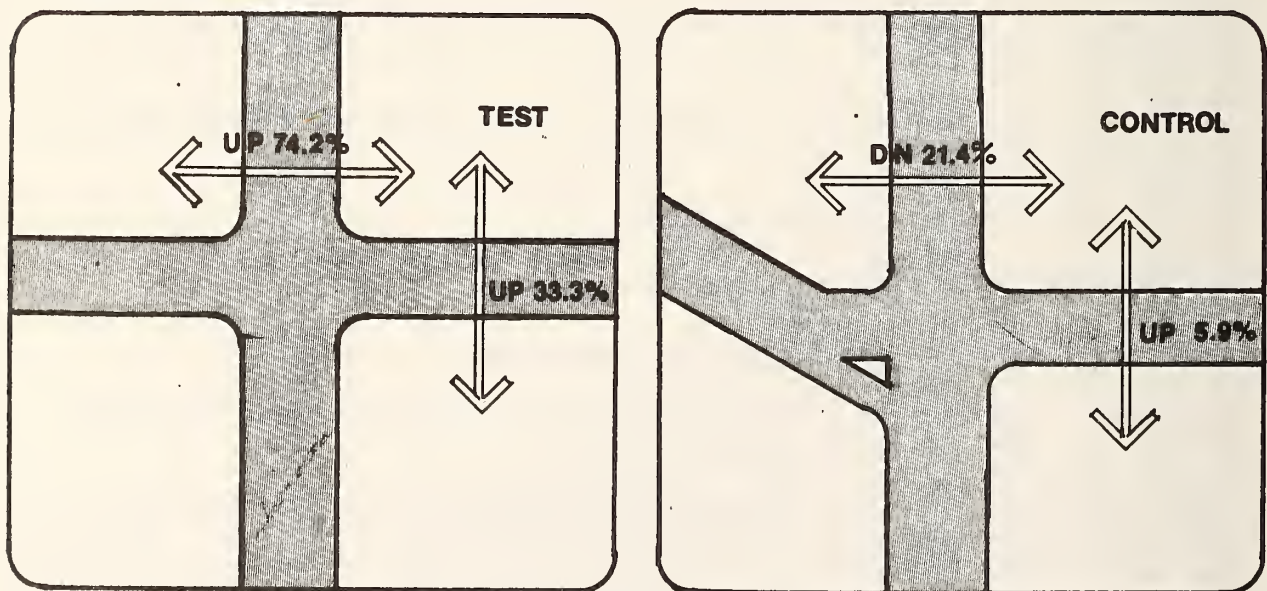


Figure 73 Baltimore: Ramped Full Quadrant: Test and Control Sites: Pedestrian Flow Changes

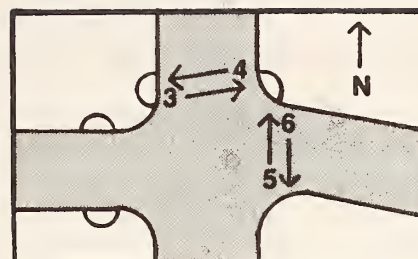
From Table 127, it can be seen that there was no significant change in the crossing times at the test site.

3. Pedestrian Use of the Ramp.

This evaluation was intended to determine the usefulness of the curb ramp to the general population, using the test intersection. The evaluation was in two parts:

TABLE 127. MEAN CROSSING TIME - SECONDS

COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	12.12	11.08	9.76	11.500	N	N
3	10.56	12.50	9.71	9.77	N	N
6	13.00	10.67	11.00	10.21	N	N
5	11.13	11.25	10.00	10.06	N	N

S = Significant

N = Not Significant

- Determining whether the ramp was used.
- Determining how the apparent difficulty of making the level change was affected by the installation of the ramp.

Table 128 illustrates the results of the evaluation of the use of the ramp.

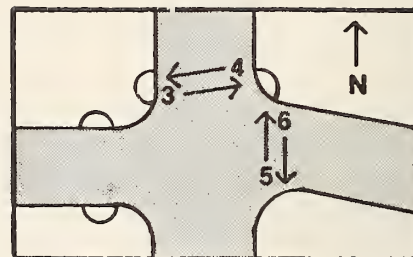
From Table 128, it can be seen that approximately 76% of the pedestrians used the ramp when entering, or leaving, the east-west crosswalk; whereas, only about 14% used the ramp from the north-south crosswalk. Only one person intentionally avoided using the ramp, and none intentionally changed direction to use it.

4. Apparent Difficulty of Level Change.

A four point scale was used to record observed difficulty of pedestrians in negotiating the change at the curb. Separate evaluations were made for entering and leaving the street. The scale ranged from a difficulty factor of 1 (equivalent to walking on a level surface, with no perceptible change in walking speed or gait), to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all—including needing to grab onto a pole for support or

TABLE 128. PEDESTRIAN USE OF CURB RAMPS

COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Used ramp	Changed direction to use ramp	Changed direction to avoid ramp
4	18/23	0/23	0/23
6	1/7	0/7.	1/7
3	10/15	0/15	0/15
8	1/7	0/7	0/7

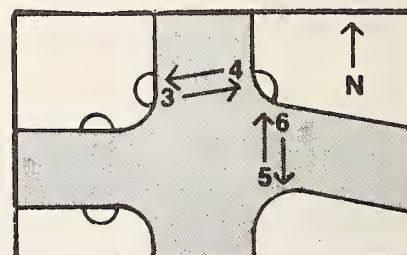
balance, or nearly falling). An average, able bodied adult, mounting or descending a typical 8-inch (20.3 cm) curb; would score 2 on the scale.

The results of this evaluation are shown in Table 129.

TABLE 129. MEAN DIFFICULTY FACTOR NEGOTIATION

LEVEL CHANGE AT THE CURB

COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant difference between columns -	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	2.125	1.200	2.059	2.154	S	N
5	2.000	1.75	2.300	2.063	N	N
4	2.24	1.30	2.00	2.22	S	N
6	2.14	2.00	2.17	2.07	N	N

S = Significant

N = Not Significant

From Table 129, it can be seen that there was a significant decrease in difficulty in negotiating the ramp from, and to, the east-west crosswalk. From the north-south crosswalk, a decrease is apparent, but it is not significant.

5. Evaluation of the Test and Control Intersections With Respect to Other Factors of Behavior Affecting Safety and Mobility.

These included evaluations of:

- a. Conflicts between crossing pedestrians and vehicles.
- b. Conflicts between crossing pedestrians and other pedestrians.
- c. Pedestrians waiting in the street for the crossing signal to change.
- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- e. Pedestrians crossing against the traffic signal.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in, or near, the intersection.

The results of this evaluation are shown in Table 130.

From Table 130, it can be seen that there was no significant change in the number of conflicts between crossing pedestrians and vehicles.

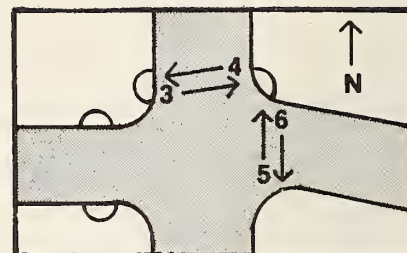
- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Table 131.

From Table 131, it can be seen that there was no significant change in the number of conflicts between pedestrians using the crosswalk.

- c. Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing, or decreasing, the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, and were therefore vulnerable to traffic.

TABLE 130. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND VEHICLES
COUNTERMEASURE NO. 11; LOCATION NO. 1

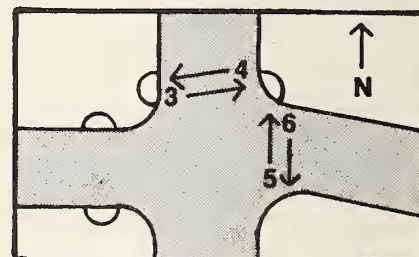


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	3/27	0/24	1/28	3/16	N	N
6	1/11	0/8	0/12	1/15	N	N
3	4/20	0/16	0/17	0/14	N	N
5	1/12	0/7	1/11	1/17	N	N

S = Significant

N = Not Significant

TABLE 131. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND OTHER PEDESTRIANS
COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	0/27	1/24	1/28	1/16	N	N
6	0/11	0/8	0/12	0/15	N	N
3	0/20	0/16	0/7	1/14	N	N
5	0/12	0/7	0/11	0/17	N	N

S = Significant

N = Not Significant

The results of this evaluation are shown in Table 132.

From Table 132, it can be seen that there was no significant change in the number of pedestrians waiting in the street for the crossing signal to change.

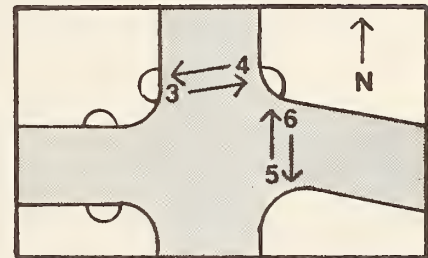


TABLE 132. PEDESTRIANS WAITING IN THE STREET

FOR THE CROSSING SIGNAL TO CHANGE

COUNTERMEASURE NO. 11; LOCATION NO. 1

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	1/27	4/24	0/28	0/16	N	N
6	1/11	0/8	0/12	0/15	N	N

S = Significant

N = Not Significant

- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

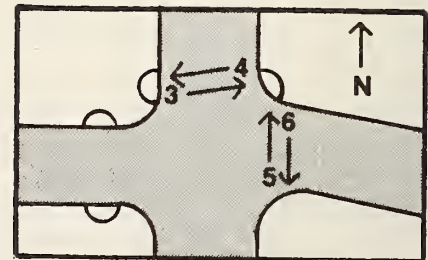
The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk, but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 133 and 134.

From Tables 133 and 134, it can be seen that there was no significant change in the number of pedestrians walking beyond the limits of the crosswalk, or leaving the crosswalk. However, because the east-west

TABLE 133. PEDESTRIANS WALKING BEYOND LIMITS
OF CROSSWALK
COUNTERMEASURE NO. 11; LOCATION NO. 1

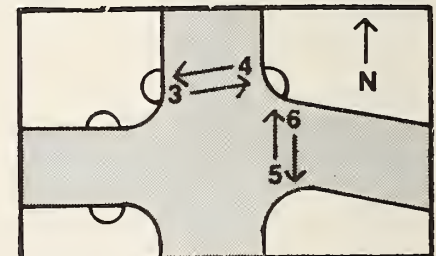


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	13/27	18/24	2/28	6/16	N	N
6	6/11	2/8	0/12	4/15	N	N
3	10/20	11/16	4/17	2/14	N	N
5	4/12	2/7	2/11	1/17	N	N

S = Significant

N = Not Significant

TABLE 134. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
3	0/20	2/16	4/17	0/14	N	N
5	1/12	2/7	1/11	0/17	N	N
4	6/27	10/24	1/28	0/16	N	N
6	2/11	1/8	0/12	0/15	N	N

S = Significant

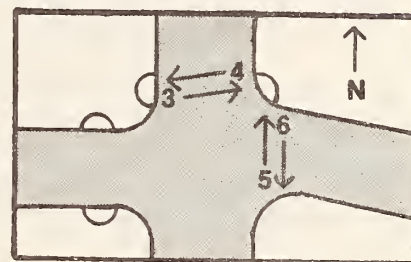
N = Not Significant

crosswalk was set at an oblique angle, a substantial proportion of the pedestrian users, before and after the installation of the countermeasure, chose to take the shortest route, at right angles to the curb.

- e. Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing, or decreasing, such behavior.

The results of this evaluation are shown in Table 135.

TABLE 135. PEDESTRIANS CROSSING AGAINST
TRAFFIC SIGNAL
COUNTERMEASURE NO. 11; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	7/27	9/24	8/28	6/16	N	N
6	2/11	4/8	2/12	6/15	N	N
3	6/20	7/16	5/17	6/14	N	N
5	3/12	3/7	1/11	7/17	N	N

S = Significant

N = Not Significant

From Table 135, it can be seen that there was no significant change in the number of pedestrians crossing against the traffic signal.

Evaluation of Countermeasure No. 1 by Special Groups of Handicapped Persons.

The procedures used in the special group evaluations were described earlier. In brief, two groups of handicapped persons were considered to be sufficiently important to the evaluation of this countermeasure to warrant the use of special evaluation procedures. These were the visually impaired, and persons using wheelchairs. Approximately 15 persons from each group were brought to the site.

They had been asked to familiarize themselves with the test locale. The test subjects used the test locations individually, following prescribed routes, while observations were made of their use of design features incorporated into those locations, and any apparent benefits or difficulties resulting from those design features. Time lapse photography was used to record the results of these evaluations. The results sought from this process were the same as sought in the observation of pedestrian behavior evaluations--the impact of the countermeasure design on pedestrian safety, and on mobility. In addition, several specific questions, relating to the needs and difficulties of these groups, were considered. These included the following:

- Visually Impaired Group.

- Was the ramp detectable by the visually impaired; could they detect the points at which the ramp meets the level sidewalk and the street.
- Was the exposed aggregate ramp surfacing useful, or hazardous, for the visually impaired.

- Wheelchair Group.

- Was the exposed aggregate ramp surfacing useful, or hazardous, for persons in wheelchairs.
- Was the ramp satisfactorily designed and located, or was it difficult to use, or hazardous.

Evaluation Results: Severely Visually Impaired Group.

1. Detectability of the ramped quadrants. The ramped quadrants, which extended the full distance back to the building frontage at the corner, were very gently sloped, at a ratio of approximately 1:32 (3.10 percent). The sideslopes were given a 1:12 (8.33 percent) slope. Both the ramped quadrants and the sideslopes were given an exposed pebble surface concrete finish. This evaluation was intended to determine if test subjects, who are visually impaired, could identify the sloped surface of the ramped quadrants; and, in particular, the points at which they met the street.

The method used for this evaluation was to observe whether, and precisely where, visually impaired subjects, who used the ramp quadrant to enter the street, stopped and waited for a safe time to cross.

There were ten crossing events, involving visually impaired test subjects using the ramped quadrants to enter the street. On nine of these, the test subjects stopped at the bottom of the ramp, just short of the street. On the remaining event, the subject stopped and waited at the midpoint of the ramp. These results appear to give substantial evidence that the visually impaired subjects were able to detect the point where the ramped quadrant joins the street.

2. Usefulness of the exposed aggregate ramp surface. The purpose of this evaluation was to determine whether the exposed pebble aggregate surfacing of the ramped quadrant was of use to the blind as a cue to assist them in negotiating their way through the pedestrian environment. The visually impaired test subjects were observed as they approached the textured area, for indications that they detected the pebble surface, and made use of that information.

All test subjects appeared to be aware of the pebble surface, either having detected it with their canes, or with their feet. A common reaction, upon reaching the pebble surface, was to slow, change direction, and seek the next cue to align with the crosswalk, as described in these narratives:

- The subject reached the textured surface of the quadrant, turned slightly to his right, having nearly stopped, and began searching with his cane for the utility pole at the curb. Finding the pole, he walked to it, found the curb, and waited on the ramped quadrant side of the pole for an opportunity to cross the street.
 - The subject reached the textured surface of the ramped quadrant, turned immediately more than 20 degrees, and sought the point at the curb where the textured surface of the quadrant joined the brushed concrete sidewalk surface.
3. Usefulness of the ramped quadrant. The ramped surface of the quadrant was designed such that it would provide direct access to both crosswalks. It was also possible to use the sideslope to enter, or leave, the crosswalks; however, persons attempting to use the curb would be beyond the limits of the crosswalk. This evaluation was intended to determine how many visually impaired test subjects chose to use the ramped surface, the sideslope, or the curb, to negotiate the level change between the sidewalk and the street. In addition, factors of difficulty were measured for all test subjects, to determine the relative ease, or difficulty, of using the curb, sideslope, or ramped surface. Table 136 shows the results of this evaluation.

Evaluation Results: Wheelchair Group.

1. Location and design of ramped surface. The ramped quadrant was designed so as to provide direct access to both crosswalks. The purpose of this evaluation was to determine whether the ramped surface, and the treatment of that surface, resulted in any difficulty to wheelchair users. The results are shown on Table 137.

TABLE 136. ASCENT AND DESCENT OF RAMPS: BLIND USERS

	Number of Events	Mean Difficulty Factor
Entered Street		
Used Curb	0	---
Used Sideslope	6	2.00
Used Ramp	6	1.00
Left Street		
Used Curb	2	2.00
Used Sideslope	3	2.67
Used Ramp	7	1.14

TABLE 137. DIFFICULTY FACTORS

ASCENT AND DESCENT OF RAMPED SURFACE - WHEELCHAIR USERS

	Wheelchair Users													Mean Difficulty Factor
	1	2	3	4	5	6	7	8	9	10	11	12	13	
	Difficulty Factors													
Enter Street	2	2		2	1	3	3	2	2	2	2	2	2	2.08
Leave Street	2	2		2	3	4	4	3	3	2	2	2	2	2.58

The gentle slope of the ramped surface (3.1 percent) gave no evidence of difficulty to any of the wheelchair users. Several wheelchair users had difficulties with the pebble surfacing of the ramped quadrant. The difficulties were similar to, but not nearly as severe, as those encountered with the double conventional ramp design, which had a similar surface (see special handicapped group evaluations, Countermeasure No. 9).

2. LOCATION NO. 2: CLEARWATER, FLORIDA

The location selected for the construction and evaluation of Countermeasure No. 11 was on a moderately used road, on an intersection located a short block (350 feet, 106.75 m) from the main business street in Clearwater, Florida (see Figure 74). Both intersection quadrants of the T-intersection were selected for construction of the countermeasure design.

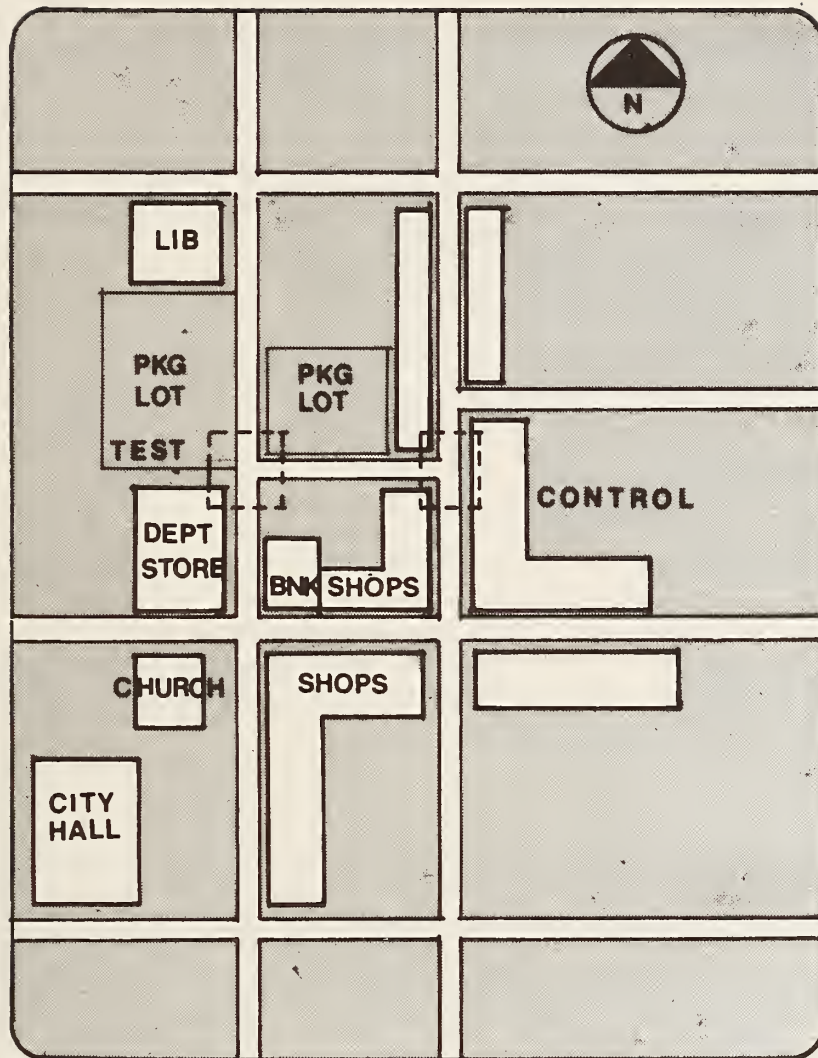


Figure 74 Clearwater: Ramped Full Quadrant: Test and Control Site Locations

The intersection is used by a large number of persons walking between public parking lots and the public library, a major bank, a nine story department store, and other shops. The city has a higher than typical proportion of elderly and handicapped persons in its population; it is in the midst of a part of the country that is attractive as a retirement and recreation resource.

The intersection's sidewalks are very narrow, approximately six feet in width. Consequently, typical curb ramp designs, such as single or double conventional ramps, are not feasible. The ramped full quadrant design, however, is adaptable to very narrow sidewalk locations; hence the selection of these quadrants for its construction and evaluation.

Pedestrian flows through the intersection were generally moderate to light, although there were some periods of relatively active use. The flow rates are described in Table 138.

TABLE 138. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
East-West Direction						
North Crosswalk	10	5	25	0	5	10
South Crosswalk	0	5	25	10	30	15
North-South Direction						
East Crosswalk	0	0	140	30	50	45

Traffic flows, passing through the intersection, are also light to moderate. They are substantially higher on the north-south street; the intersecting street is only one block in length, and receives only local traffic, primarily providing access to two parking lots. The intersection is not signalized.

Control Location

The control location, for this countermeasure (see Figure 75) is located one block to the east of the test intersection (see Figure 76). It is identical in design to the test location, except for a greater concentration of retail and professional office development, immediately adjacent to the intersection. Pedestrian flows through the control intersection were about the same as at the test section, except somewhat lower in the north-south direction. These are shown in Table 139.

TABLE 139. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
East-West Direction						
North Crosswalk	10	10	20	10	5	10
South Crosswalk	5	5	10	25	20	15
North-South Direction						
West Crosswalk	20	20	25	20	10	20

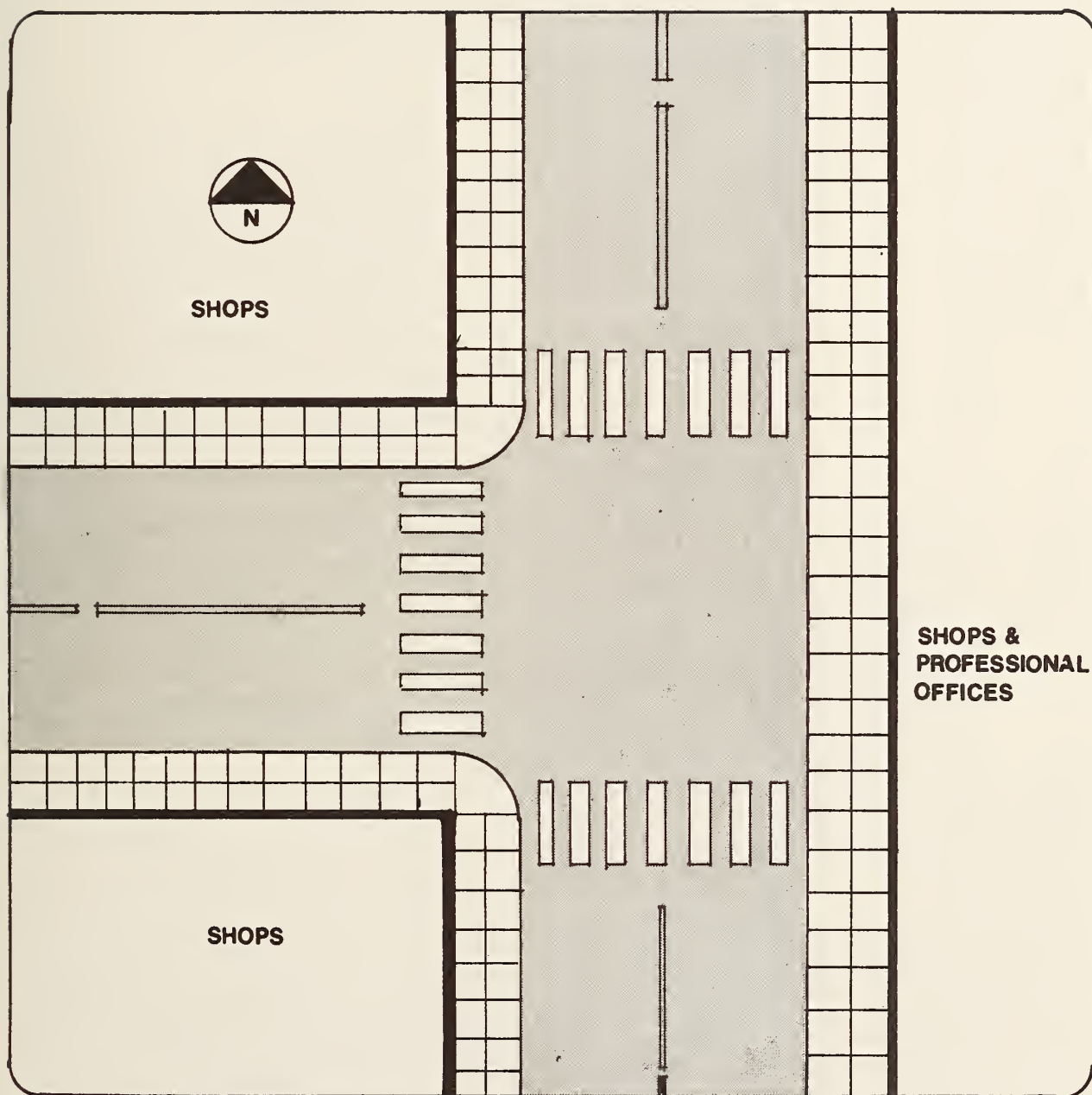


Figure 75 Clearwater: Ramped Full Quadrant: Control Site

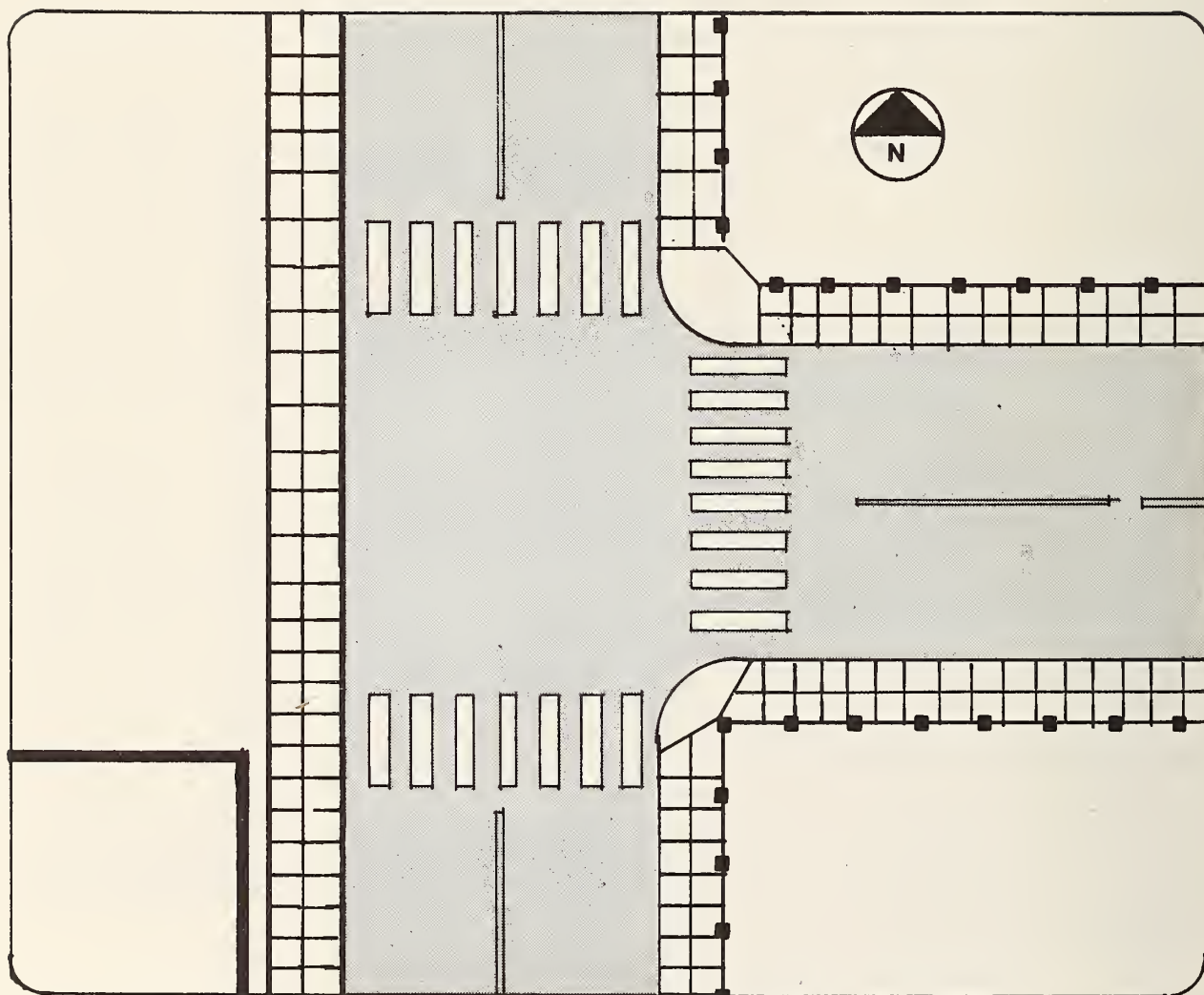


Figure 76 Clearwater: Ramped Full Quadrant: Test Site

Description of Changes Made at Location No. 2.

Figure 77 illustrates the research laboratory design for the ramped full quadrant countermeasure at this intersection. The design features included extending the curb at the intersection quadrant about 18 inches (45.72 cm), to permit a more gently sloped ramp; and chamfering (or slicing at an angle) the curb line, in order to provide reference points (where the straight line segments of the curb join at a point), for the visually impaired to use, to position and orient themselves with the crosswalks. The entire quadrant was to be ramped at a slope of approximately 1:36 (2.8 percent), with sideslopes of 1:12 (8.33 percent) to provide a transition to the level of the existing sidewalk.

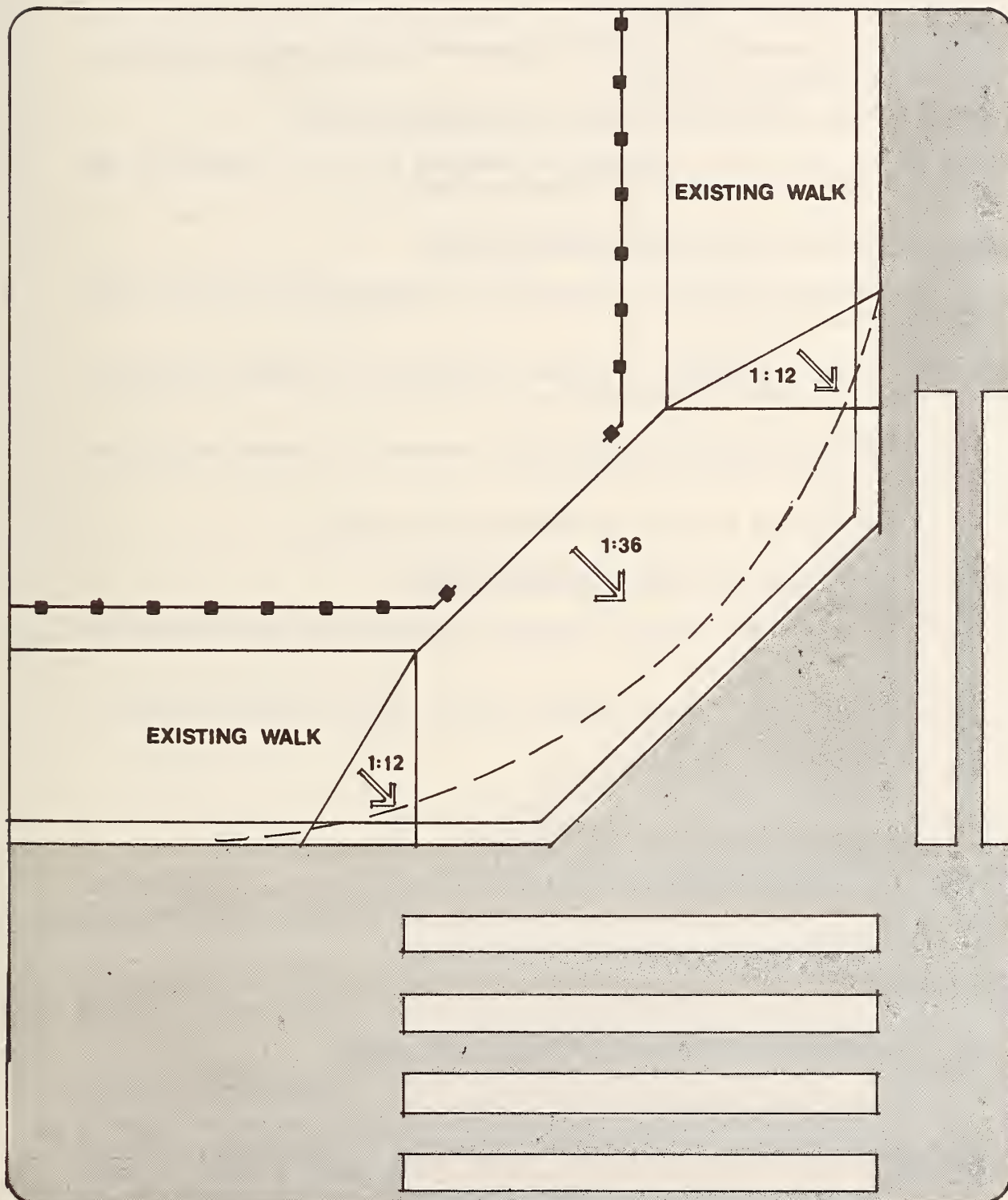


Figure 77 Clearwater: Ramped Full Quadrant: Original Design

Countermeasure Design as Constructed.

The ramps, as constructed (see Figure 78), followed the general design proposals, but without the chamfering of the curbs.

Evaluation Results: Countermeasure No. 11, Test Location No. 2.

Countermeasure No. 11 was evaluated, by observing its use by members of the general population, on the basis of the following criteria:

Criteria determining its effect upon pedestrian safety.

- Conflict between pedestrians and vehicles, for persons entering the street from the countermeasure location.
- Conflict between pedestrians and other pedestrians, for persons entering the street from the countermeasure location.
- Persons walking beyond the limits of the crosswalk, for persons entering the street from the countermeasure location.
- Person waiting in the street for the traffic signal to change.

Criteria determining its effect upon pedestrian mobility.

- Person using the ramp; using the curb—to enter the street and to leave the street.
- Persons changing direction, or veering, to use or avoid the ramp—to enter, or leave the street.
- Difficulty factor in entering the street.
- Difficulty factor in leaving the street.

In addition, the countermeasure was evaluated by special groups of handicapped persons. These evaluations are described following the evaluation descriptions of the general population.

Evaluation Results: Countermeasure No. 11.

1. Effect of Countermeasure Design Upon Pedestrian Flows.

The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians (see Figure 79). Because of the low pedestrian volumes, quite small increases in flow resulted in high percentage changes. Overall pedestrian flows through the two intersections increased by about the same amount, by about 6 percent, so it can be concluded that the countermeasures did not adversely affect the pedestrian flow.

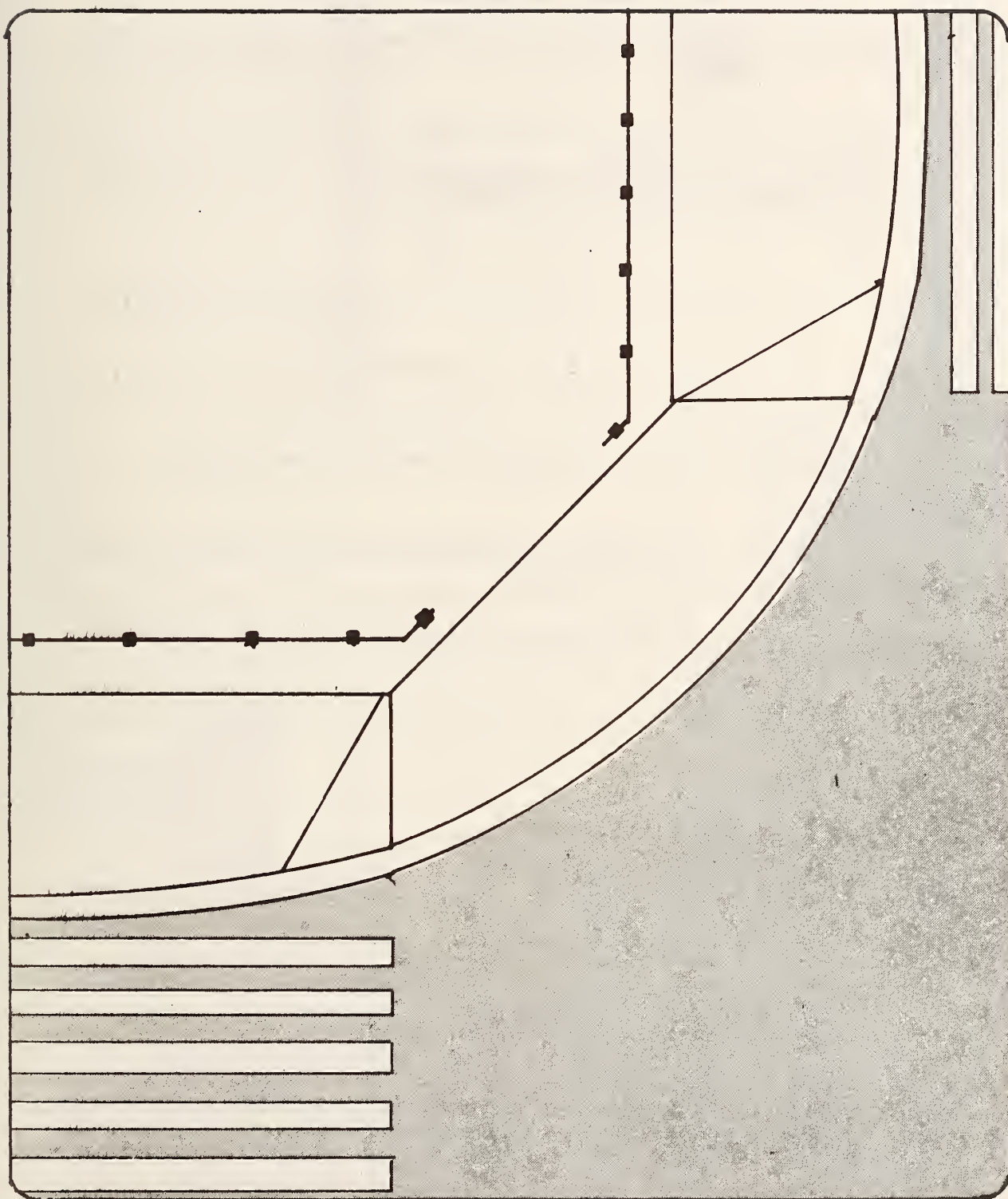


Figure 78 Clearwater: Ramped Full Quadrant: Design as Constructed

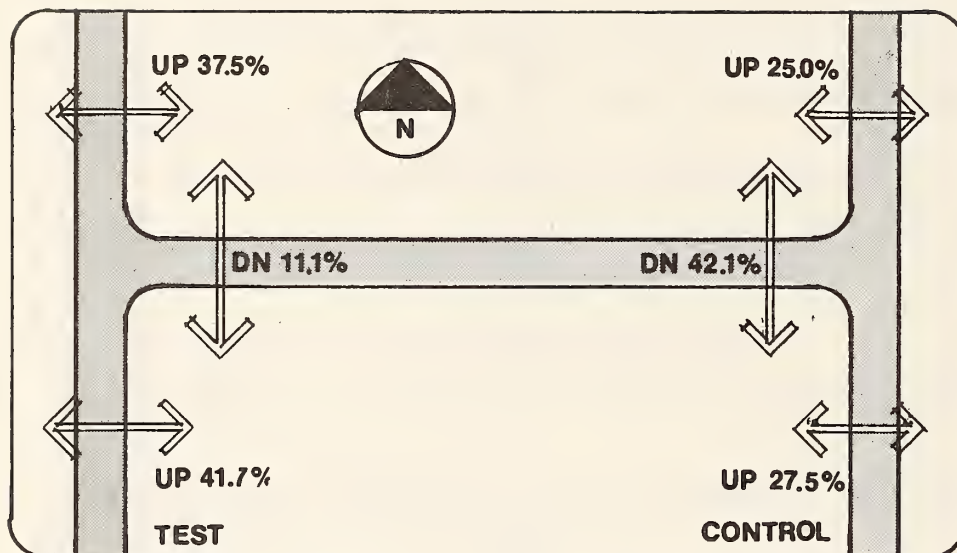


Figure 79 Clearwater: Ramped Full Quadrant: Pedestrian Flow Changes

2. Crossing Time.

Countermeasure No. 11 was evaluated to determine whether it had any effect upon the time required to cross the street

The results of the evaluation are shown in Table 140. There was no significant change in the crossing time.

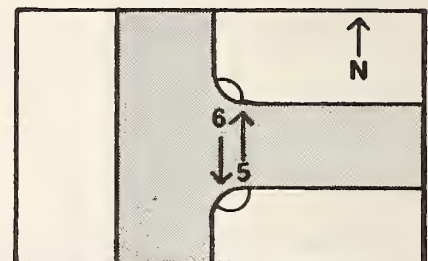


TABLE 140. MEAN CROSSING TIME - SECONDS

COUNTERMEASURE NO. 11; LOCATION NO. 2

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	9.6000	10.2727	7.500	8.4615	N	N
6	9.4783	9.7308	6.5625	8.1200	N	N

S = Significant

N = Not Significant

3. Effect of the Countermeasure Upon the Difficulty in Negotiating the Level Change at the Curb.

Curb ramps were installed as part of the countermeasure design at the test location. This evaluation was intended to determine the usefulness of these curb ramps to the general population, using the test intersection. The evaluation was in two parts:

- . Determining whether the ramps were used;
- . Determining how the apparent difficulty of making the level change was affected by the installation of the ramps.

Table 141 illustrates the results of the evaluation of the use of the ramps.

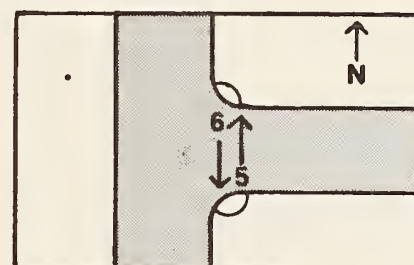


TABLE 141. PEDESTRIAN USE OF CURB RAMP
COUNTERMEASURE NO. 11; LOCATION NO. 2

Direction	Used ramp	Changed direction to use ramp	Changed direction to avoid ramp
5	50/53	4/53	0/53
6	50/52	1/52	0/52

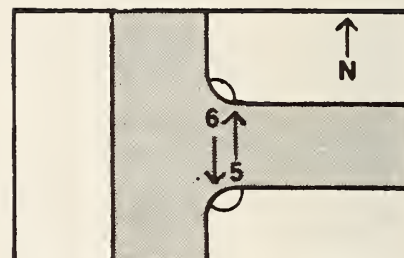
From Table 141, it can be seen that almost all the pedestrians chose to use the ramps; five even, apparently, changed direction to use them; none changed direction to avoid them.

4. Apparent Difficulty of Level Change.

A four point scale was used to record observed difficulty of pedestrians in negotiating the level change at the curb. Separate evaluations were made for entering, and leaving, the street. The scale ranged from a difficulty factor of 1 (equivalent to walking on a level surface, with no perceptible change in walking speed or gait), to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all--including needing to grab onto a pole for support or balance, or nearly falling). An average, able bodied adult, mounting, or descending a typical 8 inch (20.32 cm) curb, would score 2 on the scale.

Tables 142 and 143 present the results of this evaluation.

TABLE 142. MEAN DIFFICULTY FACTOR - NEGOTIATING
LEVEL CHANGE AT THE CURB - ENTERING STREET
COUNTERMEASURE NO. 11; LOCATION NO. 2

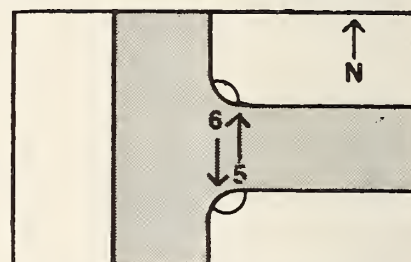


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	2.0400	1.5455	2.2857	2.5385	S	N
6	2.0435	1.3462	2.0000	1.9600	S	N

S = Significant

N = Not Significant

TABLE 143. MEAN DIFFICULTY FACTOR - NEGOTIATING
LEVEL CHANGE AT THE CURB - LEAVING STREET
COUNTERMEASURE NO. 11; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant difference between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	2.0400	1.5455	2.1429	2.0000	S	N
6	2.2174	1.3462	2.0625	2.8400	S	S

S = Significant

N = Not Significant

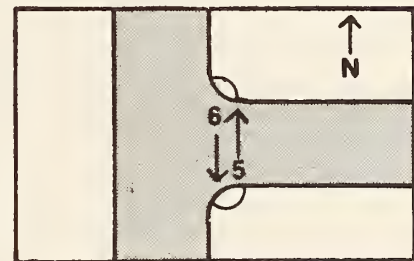
Tables 142 and 143 show that there was a significant decrease in difficulty in making level changes at the curbs of the two curb ramps.

5. Evaluation of the Test and Control Intersections With Respect to Other Factors of Behavior Affecting Safety and Mobility.

- a. Conflicts between crossing pedestrians and vehicles.
- b. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
 - a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in, or near, the intersection.

The results of this evaluation are shown in Table 144.

TABLE 144. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND VEHICLES
COUNTERMEASURE NO. 11; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	4/27	4/25	0/18	1/29	N	N
6	4/25	2/28	2/15	3/15	N	N

S = Significant

N = Not Significant

From Table 144, there was no significant change in the number of conflicts between crossing pedestrians and vehicles.

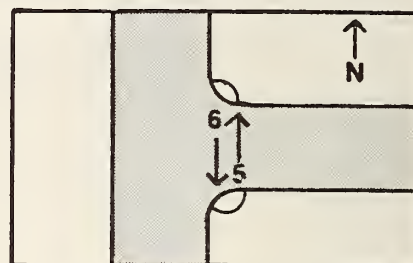
- b. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk, but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 145 and 146.

TABLE 145. PEDESTRIANS WALKING BEYOND THE
LIMITS OF THE CROSSWALK
COUNTERMEASURE NO. 11; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	5/27	0/25	0/15	0/15	N	N
6	3/25	0/28	0/18	0/29	N	N

S = Significant

N = Not Significant

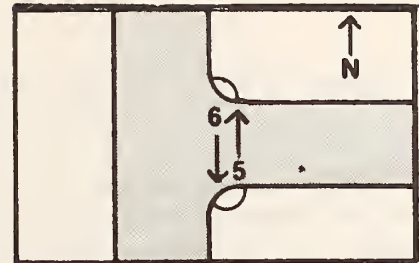
From Table 145, although there was a substantial decrease in people walking beyond the limits of the crosswalk, the changes were not quite significant.

From Table 146, there was a decrease in the number of pedestrians leaving the crosswalk while crossing the street, but this decrease was not significant.

Evaluation of Countermeasure No. 11, Location No. 2 by Special Groups of Handicapped Persons.

This evaluation was carried out, using visually impaired subjects and people using wheelchairs.

TABLE 146. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 11; LOCATION NO. 2



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
5	3/25	0/28	0/15	0/15	N	N
6	1/27	0/25	0/18	0/29	N	N

S = Significant

N = Not Significant

For the visually impaired group, the questions were:

- Is the ramp detectable; can the visually impaired detect the point at which the ramp meets the street.
- Is the ramp location and design useful, or inconvenient.

And for the wheelchair group:

- Is the ramp satisfactorily located and designed. Is it useful, or inconvenient.

Evaluation Results: Visually Impaired.

1. Detectability of the Ramp.

There were 33 crossing events involving visually impaired test subjects; 15 of these were leaving the sidewalk, and 18 were approaching. In all of these events, the subjects appeared to have no difficulty in locating the junction between the street and the ramp. Illustrating the behavior are the following descriptions.

- The subject was a 75 year old male. He approached the ramp from the street, and slowed nearly to a stop, as he detected the bottom of the ramp. He then ascended the ramp and proceeded along the sidewalk.
- The subject, a 50 year old male, crossed to the bottom of the ramp; he almost stopped, as he detected the bottom of the ramp. He then went slowly up the ramp, felt for the site boundary fence, and then set off down the sidewalk.

Evaluation Results: Wheelchair Group.

The gentle slope of the ramped surface seemed to present little difficulty to the wheelchair users, but several subjects found the flares, where the ramped quadrant met the sidewalk, to be much more difficult; these flares were, as discussed earlier, steeper. If the subject was going directly up the curb ramp and straight on down the street, then the flares presented relatively little difficulty. But if the subject had to make a turn on the ramp, then the flares were much more difficult to manage. This is illustrated by the following examples:

- The subject, a 35 year old male, stopped near the bottom of the ramp, aligned himself to approach the ramp diagonally (with his back towards the middle of the intersection). He then took a run at the ramp, turning sharply, halfway up, in order to travel to the right, down the sidewalk.
- The subject, a 50 year old male, approaching the ramped quadrant from along the sidewalk, turned his chair around and backed down the ramp into the street. He then turned his chair around and continued to cross.

Because the ramped quadrant occupied the whole corner, the wheelchair users experienced increased difficulty in travelling around the corner, and this difficulty was increased by the presence of the ramp flares. This is illustrated by the following example:

- The 55 year old female subject approached the corner in her wheelchair. As the chair tipped when it met the cross slope of the side flare, she was forced to exert considerable effort with her left arm, in order to travel around the corner.

Summary of Findings From the Two Locations.

1. The countermeasure did not adversely affect pedestrian flow; pedestrian activity increased at the test sites, while at the control site, activity decreased.
2. There were no significant changes in the crossing time at the test sites.
3. A very few pedestrians chose not to use the curb ramp when crossing the street, but only one person, at the two sites, appeared to deliberately change direction to avoid using the ramps.
4. There was a significant decrease in difficulty in making level changes at the curb in one direction, but not in the other direction, at the one site; and in both directions, at the other site.
5. There was no significant change in the number of conflicts between crossing pedestrians and vehicles.
6. There was no significant change in the number of conflicts between crossing pedestrians.

7. There was no significant change in the number of pedestrians waiting in the street for the crossing signal to change.
8. There was no significant change in the number of pedestrians walking beyond the limits of the crosswalk, or leaving the crosswalk.
9. At the oblique angle crosswalk, both before, and after, the countermeasure installations, many of the pedestrians left the crosswalk to take the most direct route.
10. There was no significant change in the number of pedestrians crossing against the traffic signal.
11. The visually impaired subjects appeared to detect the presence of the ramps in all cases.
12. The ramp, at only one of the locations, was finished with the pebble aggregate. The visually impaired subjects were able to detect the aggregate, and use it for locational information. However, the surface presented additional difficulties to wheelchair users.
13. As the swept quadrant occupies the whole corner, wheelchair users experienced increased difficulty in travelling around the corner.
14. The ramp flares, where the quadrant joins the sidewalk, were steeper than the general slope of the quadrant. At Location No. 2, where the quadrant and the sidewalks are small, these flares greatly increased the ascent difficulties of people in wheelchairs who had to change direction on the ramps.

Conclusions.

The countermeasure is reasonably effective. However, it does increase the difficulty for wheelchair users who must travel over it to turn the corner.

In principle, the flares of the swept quadrant should have the same slope on the rest of the quadrant.

countermeasure 12

add-on ramp

Countermeasure No. 12, Add-on Ramp (see Figures 80 and 81), was constructed and evaluated in the downtown area of Sioux City, Iowa, a midwestern city having a population of about 80,000 persons. The city has a very active downtown; both vehicular and pedestrian traffic volumes are quite high throughout the day.

The countermeasure was located within the downtown shopping district. Buildings at

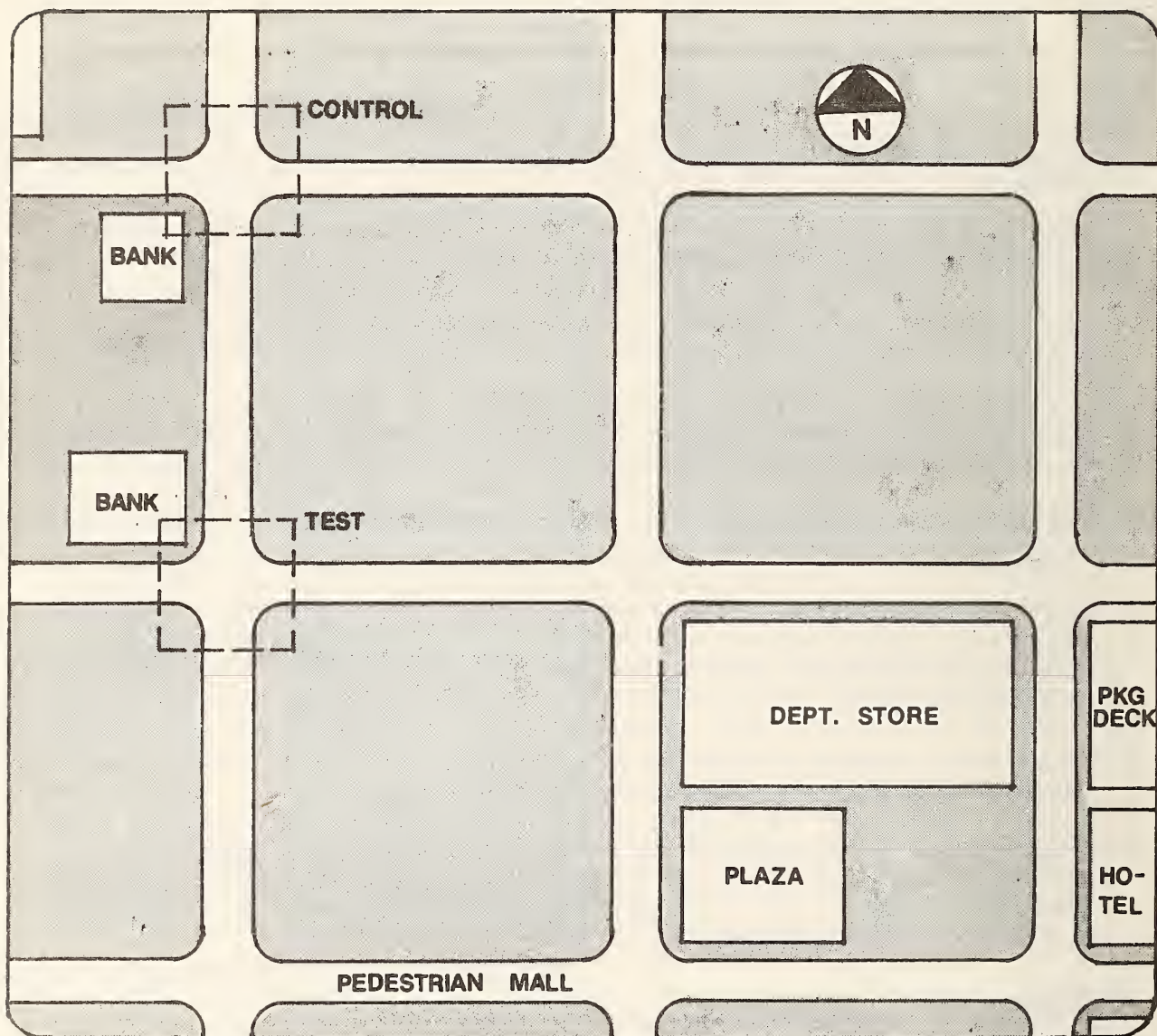


Figure 80 Sioux City: Add-On Ramp: Location Plan of Control and Test Sites

the test intersection included one and two story retail commercial structures, a bank, and a high-rise office building with ground floor retail use. Within one block of the test site are the downtown department store and the city's pedestrian mall. Also nearby, are the City Hall and several municipal parking decks.

The streets which pass through the test intersection are both one-way arterials, carrying three lanes of moving traffic, and providing parallel parking along all curbs, within a nominal 60-foot (18.3 m) width. Traffic volumes are quite heavy on both streets, averaging about 800 vehicles per hour, or 550 vehicles/lane/hour of green time on each street.

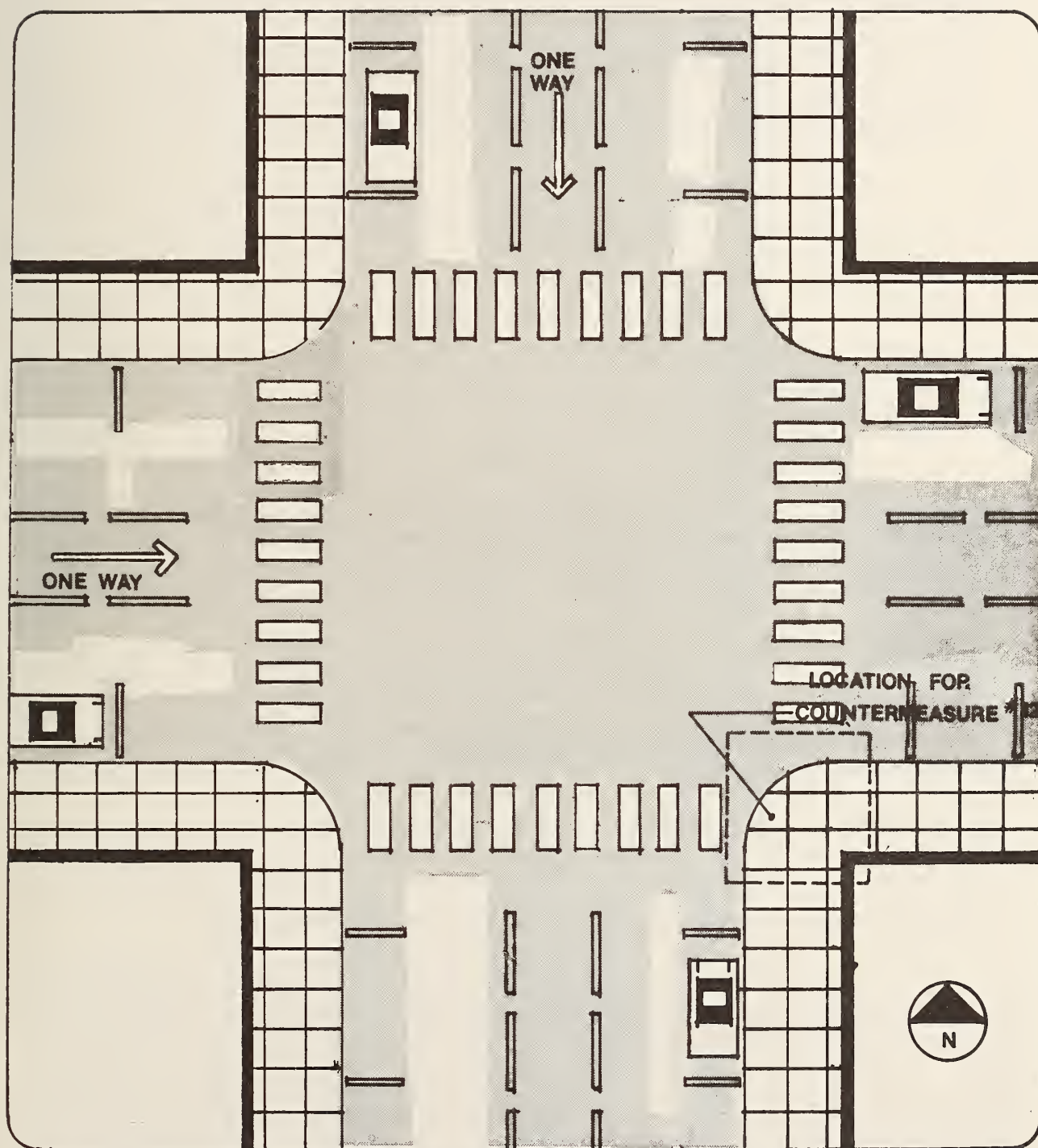


Figure 81 Sioux City: Add-On Ramp: Test Site Before Modification

Pedestrian flows are also quite high, reaching peaks of over 320 persons per hour within the most heavily used crosswalk. These pedestrian volumes are presented in Table 147:

TABLE 147. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	100	90	320	240	185	185
East-West (North Crosswalk)	110	140	125	200	170	150
(South Crosswalk)	50	55	230	205	145	135

The intersection is signalized, with right or left turn arrows as applicable, and pedestrian 'walk/don't walk' indicators. Signal timing is set to allow a short period for turning traffic at the beginning of the cycle before the pedestrian 'walk' period begins. The signal cycle is as follows:

	elapsed time
green, green 'turn' arrow	0 seconds
turn arrow off, 'walk' begins	10
flashing 'don't walk'	19
yellow, continuous 'don't walk'	30
red	33
green	66

note: signal time is equally distributed between the two directions.

The pedestrian crossing signal allowed 19 seconds of "walk" for crossing the 60 foot (18.3 m) wide street in those crosswalks which are not affected by turning traffic (the west and south crosswalks). However, because of the 10 second "turn" arrow cycle affecting the other two crosswalks (the north and east), only 9 seconds of "walk" was allowed for those crossings, requiring a walking speed of 6.7 feet (2.04 m) per second to complete crossing within the "walk" allotted time. This "walk" interval does not include the clearance time allowed for pedestrians to complete their crossing during the flashing "don't walk" phase.

Control Location for Countermeasure No. 12.

The control intersection (see Figure 82) was located one block to the north of the test site. Its design and use were essentially the same as at the test location before construction of the countermeasure. Unfortunately, the effectiveness of the location as a control site was substantially diminished when a curb ramp was constructed at the intersection before the final group of observations had been completed. Pedestrian volumes at the control intersection averaged from 10 to 30 percent lower than at the test location. These pedestrian flows are summarized in Table 148.

TABLE 148. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
Direction:						
North-South	100	120	205	185	205	165
East-West (North Crosswalk)	70	55	115	120	180	110
(South Crosswalk)	50	80	125	135	70	90

Description of Add-On Ramp Countermeasure Design.

The add-on ramp design prepared by the research laboratory is similar in design to that widely used in Denver, Colorado. It has a 1:12 slope (8.33 percent) extending from the limit of the existing sidewalk to the level of the street, and extending in width around the entire intersection quadrant, so as to provide access to both crosswalks. A transition slope was used to connect back to the existing curb line at either end of the ramp. A gently sloped gutter was also designed to aid drainage.

The ramp, as constructed, was identical to the research laboratory design, with the single exception of the addition of a painted yellow line, six inches in width, at the top of the ramp to indicate the location of the former curb. Figure 83 illustrates the location of the countermeasure design.

Evaluation Results: Countermeasure No. 12.

Countermeasure No. 12 was evaluated by observing its use by members of the general population, on the basis of the following criteria:

Criteria determining its effect upon pedestrian safety:

- Conflict between pedestrians and vehicles, for persons entering the street from the countermeasure location.

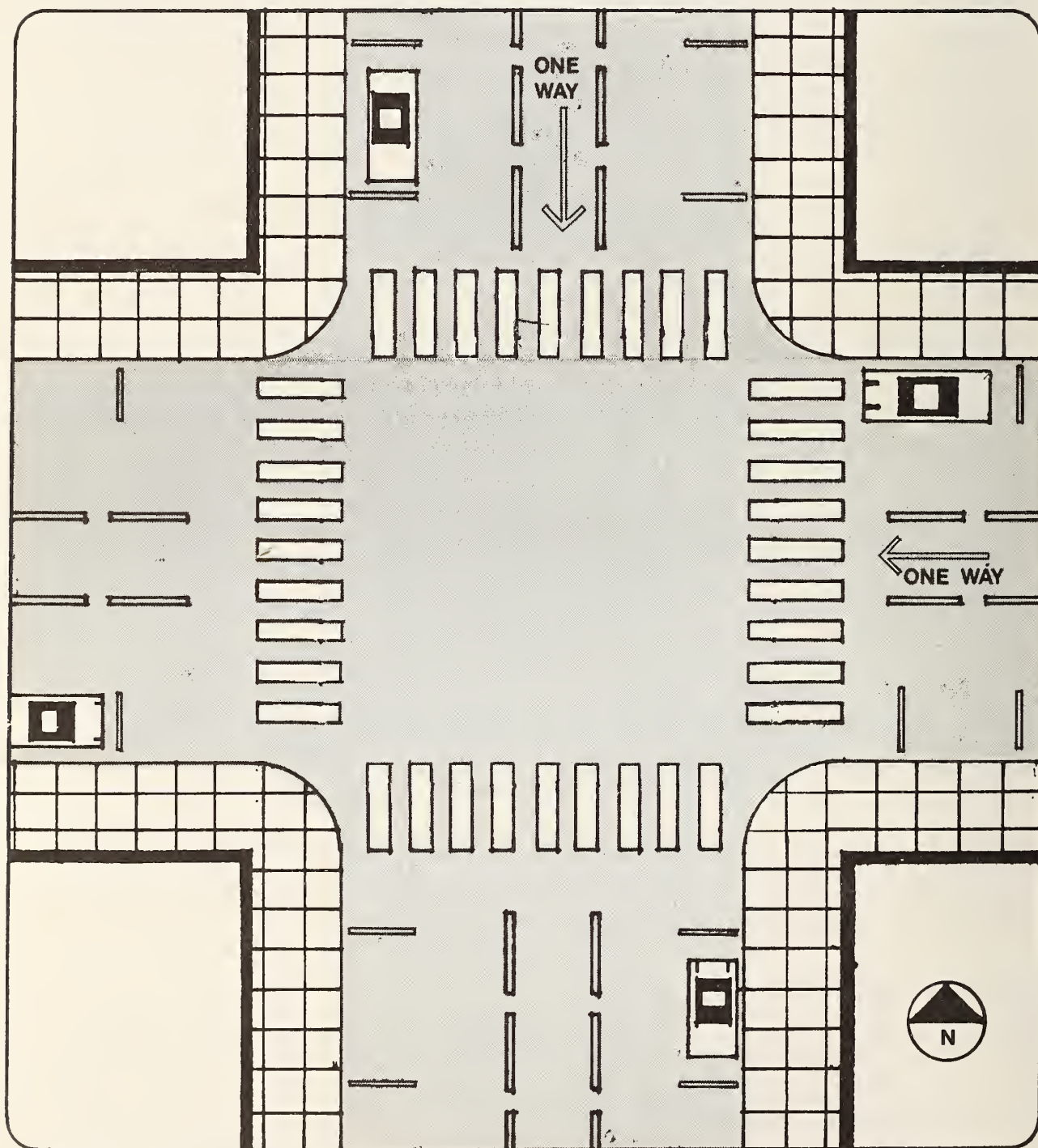


Figure 82 Sioux City: Add-On Ramp: Control Site

- Conflict between pedestrians and other pedestrians, for persons entering the street from the countermeasure location.
- Persons walking beyond the limits of the crosswalk, for persons entering the street from the countermeasure location.
- Persons waiting in the street for the traffic signal to change.

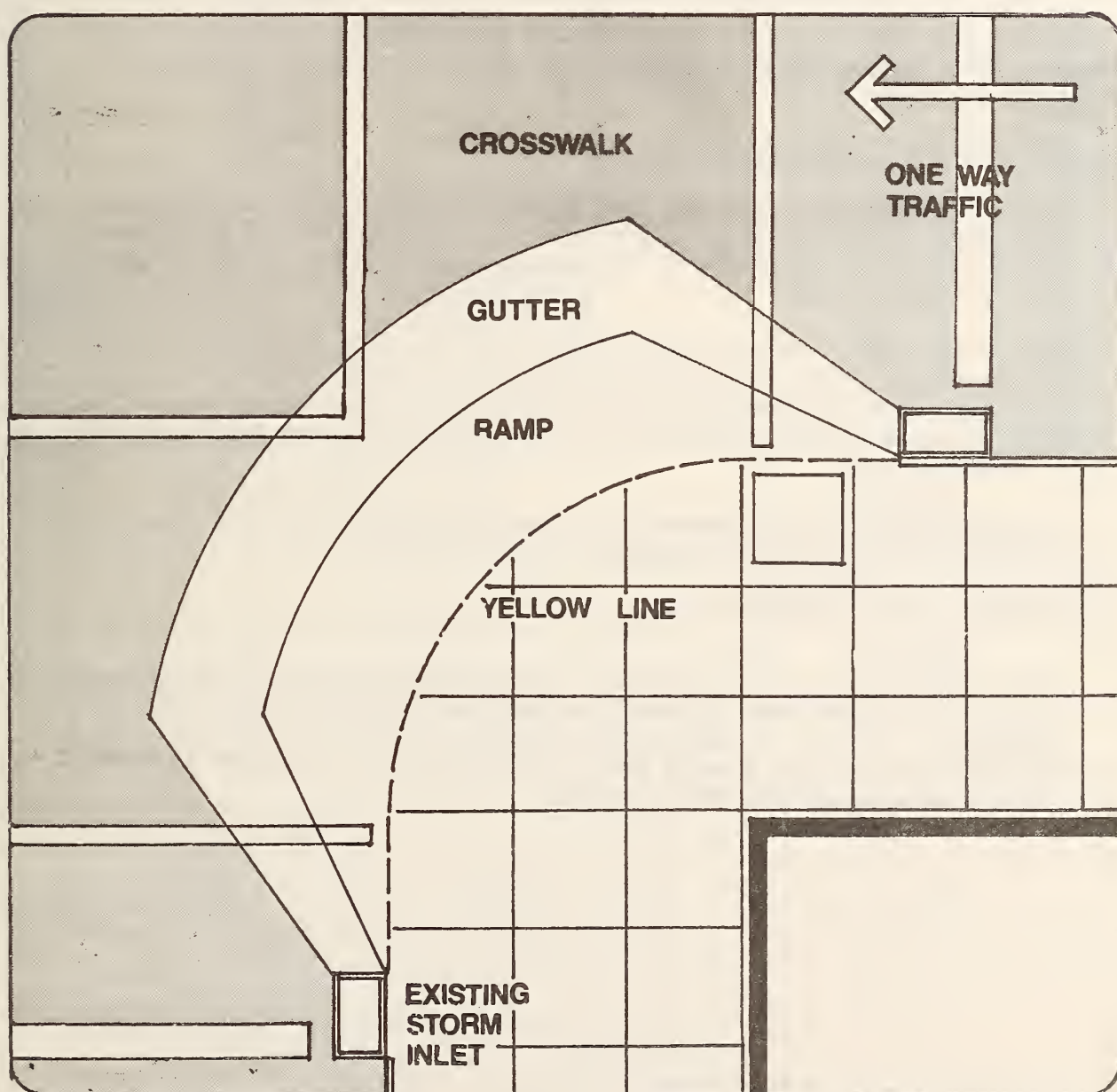


Figure 83 Sioux City: Add-On Ramp: Detailed Layout

Criteria determining its effect upon pedestrian mobility:

- Persons using the ramp; using the curb, to enter the street and to leave the street.
- Persons changing direction, or veering, to use or avoid the ramp, to enter or leave the street.
- Difficulty factor in entering the street.
- Difficulty factor in leaving the street.

In addition, the countermeasure was evaluated by special groups of handicapped persons. These evaluations are described following the evaluation descriptions of the general population.

Evaluation Results: Countermeasure No. 12.

1. Effect of countermeasure design upon pedestrian flows. The countermeasure was evaluated with respect to its possible effect upon the flow of pedestrians. The intent was to determine if the countermeasure design would have any effect on pedestrian use.

From Figure 84, it can be seen that at the test site pedestrian volumes decreased slightly after the installation of the ramp. At the control site there was a much greater decrease in the one direction and a small increase in the other. It appears that pedestrian traffic flow was not affected significantly by the countermeasure.

2. Crossing Time. Countermeasure No. 12 was evaluated to determine whether it had any effect upon the time required to cross the street.

The results of the evaluation are shown in Table 149.

From Table 149, it can be seen that, at the test site, there was a significant decrease in crossing time of pedestrians approaching the countermeasure from direction 5, but not in the reverse direction. However, there was also a significant decrease in crossing time for direction 6 at the control site, and a general reduction for the other directions.

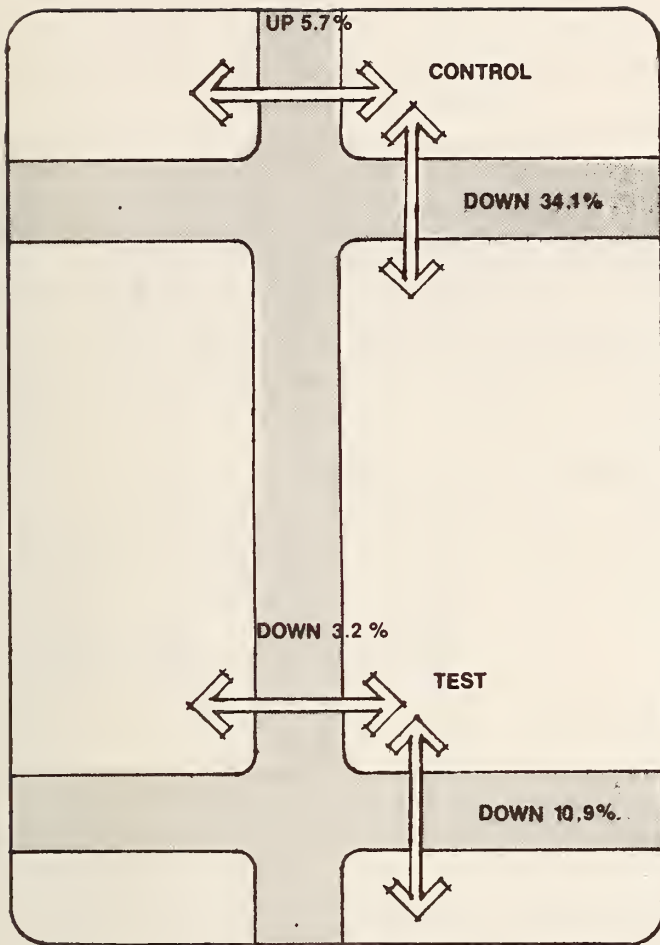


Figure 84 Sioux City: Add-On Ramp:
Changes in Pedestrian Flow

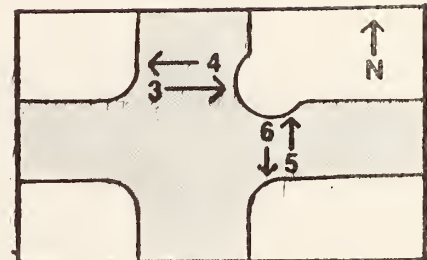


TABLE 149. PEDESTRIAN CROSSING TIME IN SECONDS
COUNTERMEASURE NO. 12; LOCATION NO. 1

Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
4	11.0909	11.7143	12.0000	10.9375	N	N
6	14.0556	12.950	14.3333	10.4091	N	S
3	13.0000	11.3889	11.3889	11.2667	N	N
5	13.9375	10.6875	13.4375	11.0000	S	N

3. Effect of the countermeasure upon the difficulty in negotiating the level change at the curb. This evaluation was intended to determine the usefulness of the curb ramp to the general population, using the test intersection. The evaluation was in two parts: one, determining whether the ramp was used; two, determining how the apparent difficulty of making the level change was affected by the installation of the ramp.

Use of ramps. Table 150 illustrates the results of the evaluations of the use of the ramps.

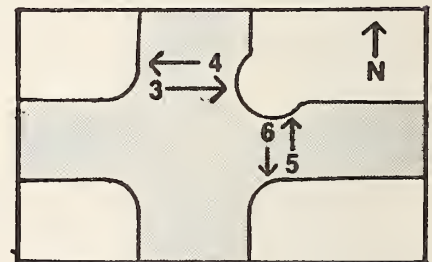


TABLE 150. PEDESTRIAN USE OF CURB RAMP

COUNTERMEASURE NO. 12; LOCATION NO. 1

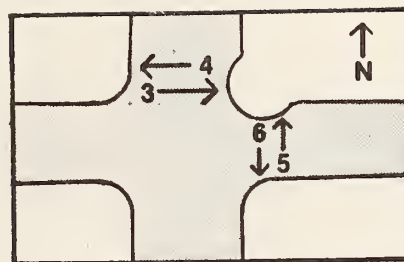
Direction	Used Ramp	Changed direction to use ramp	Changed direction to avoid ramp
4	16 / 17	1 / 17	0 / 17
6	22 / 22	0 / 22	0 / 22
3	18 / 18	1 / 18	0 / 18
5	17 / 17	0 / 17	0 / 17

From Table 150, it can be seen that all but one pedestrian used the ramp for entering or leaving the crosswalk. No pedestrian obviously avoided the ramp; and one pedestrian appeared to change direction deliberately, in order to use the ramp.

4. Apparent difficulty of level change. A four point scale was used to record observed difficulty of pedestrians in negotiating the level change at the curb. Separate evaluations were made for entering and leaving the street. The scale ranged from a difficulty factor of 1 (equivalent to walking on a level surface with no perceptible change in walking speed or gait), to a difficulty factor of 4 (could negotiate only with the greatest of difficulty, if at all—including needing to grab onto a pole for support or balance, or nearly falling). An average able bodied adult, mounting or descending a typical 8-inch (20.32 cm) curb, would score 2 on the scale.

The results of this evaluation can be seen in Table 151.

TABLE 151. MEAN DIFFICULTY FACTOR
NEGOTIATING LEVEL CHANGE AT THE CURB
COUNTERMEASURE NO. 12; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col.1 Before	Col.2 After	Col.3 Before	Col.4 After	Cols. 1&2	Cols. 3&4
4	2.1818	1.2143	1.5455	1.1825	S	N
6	2.2222	1.1000	1.6667	1.3182	S	N
3	2.0556	1.0556	1.7857	1.2000	S	S
5	2.3750	1.3125	1.8125	1.3333	S	N

S = Significant

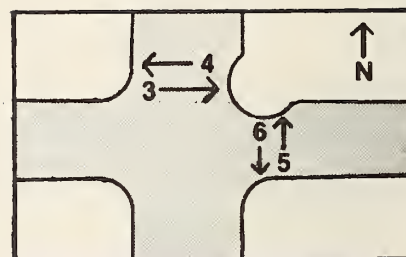
N = Not Significant

As can be seen from Table 151, there was a significant decrease in difficulty in negotiating the curb after the ramp was installed. A decrease is also evident at the control site because a curb ramp (of a different pattern) was unexpectedly built at the control site.

5. Evaluation of the test and control intersections with respect to other factors of behavior affecting safety and mobility. These included evaluations of:
 - a. Conflicts between crossing pedestrians and vehicles.
 - b. Conflicts between crossing pedestrians and other pedestrians.
 - c. Pedestrians waiting in the street for the crossing signal to change.
 - d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
 - e. Pedestrians crossing against the traffic signal.
- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation were as shown in Table 152.

TABLE 152. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND VEHICLES
COUNTERMEASURE NO. 12; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	1 / 11	1 / 19	0 / 12	1 / 16	N	N
6	2 / 18	2 / 22	0 / 15	1 / 23	N	N
3	0 / 18	1 / 18	0 / 14	0 / 15	N	N
5	2 / 17	0 / 17	0 / 17	0 / 15	N	N

S = Significant

N = Not Significant

From Table 152, it can be seen that there was no significant change in the number of conflicts between crossing pedestrians and vehicles.

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test, when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

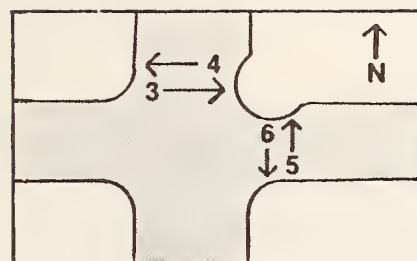
The results of this evaluation are shown on Table 153.

From Table 153, it can be seen that there was no significant change in the number of conflicts between crossing pedestrians and other pedestrians.

- c. Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing or decreasing the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, and therefore vulnerable to traffic.

The results of this evaluation are shown on Table 154.

TABLE 153. CONFLICTS BETWEEN CROSSING
PEDESTRIANS AND OTHER PEDESTRIANS
COUNTERMEASURE NO. 12; LOCATION NO. 1

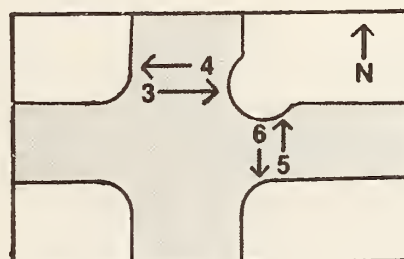


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	0 / 111	0 / 19	0 / 12	0 / 16	N	N
6	0 / 18	0 / 22	4 / 15	1 / 23	N	N
3	3 / 18	1 / 18	1 / 14	0 / 15	N	N
5	0 / 17	0 / 17	2 / 17	1 / 15	N	N

S = Significant

N = Not Significant

TABLE 154. PEDESTRIANS WAITING IN THE
STREET FOR CROSSING SIGNAL TO CHANGE
COUNTERMEASURE NO. 12; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	0 / 11	4 / 19	0 / 12	6 / 16	N	S
6	0 / 18	0 / 22	0 / 15	6 / 23	N	N

S = Significant

N = Not Significant

From Table 154, it can be seen that there was no significant change in the number of pedestrians waiting in the street at the test site for the signal to change. At the control site, the numbers increased significantly for direction 4 and substantially for direction 6.

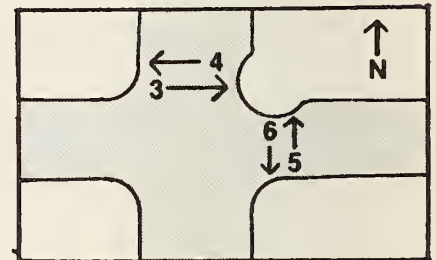
- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.

The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 155 and 156.

TABLE 155. PEDESTRIANS WALKING
BEYOND THE LIMITS OF CROSSWALK
COUNTERMEASURE NO. 12; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	1 / 11	1 / 19	0 / 12	0 / 16	N	N
6	2 / 18	0 / 22	0 / 15	2 / 23	N	N
3	0 / 18	0 / 18	0 / 14	0 / 15	N	N
5	1 / 17	0 / 17	2 / 17	0 / 15	N	N

S = Significant

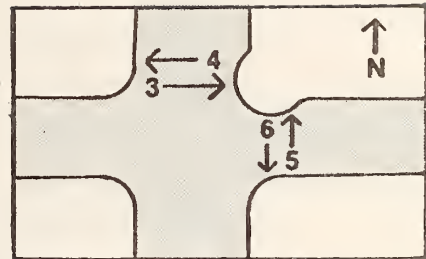
N = Not Significant

From Tables 155 and 156, it can be seen that there was no significant change in the number of pedestrians walking beyond the limits of the crosswalk or leaving the crosswalk while crossing the street at the test site.

- e. Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing or decreasing such behavior.

The results of this evaluation are shown in Table 157.

TABLE 156. PEDESTRIANS LEAVING THE
CROSSWALK WHILE CROSSING THE STREET
COUNTERMEASURE NO. 12; LOCATION NO. 1

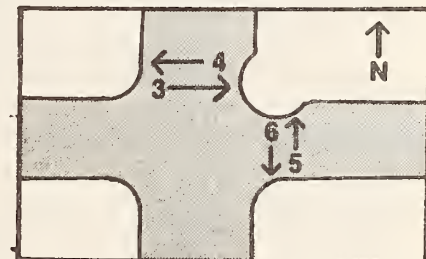


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	0 / 11	3 / 19	0 / 12	1 / 16	N	N
6	0 / 18	1 / 22	0 / 15	0 / 23	N	N
3	0 / 18	0 / 18	0 / 10	2 / 15	N	N
5	0 / 17	1 / 17	1 / 17	9 / 15	N	S

S = Significant

N = Not Significant

TABLE 157. PEDESTRIANS CROSSING
AGAINST WALKING SIGNAL
COUNTERMEASURE NO. 12; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
4	1 / 11	3 / 19	1 / 12	1 / 16	N	N
6	1 / 18	0 / 22	0 / 15	0 / 23	N	N
3	3 / 18	1 / 18	3 / 14	0 / 15	N	N
5	3 / 17	2 / 17	3 / 17	1 / 15	N	N

S = Significant

N = Not Significant

From Table 157, it can be seen that there was no significant change in the number of pedestrians crossing against the traffic signal.

Evaluation of Countermeasure No. 12 by Special Groups of Handicapped Persons.

Two groups of handicapped persons were considered to be sufficiently important to the evaluation of this countermeasure to warrant the use of special evaluation procedures. These were the visually impaired, and persons using wheelchairs. Approximately 15 persons from each group were brought to the site. They had been asked to familiarize themselves with the test locale. The test subjects used the test locations individually, following prescribed routes, while observations were made of their use of design features incorporated into those locations, and any apparent benefits or difficulties resulting from those design features. Time lapse photography was used to record the results of these evaluations. The results sought from this process were the same as sought in the observation of pedestrian behavior evaluations: the impact of the countermeasure design on pedestrian safety and on mobility. In addition, several specific questions relating to the needs and difficulties of these subgroups were considered. These included the following:

Visually Impaired Group.

- Is the ramp detectable; can the visually impaired detect the point at which the ramp meets the sidewalk and the street.
- Is the ramp location and design useful, or inconvenient.

Wheelchair Group.

- Is the ramp satisfactorily located and designed. Is it useful, or inconvenient.

Evaluation Results: Visually Impaired.

1. Detectability of ramp. Since this ramp design was an addition, or extension from the sidewalk into the street, it was important that the visually impaired be able to detect the beginning of the ramp slope so as to stand on the sidewalk, at the top of the ramp, while waiting for traffic conditions to permit them to cross the street. The ramp was constructed of concrete, and given a brush finish identical to the sidewalk finish. A 6 inch (15.24 cm) wide yellow line was painted along the top of the ramp along the line where the curb had formerly existed. Ten visually impaired subjects used the ramp, each on two occasions, to enter and cross the street. Observations were made of any change in gait which might indicate recognition of the ramp slope; however, of primary concern was where the subjects paused or stopped, to await safe crossing conditions. Table 158 summarizes this data:

TABLE 158. STOPPING LOCATIONS

BLIND SUBJECTS - ENTERING STREET

First Crossing	1	2	3	4	5	6	7	8	9	10
Before Ramp	X									
Top of Ramp		X		X	X	X	X	X		X
Bottom of Ramp			X						X	
In Street										
Second Crossing										
Before Ramp		X								
Top of Ramp	X		X	X	X	X	X	X	X	X
Bottom of Ramp										
In Street										

2. Is the ramp location and design useful, or inconvenient, for the visually impaired. Since this add-on ramp design sweeps around the intersection quadrant, to encompass the approach to both crosswalks, all persons wishing to cross the street are obliged to use the ramp. Since the visually impaired may find a change in slope to be hazardous, observations were made to determine whether the ramp caused difficulties in balance or otherwise affected mobility.

All subjects appeared to find very little or no difficulty with the ramp slope. Most ascended or descended with no apparent change in gait. Others changed their stride or reduced their walking speed briefly, then resumed their normal walking pace. The following are illustrative examples:

In 18 of the 20 crossing events, the visually impaired subjects stopped and waited—either before reaching the ramp (2 occasions), or, at the top of the ramp (16 occasions), remaining outside of the former curb line designating the beginning of the street. Of the two subjects who stood at the bottom of the ramp, in a location potentially vulnerable to traffic, one had apparently detected the ramp with his cane, before descending it; but chose to stand at the bottom of the ramp anyway. The other appeared not to have detected the ramp until his cane felt the level change at the street. The following examples are illustrative of these results:

- The subject, a 50 year old man, whose sight was limited to light perception, approached the intersection. His cane touched the ramp and he came to a nearly complete stop. He used his shoe to feel out the ramp slope before standing with his feet barely touching the yellow line at the top of the ramp.

- The subject, an 83 year old woman, able only to perceive light, walked very quickly towards the intersection. As her cane detected the ramp slope, she stopped immediately, about 12 inches (30.48 cm) short of the beginning of the ramp slope. After a one-second pause, she stepped forward, so that the fronts of her shoes rested on the beginning of the slope. She waited there, until ready to cross.
- An 81 year old male subject, able only to perceive light, was asked to approach the intersection, then turn, so as to cross the street to the left. He walked forward, until his cane detected the ramp slope. He then stopped, and began to step sideways, inching his way around the ramp to the left. The tip of his cane reached as far as the bottom of the ramp at the street, but the subject's feet never stepped beyond the beginning of the ramp slope at the top. The subject finally positioned himself at the left side of the ramp, and waited for the signal to change.
- A 29 year old female subject, with minimal light perception, slowed to half her normal walking speed, upon detecting the ramp slope. She continued forward, however, and began to descend the ramp, stopping at the bottom, at a point still on the ramp, but equal to about 4 feet (1.22 m) within the street space.
- A 27 year old subject slowed slightly, and shortened his steps as he climbed the ramp. This appeared to be an adjustment to the ramp slope rather than indicative of any difficulty.
- As an 81 year old male subject approached the ramp from the street, and his cane first detected the ramp slope, he took two very short, quick steps; then, as he felt out the ramp with his cane, he proceeded up the ramp at his normal speed.
- An 83 year old female subject slowed only slightly as her cane detected the ramp. She then picked up speed while ascending, but nearly stopped as she reached the sidewalk at the top of the ramp. She then resumed her normal walking speed.

Wheelchair Group.

3. Is the ramp satisfactorily located and designed. The ramp was designed to provide direct access to both crosswalks. This evaluation was for the purpose of determining whether the slope of the ramp caused any difficulty to wheelchair users, (see Table 159). The following are the results:

Nine of the eleven wheelchair users had no difficulty negotiating the ramp slope in either direction. Two subjects experienced substantial difficulty climbing the ramp, which had a 1:12 slope and was about 6 feet (1.83 m) long. The following examples describe the results:

TABLE 159. DIFFICULTY FACTOR - ASCENT AND DESCENT OF RAMP

	Wheelchair Users										
	1	2	3	4	5	6	7	8	9	10	11
	Difficulty Factor										
Ramp Descent	1	1	1	1	1	1	1	1	1	1	1
Ramp Ascent	1	1	1	1	3	1	1	4	1	1	1

- A 24 year old female subject, entering the street, stopped at the top of the ramp with her foot rest over the yellow line; then, on the green signal, let her chair onto the ramp slope, picked up speed, then crossed the street.
- A 42 year old male subject continually gained speed as he crossed the street. Upon reaching the ramp, he climbed it easily without slowing perceptibly.
- A 28 year old male subject, on beginning to ascend the ramp, found the slope difficult and slowed nearly to a stop near the top of the ramp.
- A 56 year old male subject came to a complete stop at the bottom of the ramp, then several additional times, as he struggled up the ramp in a series of 'pushes', holding the chair from rolling backwards between each push. The ascent of the ramp required 9 seconds of time.

Summary of Findings: Countermeasure No. 12, The Single Add-On Ramp

1. The countermeasure did not affect pedestrian flow significantly.
2. There was a significant decrease in crossing time of pedestrians approaching the countermeasure, from the street, but not in the other direction.
3. All but one pedestrian used the ramp. No pedestrians seemed to try to avoid using the ramp, and one pedestrian appeared to change direction deliberately to use the ramp.
4. There was a significant decrease in difficulty in negotiating the curb after the ramp was installed.
5. There was no significant change in the number of conflicts between crossing pedestrians and vehicles.
6. There was no significant change in the number of conflicts between crossing pedestrians and other pedestrians.
7. There was no significant change in the number of pedestrians waiting in the street for the signal to change.

8. There was no significant change in the number of pedestrians walking beyond the limits of the crosswalk, or leaving the crosswalk while crossing the street.
9. There was no significant change in the number of pedestrians crossing against the traffic signal.

It can therefore be concluded that the countermeasure did not generate any hazardous behavior, and in fact, significantly reduced pedestrian difficulty in changing the level at the curb.

One of the severely visually impaired subjects had difficulty in locating the curb. And he and another chose to wait on, but at the bottom, of the ramp before crossing. It is not clear whether the subjects knew that this behavior is potentially hazardous where the ramp projects into the street as it does in this design. Two of the eleven wheelchair users had considerable difficulty in ascending the ramp, but they were ultimately successful.

Conclusions

The countermeasure enabled both able bodied and wheelchair users to cross the streets with much less difficulty. However, the fact that two visually impaired people decided that it was safe to stand on the ramp at the bottom, suggests that in those locations where this type of ramp is not usually provided, it may be hazardous to introduce it, particularly as it was observed that vehicles quite frequently "violated" the ramp.

countermeasure 15

right-turn-on-red restriction

Recent revisions of traffic regulations for urban areas, permitting vehicles to make right turns on red, have posed new problems for all pedestrians, and in particular, the elderly and the handicapped. In particular, this is a problem at those intersections where the driver's visibility of on-coming traffic is obstructed as shown in Figure 85. In this situation, vehicles creep into the crosswalk, partially or completely obstructing the flow of pedestrians who have a green crossing signal, in order to be able to see whether on-coming traffic exists, or whether it is safe to proceed with the right turn.

Countermeasure No. 15, Right Turn Restriction, is intended for those locations in which the driver's visibility is obstructed by environmental objects such as trees, shrubbery and other low planting, walls and fences, or buildings located close to the sidewalk. The intent of Countermeasure No. 15 is to prohibit right turns on red in such locations.

Countermeasure No. 15 was evaluated based on its performance in solving the following problem:

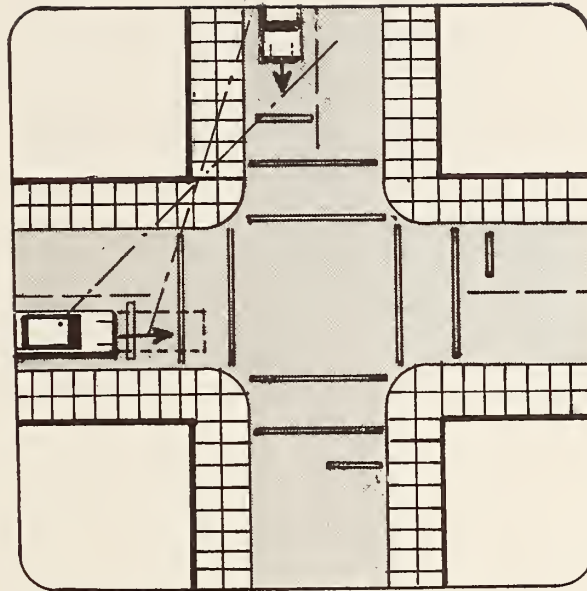


Figure 85 Driver's View Obstructed

- The inability of drivers to see on-coming traffic, causing them to intrude into the crosswalk, obstructing pedestrian movement.

In addition, evaluations were made of the general use of the locations where this countermeasure was installed, and of pedestrian behavior which might indicate problems affecting safety or mobility.

Evaluation Location.

Countermeasure 15 was evaluated at an intersection located on the main street, $1\frac{1}{2}$ miles (2.4135 km) north of, and leading to the center of Atlanta, Georgia (see Figures 86 and 87). The intersection is in the midst of a small outlying retail district, and immediately adjacent to a continuous frontage of 8-10 story office buildings and residential structures.

The main, north-south street is 60 feet (18.3 m) in width and provides for four lanes of moving traffic (two in each direction), plus curb parking. The intersecting street is 33 feet (10.065 m) in width, providing for one lane of moving traffic in each direction, plus parking along one curb.

At the test location, the countermeasure was proposed for the west crosswalk; the east crosswalk was to serve as the control location.

Traffic volumes were very heavy on the main north-south street, remaining relatively constant throughout the day at near 1200 vehicles per hour (or 300 vehicles per lane per hour). Volumes on the intersecting street were much less heavy, averaging 280 vehicles per hour (or 140 vehicles per lane per hour. These are summarized in Table 160.

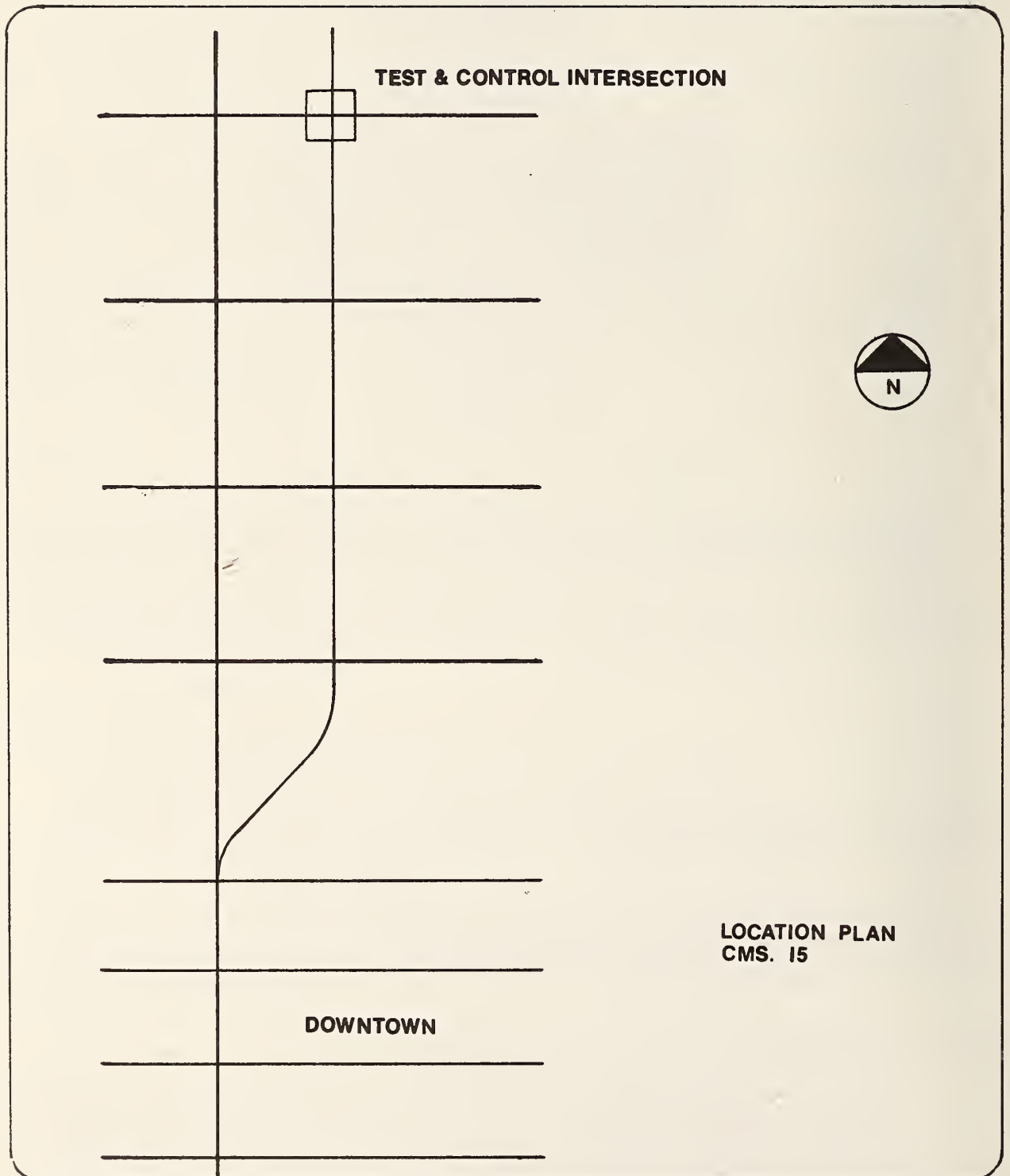


Figure 86 Atlanta: Right-Turn-On-Red Restriction: Test and Control Site Locations

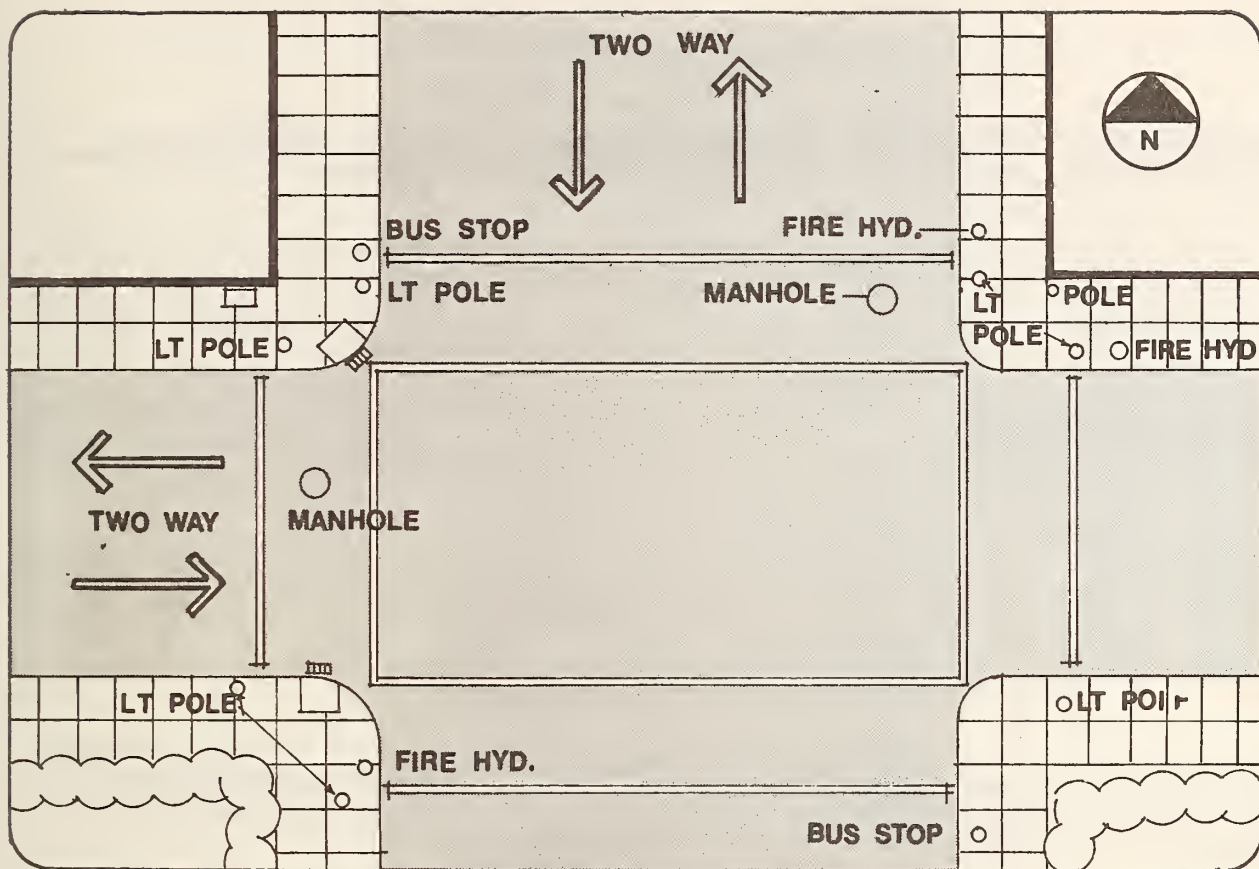


Figure 87 Atlanta: Right-Turn-On-Red Restriction:
Plan of Test and Control Sites Before Modification

TABLE 160. TRAFFIC VOLUMES - VEHICLES PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
North-South Direction						
Thru	1140	1200	1235	1225	1195	1200
All Turns	115	125	150	165	155	140
Total	1255	1325	1385	1390	1350	1340
East-West Direction						
Thru	160	140	160	180	165	160
All Turns	145	95	140	95	125	120
Total	305	235	300	275	290	280

The intersection was signalized, and was equipped with pedestrian crossing signals for the north and south crosswalks only. Signal timing was as follows:

	North-South Street	East-West Street
0:00	Green	Red
1:46	Yellow	Red
1:51	Red	Green, "Walk"
2:01	Red	"Don't Walk"
2:16	Red	Yellow
2:21	Green	Red

From Table 161, we see that pedestrian flows in the west and east crosswalks were light to moderate throughout the day, reaching a substantially higher peak during the noon hour.

TABLE 161. PEDESTRIAN FLOWS - PERSONS PER HOUR

	Morning Rush	Mid Morning	Noon	Mid Afternoon	Afternoon Rush	Avg.
West Crosswalk	215	160	290	190	200	210
East Crosswalk	200	245	515	185	215	270

Description of Changes Made at the Test Location.

The drawing, Figure 88, illustrates the change made. A sign was erected prohibiting right turns on red for traffic approaching the intersection from the west.

Evaluation Results.

1. Effect of countermeasure on traffic flows. The countermeasure was evaluated with respect to its possible effect upon the flow of traffic through the

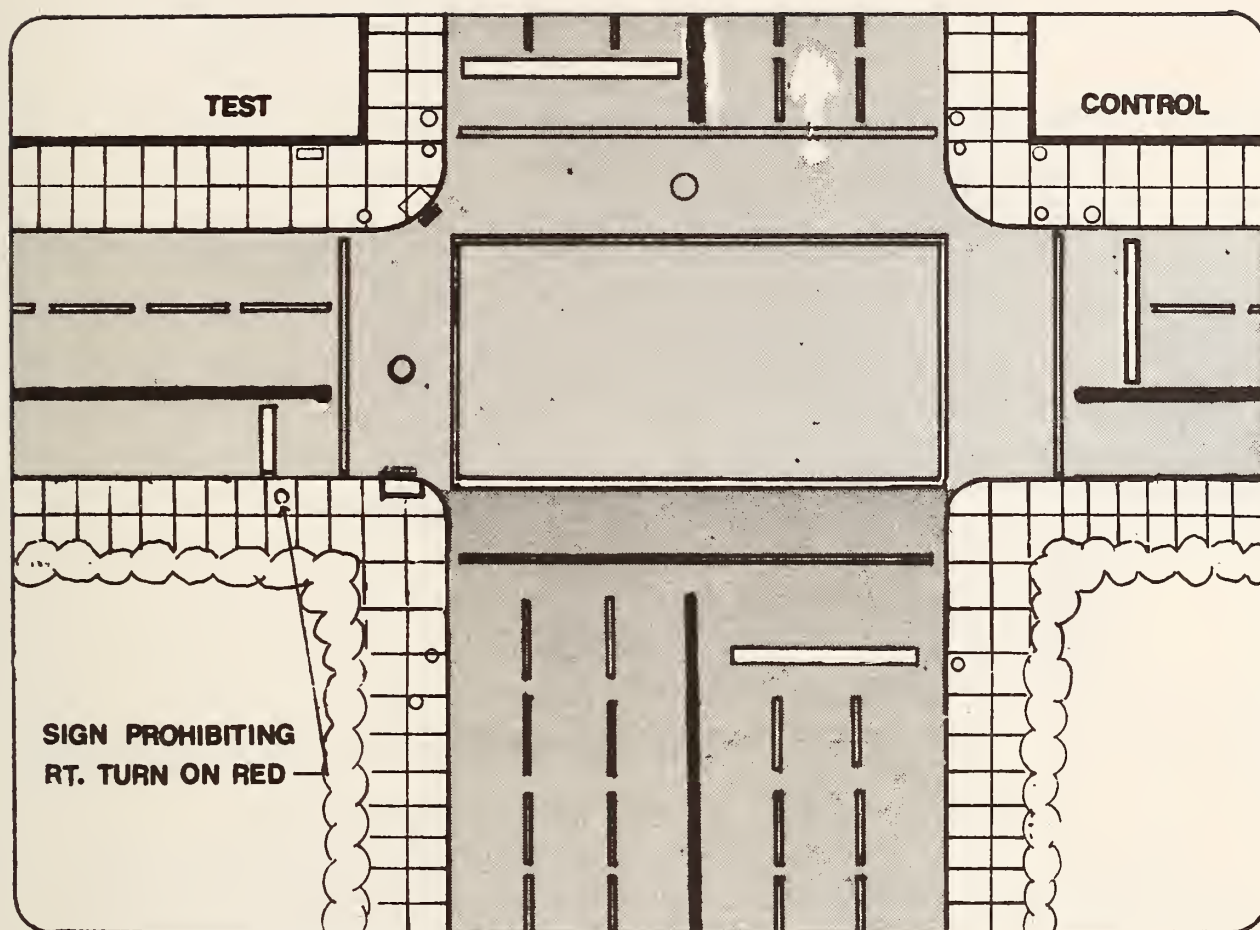


Figure 88 Atlanta: Right-Turn-On-Red Restriction:
Plan of Test and Control Sites After Modification

intersection in which they were located. The intent was to determine whether a measure such as the restriction of right turns on red would result in a reduction of traffic flows in an east-west direction.

The results of this evaluation are shown in Figure 89.

Traffic volumes increased substantially in the east-west direction while decreasing somewhat on the north-south street. Right turns showed an increase in both directions on the east-west street; however, the increase on the west side (test location) was nearly 5 times as great as on the east side (control location). The results are surprising and may result from some unknown changes—the traffic patterns in the area. Obviously, there is no logical reason to connect an improved traffic flow with the restricted turn movement at the test site.

2. The countermeasure was evaluated to determine whether there was a change in the number of vehicles stopped while waiting for a green traffic signal, and which obstructed the crosswalk. This evaluation was to determine if the right turn on red restriction had any effect in reducing these obstructions. The results were shown in Table 162.

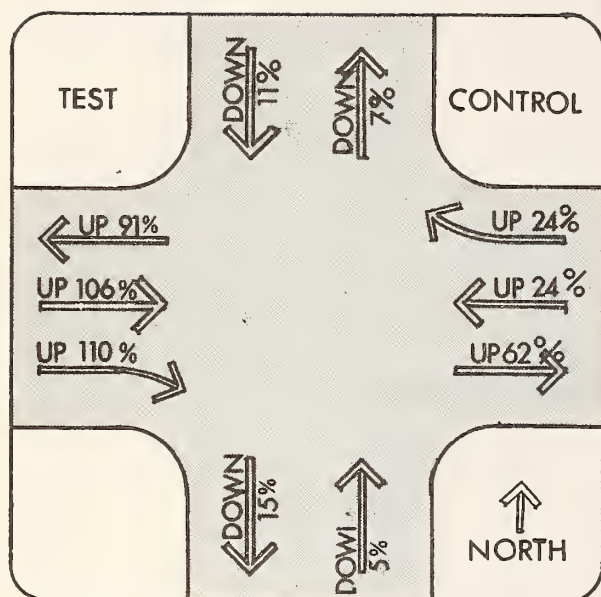


Figure 89 Atlanta: Right-Turn-On-Red Restriction: Changes in Vehicular Traffic

TABLE 162. STOPPED VEHICLES OBSTRUCTING CROSSWALK

Total of five 10-minute periods

	Before	After	Percent Reductions
West Crosswalk (Test Location)	45	25	44%
East Crosswalk (Control Location)	26	19	27%

The results indicate a substantially greater reduction in crosswalk obstructions at the test location than at the control location.

3. Effect of countermeasure design upon pedestrian flows. The countermeasure was evaluated with respect to its possible effect upon the flows of pedestrians through the intersection. The intent was to determine if the right turn on red prohibition would tend to permit higher pedestrian volumes. The results are shown in Figure 90.

The very substantial increase in pedestrian flow in the west crosswalk (test location) was more than 8 times the increase in the east crosswalk (control location).

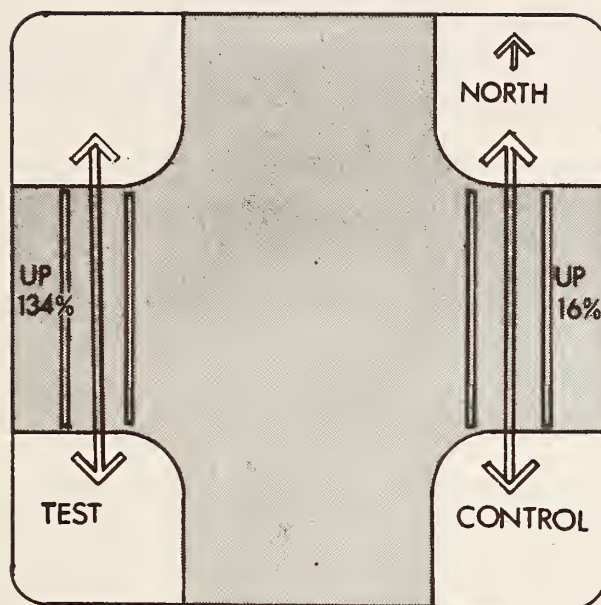
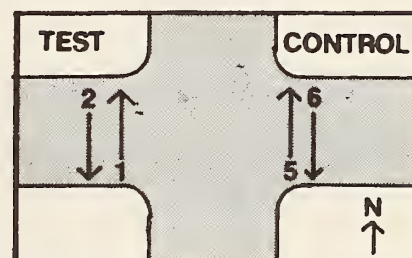


Figure 90 Atlanta: Right-Turn-On-Red Restriction: Changes in Pedestrian Traffic

4. Effect of Countermeasure 15 in crossing time. The countermeasure was evaluated with respect to its possible effect upon the time required to cross the street. The intent was to determine whether the right turn on red prohibition allowed an increase in crossing speed. The results are shown in Table 163.

TABLE 163. MEAN CROSSING TIME IN SECONDS

COUNTERMEASURE NO. 15; LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	6.526	10.000	6.714	9.769	S	S
2,6	7.4211	9.6364	7.4211	10.3529	S	S

S = Significant

N = Not Significant

From Table 163, it can be seen that there was a significant increase in mean crossing time at both the test and the control site. The reasons for this general increase are not clear, but presumably are not caused by the countermeasure.

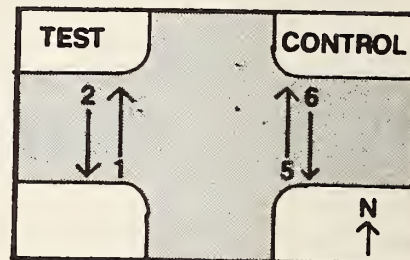
5. Evaluation of the test and control intersections with respect to other factors of behavior affecting safety and mobility. These included evaluations of:

- a. Conflicts between crossing pedestrians and vehicles.
- b. Conflicts between crossing pedestrians and other pedestrians.
- c. Pedestrians waiting in the street for the crossing signal to change.
- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk.
- e. Pedestrians crossing against the traffic signal.

- a. Conflicts between crossing pedestrians and vehicles. This evaluation measured certain types of pedestrian behavior which had the potential of resulting in accidents between pedestrians and vehicles. Specifically measured were the occurrence of incidents such as increasing crossing speed, reducing crossing speed, changing course, pausing, stopping, or reversing course, when it was apparent that these actions resulted from the presence of vehicles in or near the intersection.

The results of this evaluation are shown in Table 164.

TABLE 164. CONFLICTS BETWEEN CROSSING
PEDESTRIANS & VEHICLES
COUNTERMEASURE NO. 15, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	2 / 19	3 / 20	3 / 37	1 / 13	N	N
2,6	4 / 20	2 / 23	4 / 23	3 / 17	N	N

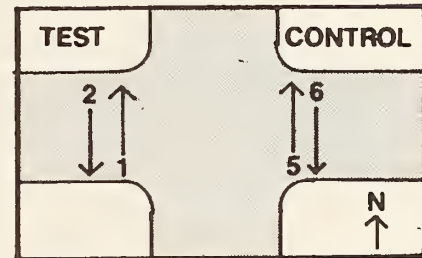
S = Significant

N = Not Significant

- b. Conflicts between crossing pedestrians and other pedestrians. Similar pedestrian behavior (to the previous evaluation) was recorded in this test when it was apparent that such actions resulted from the presence of, or density of, other pedestrians affecting those being observed.

The results of this evaluation are shown in Table 165.

TABLE 165. CONFLICTS BETWEEN CROSSING PEDESTRIANS
& OTHER PEDESTRIANS
COUNTERMEASURE NO. 15, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	3 / 19	0 / 20	1 / 37	0 / 13	N	N
2,6	3 / 20	0 / 23	1 / 23	0 / 17	N	N

S = Significant

N = Not Significant

From Table 165, it can be seen that the reduction in pedestrian/pedestrian conflicts at the test site was not significant.

- c. Pedestrians waiting in the street for the crossing signal to change. This evaluation was intended to determine whether the countermeasure design had any effect upon increasing or decreasing the numbers of pedestrians who chose to stand in the street, rather than on the sidewalk, making them vulnerable to traffic.

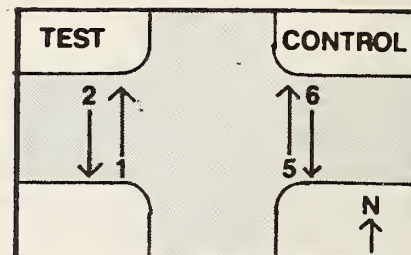
The results of this evaluation were as shown in Table 166.

From Table 166, it can be seen that at the test site there was a significant increase by those preparing to cross in a northerly direction, in the number of pedestrians waiting in the street.

- d. Pedestrians walking beyond limits of the crosswalk, or leaving the crosswalk. This evaluation was intended to determine to what extent crossing pedestrians would be inclined to remain within the limits of the crosswalk, and whether the countermeasure design would have any effect upon their doing so.

TABLE 166. PEDESTRIANS WAITING IN THE STREET FOR
THE SIGNAL TO CHANGE

COUNTERMEASURE NO. 15, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	0 / 19	5 / 20	0 / 37	0 / 13	S	N
2,6	0 / 20	4 / 23	1 / 23	1 / 17	N	N

S = Significant

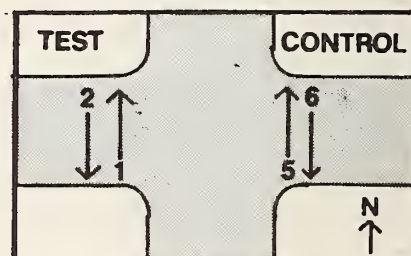
N = Not Significant

1. The first measure, walking beyond the crosswalk limits, refers to those pedestrians who entered the street outside of the crosswalk, and crossed outside of the crosswalk.
2. The second measure, pedestrians leaving the crosswalk, refers to those persons who began crossing within the crosswalk but passed beyond the crosswalk limits before completing their crossing.

The results of these evaluations are shown in Tables 167 and 168.

TABLE 167. PEDESTRIANS WALKING BEYOND THE LIMITS
OF THE CROSSWALK

COUNTERMEASURE NO. 15, LOCATION NO. 1

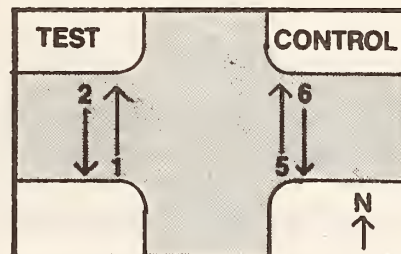


Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	2 / 19	2 / 20	3 / 37	4 / 13	N	N
2,6	6 / 20	1 / 23	6 / 23	2 / 17	N	N

S = Significant

N = Not Significant

TABLE 168. PEDESTRIANS LEAVING THE CROSSWALK
WHILE CROSSING THE STREET
COUNTERMEASURE NO. 15, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	1 / 19	1 / 20	0 / 33	0 / 13	N	N
2,6	0 / 20	4 / 23	0 / 23	2 / 17	N	N

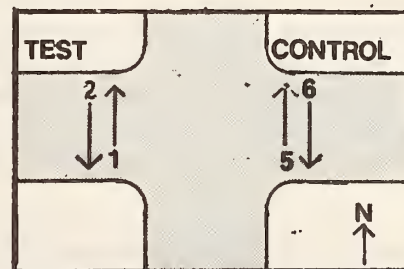
S = Significant N = Not Significant

From Tables 167 and 168, it can be seen that there was no significant change in the number of pedestrians walking beyond the limits of the crosswalk or leaving the crosswalk while crossing the street.

- e. Pedestrians crossing against the traffic signal. This evaluation measured those pedestrians who chose to cross the street while the traffic signal in their direction was red; and sought to determine whether the countermeasure design would have any effect upon increasing or decreasing such behavior.

The results of this evaluation are shown in Table 169.

TABLE 169. PEDESTRIANS CROSSING AGAINST
THE TRAFFIC SIGNAL
COUNTERMEASURE NO. 15, LOCATION NO. 1



Direction	Test Site		Control Site		Nature of significant differences between columns	
	Col. 1 Before	Col. 2 After	Col. 3 Before	Col. 4 After	Cols. 1&2	Cols. 3&4
1,5	5 / 19	3 / 20	4 / 37	2 / 13	N	N
2,6	2 / 20	3 / 23	2 / 23	4 / 17	N	N

293 S = Significant

N = Not Significant

From Table 169, it can be concluded that there was no significant change in the number of pedestrians crossing against the traffic signal.

Summary of Findings: Countermeasure No. 15, Right-Turn-On-Red Prohibition

In summary, it was found that:

- Traffic volumes were not reduced.
- Vehicular/crosswalk obstruction was reduced.
- Pedestrian flow increased.
- Mean pedestrian crossing time increased.
- Pedestrian/vehicular conflicts did not change significantly.
- Pedestrian/pedestrian conflicts did not change significantly.
- There was a significant increase in pedestrians waiting in the street preparatory to crossing in a northerly direction.
- There was no significant change in the number of pedestrians walking beyond the limits of the crosswalk or leaving the crosswalk while crossing the street.

Conclusions.

Although the number of pedestrian/vehicular conflicts did not decrease, vehicular obstruction of the crosswalk was significantly decreased without reducing traffic volumes or causing other problems. As the primary purpose of the countermeasure was to reduce vehicles from obstructing the crosswalks and impeding and causing a potential hazard to severely visually impaired people and people in wheelchairs particularly, the countermeasure is considered to be successful.

summary of conclusions and recommendations

Countermeasure No. 1: Widened Sidewalk.

For locations where the street crossings are sixty feet or more, or where sidewalks are narrow, and where parking is permitted.

From the evaluation, the countermeasure is successful for the following reasons:

- There was a significant reduction in pedestrian crossing time.

- Vehicle driver's view of pedestrians waiting to cross, and the pedestrians view of vehicles was improved.
- Increased space for benches, curb ramps and street furniture away from the sidewalk was made available.

Countermeasure No. 2: Pedestrian Refuge Island.

For crosswalk locations where there is a considerable amount of pedestrian traffic (or elderly and handicapped pedestrian traffic), and where it would be difficult or dangerous to cross the entire roadway at one time in safety. This may occur at signalized intersections to provide a refuge between traffic streams; on a multi-lane roadway; at large or irregularly shaped intersections.

From the evaluation, the countermeasure is successful for the following reasons:

- About 18% of all pedestrians used the island to avoid conflicts with vehicles.
- In one half of all crossings, people in wheelchairs paused, or stopped, on the island, using it as a refuge.

However, for visually impaired people, two changes to the design are necessary :

- The floor surface of the island must be detectable and different from that of the crosswalk, either by an appropriate change of texture or by raising the surface of the island by means of a curb cut or curb ramps.
- Where the refuge island forms a physical barrier to crossing the street at any point, then the crosswalk markings should be detectable to the visually impaired, in order to direct them through the island opening.

Countermeasure No. 3: Midblock Crosswalk Markings.

This countermeasure was intended to make midblock crosswalks more obvious to the drivers of vehicles and therefore to encourage drivers to reduce speed and respond to pedestrians at these points.

From the evaluations, the results were inconclusive. There is some evidence that vehicle drivers responded better to this type of marking than to the conventional markings; and there is good evidence that pedestrians are less likely to cross away from the crosswalk. A further evaluation of this countermeasure would be useful.

Countermeasure No. 4: Widened Crosswalks.

This countermeasure was intended as a device to reduce the effects of pedestrian congestion where people with poor balance risk being jostled or bumped. Secondly, it is intended as a device to reduce the extent of vehicles stopping within the crosswalk.

From the three evaluations, it is concluded that for the level of pedestrian and vehicular activity present at the test sites, the countermeasures were ineffective.

Countermeasure No. 5: The Textured Strip.

This countermeasure is intended as a device to indicate to visually handicapped people a direct path on sidewalks that avoids street furniture and other obstructions.

From the evaluation, the countermeasure is effective.

Countermeasure No. 6: The Raised Box.

This countermeasure is intended for locations where the sidewalks are too narrow to accept curb ramps.

The countermeasure was not evaluated.

Countermeasure No. 7: The Ramped Sidewalk.

This countermeasure is intended for location where the sidewalks are too narrow to accept curb ramps.

The countermeasure was not evaluated.

Countermeasure No. 8: The Single Conventional (Recessed) Curb Ramp.

This type of curb ramp is widely used, not always in suitable locations. For the evaluation, a common type of location was chosen.

From the evaluation, this particular layout is considered to be potentially hazardous because wheelchair users were forced to travel into the path of moving vehicles. However, in other locations where there is at least four feet between the bottom of the curb ramp and the nearest crosswalk line, the design may be effective.

Countermeasure No. 9: Double Conventional (Recessed) Curb Ramps.

This solution is also frequently used and is intended to avoid the problems associated with the single conventional curb ramp, or for locations where there is a drain at the corner.

From the evaluations, the countermeasure is successful.

Countermeasure No. 10: The Ramped Quadrant Curb Ramp.

This countermeasure is intended for wide sidewalks, as an improved version of the conventional curb ramp; wheelchair users are not forced into a path outside the crosswalk.

The countermeasure was not evaluated.

Countermeasure No. 11: Ramped Full Quadrant Curb Ramp.

This countermeasure is intended for narrow sidewalks.

From the evaluations, the countermeasure is reasonably effective; however, it does increase the difficulty for wheelchair users who must travel over it when rounding the corner. To minimize this difficulty, the flares of the quadrant should have the same slope as the rest of the quadrant.

Countermeasure No. 12: The Single Add-On Curb Ramp.

This countermeasure is intended for locations with very narrow sidewalks.

From the evaluations, it is concluded that it may be unsafe for visually impaired people to introduce this type of ramp in cities where this is not the usual design used. In cities where this design is commonly used, the design seems to be effective.

Countermeasure No. 13: The Double Add-On Ramp.

This countermeasure is intended for locations where the single add-on ramp cannot be easily constructed; where there is a corner drain, for example.

This countermeasure was not evaluated.

Countermeasure No. 14: Painted Manhole Covers and Gratings.

This countermeasure is intended to draw attention to manhole covers, gratings, etc., that may be tripping or slipping hazards.

The countermeasure was not evaluated.

Countermeasure No. 15: Right-Turn-On-Red Prohibition.

This countermeasure is intended for locations where the drivers view to the left is obscured by buildings, planting, etc.

From the evaluations, this countermeasure is effective in that vehicular obstruction of the crosswalk was significantly reduced without reducing traffic volumes.

Other conclusions.

During the evaluation of the countermeasures, several events were observed from which conclusions can be drawn and recommendations made.

- Lips, or small steps, at the bottom of curb ramps cause considerable difficulty

to people in wheelchairs and may, in fact, be hazardous. These lips should not be constructed.

- Crosswalks should be constructed of materials that are different from that of the curb ramps (and sidewalks, if these are flush with the crosswalk), so that severely visually impaired people may detect the surface change.
- Where planters, or other barriers to access to sidewalks from the street (chains, for example) are used, and a crosswalk passes through the barrier, then the edges of the crosswalk should be constructed to be detectable to the visually impaired; and these crosswalk definition lines should lead to the edges of the barrier or, at least, end at points within the opening in the barrier.

The implications of these studies have been incorporated into an implementation package entitled, "A Manual for Developing a Priority Accessible Network".

References

1. Dixon, Charles E. A study to determine the Specifications of Wheelchair Ramps, Unpublished Ph.D. Dissertation, University of Illinois, Champagne, Urbana, 1957.
2. American National Standard Specifications for Making Buildings and Facilities Accessible To and Usable By Physically Handicapped People. American National Standards Institute, Inc., A117.1. Syracuse University, Syracuse, New York, ANSI, 1977.
3. Walter, Felix. Four Architectural Movement Studies for the Wheelchair and Ambulant Disabled, Disabled Living Foundation, London, 1971.
4. Herms, Bruce F.; Elias, Helen; Robbins, Donald O. Guidestrips for Visually Handicapped Pedestrians, Metropolitan Association of Urban Designers and Environmental Planners, Third National Seminar, San Diego, California, December 1976.
5. Snyder, Monroe B.; Knoblauch, Robert L., Pedestrian Safety: The Identification of Precipitating Factors and Possible Countermeasures, Volume 1, U.S. Department of Transportation, January, 1971.

GEORGIA INSTITUTE OF TECHNOLOGY College of Architecture

I hereby consent to act as an experimental subject in the architectural research program of the Georgia Institute of Technology. I am a physically handicapped person. My participation is on a voluntary basis and is in consideration of the contributions of this program to the advancement of knowledge in the area of improved safety in the movement of handicapped persons.

1. I understand that I may be asked to go up and down 10-15 ramps that are 10 feet long or less, and that I will be asked to rate each ramp for ease of use. Two ramps will include gutter lips, of $\frac{1}{2}$ to 1 inch respectively. I understand that I may be asked to go across the slope of several ramps and to rate the degree of difficulty.
2. I understand that I may be asked to travel along 12 approximately level concrete sidewalks, and I may be asked to attempt to perceive any contrasting surface in the sidewalks.
3. I understand that although there will be no traffic, I will be asked to exercise at all times the same degree of caution that I would in an existing urban environment, and that I should not attempt any test which I feel is hazardous to myself or to my equipment, or which will be too physically demanding on me. I also understand that I will be allowed to rest at any time upon request.
4. In connection with this study, I agree that photographs may be taken of me.
5. I agree that the data that I provide may be used for analysis and publication.
6. I understand that this study is being done to benefit science and research education, and agree that any photographs or research findings may be used for that purpose including publication in scientific journals.
7. I understand that I may make any inquiries concerning the procedure and that I am free to withdraw my consent and to discontinue participation in the project at any time.
8. I understand that proper safety precautions have been taken, but in the event of an accident, I do hereby release, and forever discharge the Georgia Institute of Technology, all employees or agents of the Georgia Institute of Technology, and the Regents of the University System of Georgia of any and from all claims, demands, rights, and causes of action of whatsoever kind and nature, foreseen and unforeseen, bodily injuries, damage to property, and the consequences thereof, resulting from my participation in this architectural research program.

Subject's Signature _____

Investigator _____

Informed Consent Form

INFORMATION FORM
Georgia Institute of Technology

Name: _____

Address: _____

Phone No. _____ Age _____ Sex _____

Agency _____

Date _____ Session Time _____

Weather _____ Temperature _____

Handicap Diagnosis _____

How long have you had handicap? _____

Have you had mobility training? Yes _____ No _____

What travel aids do you use?

Guide Dog _____ Cane, White _____

Walker _____ Cane, Support _____

Prosthetic Limbs _____ Crutches _____

Leg Brace _____ No Aids _____

Wheelchair: E _____ M _____ Other _____

Make: _____

Dimension: Width _____

Tip of Foot Rest to rear wheel ground contact _____

Type of shoes _____ Sole _____ Heel _____

OTHER _____

PROTOCOL FOR PEDRAMP AND PEDTEX ELDERLY VOLUNTEERS

This test site has many ramps, curbs and textures on it, and so you should not attempt to walk about on it without instructions or assistance from one of the project staff members.

SURFACE VISUAL CUE

OPTIONAL—USE ONLY IF THE TESTER HAS REASON TO BELIEVE THE ELDERLY VOLUNTEER HAS A SERIOUS VISUAL HANDICAP.

This test is to examine your ability to see colors on the surface of the walkway.

You are on a walkway. There may be a painted line on the walkway surface. When I tell you to, please walk forward, avoiding the guide ropes marking the edge of the walkway, and stop on or as close to the painted line as you can, and wait for instructions.

If you do not see a line, proceed until I tell you to stop.

Do you have any questions?

IF THE VOLUNTEER INSISTS THAT BECAUSE OF THE NATURE OF THE HANDICAP, SHE/HE CANNOT POSSIBLY DETECT ANY LINE, RECORD THIS FACT AND MOVE ON TO THE NEXT TEST.

Walk forward now.....stop.

RECORD: YES NO

RAMP EVALUATION (and/or Ramp Detection)

IF AN ELDERLY VOLUNTEER HAS A SERIOUS VISUAL HANDICAP, HAVE HIM DO BOTH RAMP DETECTION AND RAMP EVALUATION: IF NO SERIOUS VISUAL HANDICAP, DO RAMP EVALUATION ONLY.

Ramp DetectionAscent: Optional; the test is to start from the 5' mark)

This next test is to find out how difficult it is for you to detect and ascend various types of curb ramps.

Imagine you are crossing a street. In front of you, the pavement may continue level, or you may encounter a wheelchair ramp. When I tell you to, please walk forward. If you detect a ramp in front of you, stop immediately and wait for further instructions. If you do not encounter a ramp, please proceed until I tell you to stop.

Do you have any questions?

Please walk forward now.....stop.

RECORD YES NO DISTANCE FOR DETECTION

IF VOLUNTEER DID DETECT RAMP, THEN ASK:

Please indicate to me where the ramp begins.

RECORD YES NO

Please indicate to me where the ramp ends.

RECORD: YES NO

RAMP EVALUATION--ASCENT

The purpose of this test is to evaluate your ability to go up this ramp. If you feel that the ramp is too steep for you to use in safety, please do not attempt it.

NOTE: IF THE TEST IS REFUSED, RECORD THIS FACT AND PROCEED TO THE NEXT TEST.

If you are uncertain whether you will manage to ascend the whole distance, we would like you to attempt it anyway, so that we can see how far you reach.

Please position yourself at a distance from the ramp that will give you sufficient momentum.

When I tell you to, please move up the ramp at the speed you would normally use. When you reach the level surface again, please stop.

Do you have any questions?

Please move forward now.....stop.

How difficult was it for you to ascend the ramp:

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

THE OBSERVER MUST ALSO NOTE THE VOLUNTEER'S STABILITY AND BALANCE

1. No noticeable affect
2. Noticeable affect
3. A marked loss of control
4. A complete loss of control

Ramp Detection--Descent (Optional, test to start from the 5' mark)

The next test is to examine how difficult it is for you to detect and descend various types of curb ramps.

You are on a sidewalk proceeding towards a street that you wish to cross. In front of you there may be a wheelchair ramp down the street, or the sidewalk may continue level.

When I tell you to, please walk forward. If you detect a ramp in front of you, please stop immediately. If you do not encounter a ramp, please proceed until I tell you to stop.

Do you have any questions?

Please walk forward now.....stop.

RECORD: YES NO DISTANCE FOR DETECTION

IF VOLUNTEER DID DETECT THE RAMP, THEN ASK:

Please indicate to me the beginning of the ramp.

RECORD: YES NO

Please indicate to me where the ramp ends.

RECORD: YES NO

RAMP EVALUATION DESCENT

The purpose of this test is to evaluate your ability to go down this ramp. If you feel that this ramp is too steep for you to use in safety, please do not attempt it.

NOTE: IF TEST IS REFUSED, RECORD THIS FACT AND PROCEED TO THE NEXT TEST.

Please position yourself at the top of the ramp.

When I tell you to, please move down the ramp at the speed you would usually use. When you reach the level surface again, please stop as soon as you can.

Do you have any questions?

Please move forward now.....stop.

How difficult was it for you to descend the ramp?

1. Not at all difficult
2. A little difficult
3. Difficult
4. Very difficult

THE OBSERVER MUST ALSO NOTE THE VOLUNTEER'S BALANCE AND STABILITY.

1. No noticeable affect
2. Noticeable affect
3. A marked loss of control
4. A complete loss of control

CROSS SLOPE TEST

This test is to find out your reaction to crossing ramps that slope down across the direction that you will be moving. We wish to find out whether these ramps are comfortable for you to use or not.

When I tell you to, please move forward and stop after you have crossed the ramp.

Do you have any questions?

Please move forward now.....stop.

How did the cross slope affect your comfort and your stability and balance?

1. Not at all
2. A little
3. Moderately
4. Very much

THE OBSERVER MUST ALSO NOTE THE VOLUNTEER'S STABILITY AND BALANCE ON THE CROSS SLOPE.

1. No noticeable affect
2. Noticeable affect
3. Moderately
4. Very much

HALF-CURB EVALUATION TEST

This next test is to find out how difficult it is for you to detect and ascend various types of curbs and your ability to cross ramps and their cross slopes.

Imagine you are crossing a street. In front of you, you will encounter a full curb or a curb that is part of a ramp's side slopes. When I tell you to, please walk forward, cross the curb and proceed across the ramp's cross slopes and stop when I tell you.

Do you have any questions?

Please walk forward now.....stop.

How would you say the cross slope you have just crossed affected you?

1. None at all
2. A little

3. Quite a lot
4. Lost balance completely

THE OBSERVER MUST ALSO RATE THE VOLUNTEER'S STABILITY ON THE CROSS SLOPE.

1. No noticeable affect
2. A noticeable but minor gait change
3. A marked loss of balance
4. A complete loss of balance

TEXTURE ORIENTATION (Optional; use only if the elderly volunteer has a serious visual handicap. There are 9 surfaces).

THE OBSERVER WILL LEAD THE TEST SUBJECT TO THE FIRST APPROACH PATH AND INSTRUCT HIM THAT AHEAD OF HIM ON THE APPROACH PATH THERE WILL BE A SURFACE CHANGE. IF HE DETECTS A SURFACE CHANGE WHILE BEING LED DOWN THE APPROACH PATH, HE SHOULD STOP IMMEDIATELY. THE SUBJECT WILL THEN BE LED ALONG THE APPROACH PATH AND OVER THE SURFACE CHANGE. THE OBSERVER WILL RECORD YES OR NO AS TO THE DETECTION OF THE SURFACE CHANGE. THE SUBJECT WILL THEN BE INSTRUCTED TO STUDY THE SURFACE FOR A 2-MINUTE PERIOD (MAXIMUM) AND DEVELOP A LEVEL OF CONFIDENCE IN RECOGNIZING THAT SURFACE, ONLY AS A SURFACE CHANGE, AT A LATER TIME. THE OBSERVER WILL THEN ASK IF THE SUBJECT HAS ATTAINED THAT LEVEL OF CONFIDENCE, YES OR NO (NOT TO BE RECORDED). IN EITHER CASE, THE SUBJECT WILL THEN BE LED THROUGH THE REMAINING EIGHT SURFACE CHANGES. UPON COM-

PLETION OF THIS ORIENTATION, TESTING WILL BEGIN.

TEXTURE DETECTION (Optional; use only if the elderly volunteer has a serious visuual handicap. From the ramdom start location).

On the level walkway in front of you, you may encounter a change in the walkway surface. The purpose of the next tests is to see whether you can detect changes in the walkway surfaces.

When I tell you to, please walk forward at your usual walking speed. If you encounter a surface material that is distinctly and definitely different from the one you are now on, not found on a normal sidewalk, please stop immediately.

If you do not detect any change, please proceed until I ask you to stop.

Do you have any questions?

Please walk forward now.....stop.

IF THE VOLUNTEER STOPPED WITHOUT INSTRUCTIONS, THEN SAY:

How easy or difficult was it to detect the sidewalk surface?

1. Very easy
2. Easy
3. Somewhat difficult
4. Very difficult

TEXTURE EVALUATION

This test is to find out your reactions to various floor finishes. We wish to find out whether these surfaces are comfortable for you to use or not.

When I tell you to, please move forward and stop after you have crossed the test surface in front of you.

Do you have any questions?

How did the surface affect your comfort and your stability and balance?

1. Not at all
2. A little
3. Moderately
4. Very much

RAMP EVALUATION

#/ DIRECTION						
START/STOP						
EXECUTION	Y			N	Y	
RATING SCALE	1	2	3	4	1	2
RATING SCALE	1	2	3	4	1	2

#/ DIRECTION						
START/STOP						
EXECUTION	Y			N	Y	
RATING SCALE	1	2	3	4	1	2
RATING SCALE	1	2	3	4	1	2

#/ DIRECTION						
START/STOP						
EXECUTION	Y			N	Y	
RATING SCALE	1	2	3	4	1	2
RATING SCALE	1	2	3	4	1	2

#/ DIRECTION						
START/STOP						
EXECUTION	Y			N	Y	
RATING SCALE	1	2	3	4	1	2
RATING SCALE	1	2	3	4	1	2

RAMP DETECTION

#/ DIRECTION	4	UP	4	DN	10/6	UP	10/6	DN	1/8	UP	1/8	DN
DETECTION	Y		N	Y		N	Y		N	Y		N
BOTTOM		Y	N		Y	N		Y	N		Y	N
TOP		Y	N		Y	N		Y	N		Y	N

#/ DIRECTION	2	UP	2	DN	11	UP	11	DN	MOUNTABLE CURB			
DETECTION	Y		N	Y		N	Y		N	DETECTION		
BOTTOM		Y	N		Y	N		Y	N	UP		
TOP		Y	N		Y	N		Y	N	DN		

CROSS SLOPE

CROSS SLOPE #	10CS	4CS	3CS
RATING SCALE	1	2	3
RATING SCALE	1	2	3

HALF CURE

HALF CURE #	3HC	7HC
RATING SCALE	1	2
RATING SCALE	1	2

REMARKS: * Experimenter's Evaluation

NAME

INVESTIGATOR

TEXTURE ORIENTATION

#/ DETECTION METHOD	3 Y N V T	10 Y N V T	4 Y N V T	9 Y N V T	1 Y N V T	7 Y N V T
#/ DETECTION METHOD	2 Y N V T	6 Y N V T	8 Y N V T			

TEXTURE DETECTION

#/ METHOD		V T		V T		V T		V T		V T		V T
DISTANCE/TIME												
DETECTION	Y		N	Y		N	Y		N	Y		N
RATING SCALE	1	2	3	4	1	2	3	4	1	2	3	4

#/ METHOD		V T		V T		V T		V T		V T		V T
DISTANCE/TIME												
DETECTION	Y		N	Y		N	Y		N	Y		N
RATING SCALE	1	2	3	4	1	2	3	4	1	2	3	4

#/ METHOD		V T		
DISTANCE/TIME				
DETECTION	Y	N		
RATING SCALE	1	2	3	4

TEXTURE EVALUATION

TEXTURE #	1	2	3	4	6	7		
RATING SCALE	1	2	3	4	1	2	3	4

TEXTURE #	8	9	10	
RATING SCALE	1	2	3	4

LINE

TIME

NAME

YELLOW	Y	N
WHITE	Y	N

____ MIN. ____ SEC.

INVESTIGATOR

REMARKS:

TE 662 .A3 n
79-3
Templer, John
Provisions for
handicapped

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Form DOT F 1720.2
FORMERLY FORM DOT F

FEDERALLY COORDINATED PROGRAM (FCP) OF HIGHWAY RESEARCH AND DEVELOPMENT

The Offices of Research and Development (R&D) of the Federal Highway Administration (FHWA) are responsible for a broad program of staff and contract research and development and a Federal-aid program, conducted by or through the State highway transportation agencies, that includes the Highway Planning and Research (HP&R) program and the National Cooperative Highway Research Program (NCHRP) managed by the Transportation Research Board. The FCP is a carefully selected group of projects that uses research and development resources to obtain timely solutions to urgent national highway engineering problems.*

The diagonal double stripe on the cover of this report represents a highway and is color-coded to identify the FCP category that the report falls under. A red stripe is used for category 1, dark blue for category 2, light blue for category 3, brown for category 4, gray for category 5, green for categories 6 and 7, and an orange stripe identifies category 0.

FCP Category Descriptions

1. Improved Highway Design and Operation for Safety

Safety R&D addresses problems associated with the responsibilities of the FHWA under the Highway Safety Act and includes investigation of appropriate design standards, roadside hardware, signing, and physical and scientific data for the formulation of improved safety regulations.

2. Reduction of Traffic Congestion, and Improved Operational Efficiency

Traffic R&D is concerned with increasing the operational efficiency of existing highways by advancing technology, by improving designs for existing as well as new facilities, and by balancing the demand-capacity relationship through traffic management techniques such as bus and carpool preferential treatment, motorist information, and rerouting of traffic.

3. Environmental Considerations in Highway Design, Location, Construction, and Operation

Environmental R&D is directed toward identifying and evaluating highway elements that affect

the quality of the human environment. The goals are reduction of adverse highway and traffic impacts, and protection and enhancement of the environment.

4. Improved Materials Utilization and Durability

Materials R&D is concerned with expanding the knowledge and technology of materials properties, using available natural materials, improving structural foundation materials, recycling highway materials, converting industrial wastes into useful highway products, developing extender or substitute materials for those in short supply, and developing more rapid and reliable testing procedures. The goals are lower highway construction costs and extended maintenance-free operation.

5. Improved Design to Reduce Costs, Extend Life Expectancy, and Insure Structural Safety

Structural R&D is concerned with furthering the latest technological advances in structural and hydraulic designs, fabrication processes, and construction techniques to provide safe, efficient highways at reasonable costs.

6. Improved Technology for Highway Construction

This category is concerned with the research, development, and implementation of highway construction technology to increase productivity, reduce energy consumption, conserve dwindling resources, and reduce costs while improving the quality and methods of construction.

7. Improved Technology for Highway Maintenance

This category addresses problems in preserving the Nation's highways and includes activities in physical maintenance, traffic services, management, and equipment. The goal is to maximize operational efficiency and safety to the traveling public while conserving resources.

0. Other New Studies

This category, not included in the seven-volume official statement of the FCP, is concerned with HP&R and NCHRP studies not specifically related to FCP projects. These studies involve R&D support of other FHWA program office research.

* The complete seven-volume official statement of the FCP is available from the National Technical Information Service, Springfield, Va. 22161. Single copies of the introductory volume are available without charge from Program Analysis (HRD-3), Offices of Research and Development, Federal Highway Administration, Washington, D.C. 20590.

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